

Large drum granulators are widely used in the manufacture of fertilizer pellets and can be designed for plant capacities up to the order of a thousand metric tons per hour. Granulation drums typically have 1- to 5-minute residence time, depending on the ease of agglomeration, and the solids holdup is usually between 10% and 20% of the internal volume. Smaller agitated granulators are used for making various bead products that must be approximately spherical and typically operate at a few metric tons per hour. These devices are often operated in batch mode. Some powdered foods are also made by granulation, for example, granulated sugar.

Fluidized beds can be used for granulation and also permit addition of a slurry of smaller particles. Care must be taken in a fluidized-bed granulator to ensure that the secondary feed is not elutriated before it can coat the particles of the bed. The design of fluidized beds is discussed in [Section 18.2.2](#).

Spray drying and prilling

Spray drying was described in [Section 18.7.2](#) and illustrated in [Fig. 18.64](#). Prilling is essentially the same operation as spray drying but forms a larger particle, usually by crystallization combined with drying.

Spray drying and flash drying tend to form small, low-density particles. Particle sizes between ~ 10 and ~ 500 microns are typical. The particle shape can be quite variable and can range from spheres to flakes. Prilling forms larger particles, up to a few millimeters in diameter, and tends to form spheroids. The size of particles formed is strongly dependent on the spray nozzles used, and a wide range of proprietary designs are available from nozzle vendors. If a drying gas is used in countercurrent flow to the falling solids, then fines will be elutriated and must be collected from the effluent gas and recycled to the process.

Prilling is widely used for making fertilizers such as urea and ammonium nitrate. Spray drying is used to make catalysts, powdered foods, and detergent powders.

Crystallization

Crystallization can be used to form free-flowing solid particles from materials that will crystallize. Crystallization and precipitation are described in detail in [Section 16.5](#), together with other processes for removing dissolved solids from a liquid.

18.8.3 Postforming processes

The first step after particle forming is usually to screen the particles to reject those that are undersized or oversized. Undersized material can usually be recycled to the forming step. Oversized material may need to be reground before it can be recycled.

Once the desired size and shape particles have been formed, they will often be dried or heat treated to reduce their plasticity and ensure that the shape is retained. In some cases, extreme heat treatment known as calcining will be carried out to drive off or burn off the binder.

Many solids products will subsequently be coated. Various types of coatings are applied; for example, pharmaceutical tablets may be coated with sugar or gels to make them more palatable or easier to swallow, while many solids are coated with waxes to suppress dust formation during handling. Coatings are usually applied in a tumbler or spray coating drum that is essentially the same as a rotary agglomerator. The particles must be kept moving during coating to ensure a uniform coating and prevent unwanted agglomeration. If the coating is applied in a solvent, then a rotary dryer is sometimes used as the coating drum. The selection of a coating material must consider the impact of the coating on the end-use application of the product.

18.9 Particle size reduction (comminution)

Crushing is the first step in the process of size reduction, reducing large lumps to manageable sized pieces. For some processes crushing is sufficient, but for chemical processes, it is usually followed by grinding to produce a fine-sized powder. Though many articles have been published on comminution, and [Marshall \(1974\)](#) mentions over 4000, the subject remains essentially empirical. The designer must rely on experience and the advice of the equipment manufacturers when selecting and sizing crushing and grinding equipment. The power requirements are best estimated based on grinding tests carried out on existing plant or in consultation with an equipment vendor. For a fuller treatment of the subject, the reader should refer to the books by [Lowrison \(1974\)](#), [Prasher \(1987\)](#), and [Fuerstenau and Han \(2003\)](#).

18.9.1 Crushing and grinding theory

Several models have been proposed for the calculation of the energy consumed in size reduction. None of these apply particularly well in practice, but they are suitable for making initial estimates of energy input for the purposes of undergraduate design projects. The energy required to overcome bonding forces in the solid and create new surface area is generally less than 1% of the energy consumed in a real mill, which is usually less than the error in closing an energy balance on the mill (Austin & Trass, 1997). For realistic estimates, the designer should always carry out trials in consultation with an equipment manufacturer.

Size reduction can occur through brittle fracture, shearing (cutting), and surface abrasion. The importance of different mechanisms depends on the material type and can also vary with particle size and temperature. For example, for rubber particles ground at low temperatures, brittle fracture will dominate, whereas at higher temperatures, the particles are more ductile and shearing is more important.

The most useful simple model of energy consumption is that developed by Bond (1952):

$$E = 100 E_i \left(\frac{1}{\sqrt{d_2}} - \frac{1}{\sqrt{d_1}} \right) \quad (18.33)$$

where:

E = work done in grinding per unit mass feed (kWh/ton)

E_i = a constant known as the Bond work index (kWh/ton)

d_1 = initial particle size (m)

d_2 = final particle size (m)

The Bond work index is a constant and is equal to the amount of energy in kWh used to reduce a short ton (2000 lb) of solids from an initial theoretical infinite size to 80% passing 100 μm . In the original paper, Bond gave equations to adapt the work index for different types of equipment, but these equations have since become dated due to improvements in device and driver design. Work index values found in the literature should generally be treated with suspicion, as the device is often not specified, and the work index can vary substantially from one device to another. Average values of the work index can be misleading due to this device to device variation. Some average values of the Bond work index are given in Table 18.2, which can be used with Equation 18.33 for preliminary estimates of grinding work for the purposes of undergraduate design projects.

More sophisticated models of size reduction processes are usually formulated as population balance models that predict the rate of destruction and formation of particles in discretized size ranges. A breakage function can be used to predict the distribution of smaller particle sizes formed from grinding each particle size range. Rate of breakage functions can be determined experimentally in smaller mills and used to predict the required residence time and power to produce a desired particle size distribution from a given starting material in a larger mill. These models are usually proprietary.

18.9.2 Wet and dry grinding

In wet grinding, the solids are ground in a slurry, usually with the addition of surfactants known as grinding aids that modify the slurry viscosity and prevent the particles from agglomerating. Smaller particles can be formed in wet grinding than in dry grinding—sometimes as much as an order of magnitude smaller.

The work required in wet grinding is less than is needed in dry grinding for the same change in particle size. For wet grinding, the work index is typically about 75% of the work index for dry grinding.

Wet grinding causes substantially less dust formation than dry grinding. In some cases, use of gas cleaning equipment can be avoided, with resulting capital and operating cost savings.

Wet grinding tends to cause higher wear of the grinding equipment. The wet environment can cause corrosion of the grinding surfaces or grinding media, whereas abrasion by the solids prevents formation of a corrosion-resistant oxidized layer on the surface. Wet grinding equipment consequently requires more preventive maintenance.

The selection of wet or dry grinding is mainly determined by the downstream processing operations. If the solids are initially wet or if the liquid used for grinding is compatible with downstream process operations, then wet grinding is usually preferred. If surfactants or other additives are introduced, then these must also be compatible with downstream process operations.

18.9.3 Crushing and grinding (comminution) equipment

The main factors to be considered when selecting equipment for crushing and grinding are:

1. The size of the feed
2. The size reduction ratio
3. The required particle size distribution of the product
4. The throughput
5. The properties of the material: hardness, abrasiveness, stickiness, density, toxicity, flammability
6. Whether wet grinding is permissible

The selection guides given by [Lowrison \(1974\)](#) and [Marshall \(1974\)](#), which are reproduced in [Fig. 18.69](#) and [Table 18.14](#), can be used to make a preliminary selection based on particle size and material hardness. Detailed descriptions of most of the equipment listed in these tables are given in [Chhabra and Gurappa \(2019\)](#), [Green and Southard \(2018\)](#), [Hiorns \(1970\)](#), and [Lowrison \(1974\)](#); see also [Austin and Trass \(1997\)](#).

Crushing refers to coarse size reduction and is usually applied to large lumps of minerals entering a process. The most commonly used devices for coarse size reduction are jaw crushers and gyratory crushers. A jaw crusher crunches the material between a vertical fixed plate and a moving plate ([Fig. 18.70a](#)). A gyratory crusher is essentially a giant pestle and mortar in which the solids are crushed between an oscillating inner cone and a stationary outer cone ([Fig. 18.70b](#)). In both types of crusher, the size of the outlet opening determines the maximum particle dimension that is produced. If a finer particle size is required, then additional grinding machinery must be used downstream of the crusher. Sometimes two crushers will be used in series, with the downstream crusher having a narrower outlet to produce smaller particles.

Finer grinding is most often carried out in ball mills or their variants: pebble, rod, and tube mills. A ball mill is a cylindrical drum that contains a grinding medium such as metal balls. The drum is rotated and the feed material is crushed by impacts with the grinding medium. If flint, ceramic, or stone balls are used as a grinding medium, then it is referred to as a pebble mill, and if metal rods are used, then it is a *rod mill*. The diameter of the drum is usually less than the length, but for finer grinding a longer cylinder can be used, in which case the mill is called a tube mill. The grinding media are retained inside the drum by an inner screen, and feed and product are usually removed continuously. A schematic diagram of a typical ball mill is shown in [Fig. 18.70\(c\)](#). For very small particle sizes, a jet mill is used; see [Liu \(2017\)](#).

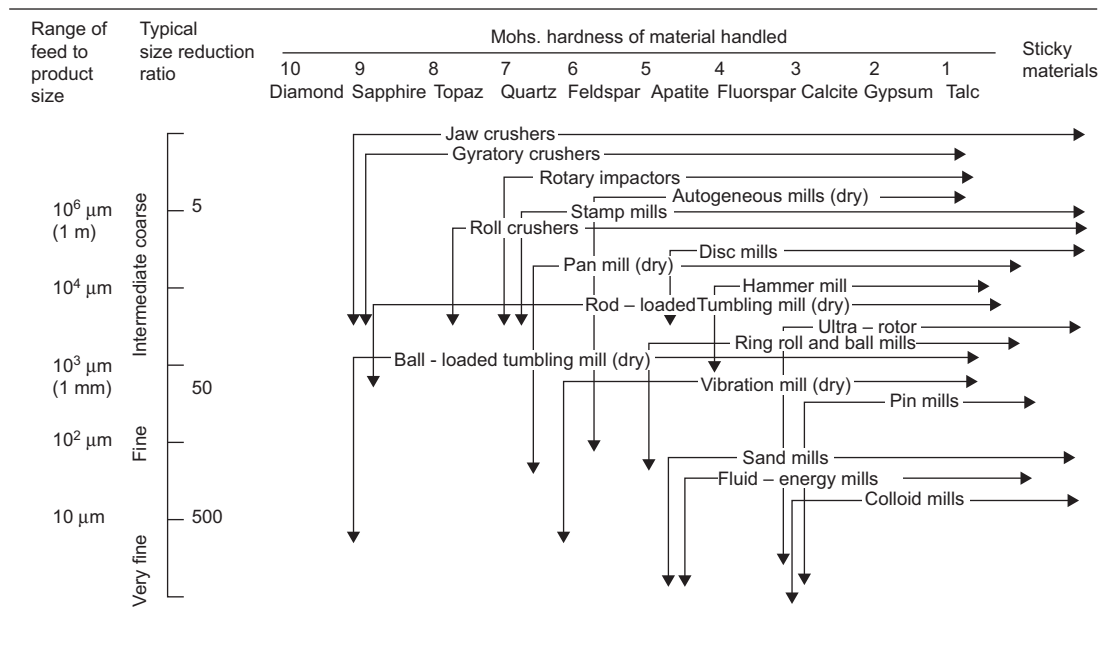


FIG. 18.69 Selection of comminution equipment. (From Marshall, V. C. [1974]. *Comminution*. Institute of Chemical Engineers.)

TABLE 18.14 Selection of comminution equipment for various materials

Material class no.	Material classification	Typical materials in class	Suitable equipment for product size classes			Remarks
			Down to 5 mesh	Between 5 and 300 mesh	Less than 300 mesh	
1	Hard and tough	Mica Scrap and powdered metals	Jaw crushers Gyratory crushers Cone crushers Autogeneous mills	Ball, pebble, rod and cone mills Tube mills Vibration mills	Ball, pebble and cone mills Tube mills Vibration and vibro-energy mills Fluid-energy mills	Mohs hardness 5–10, but includes other tough materials of lower hardness
2	Hard, abrasive and brittle	Coke, quartz, granite	Jaw crushers Gyratory and cone crushers Roll crushers	Ball, pebble, rod and cone mills Vibration mills Roller mills	Ball, pebble and cone mills Tube mills Vibration and vibro-energy mills Fluid-energy mills	Mohs hardness 5–10 High wear rate/ contamination in high-speed machinery Use machines with abrasion resistant linings
3	Intermediate hard, and friable	Barytes, fluorspar, limestone	Jaw crushers Gyratory crushers Roll crushers Edge runner mills Impact breakers Autogeneous mills Cone crushers	Ball, pebble, rod and cone mills Tube mills Ring roll mills Ring ball mills Roller mills Peg and disc mills Cage mills Impact breakers Vibration mills	Ball, pebble and cone mills Tube mills Perl mills Vibration and vibro-energy mills Fluid-energy mills	Mohs hardness 3–5
4	Fibrous, low abrasion and possibly tough	Wood, asbestos	Cone crushers Roll crushers Edge runner mills Autogeneous mills Impact breakers	Ball, pebble, rod and cone mills Tube mills Roller mills Peg and disc mills Cage mills Impact breakers Vibration mills Rotary cutters and dicers	Ball, pebble and cone mills Tube mills Sand mills Perl mills Vibration and vibro-energy mills mills cone mills	Wide range of hardness Low-temperature, liquid nitrogen, useful to embrittle soft but tough materials

5	Soft and friable	Sulfur, gypsum, rock salt	Cone crushers Roll crushers Edge runner mills Impact breakers Autogeneous mills	Ball, pebble and cone mills Tube mills Ring roll mills Ring ball mills Roller mills Peg and disc mills Cage mills Impact breakers Vibration mills	Ball, pebble and cone mills Tube mills Sand mills Perl mills Vibration and vibro-energy mills Colloid mills Fluid-energy mills Peg and disc mills	Mohs hardness 1–3
6	Sticky	Clays, certain organic pigments	Roll crushers Impact breakers Edge runner mills	Ball, pebble, rod and cone mills* Tube mills* Peg and disc mills Cage mills Ring roll mills Colloid mills	Ball, pebble and cone mills* Tube mills* Sand mills Perl mills Vibration and vibro-energy mills	Wide range of Mohs hardness although mainly less than 3 Tends to clog *Wet grinding employed except for certain exceptional cases

Note: Moh's scale of hardness is given in Fig. 18.69.

* All ball, pebble, rod, and cone mills; edge runner mills; tube mills; vibration mills; and some ring ball mills may be used wet or dry except where stated. The perl mills, sand mills, and colloid mills may be used for wet milling only. From Marshall (1974).

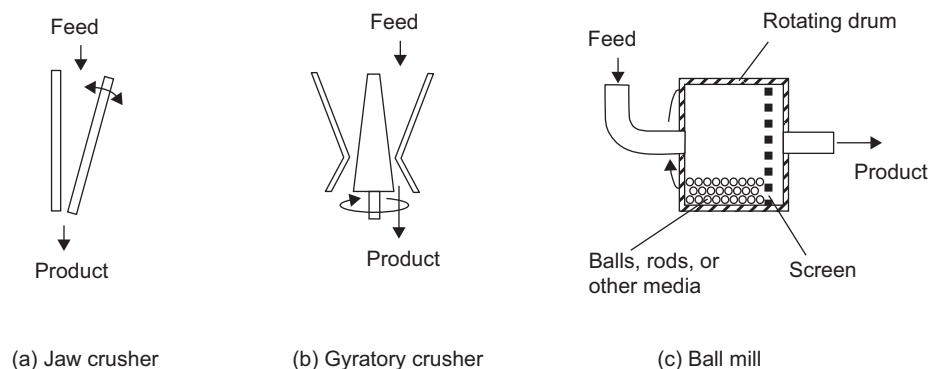


FIG. 18.70 Crushing and grinding equipment.

Ball mills are usually lined with a rubber liner to prevent wear of the casing. Abrasion or wear of the grinding media can be compensated by periodic continuous make-up of additional balls or by replacement of the media. Ball mills can be used in wet or dry grinding. The outer casing can be heated or cooled if necessary.

Ball mills are relatively inexpensive to install and operate. A preliminary estimate of the capital cost for a ball mill can be made using the equation given in Table 7.2.

18.9.4 Grinding cellular material

Grinding is often used as the initial step in recovery of biological products from cells. Grinding is used to break the cell walls so that components such as proteins can be extracted from the cell contents. Yeasts, molds, bacteria, algal, and mammalian cells have different grinding requirements because of differences in the cell wall strength and differences in the growth conditions.

The simplest method of cell disruption is to induce lysis (cell wall breakage) by changing the osmotic potential of the solution, adding a surfactant that disrupts the cell wall or by adding an enzyme such as lysozyme, lysostaphin, or cellulase. These methods are widely used in small-scale preparations but can be difficult and expensive at commercial scale. For large-scale harvesting of cellular products, the cells are usually disrupted in a high-shear device known as a valve homogenizer or in a high-speed continuous bead mill. The beads are usually glass, ceramic, or steel balls, typically less than 1 mm in diameter. Enzymes or surfactants are often added to reduce the specific energy input and hence reduce potential product loss. Cooling of the mill may also be necessary to prevent thermal degradation of products.

The energy input required for grinding cellular material is difficult to predict from theory. Experiments or plant trials are usually carried out to optimize the extent of product recovery that can be obtained as a function of energy input and residence time.

18.9.5 Process design and safety considerations in crushing and grinding

Size reduction equipment is usually used in combination with particle classification equipment to achieve a desired particle size. The crushing or grinding equipment is followed by sieving or another classification process (see Section 18.4), with recycle of oversize material to the grinder. It is common to use two or more stages of grinding with intermediate recycles of oversize material when the goal is to produce a fine powder for further processing. The combination of a mill, classification process, and recycle is known as a mill circuit (Fig. 18.71).

Milling can cause considerable heating of the solids. If this reduces mill efficiency, it may be necessary to circulate the solids through a cooler or blow cooling air (or inert gas) through the grinder. If gas circulation is used, a gas cleaning system must be added to prevent emissions of particulate matter, as illustrated in Fig. 18.71.

Heating of the solids can be a significant hazard during comminution, particularly if the material is flammable or if dust formed by grinding can form an explosive mixture. When there is a concern about flammability or ignition, then grinding should be carried out in an inert atmosphere. The inert gas can be recycled through a cooler after the gas-cleaning step, if desired.

Wet grinding reduces the level of dust formation, but can also lead to hazards if the exposed surface of the solids is reactive with water or any other component of the grinding fluid.

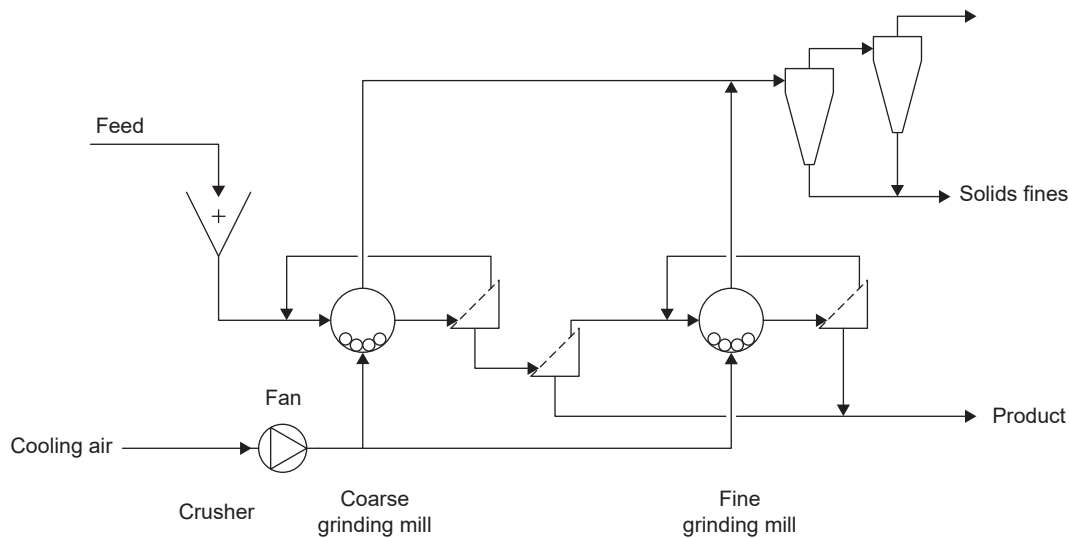


FIG. 18.71 Typical mill circuit.

Large crushers pose mechanical hazards in addition to the hazards listed earlier. With larger equipment it may be possible for workers to reach into or even fall into the machinery. Care should be taken to ensure that proper machine guarding is in place to prevent workers from entering the equipment. If blockages or jams occur, the equipment must be isolated and locked out from the power source before workers enter the equipment to clear the blockage.

Example 18.5

A coal gasification process is to be fed with 15 metric tons per hour of limestone to capture sulfur from the coal gas. The limestone is supplied with average dimension 1.5 cm and must be reduced to 20 mesh before feeding to the gasification process. Recommend equipment to achieve the desired particle size and estimate the work required.

Solution

From Table 18.1, 20 mesh corresponds to a sieve opening size of 0.85 mm. Assuming that the feed dimensions are on the same basis, the size reduction ratio is $15/0.85 = 17.6$.

From Fig. 18.69, for a final particle size of ~ 1 mm, with a Moh hardness of 4 (from Table 18.2), we could select a ball mill, rod mill, hammer mill, disc mill, or any of the crushers. In Table 18.14, limestone falls into material class 3, and for a product particle size range between 5 and 300 mesh, we would recommend a ball mill, rod mill, tube mill, roller mill, cage mill, impact breaker, or vibration mill. Because the size reduction ratio is not that large, it looks like a single-stage media mill such as a rod mill or ball mill will be adequate for this duty.

To make an estimate of the duty use Equation 18.33:

$$E = 100 E_i \left(\frac{1}{\sqrt{d_2}} - \frac{1}{\sqrt{d_1}} \right) \quad (18.33)$$

From Table 18.2, E_i for limestone = 11.6, so:

$$\begin{aligned} E &= 100 \times 11.6 \left(\frac{1}{\sqrt{0.00085}} - \frac{1}{\sqrt{0.015}} \right) \\ &= 30.3 \times 10^3 \text{ kWh/ton} \end{aligned}$$

1 short ton = 2000 lb and 1 metric ton = 2200 lb, so energy requirement = $1.1 \times 30.3 = 33.3$ MWh/metric ton.

A quick search on the Internet for rod mills can identify several devices that are suitable. For example, ESONG (a Chinese mining equipment supplier) makes a mill MBS2130 with diameter 2.1 m and length 3.6 m that can grind material < 2.5 cm to a size range of 0.83 to 0.147 mm with capacity 14 to 35 metric tons/h and power consumption 155 kW. If this machine were operated for 1 hour and processed 15 metric tons of limestone, the specific power consumption would be $155 \times 3600 / 15 = 37.2$ MWh/metric ton, so the value from Equation 18.33 is not too bad an estimate. At this point the designer would

be ready to begin searching for vendors that had equipment that appeared to meet the specifications and could begin the process of qualifying vendors through test runs or reference cases.

18.10 Heat transfer to flowing solid particles

Particulate solids streams can have substantial heat capacity and hence significant heating or cooling requirements. It is often necessary to heat or cool a flowing solid stream at the entry or exit to a reactor or other process operation. When the heating or cooling requirement of the solids is small, the heat needed can be provided by additional heating or cooling of the fluid process streams that will contact the solids. When this is not feasible, a solids heater or cooler must be specified.

When the solids can be entrained in a liquid stream to form a slurry, then heat transfer can be carried out using conventional heat transfer equipment; see [Chapter 19](#). Care must be taken to select exchanger designs that minimize dead spots where the slurry could settle and cause fouling. Slurries are usually fed on the tube side of shell and tube heat exchangers for this reason. The design must also allow for removal of the tube bundle or plates for cleaning.

Flowing solids in gas are usually heated or cooled by direct heat transfer with a hot or cold gas. Direct heat transfer can be accomplished in equipment similar to the continuous drying devices described in [Section 18.7.2](#); the most common designs are the fluidized bed, rotary dryer, and flash dryer designs. Steam, flue gas from a burner, heated air, or heated process gases are used as heat sources, and cold air and nitrogen are the most commonly used coolants. Direct heat transfer between the gas and solid streams is very rapid in fluidized beds that have back-mixing of the solids. In other devices the heat transfer coefficient between the solids and fluid depends on the device geometry, flow pattern, and contact time. Few reliable correlations are available in the open literature, and the extent of heating or cooling achievable is best determined from plant trials or in consultation with an equipment vendor.

Direct heat transfer can also be accomplished by combustion of material dispersed on the solid surface. For example, in fluid catalytic cracking processes, the catalyst is heated by combustion of coke that forms as a by-product on the catalyst surface. Combustion of the coke regenerates the catalyst activity, as well as providing the heat needed for reaction and feed heating.

Indirect heat transfer to solids can be accomplished in any piece of solids handling equipment that can be fitted with a heating or cooling jacket or coil. The area available for heat transfer is generally low. If the solids are moving slowly, the heat transfer coefficient on the process side can be predicted from the correlations developed by [Leva \(1950\)](#) for heat transfer to packed or moving beds:

$$h = 0.813 \frac{k}{D_T} e^{-6d_p/D_T} \left(\frac{d_p G}{\mu} \right)^{0.9} \quad \text{for } \frac{d_p}{D_T} < 0.35 \quad (18.34)$$

$$h = 0.125 \frac{k}{D_T} \left(\frac{d_p G}{\mu} \right)^{0.75} \quad \text{for } 0.35 < \frac{d_p}{D_T} < 0.6 \quad (18.35)$$

where:

- h = heat transfer coefficient
- k = fluid thermal conductivity
- d_p = particle diameter
- D_t = vessel diameter
- G = superficial mass velocity
- μ = fluid viscosity

The heat transfer coefficient on the utility side can be found using the correlations for heat transfer to jacketed vessels given in [Section 19.18](#).

Indirect heat transfer to or from a particulate solid flowing in a gas is less common and is usually accomplished by immersing heat transfer tubes into a fluidized bed of the solids. Heat transfer to and from fluidized beds has been the subject of much research, but there is significant variation in the measured data and consequently disagreement on

the best correlations to use. Heat transfer to and from fluidized beds is discussed in detail in [Zenz and Othmer \(1960\)](#). [Green and Southard \(2018\)](#) also give correlations for external walls, immersed tubes, and tube bundles.

18.11 Hazards of solids processing

In addition to the mechanical hazards inherent in solids-handling machinery, many solids-handling operations have the potential to form or release fine particulate material in the form of dust. Dust can pose a health hazard to workers who might inhale the dust, as well as creating the potential for dust explosions. Situations where dust can form are usually addressed at the design stage by the addition of engineering controls to mitigate dust formation and prevent the formation of toxic or explosive atmospheres.

18.11.1 Health impacts of dust inhalation

As a result of inadequate attention to worker safety in the early years of industrialization, the toxicological effects of worker exposure to many different types of dust have been well documented. Solids that have low inherent toxicity can nonetheless cause chronic health problems when absorbed into the bronchioles or alveoli as fine particulates. When dust reaches the lungs, it is absorbed by macrophages, which are subsequently swept out of the lungs by tiny hairs known as cilia. If the loading of dust is high or if the action of the cilia is impeded (for example, by lifestyle choices such as tobacco smoking), the macrophage system can be overwhelmed. Dead macrophages then accumulate in the lungs, releasing toxic substances that form scar tissue, leading to a condition known as *fibrosis*, in which the lungs lose their elasticity. [Table 18.15](#) lists some of the chronic afflictions that have been traced to dust inhalation in different industries. These conditions are collectively known as *pneumoconiosis*, meaning “dusty lung.” Health impacts of dust inhalation are discussed in more detail by [Kaye \(1997\)](#) and [Baxter et al. \(2010\)](#).

Whenever there are concerns over the toxicity or industrial hygiene impacts of dust formation, the designer should include appropriate engineering controls to reduce or eliminate worker exposure to dust. When feasible, use of wet process conditions will dramatically reduce dust formation. The primary engineering control on dust exposure is the plant itself. Plants that are susceptible to dust formation should be designed to operate under a slight vacuum so that dust cannot escape from the plant. A suction compressor is used to pull air from the plant and create a slight vacuum so that air flows into the equipment through any leaks but dust cannot flow out. Cyclones, filters, or other gas cleaning equipment are usually placed upstream of the suction compressor so that dust is not blown out to the environment; see [Section 18.5](#). If it is not safe or feasible to pull air into the process (for example, if the dust is flammable), then the dust-forming equipment can be placed in a housing that is ventilated with an inert gas such as nitrogen.

A secondary control on dust formation is the building in which the plant is housed. Adequate ventilation should be provided to maintain dust concentrations below the limits required by health and safety legislation. Equipment should be cleaned regularly to prevent accumulations of dust, and cleaning should be carried out by washing or using approved vacuum cleaners rather than using brushes or brooms that would resuspend the dust in air. Finally,

TABLE 18.15 Health hazards of dust inhalation

Health problem	Cause (dust exposure)	Symptoms and effects
Asbestosis (“white lung”)	Asbestos	Fibrosis, emphysema, mesothelioma (lung cancer)
Bagassosis	Moldy sugar cane	Fibrosis
Beryllium disease	Beryllium dust	Fibrosis
“Black lung” (coal pneumoconiosis)	Coal dust	Fibrosis, emphysema, lung cancer
“Brown lung” (byssinosis)	Cotton dust	Fibrosis, bronchitis, emphysema
“Farmer’s lung”	Moldy hay, straw, or grain	Asthma, fibrosis, histoplasmosis
Silicosis	Silica, quartz, sand	Fibrosis, emphysema

workers may need to wear respirators or breathing masks when working around dust. The use of personal protective equipment, though often necessary, should never be viewed as a substitute for proper engineering design of the plant.

18.11.2 Dust explosions

If fine particles of a flammable material are suspended in air at the right concentration, an explosive mixture can be formed. If this mixture is ignited, the dust burns, releasing heat that can ignite more dust and hence propagate a flame front. The combustion of the dust leads to a large increase in molar volume and temperature and thereby creates a destructive pressure wave. Technically a dust explosion is usually a deflagration; however, if it occurs in a confined space such as inside a building, it can propagate into a detonation. Dust explosions that occur around solids-handling equipment also have the potential to disturb and suspend accumulated dust on or around the equipment, leading to secondary explosions that can be larger than the initial event.

Ignition sources for dust explosions can include hot equipment in grinders and driers, open flames, heat released by oxidation of exposed solid surfaces due to attrition or grinding, and electrostatic discharge sparks caused by the accumulation of static electricity on flowing powders.

During plant design, the design engineer should, whenever possible, seek conditions that minimize the likelihood of a dust explosion occurring. The design should avoid dust-forming operations, use wet processing when safe and feasible, eliminate ignition sources as much as possible, avoid dust accumulation, and make use of inert gas ventilation ("inerting"). The use of an inert sweep gas such as nitrogen in equipment that forms flammable dust is the most common method of explosion prevention, although some metal dusts will even react with nitrogen, in which case a noble gas such as argon must be used instead.

Dust explosion should be considered as a pressure relief scenario whenever flammable dusts are present. Methods for plant design to prevent or suppress explosions are discussed in more detail in [Section 10.3.6](#) and in the National Fire Protection Association standards NFPA 61, NFPA 68, NFPA 69, NFPA 654, and NFPA 664. See also [Mannan \(2012\)](#), [Field \(1982\)](#), [Cross and Farrer \(1982\)](#), [Grossel \(1997\)](#), [Barton \(2001\)](#), [Eckhoff \(2003\)](#), and BS EN 1127.

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18.13 Nomenclature

		Dimensions in MLT
A	Gas–solids contact area in a dryer	L^2
A_i	Area of interface	L^2
A_{ps}	Projected area of a particle	L^2
A_s	Surface area of cyclone	L^2
A_1	Area of cyclone inlet duct	L^2
C	Constant in Beverloo correlation (Equation 18.20)	—
C'	Constant in Nedderman version of the Beverloo correlation (Equation 18.21)	—
C_{ps}	Specific heat capacity of solids	$L^2 T^{-2} \theta^{-1}$
C_{pg}	Specific heat capacity of gas	$L^2 T^{-2} \theta^{-1}$
c	cohesion	$ML^{-1} T^{-2}$
D	Vessel diameter	L
D_c	Cyclone diameter	L
D_{c1}	Diameter of standard cyclone	L
D_{c2}	Diameter of proposed cyclone design	L

Continued

		Dimensions in MLT
D_o	Diameter of the opening at the base of a vessel	L
D_T	Tank or vessel diameter	L
d	Particle diameter	L
d_a	Diameter of a circle with same projected area as a particle	L
$\bar{d}_a$	Arithmetic mean diameter	L
$\bar{d}_g$	Geometric mean diameter	L
d_h	Effective hydraulic diameter of a particle	L
d_i	Diameter of particle i	L
d_p	Particle "diameter"	L
d_{pr}	Diameter of a circle with the same perimeter as a particle	L
d_s	Diameter of solid particle removed in a centrifuge	L
d_{st}	Stokes diameter of a particle	L
$\bar{d}_{vs}$	Volume surface mean diameter	L
d_1	Initial particle size in grinding	L
d_1	Mean diameter of particles separated in cyclone under standard conditions	L
d_2	Final particle size in grinding	L
d_2	Mean diameter of particles separated in proposed cyclone design	L
d_{50}	Particle diameter for which cyclone is 50% efficient	L
E	Work done in grinding per unit mass feed	L^2T^{-2}
E_i	Bond work index	L^2T^{-2}
f_c	Friction factor for cyclones	—
G	Superficial mass velocity	$ML^{-2}T^{-1}$
g	Gravitational acceleration	LT^{-2}
H	Depth of solids in a bin or hopper	L
h	Heat transfer coefficient	$MT^{-3}\theta^{-1}$
K_G	Mass transfer coefficient	LT^{-1}
k	Fluid thermal conductivity	$MLT^{-3}\theta^{-1}$
k	Shape parameter in Beverloo correlation (Equation 18.20)	—
L	Liquid volumetric flow rate to a hydrocyclone	L^3T^{-1}
L	Particle length (longest dimension)	L
L_b	Length of a bed of solids	L
L_c	Conveying distance	L
M_{wG}	Average molecular weight of the gas	M
M_{wL}	Molecular weight of the liquid	M
m_e	Mass flow rate of solvent or water in entering gas	MT^{-1}
m_g	Mass flow rate of dry gas	MT^{-1}
m_o	Mass flow rate of moisture at the outlet of a dryer	MT^{-1}
m_s	Mass flow rate of bone dry solids	MT^{-1}
m	Mass flow rate of solids	MT^{-1}
N	Number of particles in a sample	—
P	Pressure	$ML^{-1}T^{-2}$
P_{ps}	Perimeter of a particle	L
ΔP	Pressure differential (pressure drop)	$ML^{-1}T^{-2}$
p	Power consumption	ML^2T^{-3}
p_s	Vapor pressure of liquid at the surface	$ML^{-1}T^{-2}$
p_b	Vapor pressure of liquid in the bulk gas	$ML^{-1}T^{-2}$
Q	Volumetric flow rate of liquid through a centrifuge	L^3T^{-1}
Q_1	Standard flow rate in cyclone	L^3T^{-1}
Q_2	Proposed flow rate in cyclone	L^3T^{-1}
r_e	Radius of cyclone exit pipe	L

		Dimensions in MLT
r_t	Radius of circle to which center line of cyclone inlet duct is tangential	L
T	Temperature	θ
T_g	Temperature of the bulk gas	θ
T_i	Gas inlet temperature	θ
T_o	Gas outlet temperature	θ
T_s	Temperature at the gas-liquid interface	θ
T_{si}	Solids inlet temperature	θ
T_{so}	Solids outlet temperature	θ
ΔT	Temperature difference between the gas and solids	θ
t	Time	T
t	Particle thickness	L
U	Fluid superficial velocity	LT^{-1}
U_{mf}	Fluid superficial velocity at minimum fluidization	LT^{-1}
u_g	Terminal velocity of solid particles settling under gravity	LT^{-1}
u_1	Velocity in cyclone inlet duct	LT^{-1}
u_2	Velocity in cyclone exit duct	LT^{-1}
W	Particle width	L
X	Moisture content	—
X_c	Critical moisture content during drying	—
X_e	Moisture content in equilibrium with gas entering a dryer	—
X_i	Inlet moisture content	—
X_o	Outlet moisture content	—
α	Angle of internal friction	—
α_w	Angle of wall friction	—
B	Angle of repose	—
ε	Bed void fraction	—
ε_{mf}	Void fraction at minimum fluidization	—
η	Separating efficiency of a centrifuge	—
η	Efficiency of a cyclone in separating any particle of diameter d	—
λ	latent heat of vaporization	L^2T^{-2}
μ	Liquid (or fluid) viscosity	$ML^{-1}T^{-1}$
μ_c	Viscosity of continuous phase	$ML^{-1}T^{-1}$
μ_1	Cyclone test fluid viscosity	$ML^{-1}T^{-1}$
μ_2	Viscosity of fluid in proposed cyclone design	$ML^{-1}T^{-1}$
ρ_b	Bulk density of particulate solids	ML^{-3}
ρ_{bf}	Flowing bulk density of a particulate solid	ML^{-3}
ρ_c	Density of the continuous phase	ML^{-3}
ρ_d	Density of the dispersed phase	ML^{-3}
ρ_f	Fluid density	ML^{-3}
ρ_L	Liquid density	ML^{-3}
ρ_p	Particle (piece) density of a solid	ML^{-3}
ρ_s	Density of solid	ML^{-3}
ρ_t	True density of a solid particle	ML^{-3}
ρ_w	Wetted particle density	ML^{-3}
$\Delta\rho$	Difference in density between solid and liquid	ML^{-3}
$\Delta\rho_1$	Density difference under standard conditions in standard cyclone	ML^{-3}
$\Delta\rho_2$	Density difference in proposed cyclone design	ML^{-3}
Σ	Sigma value for centrifuges, defined by equation 10.6	L^2
σ	Normal stress (applied load)	$ML^{-1}T^{-2}$
σ_w	Normal stress at the wall	$ML^{-1}T^{-2}$
τ	Shear stress	$ML^{-1}T^{-2}$

Continued

		Dimensions in MLT
τ_w	Shear stress at the wall	$\text{ML}^{-1}\text{T}^{-2}$
ϕ	Factor in Fig. 18.35	—
X	Internal pore volume fraction of a particle	—
Ψ	Parameter in Fig. 18.35	—
ψ	Particle sphericity in Equation 18.4	—
Dimensionless numbers		
Ar	Archimedes number (defined in Equation 18.17)	—
Re	Reynolds number	—
Re_{mf}	Reynolds number at minimum fluidization	—

18.14 Problems

- 18.1.** Design a storage and feed system for the grinder introduced in Example 18.5. It is desired to control the mass flow rate to the gasification process to within $\pm 5\%$ of set point.
- 18.2.** A proposed belt dryer design has a 2-m-wide belt that will hold a 50-mm-thick layer of solids. The solid particles have average diameter 3 mm and piece density 730 kg/m^3 . Hot air at 150°C is blown up through the dryer belt at a volumetric flow rate of $7200 \text{ m}^3/\text{h}$ per m^2 of belt area. Will the bed of solids on the belt fluidize?
- 18.3.** The product from a crystallizer is to be separated from the liquor using a centrifuge. The concentration of the crystals is 6.5% and the slurry feed rate to the centrifuge will be $5.0 \text{ m}^3/\text{h}$. The density of the liquor is 995 kg/m^3 and that of the crystals 1500 kg/m^3 . The viscosity of the liquor is $0.7 \text{ mN m}^{-2}\text{s}$. The cut-off crystal size required is $5 \text{ }\mu\text{m}$.
Select a suitable type of centrifuge to use for this duty.
- 18.4.** Dissolved solids in the tar from the bottom of a distillation column are precipitated by quenching the hot tar in oil. The solids are then separated from the oil and burnt. The density of the solids is 1100 kg/m^3 . The density of the liquid phase after addition of the tar is 860 kg/m^3 and its viscosity at the temperature of the mixture is $1.7 \text{ mN m}^{-2}\text{s}$. The solid content of the oil and tar mixture is 10% and the flow rate of the liquid phase leaving the separator will be 1000 kg/h . The cut-off particle size required is 0.1 mm .
List the types of separators that could be considered for separating the solids from the liquid. Bearing in mind the nature of the process, what type of separator would you recommend for this duty?
- 18.5.** The solids from a dilute slurry are to be separated using hydrocyclones. The density of the solids is 2900 kg/m^3 and the liquid is water. A recovery of 95% of particles greater than $100 \text{ }\mu\text{m}$ is required. The minimum operating temperature will be 10°C and the maximum 30°C . Design a hydrocyclone system to handle 1200 liters/min of this slurry.
- 18.6.** A fluidized bed is used in the production of aniline by the hydrogenation of nitrobenzene. Single-stage cyclones, followed by candle filters, are used to remove fines from the gases leaving the fluidized bed.
The reactor operates at a temperature of 270°C and a pressure of 2.5 bara. The reactor diameter is 10 m. Hydrogen is used in large excess in the reaction, and for the purposes of this exercise the properties of the gas may be taken as those of hydrogen at the reactor conditions. The density of the catalyst particles is 1800 kg/m^3 . The estimated particle size distribution of the fines is:

Particle size, μm	50	40	30	20	10	5	2
Weight less than	100	70	40	20	10	5	2

A 70% recovery of the solids is required in the cyclones.

For a gas flow rate of $100,000 \text{ m}^3/\text{h}$ at the reactor conditions, determine how many cyclones operating in parallel are needed and design a suitable cyclone. Estimate the size distribution of the particles entering the filters.

18.7. What dryer would you recommend to process each of the following streams:

1. A continuous stream of shaped (extruded) breakfast cereal with flow rate 50 kg/h bone-dry mass and initial moisture content $150 \text{ wt\%}$ moisture.
2. A 10-kg batch of an enzyme.
3. A continuous flow of 100 dry kg/min of granulated sugar that contains 20% moisture and must be reduced to $<1\%$ moisture.
4. A slurry of heat-sensitive crystals with flow rate of 10 kg/h of solids and 30 kg/h of water.

Heat transfer equipment

KEY LEARNING OBJECTIVES

- How to specify and design a shell and tube heat exchanger
- How to design boilers, thermosiphon reboilers, and condensers
- How to design a plate heat exchanger
- How to design air coolers and fired heaters
- How to determine whether a reactor can be heated or cooled using a jacket or internal coil

19.1 Introduction

The transfer of heat to and from process fluids is an essential part of most chemical processes. The most commonly used type of heat transfer equipment is the shell and tube heat exchanger, the design of which is the main subject of this chapter.

The fundamentals of heat transfer theory are covered in [Chhabra and Shankar \(2017\)](#) and in many other textbooks: [Holman \(2009\)](#), [Ozisik \(1985\)](#), [Rohsenow et al. \(1998\)](#), [Kreith and Manglik \(2017\)](#), and [Bergman et al. \(2011\)](#).

Several useful books have been published on the design of heat-exchange equipment. These should be consulted for more details on the construction of equipment and design methods than can be given in this book. A selection of the more useful texts is listed in the bibliography at the end of this chapter. The compilation edited by Schlünder (1983)—see also the edition by [Hewitt \(2009\)](#)—is probably the most comprehensive work on heat exchanger design methods available in the open literature. The book by [Saunders \(1988\)](#) is recommended as a good source of information on heat exchanger design, especially for shell and tube exchangers.

As with distillation, work on the development of reliable design methods for heat exchangers has been dominated in recent years by commercial research organizations: Heat Transfer Research, Inc. (HTRI) in the United States and Heat Transfer and Fluid Flow Service (HTFS) in the United Kingdom. The HTFS program was developed by the United Kingdom Atomic Energy Authority and the National Physical Laboratory, but is now available from Aspen Technology, Inc., and as part of the Honeywell UniSim Design Suite; see [Chapter 4](#), Table 4.1. Their proprietary methods are not available in the open literature. They will, however, be available to design engineers in the major operating and contracting companies, whose companies subscribe to these organizations.

The principal types of heat exchangers used in the process industries, which will be discussed in this chapter, are listed here:

1. Double-pipe exchanger: the simplest type, used for cooling and heating
2. Shell and tube exchangers: used for all applications
3. Plate and frame exchangers (plate heat exchangers): used for heating and cooling

4. Plate-fin exchangers
5. Spiral heat exchangers
6. Air cooled: used for coolers and condensers
7. Direct contact: used for cooling and quenching
8. Fired heaters
9. Agitated vessels

The word “exchanger” really applies to all types of equipment in which heat is exchanged, but is often used specifically to denote equipment in which heat is exchanged between two process streams. Exchangers in which a process fluid is heated or cooled by a plant service stream are referred to as heaters or coolers, respectively. If the process stream is vaporized, the exchanger is called a vaporizer if the stream is essentially completely vaporized; a *reboiler* if associated with a distillation column; and an *evaporator* if used to concentrate a solution (see [Chapter 16](#)). The terms *fired exchanger* and *fired heater* are used for exchangers heated by combustion gases, such as boilers; other exchangers are referred to as unfired exchangers.

19.2 Basic design procedure and theory

The general equation for heat transfer across a surface is:

$$Q = UA\Delta T_m \quad (19.1)$$

where Q = heat transferred per unit time, W

U = the overall heat transfer coefficient, $W/m^2\text{ }^\circ\text{C}$

A = heat transfer area, m^2

ΔT_m = the mean temperature difference, the temperature driving force, $^\circ\text{C}$

The prime objective in the design of an exchanger is to determine the surface area required for the specified duty (rate of heat transfer) using the temperature differences available.

The overall coefficient is the reciprocal of the overall resistance to heat transfer, which is the sum of several individual resistances. For heat exchange across a typical heat exchanger tube, the relationship between the overall coefficient and the individual coefficients, which are the reciprocals of the individual resistances, is given by:

$$\frac{1}{U_o} = \frac{1}{h_o} + \frac{1}{h_{od}} + \frac{d_o \ln\left(\frac{d_o}{d_i}\right)}{2k_w} + \frac{d_o}{d_i} \times \frac{1}{h_{id}} + \frac{d_o}{d_i} \times \frac{1}{h_i} \quad (19.2)$$

where U_o = the overall coefficient based on the outside area of the tube, $W/m^2\text{ }^\circ\text{C}$

h_o = outside fluid film coefficient, $W/m^2\text{ }^\circ\text{C}$

h_i = inside fluid film coefficient, $W/m^2\text{ }^\circ\text{C}$

h_{od} = outside dirt coefficient (fouling factor), $W/m^2\text{ }^\circ\text{C}$

h_{id} = inside dirt coefficient, $W/m^2\text{ }^\circ\text{C}$

k_w = thermal conductivity of the tube wall material, $W/m\text{ }^\circ\text{C}$

d_i = tube inside diameter, m

d_o = tube outside diameter, m

The magnitude of the individual coefficients depends on the nature of the heat transfer process (conduction, convection, condensation, boiling, or radiation), on the physical properties of the fluids, on the fluid flow rates, and on the physical arrangement of the heat transfer surface. As the physical layout of the exchanger cannot be determined until the area is known, the design of an exchanger is, of necessity, a trial-and-error procedure. The steps in a typical design procedure are as follows:

1. Define the duty: heat transfer rate, fluid flow rates, and temperatures.
2. Collect the fluid physical properties required: density, viscosity, heat capacity, and thermal conductivity.
3. Decide on the type of exchanger to be used.
4. Select a trial value for the overall coefficient, U .
5. Calculate the mean temperature difference, ΔT_m .
6. Calculate the area required from Equation 19.1.
7. Decide the exchanger layout.

8. Calculate the individual coefficients.
9. Calculate the overall coefficient and compare with the trial value. If the calculated value differs significantly from the estimated value, substitute the calculated for the estimated value and return to step 6.
10. Calculate the exchanger pressure drop; if unsatisfactory, return to steps 7 or 4 or 3, in that order of preference.
11. Optimize the design: repeat steps 4 to 10, as necessary, to determine the cheapest exchanger that will satisfy the duty. Usually this will be the one with the smallest area that satisfies the pressure drop constraint.

Procedures for estimating the individual heat transfer coefficients and the exchanger pressure drops are given in this chapter.

19.2.1 Heat exchanger analysis: The effectiveness–NTU method

The *effectiveness–NTU* method is a procedure for evaluating the performance of heat exchangers, which has the advantage that it does not require the evaluation of the mean temperature differences. *NTU* stands for the number of transfer units and is analogous with the use of transfer units in mass transfer; see [Chapter 17](#).

The principal use of this method is in the rating of an existing exchanger. It can be used to determine the performance of the exchanger when the heat transfer area and construction details are known. The method has an advantage over the use of the design procedure outlined earlier, as an unknown stream outlet temperature can be determined directly, without the need for iterative calculations. It makes use of plots of the exchanger *effectiveness* versus *NTU*. The effectiveness is the ratio of the actual rate of heat transfer to the maximum possible rate.

The *effectiveness–NTU* method will not be covered in this book, as it is more useful for rating than design. The method is covered in books by Bergman et al. (2011), Ozisik (1985), and Hewitt et al. (1994). The method is also covered by the Engineering Sciences Data Unit in their Design Guides 98003 to 98007 (1998). These guides give large clear plots of effectiveness versus NTU and are recommended for accurate work.

19.3 Overall heat transfer coefficient

Typical values of the overall heat transfer coefficient for various types of heat exchangers are given in [Table 19.1](#). More extensive data can be found in the books by [Green and Southard \(2018\)](#), TEMA (2019), and [Coker \(2015\)](#).

[Fig. 19.1](#), which is adapted from a similar nomograph given by [Frank \(1974\)](#), can be used to estimate the overall coefficient for tubular exchangers (shell and tube). The film coefficients given in [Fig. 19.1](#) include an allowance for fouling.

The values given in [Table 19.1](#) and [Fig. 19.1](#) can be used for the preliminary sizing of equipment for process evaluation and as trial values for starting a detailed thermal design.

19.4 Fouling factors (dirt factors)

Most process and service fluids will foul the heat transfer surfaces in an exchanger to a greater or lesser extent. The deposited material will normally have a relatively low thermal conductivity and will reduce the overall coefficient. It is therefore necessary to oversize an exchanger to allow for the reduction in performance during operation. The effect of fouling is allowed for in design by including the inside and outside fouling coefficients in Equation 19.2. Fouling factors are usually quoted as heat transfer resistances rather than coefficients. They are difficult to predict and are usually based on past experience. Estimating fouling factors introduces a considerable uncertainty into exchanger design; the value assumed for the fouling factor can overwhelm the accuracy of the predicted values of the other coefficients. Fouling factors are often wrongly used as factors of safety in exchanger design. Some work on the prediction of fouling factors has been done by HTRI; see [Taborek et al. \(1972\)](#). Fouling is the subject of books by [Bott \(1990\)](#) and [Garrett-Price \(1985\)](#) and is also discussed by [Zhu \(2014\)](#).

Typical values for the fouling coefficients and factors for common process and service fluids are given in [Table 19.2](#). These values are for shell and tube exchangers with plain (not finned) tubes. More extensive data on fouling factors are given in the TEMA standards (2019) and by [Coker \(2015\)](#).

The selection of the design fouling coefficient will often be an economic decision. The optimum design will be obtained by balancing the extra capital cost of a larger exchanger against the savings in operating cost obtained from the longer operating time between cleaning that the larger area will give. Duplicate exchangers should be

TABLE 19.1 Typical overall coefficients

Shell and tube exchangers		
Hot fluid	Cold fluid	U (W/m ² °C)
<i>Heat exchangers</i>		
Water	Water	800–1500
Organic solvents	Organic solvents	100–300
Light oils	Light oils	100–400
Heavy oils	Heavy oils	50–300
Gases	Gases	10–50
<i>Coolers</i>		
Organic solvents	Water	250–750
Light oils	Water	350–900
Heavy oils	Water	60–300
Gases	Water	20–300
Organic solvents	Brine	150–500
Water	Brine	600–1200
Gases	Brine	15–250
<i>Heaters</i>		
Steam	Water	1500–4000
Steam	Organic solvents	500–1000
Steam	Light oils	300–900
Steam	Heavy oils	60–450
Steam	Gases	30–300
Dowtherm	Heavy oils	50–300
Dowtherm	Gases	20–200
Flue gases	Steam	30–100
Flue	Hydrocarbon vapors	30–100
<i>Condensers</i>		
Aqueous vapors	Water	1000–1500
Organic vapors	Water	700–1000
Organics (some noncondensables)	Water	500–700
Vacuum condensers	Water	200–500
<i>Vaporizers</i>		
Steam	Aqueous solutions	1000–1500
Steam	Light organics	900–1200
Steam	Heavy organics	600–900
<i>Air-cooled exchangers</i>		
<i>Process fluid</i>		
Water		300–450

TABLE 19.1 Typical overall coefficients—cont'd

Air-cooled exchangers		
Process fluid		
Light organics		300–700
Heavy organics		50–150
Gases, 5–10 bar		50–100
10–30 bar		100–300
Condensing hydrocarbons		300–600
Immersed coils		
Coil	Pool	
Natural circulation		
Steam	Dilute aqueous solutions	500–1000
Steam	Light oils	200–300
Steam	Heavy oils	70–150
Water	Aqueous solutions	200–500
Water	Light oils	100–150
Immersed coils		
Coil	Pool	U (W/m ² °C)
Agitated		
Steam	Dilute aqueous solutions	800–1500
Steam	Light oils	300–500
Steam	Heavy oils	200–400
Water	Aqueous solutions	400–700
Water	Light oils	200–300
Jacketed vessels		
Jacket	Vessel	
Steam	Dilute aqueous solutions	500–700
Steam	Light organics	250–500
Water	Dilute aqueous solutions	200–500
Water	Light organics	200–300
Gasketed-plate exchangers		
Hot fluid	Cold fluid	
Light organic	Light organic	2500–5000
Light organic	Viscous organic	250–500

Continued

TABLE 19.1 Typical overall coefficients—cont'd

Gasketed-plate exchangers		
Hot fluid	Cold fluid	
Viscous organic	Viscous organic	100–200
Light organic	Process water	2500–3500
Viscous organic	Process water	250–500
Light organic	Cooling water	2000–4500
Viscous organic	Cooling water	250–450
Condensing steam	Light organic	2500–3500
Condensing steam	Viscous organic	250–500
Process water	Process water	5000–7500
Process water	Cooling water	5000–7000
Dilute aqueous solutions	Cooling water	5000–7000
Condensing steam	Process water	3500–4500

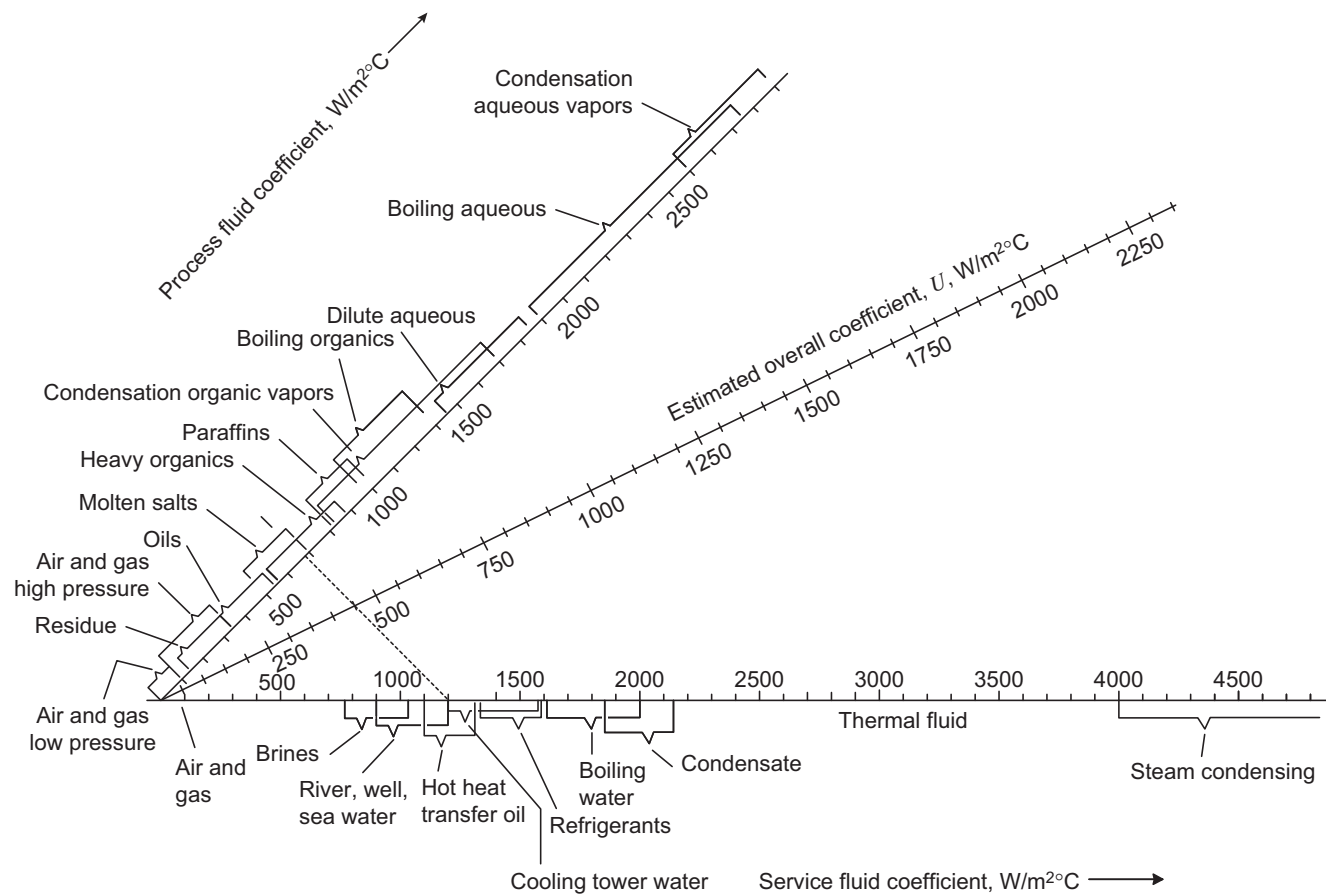


FIG. 19.1 Overall coefficients (join process side duty to service side and read *U* from center scale).

TABLE 19.2 Fouling factors (coefficients), typical values

Fluid	Coefficient ($\text{W/m}^2\text{ }^\circ\text{C}$)	Factor (resistance) ($\text{m}^2\text{ }^\circ\text{C/W}$)
River water	3000–12,000	0.0003–0.0001
Sea water	1000–3000	0.001–0.0003
Cooling water (towers)	3000–6000	0.0003–0.00017
Towns water (soft)	3000–5000	0.0003–0.0002
Towns water (hard)	1000–2000	0.001–0.0005
Steam condensate	1500–5000	0.00067–0.0002
Steam (oil free)	4000–10,000	0.0025–0.0001
Steam (oil traces)	2000–5000	0.0005–0.0002
Refrigerated brine	3000–5000	0.0003–0.0002
Air and industrial gases	5000–10,000	0.0002–0.0001
Flue gases	2000–5000	0.0005–0.0002
Organic vapors	5000	0.0002
Organic liquids	5000	0.0002
Light hydrocarbons	5000	0.0002
Heavy hydrocarbons	2000	0.0005
Boiling organics	2500	0.0004
Condensing organics	5000	0.0002
Heat transfer fluids	5000	0.0002
Aqueous salt solutions	3000–5000	0.0003–0.0002

considered for severely fouling systems so that one exchanger can be taken offline for cleaning while the plant continues to operate using the other exchanger.

When the design engineer adds area to allow for fouling, care must be taken to ensure that the velocity of the fluid is not reduced; otherwise, the fouling could be accelerated. For example, if more tubes are added to a shell and tube heat exchanger then the tube-side flow rate per tube is reduced. Lower tube-side velocity reduces the shear inside the tubes and increases the rate of tube-side fouling. An alternative method of increasing area would be to increase tube length, which comes at the expense of higher pressure drop.

19.5 Shell and tube exchangers: Construction details

The shell and tube exchanger is by far the most common type of heat transfer equipment used in the process industries. The advantages of this type are:

1. The configuration gives a large surface area in a small volume.
2. Good mechanical layout: a good shape for pressure operation.
3. Uses well-established fabrication techniques.
4. Can be constructed from a wide range of materials.
5. Easily cleaned.
6. Well-established design procedures.

Essentially, a shell and tube exchanger consists of a bundle of tubes enclosed in a cylindrical shell. The ends of the tubes are fitted into tubesheets, which separate the shell-side and tube-side fluids. Baffles are provided in the shell to direct the fluid flow and support the tubes. The assembly of baffles and tubes is held together by support rods and spacers (Fig. 19.2).

Exchanger types

The principal types of shell and tube exchangers are shown in Figs. 19.3 to 19.8. Diagrams of other types and full details of their construction can be found in the heat exchanger standards (see Section 19.5.1). The standard

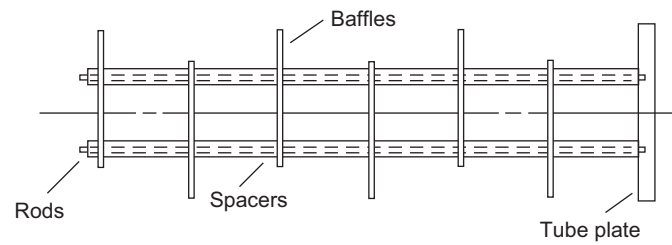


FIG. 19.2 Baffle spacers and tie rods.

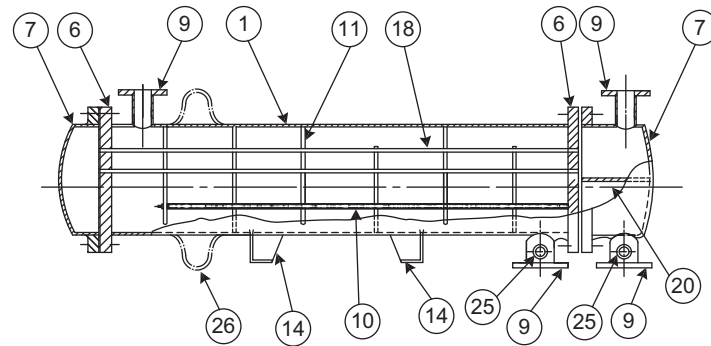


FIG. 19.3 Fixed-tube plate, type BEM. Based on figures from BS 3274: 1960.

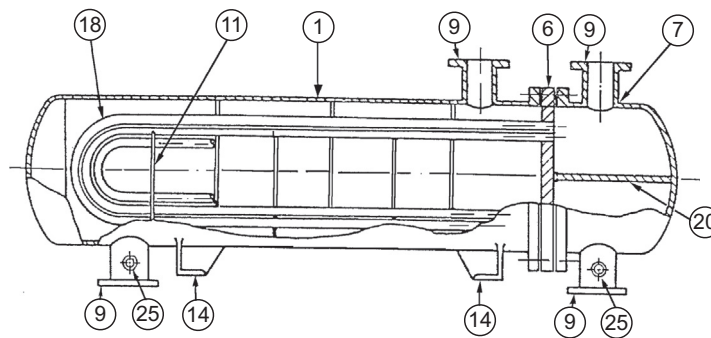


FIG. 19.4 U-tube, type BEU. Based on figures from BS 3274: 1960.

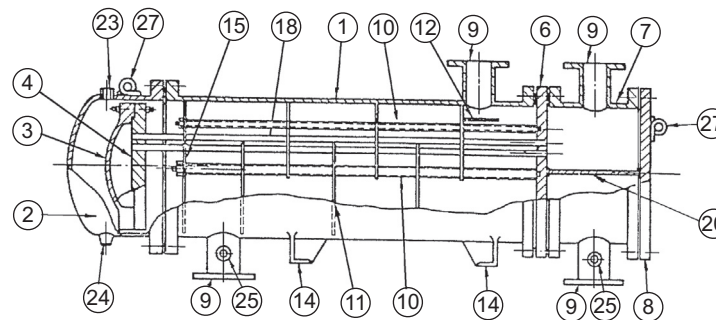


FIG. 19.5 Internal floating head without clamp ring, type AET. Based on figures from BS 3274: 1960.

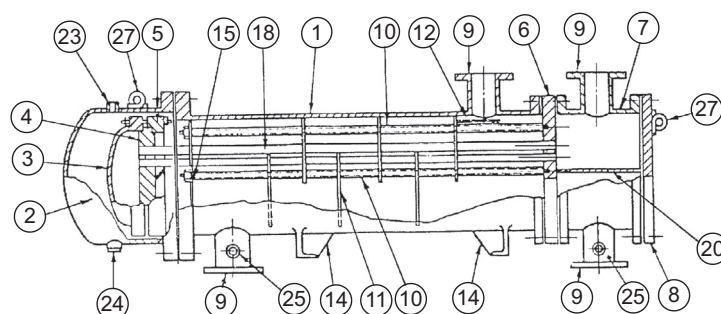


FIG. 19.6 Internal floating head with clamp ring, type AES. Based on figures from BS 3274: 1960.

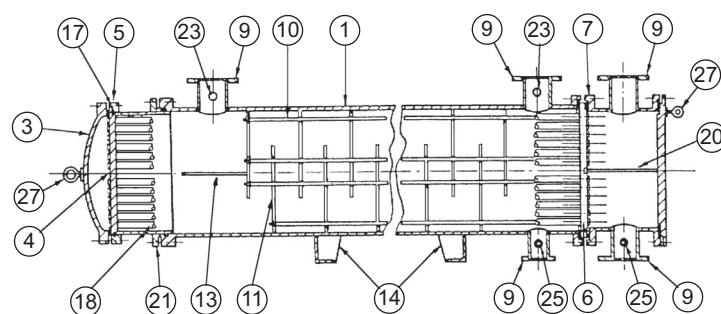


FIG. 19.7 External floating head, packed gland, type AEP. Based on figures from BS 3274: 1960.

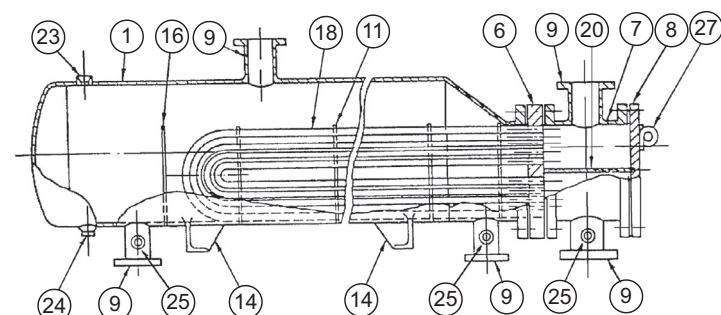


FIG. 19.8 Kettle reboiler with U-tube bundle, type AKU. Based on figures from BS 3274: 1960.

nomenclature used for shell and tube exchangers is given next; the numbers refer to the features shown in Figs. 19.3 to 19.8.

Nomenclature

Part number:

1. Shell
2. Shell cover
3. Floating-head cover
4. Floating-tube plate
5. Clamp ring
6. Fixed-tubesheet (tube plate)
7. Channel (end-box or header)

8. Channel cover
9. Branch (nozzle)
10. Tie rod and spacer
11. Cross baffle or tube-support plate
12. Impingement baffle
13. Longitudinal baffle
14. Support bracket
15. Floating-head support
16. Weir
17. Split ring
18. Tube
19. Tube bundle
20. Pass partition
21. Floating-head gland (packed gland)
22. Floating-head gland ring
23. Vent connection
24. Drain connection
25. Test connection
26. Expansion bellows
27. Lifting ring

The simplest and cheapest type of shell and tube exchanger is the fixed tubesheet design shown in Fig. 19.3 (TEMA type BEM). The main disadvantages of this type are that the tube bundle cannot be removed for cleaning and there is no provision for differential expansion of the shell and tubes. As the shell and tubes will be at different temperatures and may be of different materials, the differential expansion can be considerable, and the use of this type is limited to temperature differences up to about 80 °C. Some provision for expansion can be made by including an expansion loop in the shell (shown dotted on Fig. 19.3), but their use is limited to low shell pressure; up to about 8 bar. In the other types, only one end of the tubes is fixed and the bundle can expand freely.

The U-tube (U-bundle) type shown in Fig. 19.4 requires only one tubesheet and is cheaper than the floating-head types. This is the TEMA type BEU exchanger, which is widely used but is limited in use to relatively clean fluids, as the tubes and bundle are difficult to clean. It is also more difficult to replace a tube in this type.

Exchangers with an internal floating head (Figs. 19.5 and 19.6; TEMA types AET and AES) are more versatile than fixed-head and U-tube exchangers. They are suitable for high temperature differentials and, as the tubes can be rodded from end to end and the bundle removed, are easier to clean and can be used for fouling liquids. A disadvantage of the pull-through design (see Fig. 19.5) is that the clearance between the outermost tubes in the bundle and the shell must be made greater than in the fixed and U-tube designs to accommodate the floating-head flange, allowing fluid to bypass the tubes. The clamp ring (split flange design; see Fig. 19.6) is used to reduce the clearance needed. There will always be a danger of leakage occurring from the internal flanges in these floating-head designs.

In the external floating-head designs (Fig. 19.7; TEMA type AEP), the floating-head joint is located outside the shell and the shell sealed with a sliding gland joint employing a stuffing box. Because of the danger of leaks through the gland, the shell-side pressure in this type is usually limited to about 20 bar, and flammable or toxic materials should not be used on the shell side.

The kettle reboiler with U-tubes (TEMA type AKU) shown in Fig. 19.8 is commonly used for reboilers and evaporators that are heated with steam, as steam is a nonfouling service. TEMA type BKU without a removable channel cover is also widely used for kettle reboilers.

19.5.1 Heat exchanger standards and codes

The mechanical design features, fabrication, materials of construction, and testing of shell and tube exchangers are covered by the standards of the Tubular Heat Exchanger Manufacturers Association (TEMA). The TEMA standards cover three classes of exchangers: class R covers exchangers for the generally severe duties of the petroleum and related industries, class C covers exchangers for moderate duties in commercial and general process applications, and class B covers exchangers for use in the chemical process industries. The TEMA standards should be

consulted for full details of the mechanical design features of shell and tube exchangers; only brief details will be given in this chapter. Internationally, BS 3274 is also sometimes used, although the TEMA codes are most often followed.

The TEMA standards identify heat exchanger type by a three-letter code. The first letter denotes the tube-side head type, also known as the front end; the second letter identifies the shell type; and the third letter defines the rear end. Fig. 19.9 illustrates the TEMA nomenclature.

The standards give the preferred shell and tube dimensions; the design and manufacturing tolerances; corrosion allowances; and the recommended design stresses for materials of construction. The shell of an exchanger is a

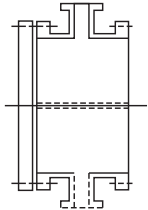
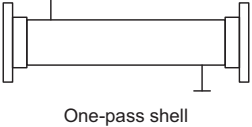
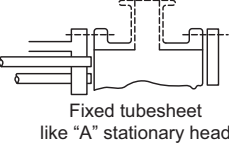
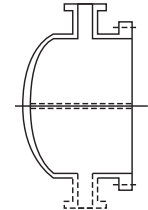
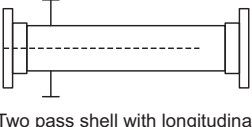
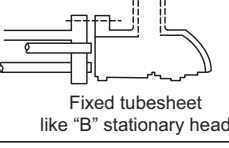
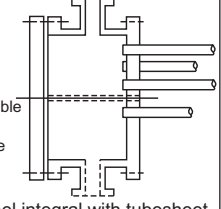
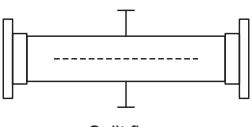
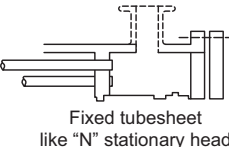
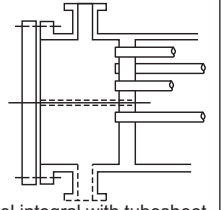
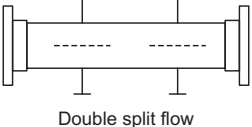
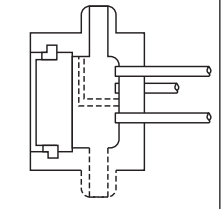
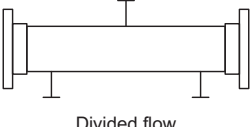
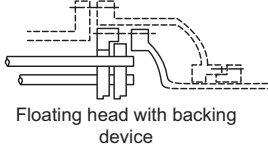
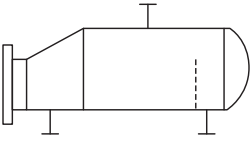
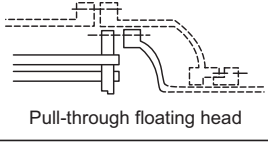
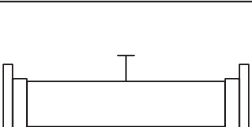
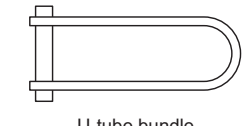
	Front-end stationary head types		Shell types		Rear-end head types
A	 Channel and removable cover	E	 One-pass shell	L	 Fixed tubesheet like "A" stationary head
B	 Bonnet (integral cover)	F	 Two pass shell with longitudinal baffle	M	 Fixed tubesheet like "B" stationary head
C	 Removable tube bundle only Channel integral with tubesheet and removable cover	G	 Split flow	N	 Fixed tubesheet like "N" stationary head
N	 Channel integral with tubesheet and removable cover	H	 Double split flow	P	 Outside packed floating head
D	 Special high-pressure closure	J	 Divided flow	S	 Floating head with backing device
		K	 Kettle type reboiler	T	 Pull-through floating head
		X	 Cross flow	U	 U-tube bundle
				W	 Externally sealed floating tubesheet

FIG. 19.9 TEMA designations for shell and tube heat exchangers. Reproduced with permission from the Thermal Exchanger Manufacturers Association.

pressure vessel and will be designed in accordance with the appropriate national pressure vessel code or standard; see [Chapter 14, Section 14.2](#). The dimensions of standard flanges for use with heat exchangers are given in the TEMA standards.

In the TEMA standards dimensions are given in feet and inches, so these units have been used in this chapter with the equivalent values in SI units given in brackets.

19.5.2 Tubes

Dimensions

The TEMA design standard allows tube diameters between $\frac{1}{4}$ in. (6.4 mm) and 2 in. (50 mm), but tube diameters in the range $\frac{5}{8}$ in. (16 mm) to 2 in. (50 mm) are most often used. The smaller diameters $\frac{5}{8}$ to 1 in. (16 to 25 mm) are preferred for most duties, as they will give more compact, and therefore cheaper, exchangers. Larger tubes are easier to clean by mechanical methods and are selected for heavily fouling fluids.

The tube thickness (gauge) is selected to withstand the internal and external (shell-side) pressure and give an adequate corrosion allowance. TEMA standard tubing dimensions are given in table D-7 of the TEMA standards (D7-M in metric units), which is reproduced in *Perry's Chemical Engineers Handbook* ([Green & Southard, 2018](#)). The most commonly used thicknesses correspond to even-numbered Birmingham Wire Gauge (BWG) units. Standard diameters and wall thicknesses for steel tubes are given in [Table 19.3](#).

The preferred lengths of tubes for heat exchangers are 6 ft. (1.83 m), 8 ft (2.44 m), 12 ft (3.66 m), 16 ft (4.88 m), 20 ft (6.10 m), and 24 ft (7.32 m). For a given surface area, the use of longer tubes will reduce the shell diameter. This will generally result in a lower-cost exchanger, particularly for high shell pressures, but will lead to an increase in pressure drop and pump work. The optimum tube length-to-shell diameter ratio will usually fall within the range of 5 to 10.

If U-tubes are used, the tubes on the outside of the bundle will be longer than those on the inside. The average length needs to be estimated for use in the thermal design. U-tubes will be bent from standard tube lengths and cut to size.

The tube size is often determined by the plant maintenance department standards, as clearly it is an advantage to reduce the number of sizes that have to be held in stores for tube replacement.

As a guide, $\frac{3}{4}$ in. (19 mm) is a good trial diameter with which to start design calculations.

Tube arrangements

The tubes in an exchanger are usually arranged in an equilateral triangular, square, or rotated square pattern ([Fig. 19.10](#)).

The triangular and rotated square patterns give higher heat transfer rates, but at the expense of a higher pressure drop than the square pattern. A square or rotated square arrangement is used for heavily fouling fluids, where it is necessary to mechanically clean the outside of the tubes. The recommended tube pitch (distance between tube centers) is 1.25 times the tube outside diameter; this will normally be used unless process requirements dictate otherwise. Where a square pattern is used for ease of cleaning, the recommended minimum clearance between the tubes is 0.25 in. (6.4 mm).

TABLE 19.3 Standard dimensions for steel tubes

Outside diameter (mm)		Wall thickness (mm)			
16	1.2	1.7	2.1	—	—
19	—	1.7	2.1	2.8	—
25	—	1.7	2.1	2.8	3.4
32	—	1.7	2.1	2.8	3.4
38	—	—	2.1	2.8	3.4
50	—	—	2.1	2.8	3.4

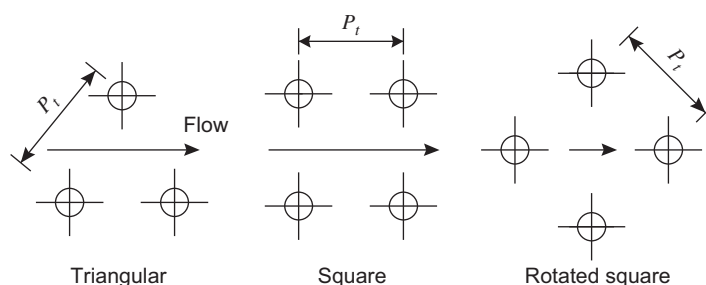


FIG. 19.10 Tube patterns.

Tube-side passes

The fluid in the tube is usually directed to flow back and forth in a number of “passes” through groups of tubes arranged in parallel to increase the length of the flow path. The number of passes is selected to give the required tube-side design velocity. Exchangers are built with from 1 to up to about 16 tube passes. The tubes are arranged into the number of passes required by dividing up the exchanger headers (channels) with partition plates (pass partitions). The arrangement of the pass partitions for two, four, and six tube passes are shown in Fig. 19.11. The layouts for higher numbers of passes are given by [Saunders \(1988\)](#).

19.5.3 Shells

The TEMA standards cover exchangers up to 60 in. (1520 mm) in diameter. Up to about 24 in. (610 mm), shells are normally constructed from standard, close tolerance pipe; above 24 in. (610 mm) they are rolled from plate.

For high-pressure applications, the shell thickness would be sized according to the pressure vessel design standards; see [Chapter 14](#). The minimum allowable shell thickness is given in the TEMA standards. The values, converted to SI units and rounded, are given next.

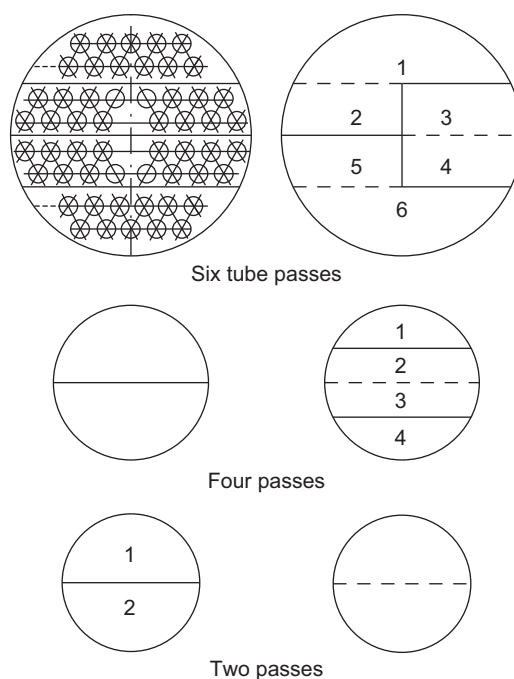


FIG. 19.11 Tube arrangements showing pass partitions in headers.

Minimum shell thickness (mm)

Nominal shell dia., mm	Carbon steel pipe	Alloy plate	Steel
150	7.1	—	3.2
200–300	9.3	—	3.2
330–580	9.5	7.9	3.2
610–740	—	7.9	4.8
760–990	—	9.5	6.4
1010–1520	—	11.1	6.4
1550–2030	—	12.7	7.9
2050–2540	—	12.7	9.5

The shell diameter must be selected to give as close a fit to the tube bundle as is practical to reduce bypassing around the outside of the bundle; see Section 19.9. The clearance required between the outermost tubes in the bundle and the shell inside diameter will depend on the type of exchanger and the manufacturing tolerances; typical values are given in Fig. 19.12.

19.5.4 Tubesheet layout (tube count)

The bundle diameter depends not only on the number of tubes but also on the number of tube passes, as spaces must be left in the pattern of tubes on the tubesheet to accommodate the pass partition plates.

An estimate of the bundle diameter D_b can be obtained from Equation 19.3b, which is an empirical equation based on standard tube layouts. The constants for use in this equation for triangular and square patterns are given in Table 19.4.

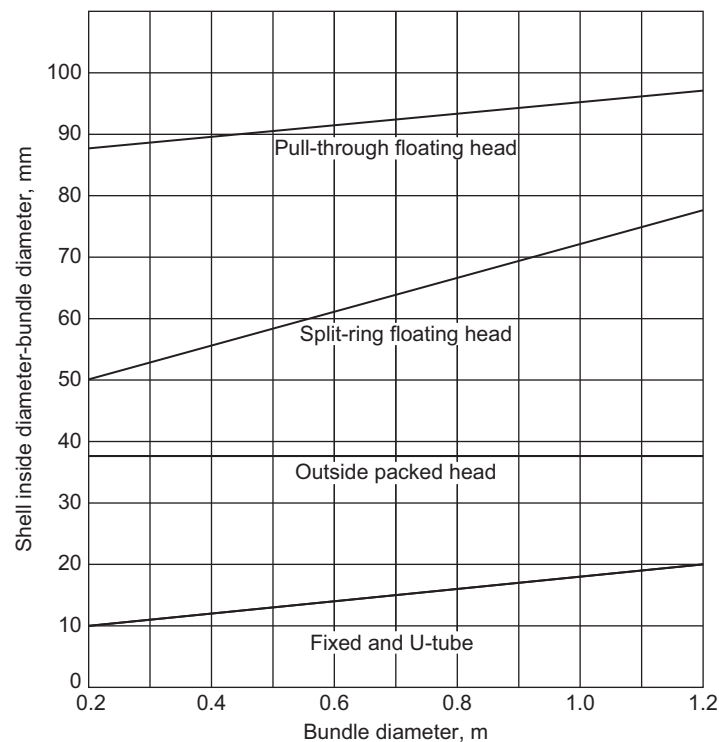


FIG. 19.12 Shell bundle clearance.

TABLE 19.4 Constants for use in Equation 19.3

Triangular pitch, $p_t = 1.25d_o$					
No. passes	1	2	4	6	8
K_1	0.319	0.249	0.175	0.0743	0.0365
n_1	2.142	2.207	2.285	2.499	2.675
Square pitch, $p_t = 1.25d_o$					
No. passes	1	2	4	6	8
K_1	0.215	0.156	0.158	0.0402	0.0331
n_1	2.207	2.291	2.263	2.617	2.643

$$N_t = K_1 \left(\frac{D_b}{d_o} \right)^{n_1}, \quad (19.3a)$$

$$D_b = d_o \left(\frac{N_t}{K_1} \right)^{1/n_1}, \quad (19.3b)$$

where N_t = number of tubes

D_b = bundle diameter, mm

d_o = tube outside diameter, mm

If U-tubes are used, the number of tubes will be slightly less than that given by Equation 19.3a, as the spacing between the two central rows will be determined by the minimum allowable radius for the U-bend. The minimum bend radius will depend on the tube diameter and wall thickness. It will range from 1.5 to 3.0 times the tube outside diameter. The tighter bend radius will lead to some thinning of the tube wall.

An estimate of the number of tubes in a U-tube exchanger (twice the actual number of U-tubes) can be made by reducing the number given by Equation 19.3a by one center row of tubes.

The number of tubes in the center row, the row at the shell equator, is given by:

$$\text{Tubes in center row} = \frac{D_b}{p_t}$$

where p_t = tube pitch, mm.

The tube layout for a particular design will normally be planned with the aid of computer programs. These will allow for the spacing of the pass partition plates and the position of the tie rods. Also, one or two rows of tubes may be omitted at the top and bottom of the bundle to increase the clearance and flow area opposite the inlet and outlet nozzles.

Tube count tables that give an estimate of the number of tubes that can be accommodated in standard shell sizes for commonly used tube sizes, pitches, and number of passes can be found in several books: Kern (1950), Coker (2015), Green and Southard (2018), and Saunders (1988).

Some typical tube arrangements are shown in Appendix H, available in the online material at www.elsevier.com/books-and-journals/book-companion/9780128211793.

19.5.5 Shell types (passes)

The principal shell arrangements are shown in Fig. 19.9. The letters E, F, G, H, and J are those used in the TEMA standards to designate the various types. The E shell is the most commonly used arrangement.

Two shell passes (F shell) are occasionally used where the shell- and tube-side temperature differences are unsuitable for a single pass (see Section 19.6); however, it is difficult to obtain a satisfactory seal with a shell-side baffle, and the same flow arrangement can be achieved by using two shells in series.

The divided flow and split-flow arrangements (G and J shells) are used to reduce the shell-side pressure drop where pressure drop, rather than heat transfer, is the controlling factor in the design.

19.5.6 Shell and tube designation

A common method of describing an exchanger is to designate the number of shell and tube passes: m/n or $m:n$; where m is the number of shell passes and n the number of tube passes. So 1/2 or 1:2 describes an exchanger with one shell pass and two tube passes, and 2/4 an exchanger with two shell passes and four tube passes.

19.5.7 Baffles

Baffles are used in the shell to direct the fluid stream across the tubes to increase the fluid velocity and thus improve the rate of heat transfer. The most commonly used type of baffle is the single segmental baffle shown in Fig. 19.13(a); other types are shown in Fig. 19.13(b, c, and d).

Only the design of exchangers using single segmental baffles will be considered in this chapter.

If the arrangement shown in Fig. 19.13(a) were used with a horizontal condenser, the baffles would restrict the condensate flow. This problem can be overcome either by rotating the baffle arrangement through 90 degrees or by trimming the base of the baffle (Fig. 19.14).

The term “baffle cut” is used to specify the dimensions of a segmental baffle. The baffle cut is the height of the segment removed to form the baffle, expressed as a percentage of the baffle disc diameter. Baffle cuts from 15% to 45% are used. Generally, a baffle cut of 20% to 25% will be the optimum, giving good heat transfer rates without excessive pressure drop. There will be some leakage of fluid round the baffle, as a clearance must be allowed for assembly. The clearance needed depends on the shell diameter; typical values and tolerances are given in Table 19.5.

Another leakage path occurs through the clearance between the tube holes in the baffle and the tubes. The maximum design clearance will normally be $1/32$ in. (0.8 mm).

The minimum thickness to be used for baffles and support plates is given in the standards. The baffle spacings used range from 0.2 to 1.0 shell diameters. A close baffle spacing will give higher heat transfer coefficients, but at the expense of higher pressure drop. The optimum spacing will usually be between 0.3 and 0.5 times the shell diameter.

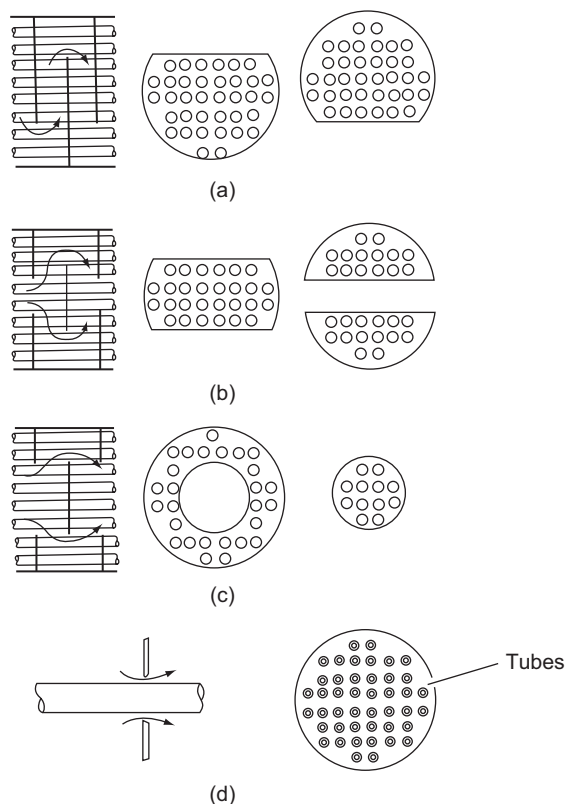


FIG. 19.13 Types of baffle used in shell and tube heat exchangers. (a) Segmental. (b) Segmental and strip. (c) Disc and doughnut. (d) Orifice.

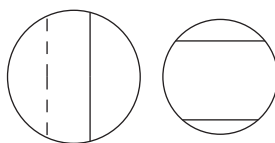


FIG. 19.14 Baffles for condensers.

TABLE 19.5 Typical baffle clearances and tolerances

Shell diameter, D_s	Baffle diameter	Tolerance
<i>Pipe shells</i>		
6–25 in. (152–635 mm)	$D_s - \frac{1}{16}$ in. (1.6 mm)	$+\frac{1}{32}$ in. (0.8 mm)
<i>Plate shells</i>		
6–25 in. (152–635 mm)	$D_s - \frac{1}{8}$ in. (3.2 mm)	$+0, -\frac{1}{32}$ in. (0.8 mm)
27–42 in. (686–1067 mm)	$D_s - \frac{3}{16}$ in. (4.8 mm)	$+0, -\frac{1}{16}$ in. (1.6 mm)

19.5.8 Support plates and tie rods

Where segmental baffles are used, some will be fabricated with closer tolerances, $\frac{1}{64}$ in. (0.4 mm), to act as support plates. For condensers and vaporizers, where baffles are not needed for heat transfer purposes, a few will be installed to support the tubes.

The minimum spacings to be used for support plates are given in the standards. The spacing ranges from around 1 m for 16-mm tubes to 2 m for 25-mm tubes.

The baffles and support plate are held together with tie rods and spacers. The number of rods required depends on the shell diameter and ranges from four 16-mm-diameter rods, for exchangers under 380 mm in diameter; to eight 12.5-mm rods, for exchangers of 1-m diameter. The recommended number for a particular diameter can be found in the standards.

19.5.9 Tubesheets (plates)

In operation, the tubesheets are subjected to the differential pressure between shell and tube sides. The design of tubesheets as pressure vessel components is covered by the ASME BPV Code. Design formulae for calculating tube-sheet thicknesses are also given in the TEMA standards.

The joint between the tubes and tubesheet is normally made by expanding the tube by rolling with special tools (Fig. 19.15). Tube rolling is a skilled task; the tube must be expanded sufficiently to ensure a sound leakproof joint, but not overthinned, thereby weakening the tube. The tube holes are normally grooved (Fig. 19.16a) to lock the tubes more firmly in position and to prevent the joint from being loosened by the differential expansion of the shell and

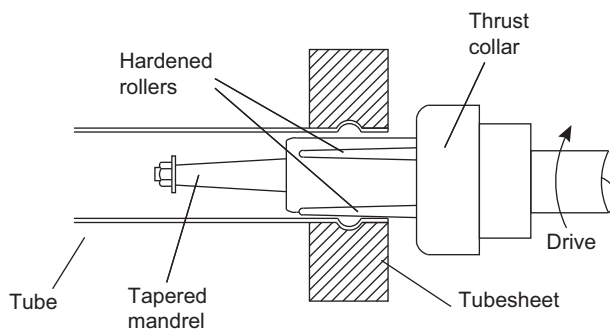


FIG. 19.15 Tube rolling.

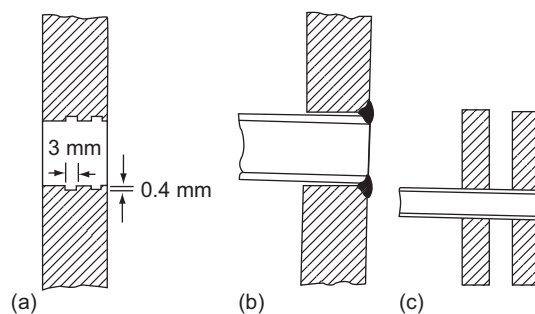


FIG. 19.16 Tube/tubesheet joints.

tubes. When it is essential to guarantee a leakproof joint, the tubes can be welded to the sheet (see Fig. 19.16b). This adds to the cost of the exchanger not only due to the cost of welding but also because a wider tube spacing is needed.

The tubesheet forms the barrier between the shell and tube fluids, and where it is essential for safety or process reasons to prevent any possibility of intermixing due to leakage at the tubesheet joint, double tubesheets can be used, with the space between the sheets vented (see Fig. 19.16c).

To allow sufficient thickness to seal the tubes, the tubesheet thickness should not be less than the tube outside diameter, up to about 25 mm diameter. Recommended minimum plate thicknesses are given in the standards.

The thickness of the tubesheet will reduce the effective length of the tube slightly, and this should be allowed for when calculating the area available for heat transfer. As a first approximation, the length of the tubes can be reduced by 25 mm for each tubesheet.

19.5.10 Shell and header nozzles (branches)

Standard pipe sizes are used for the inlet and outlet nozzles. It is important to avoid flow restrictions at the inlet and outlet nozzles to prevent excessive pressure drop and flow-induced vibration of the tubes. As well as omitting some tube rows (see Section 19.5.4), the baffle spacing is usually increased in the nozzle zone to increase the flow area. For vapors and gases, where the inlet velocities will be high, the nozzle may be flared, or special designs used, to reduce the inlet velocities (Fig. 19.17a and b). The extended shell design shown in Fig. 19.19(b) also serves as an impingement plate. Impingement plates are used when the shell-side fluid contains liquid drops or for high-velocity fluids containing abrasive particles.

19.5.11 Flow-induced tube vibrations

Premature failure of exchanger tubes can occur through vibrations induced by the shell-side fluid flow. Care must be taken in the mechanical design of large exchangers where the shell-side velocity is high, say greater than 3 m/s, to ensure that tubes are adequately supported.

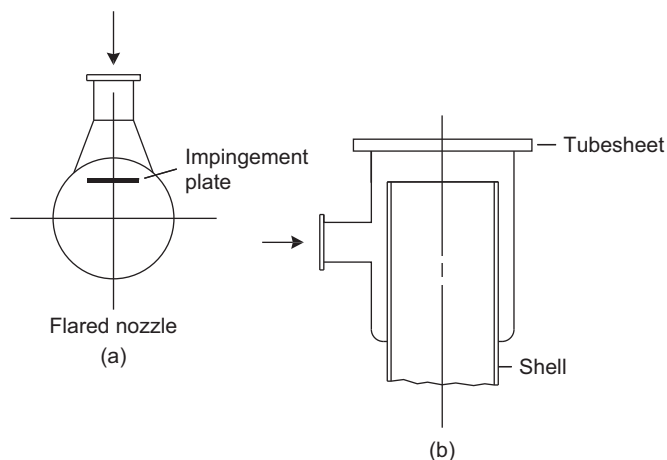


FIG. 19.17 Inlet nozzle designs.

The vibration induced by the fluid flowing over the tube bundle is caused principally by vortex shedding and turbulent buffeting. As fluid flows over a tube, vortices are shed from the downstream side, causing disturbances in the flow pattern and pressure distribution around the tube. Turbulent buffeting of tubes occurs at high flow rates due to the intense turbulence at high Reynolds numbers.

The buffeting caused by vortex shedding or by turbulent eddies in the flow stream will cause vibration, but large-amplitude vibrations will normally only occur above a certain critical flow velocity. Above this velocity, the interaction with the adjacent tubes can provide a feedback path that reinforces the vibrations. Resonance will also occur if the vibrations approach the natural vibration frequency of the unsupported tube length. Under these conditions, the magnitude of the vibrations can increase dramatically, leading to tube failure. Failure can occur either through the impact of one tube on another or through wear on the tube where it passes through the baffles.

For most exchanger designs, following the recommendations on support sheet spacing given in the standards will be sufficient to protect against premature tube failure from vibration. For large exchangers with high velocities on the shell side, the design should be analyzed to check for possible vibration problems. The computer-aided design programs for shell and tube exchanger design available from commercial organizations, such as HTFS and HTRI (see Section 19.1), include programs for vibration analysis.

Much work has been done on tube vibration over the past 20 years due to an increase in the failure of exchangers as larger sizes and higher flow rates have been used. Discussion of this work is beyond the scope of this book; for review of the methods used, see [Saunders \(1988\)](#) and [Singh and Soler \(1992\)](#). See also the Engineering Science Data Unit Design Guide ESDU 87019, which gives a clear explanation of mechanisms causing tube vibration in shell and tube heat exchangers and their prediction and prevention.

19.6 Mean temperature difference (temperature driving force)

Before Equation 19.1 can be used to determine the heat-transfer area required for a given duty, an estimate of the mean temperature difference, ΔT_m , must be made. This will normally be calculated from the terminal temperature differences: the difference in the fluid temperatures at the inlet and outlet of the exchanger. The well-known “logarithmic mean” temperature difference is only applicable to sensible heat transfer in true co-current or counter-current flow, with linear temperature–enthalpy curves. This situation occurs when the heat capacities of both streams are constant and there is no phase change or if there is a phase change at constant pressure for a stream that contains a single component. These conditions are only approximated in reality. For countercurrent flow ([Fig. 19.18a](#)), the logarithmic mean temperature difference is given by:

$$\Delta T_{\text{lm}} = \frac{(T_1 - t_2) - (T_2 - t_1)}{\ln \frac{(T_1 - t_2)}{(T_2 - t_1)}} \quad (19.4)$$

where ΔT_{lm} = log mean temperature difference

T_1 = hot fluid temperature, inlet

T_2 = hot fluid temperature, outlet

t_1 = cold fluid temperature, inlet

t_2 = cold fluid temperature, outlet

The equation is the same for co-current flow, but the terminal temperature differences will be $(T_1 - t_1)$ and $(T_2 - t_2)$. Strictly, Equation 19.4 will only apply when there is no change in the specific heats, the overall heat transfer coefficient is constant, and there are no heat losses. In design, these conditions can be assumed to be satisfied providing the temperature change in each fluid stream is not large.

In most shell and tube exchangers, the flow will be a mixture of co-current, countercurrent and, cross flow. [Fig. 19.18\(b and c\)](#) show typical temperature profiles for an exchanger with one shell pass and two tube passes (a 1:2 exchanger). [Fig. 19.18\(c\)](#) shows two different cases of temperature cross, where the outlet temperature of the cold stream is above that of the hot stream.

The usual practice in the design of shell and tube exchangers is to estimate the “true temperature difference” from the logarithmic mean temperature by applying a correction factor to allow for the departure from true countercurrent flow:

$$\Delta T_m = F_t \Delta T_{\text{lm}} \quad (19.5)$$

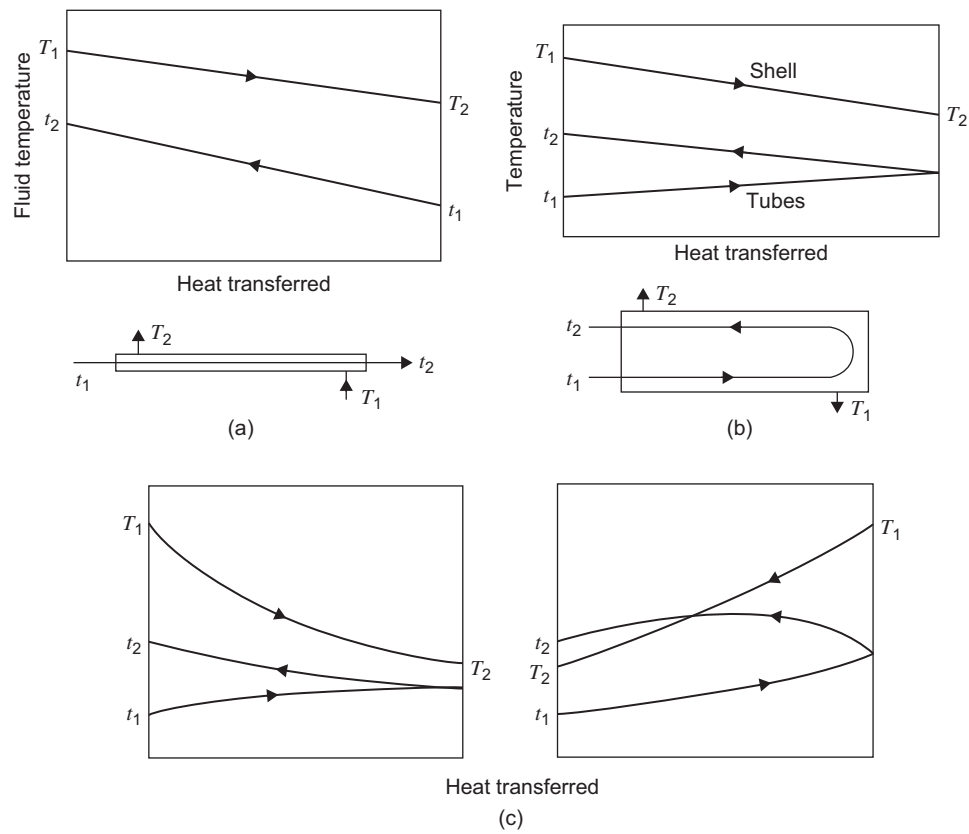


FIG. 19.18 Temperature profiles. (a) Countercurrent flow. (b) 1:2 exchanger. (c) Temperature cross.

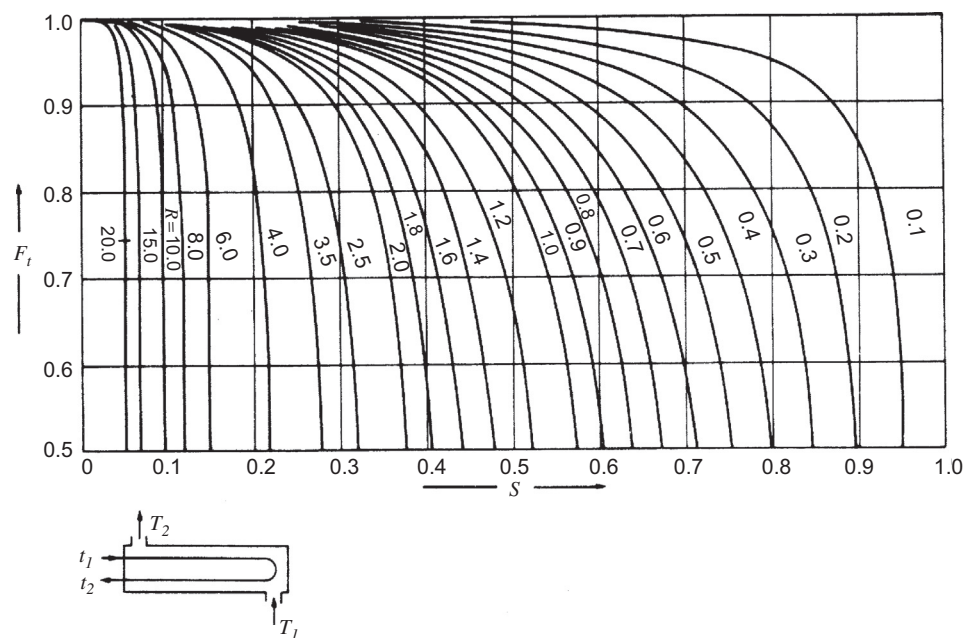


FIG. 19.19 Temperature correction factor: one-shell pass; two or more even-tube passes.

where ΔT_m = true temperature difference, the mean temperature difference for use in the design in Equation 19.1

F_t = the temperature correction factor

The correction factor is a function of the shell and tube fluid temperatures and the number of tube and shell passes. It is normally correlated as a function of two dimensionless temperature ratios:

$$R = \frac{(T_1 - T_2)}{(t_2 - t_1)} \quad (19.6)$$

and

$$S = \frac{(t_2 - t_1)}{(T_1 - t_1)} \quad (19.7)$$

R is equal to the shell-side fluid flow rate times the fluid mean specific heat divided by the tube-side fluid flow rate times the tube-side fluid specific heat.

S is a measure of the temperature efficiency of the exchanger.

For a 1 shell:2 tube pass exchanger, the correction factor is given by:

$$F_t = \frac{\sqrt{(R^2 + 1)} \ln \left[\frac{(1 - S)}{(1 - RS)} \right]}{(R - 1) \ln \left[\frac{2 - S[R + 1 - \sqrt{(R^2 + 1)}]}{2 - S[R + 1 + \sqrt{(R^2 + 1)}]} \right]} \quad (19.8)$$

The derivation of Equation 19.8 is given by Kern (1950). The equation for a 1 shell:2 tube pass exchanger can be used for any exchanger with an even number of tube passes and is plotted in Fig. 19.19. The correction factor for two shell passes and four, or multiples of four, tube passes is shown in Fig. 19.20 and that for divided and split-flow shells in Figs. 19.21 and 19.22.

Temperature correction factor plots for other arrangements can be found in the TEMA standards and the books by Kern (1950) and Coker (2015). Mueller (1973) gives a comprehensive set of figures for calculating the log mean temperature correction factor, which includes figures for cross-flow exchangers.

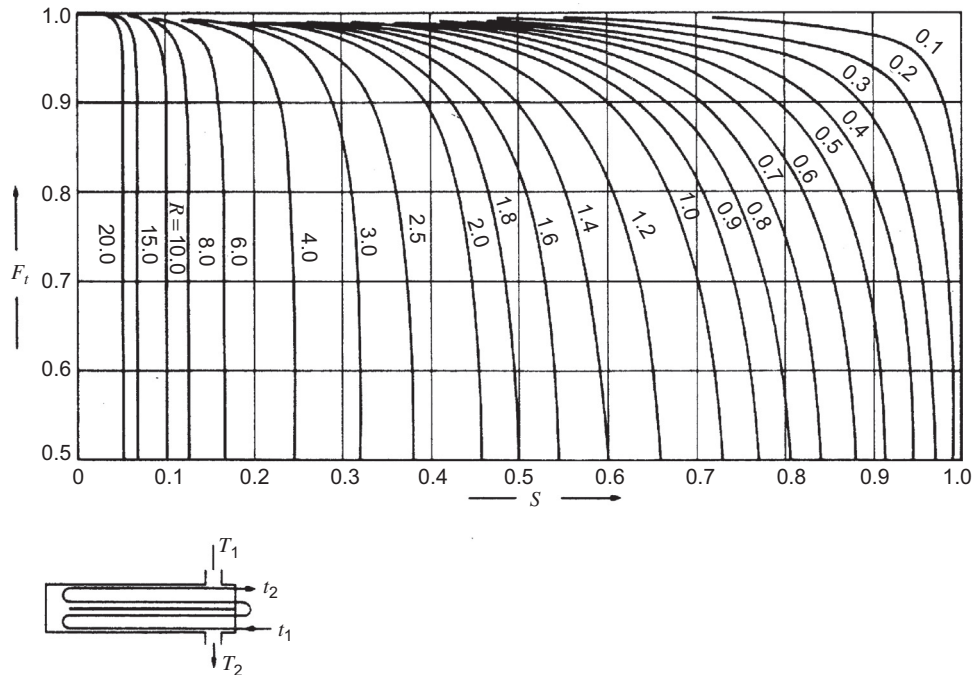


FIG. 19.20 Temperature correction factor: two-shell passes; four or multiples of four tube passes.

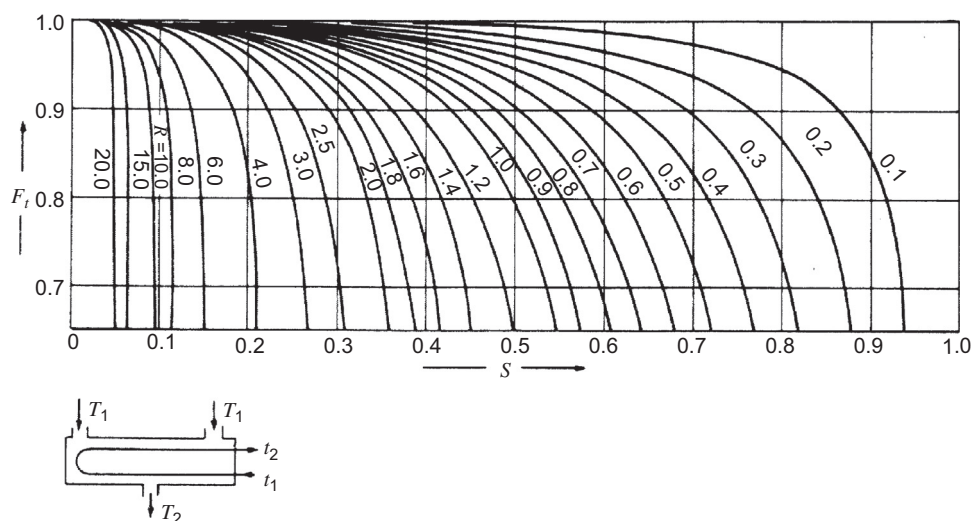


FIG. 19.21 Temperature correction factor: divided-flow shell; two or more even-tube passes.

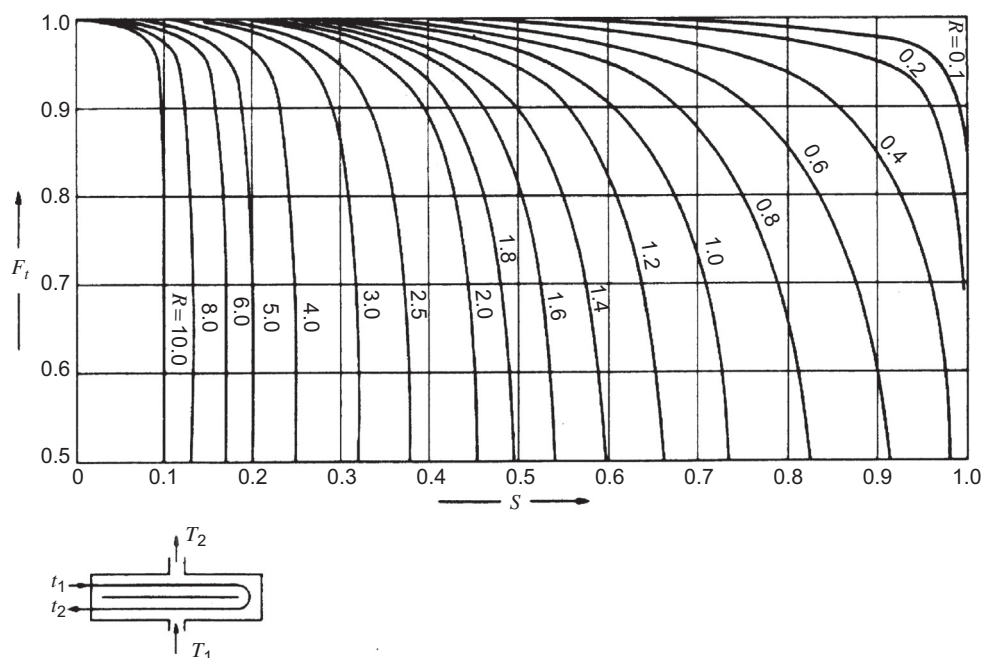


FIG. 19.22 Temperature correction factor, split-flow shell, two-tube pass.

The following assumptions are made in the derivation of the temperature correction factor, F_t , in addition to those made for the calculation of the log mean temperature difference:

1. Equal heat transfer areas in each pass.
2. A constant overall heat transfer coefficient in each pass.
3. The temperature of the shell-side fluid in any pass is constant across any cross-section.
4. There is no leakage of fluid between shell passes.

Though these conditions will not be strictly satisfied in practical heat exchangers, the F_t values obtained from the curves will give an estimate of the “true mean temperature difference” that is sufficiently accurate for most designs. [Mueller \(1973\)](#) discusses these assumptions and gives F_t curves for conditions when all the assumptions are not met; see also [Butterworth \(1973\)](#) and [Emerson \(1973\)](#). Values of F_t are calculated for heat exchangers in most process simulation programs, as described in [Chapter 4](#).

The shell-side leakage and bypass streams (see [Section 19.9](#)) will affect the mean temperature difference, but are not normally taken into account when estimating the correction factor F_t . [Fisher and Parker \(1969\)](#) give curves that show the effect of leakage on the correction factor for a one shell pass:two tube pass exchanger.

The value of F_t will be close to 1 when the terminal temperature differences are large, but will appreciably reduce the logarithmic mean temperature difference when the temperatures of shell and tube fluids approach each other; it will fall drastically when there is a temperature cross. A temperature cross will occur if the outlet temperature of the cold stream is greater than the outlet temperature of the hot stream (see [Fig. 19.18c](#)).

Where the F_t curve is near vertical, values cannot be read accurately, which will introduce a considerable uncertainty into the design.

An economic exchanger design cannot normally be achieved if the correction factor F_t falls below about 0.75. In these circumstances, an alternative type of exchanger should be considered that gives a closer approach to true countercurrent flow. The use of two or more shells in series, or multiple shell-side passes, will give a closer approach to true countercurrent flow and should be considered where a temperature cross is likely to occur.

When both sensible and latent heat are transferred, it will be necessary to divide the temperature profile into sections and calculate the mean temperature difference for each section. The overall heat transfer coefficient should also be different in each section.

19.7 Shell and tube exchangers: General design considerations

19.7.1 Fluid allocation: Shell or tubes

Where no phase change occurs, the following factors determine the allocation of the fluid streams to the shell or tubes.

Corrosion. The more corrosive fluid should be allocated to the tube side. This will reduce the cost of expensive alloy or clad components.

Fouling. The fluid that has the greatest tendency to foul the heat transfer surfaces should be placed in the tubes. This gives better control over the design fluid velocity, and the higher allowable velocity in the tubes will reduce fouling. Also, the tubes will be easier to clean.

Fluid temperatures. If the temperatures are high enough to require the use of special alloys, placing the higher-temperature fluid in the tubes will reduce the overall cost. At moderate temperatures, placing the hotter fluid in the tubes will reduce the shell surface temperatures and hence the need for lagging to reduce heat loss, or for safety reasons.

Operating pressures. The higher-pressure stream should be allocated to the tube side. High-pressure tubes will be cheaper than a high-pressure shell. The required tube thickness is less for high internal pressure than high external pressure, and an expensive high-pressure shell may be avoided.

Pressure drop. For the same pressure drop, higher heat transfer coefficients will be obtained on the tube side than the shell side, and fluid with the lowest allowable pressure drop should be allocated to the tube side.

Viscosity. Generally, a higher heat transfer coefficient will be obtained by allocating the more viscous material to the shell side, provided the flow is turbulent. The critical Reynolds number for turbulent flow in the shell is in the region of 200. If turbulent flow cannot be achieved in the shell, it is better to place the fluid in the tubes, as the tube-side heat transfer coefficient can be predicted with more certainty.

Stream flow rates. Allocating the fluids with the lowest flow rate to the shell side will normally give the most economical design.

19.7.2 Shell and tube fluid velocities

High velocities will give high heat transfer coefficients but also a high pressure drop. The velocity must be high enough to prevent any suspended solids settling, but not so high as to cause erosion. High velocities will reduce fouling. Plastic inserts are sometimes used to reduce erosion at the tube inlet. Typical design velocities are given next.

Liquids

Tube-side process fluids: 1 to 2 m/s, maximum 4 m/s if required to reduce fouling; water: 1.5 to 2.5 m/s. Shell side: 0.3 to 1 m/s.

Vapors

For vapors, the velocity used will depend on the operating pressure and fluid density; the lower values in the ranges given here will apply to high-molecular-weight materials.

Vacuum	50 to 70 m/s
Atmospheric pressure	10 to 30 m/s
High pressure	5 to 10 m/s

19.7.3 Stream temperatures

The closer the temperature approach used (the difference between the temperatures of the two streams at a given point, usually calculated at the two ends of the exchanger), the larger the heat transfer area required for a given duty. The optimum value depends on the application and can only be determined by making an economic analysis of alternative designs. As a general guide, the optimum temperature approach will usually be in the range of 10 °C to 30 °C for heat exchange between process streams. Lower-temperature approaches are used for coolers, and a temperature approach of 5 °C to 7 °C for coolers using cooling water and 3 °C to 5 °C using refrigerated brines is common. The maximum temperature rise in recirculated cooling water is limited to around 30 °C. Care should be taken to ensure that cooling media temperatures are kept well above the freezing point of the process materials. Temperature approaches as low as 1 °C or 2 °C are used in very low-temperature subambient processes, such as air separation and natural gas liquefaction. When the heat exchange is between process fluids for heat recovery, the optimum approach temperature can be determined by pinch analysis, as described in [Chapter 3](#). The optimum temperature approach for heat recovery depends on the trade-off between capital and energy costs (see Fig. 3.16). The numerical optimum temperature approach is seldom lower than 20 °C, but lower values are often used, as lower-temperature approaches lead to more conservative designs with more exchanger area and greater potential heat recovery.

19.7.4 Pressure drop

In many applications, the pressure drop available to drive the fluids through the exchanger will be set by the process conditions, and the available pressure drop will vary from a few millibars in vacuum service to several bars in pressure systems.

When the designer is free to select the pressure drop, an economic analysis can be made to determine the exchanger design that gives the lowest operating costs, taking into consideration both capital and pumping costs; however, a full economic analysis will only be justified for very large, expensive exchangers. The values suggested below can be used as a general guide and will normally give designs that are near the optimum.

Liquids

Viscosity	Allowable pressure drop
<1 mN s/m ²	35 kN/m ²
1–10 mN s/m ²	50–70 kN/m ²

Gas and vapors

High vacuum	0.4–0.8 kN/m ²
Medium vacuum	0.1 × absolute pressure
1–2 bar	0.5 × system gauge pressure
Above 10 bar	0.1 × system gauge pressure

When a high pressure drop is used, care must be taken to ensure that the resulting high fluid velocity does not cause erosion or flow-induced tube vibration.

19.7.5 Fluid physical properties

The fluid physical properties required for heat exchanger design are density, viscosity, thermal conductivity, and temperature–enthalpy correlations (specific and latent heats). Physical properties are usually obtained from a process simulation model; see [Chapter 4](#). The thermal conductivities of commonly used tube materials are given in [Table 19.6](#).

In the correlations used to predict heat transfer coefficients, the physical properties are usually evaluated at the mean stream temperature. This is satisfactory when the temperature change is small, but can cause a significant error when the change in temperature is large. In these circumstances, a simple and safe procedure is to evaluate the heat transfer coefficients at the stream inlet and outlet temperatures and use the lower of the two values. Alternatively, the method suggested by [Frank \(1978\)](#) can be used, in which Equations 19.1 and 19.3 are combined:

$$Q = \frac{A[U_2(T_1 - t_2) - U_1(T_2 - t_1)]}{\ln \left[\frac{U_2(T_1 - t_2)}{U_1(T_2 - t_1)} \right]} \quad (19.9)$$

where U_1 and U_2 are evaluated at the ends of the exchanger. Equation 19.9 is derived by assuming that the heat transfer coefficient varies linearly with temperature.

If the variation in the physical properties is too large for these simple methods to be used, it will be necessary to divide the temperature–enthalpy profile into sections and evaluate the heat transfer coefficients and area required for each section.

TABLE 19.6 Conductivity of metals

Metal	Temperature (°C)	k_w (W/m°C)
Aluminum	0	202
	100	206
Brass (70 Cu, 30 Zn)	0	97
	100	104
	400	116
Copper	0	388
	100	378
Nickel	0	62
	212	59
Cupro-nickel (10% Ni)	0–100	45
Monel	0–100	30
Stainless steel (18/8)	0–100	16
Carbon steel	40	60
	100	58
	260	51
Titanium	0–100	16

19.8 Tube-side heat transfer coefficient and pressure drop (single phase)

19.8.1 Heat transfer

Turbulent flow

Heat transfer data for turbulent flow inside conduits of uniform cross-section are usually correlated by an equation of the form:

$$\text{Nu} = C \text{Re}^a \text{Pr}^b \left(\frac{\mu}{\mu_w} \right)^c \quad (19.10)$$

where Nu = Nusselt number = $(h_i d_e / k_f)$
 Re = Reynolds number = $(\rho u_t d_e / \mu) = (G_t d_e / \mu)$
 Pr = Prandtl number = $(C_p \mu / k_f)$
 h_i = inside coefficient, $\text{W/m}^2\text{°C}$
 d_e = equivalent (or hydraulic mean) diameter, m

$$d_e = \frac{4 \times \text{cross-sectional area for flow}}{\text{wetted perimeter}} = d_i \text{ for tubes,}$$

u_t = fluid velocity, m/s
 k_f = fluid thermal conductivity, $\text{W/m}^{\circ}\text{C}$
 G_t = mass velocity, mass flow per unit area, $\text{kg/m}^2\text{s}$
 μ = fluid viscosity at the bulk fluid temperature, Ns/m^2
 μ_w = fluid viscosity at the wall
 C_p = fluid-specific heat, heat capacity, $\text{J/kg}^{\circ}\text{C}$

The index for the Reynolds number, a , is generally taken as 0.8. That for the Prandtl number, b , can range from 0.3 for cooling to 0.4 for heating. The index for the viscosity factor, c , is normally taken as 0.14 for flow in tubes, from the work of [Sieder and Tate \(1936\)](#), but some workers report higher values. A general equation that can be used for exchanger design is:

$$\text{Nu} = C \text{Re}^{0.8} \text{Pr}^{0.33} \left(\frac{\mu}{\mu_w} \right)^{0.14} \quad (19.11)$$

where $C = 0.021$ for gases
 $= 0.023$ for nonviscous liquids
 $= 0.027$ for viscous liquids

It is not possible to find values for the constant and indices to cover the complete range of process fluids, from gases to viscous liquids, but the values predicted using Equation 19.11 should be sufficiently accurate for design purposes. The uncertainty in the prediction of the shell-side coefficient and fouling factors will usually far outweigh any error in the tube-side value. Where a more accurate prediction than that given by Equation 19.11 is required, and justified, the data and correlations given in the Engineering Science Data Unit (ESDU) reports are recommended: ESDU 92003 and 93018 (1998).

[Butterworth \(1977\)](#) gives the following equation, which is based on the ESDU work:

$$\text{St} = E \text{Re}^{-0.205} \text{Pr}^{-0.505} \quad (19.12)$$

where St = Stanton number = $(\text{Nu} / \text{RePr}) = (h_i / \rho u_t C_p)$
 and $E = 0.0225 \exp(-0.0225(\ln \text{Pr})^2)$

Equation 19.12 is applicable at Reynolds numbers greater than 10,000.

Hydraulic mean diameter

In some texts the equivalent (hydraulic mean) diameter is defined differently for use in calculating the heat transfer coefficient in a conduit or channel than for calculating the pressure drop. The perimeter through which the heat is transferred is used in place of the total wetted perimeter. In practice, the use of d_e calculated either

way will make little difference to the value of the estimated overall coefficient, as the film coefficient is only roughly proportional to $d_e^{-0.2}$.

It is the full wetted perimeter that determines the flow regime and the velocity gradients in a channel. So, in this book, d_e determined using the full wetted perimeter will be used for both pressure drop and heat transfer calculations. The actual area through which the heat is transferred should, of course, be used to determine the rate of heat transfer (Equation 19.1).

Laminar flow

Below a Reynolds number of about 2000, the flow in pipes will be laminar. Providing the natural convection effects are small, which will normally be so in forced convection, the following equation can be used to estimate the film heat transfer coefficient:

$$\text{Nu} = 1.86 (\text{RePr})^{0.33} \left(\frac{d_e}{L} \right)^{0.33} \left(\frac{\mu}{\mu_w} \right)^{0.14} \quad (19.13)$$

where L is the length of the tube in meters.

If the Nusselt number given by Equation 19.13 is less than 3.5, it should be taken as 3.5.

In laminar flow, the length of the tube can have a marked effect on the heat transfer rate for length-to-diameter ratios less than 500.

Transition region

In the flow region between laminar and fully developed turbulent flow, heat transfer coefficients cannot be predicted with certainty, as the flow in this region is unstable. The transition region should be avoided in exchanger design. If this is not practicable, the coefficient should be evaluated using both Equations 19.11 and 19.13 and the lower value taken.

Heat transfer factor, j_h

It is often convenient to correlate heat transfer data in terms of a heat transfer “ j ” factor, which is similar to the friction factor used for pressure drop. The heat transfer factor is defined by:

$$j_h = \text{StPr}^{0.67} \left(\frac{\mu}{\mu_w} \right)^{-0.14} \quad (19.14)$$

The use of the j_h factor allows data for laminar and turbulent flow to be represented on the same graph (Fig. 19.23). The j_h values obtained from Fig. 19.23 can be used with Equation 19.14 to estimate the heat transfer coefficients for heat exchanger tubes and commercial pipes. The coefficient estimated for pipes will normally be conservative (on the low side), as pipes are rougher than the tubes used for heat exchangers, which are finished to closer tolerances. Equation 19.14 can be rearranged to a more convenient form:

$$\frac{h_i d_i}{k_f} = j_h \text{RePr}^{0.33} \left(\frac{\mu}{\mu_w} \right)^{0.14} \quad (19.15)$$

Note. Kern (1950) and others define the heat transfer factor as:

$$j_H = \text{NuPr}^{-1/3} \left(\frac{\mu}{\mu_w} \right)^{-0.14}$$

The relationship between j_h and j_H is given by:

$$j_H = j_h \text{Re}$$

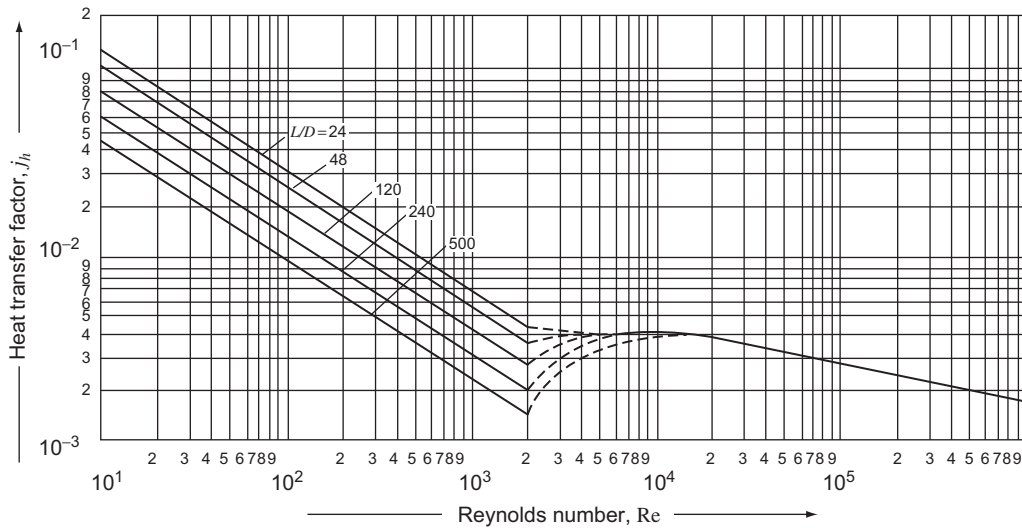


FIG. 19.23 Tube-side heat transfer factor.

Viscosity correction factor

The viscosity correction factor will normally only be significant for viscous liquids. To apply the correction, an estimate of the wall temperature is needed. This can be made by first calculating the coefficient without the correction and using the following relationship to estimate the wall temperature:

$$h_i(t_w - t) = U(T - t) \quad (19.16)$$

where t = tube-side bulk temperature (mean)

t_w = estimated wall temperature

T = shell-side bulk temperature (mean)

Usually an approximate estimate of the wall temperature is sufficient, but trial-and-error calculations can be made to obtain a better estimate if the correction is large.

Coefficients for water

Though Equations 19.11 and 19.13 and Fig. 19.23 may be used for water, a more accurate estimate can be made by using equations developed specifically for water. The physical properties are conveniently incorporated into the correlation. The following equation has been adapted from data given by Eagle and Ferguson (1930):

$$h_i = \frac{4200(1.35 + 0.02t)u_t^{0.8}}{d_i^{0.2}} \quad (19.17)$$

where h_i = inside coefficient for water, $\text{W/m}^2\text{°C}$

t = water temperature, °C

u_t = water velocity, m/s

d_i = tube inside diameter, mm

19.8.2 Tube-side pressure drop

There are two major sources of pressure loss on the tube side of a shell and tube exchanger: the friction loss in the tubes and the losses due to the sudden contraction and expansion and flow reversals that the fluid experiences in flow through the tube arrangement.

The tube friction loss can be calculated using the familiar equations for pressure drop in pipes (see [Chapter 20](#)). The basic equation for isothermal flow in pipes (constant temperature) is:

$$\Delta P = 8j_f \left(\frac{L'}{d_i} \right) \frac{\rho u_t^2}{2} \quad (19.18)$$

where j_f is the dimensionless friction factor and L' is the effective pipe length.

The flow in a heat exchanger is clearly not isothermal, and this is allowed for by including an empirical correction factor to account for the change in physical properties with temperature. Normally only the change in viscosity is considered:

$$\Delta P = 8j_f (L' / d_i) \rho \frac{u_t^2}{2} \left(\frac{\mu}{\mu_w} \right)^{-m} \quad (19.19)$$

where $m = 0.25$ for laminar flow, $Re < 2100$
 $= 0.14$ for turbulent flow, $Re > 2100$

Values of j_f for heat exchanger tubes can be obtained from [Fig. 19.24](#); commercial pipes are given in [Chapter 20](#).

The pressure losses due to contraction at the tube inlets, expansion at the exits, and flow reversal in the headers can be a significant part of the total tube-side pressure drop. There is no entirely satisfactory method for estimating these losses. [Kern \(1950\)](#) suggests adding four velocity heads per pass. [Frank \(1978\)](#) considers this to be too high and recommends 2.5 velocity heads. [Butterworth \(1978\)](#) suggests 1.8. [Lord et al. \(1970\)](#) take the loss per pass as equivalent to a length of tube equal to 300 tube diameters for straight tubes and 200 for U-tubes, whereas [Evans \(1980\)](#) appears to add only 67 tube diameters per pass.

The loss in terms of velocity heads can be estimated by counting the number of flow contractions, expansions, and reversals and using the factors for pipe fittings to estimate the number of velocity heads lost. For two tube passes, there will be two contractions, two expansions, and one flow reversal. The head loss for each of these effects is contraction 0.5, expansion 1.0, and 180-degree bend 1.5, so for two passes, the maximum loss will be:

$$\begin{aligned} 2 \times 0.5 + 2 \times 1.0 + 1.5 &= 4.5 \text{ velocity heads} \\ &= \underline{\underline{2.25 \text{ per pass}}} \end{aligned}$$

From this, it appears that Frank's recommended value of 2.5 velocity heads per pass is the most realistic value to use. Combining this factor with Equation 19.19 gives:

$$\Delta P_t = N_p \left[8j_f \left(\frac{L}{d_i} \right) \left(\frac{\mu}{\mu_w} \right)^{-m} + 2.5 \right] \frac{\rho u_t^2}{2} \quad (19.20)$$

where ΔP_t = tube-side pressure drop, N/m² (Pa)

N_p = number of tube-side passes

u_t = tube-side velocity, m/s

L = length of one tube

Another source of pressure drop is the flow expansion and contraction at the exchanger inlet and outlet nozzles. This can be estimated by adding 1 velocity head for the inlet and 0.5 for the outlet, based on the nozzle velocities.

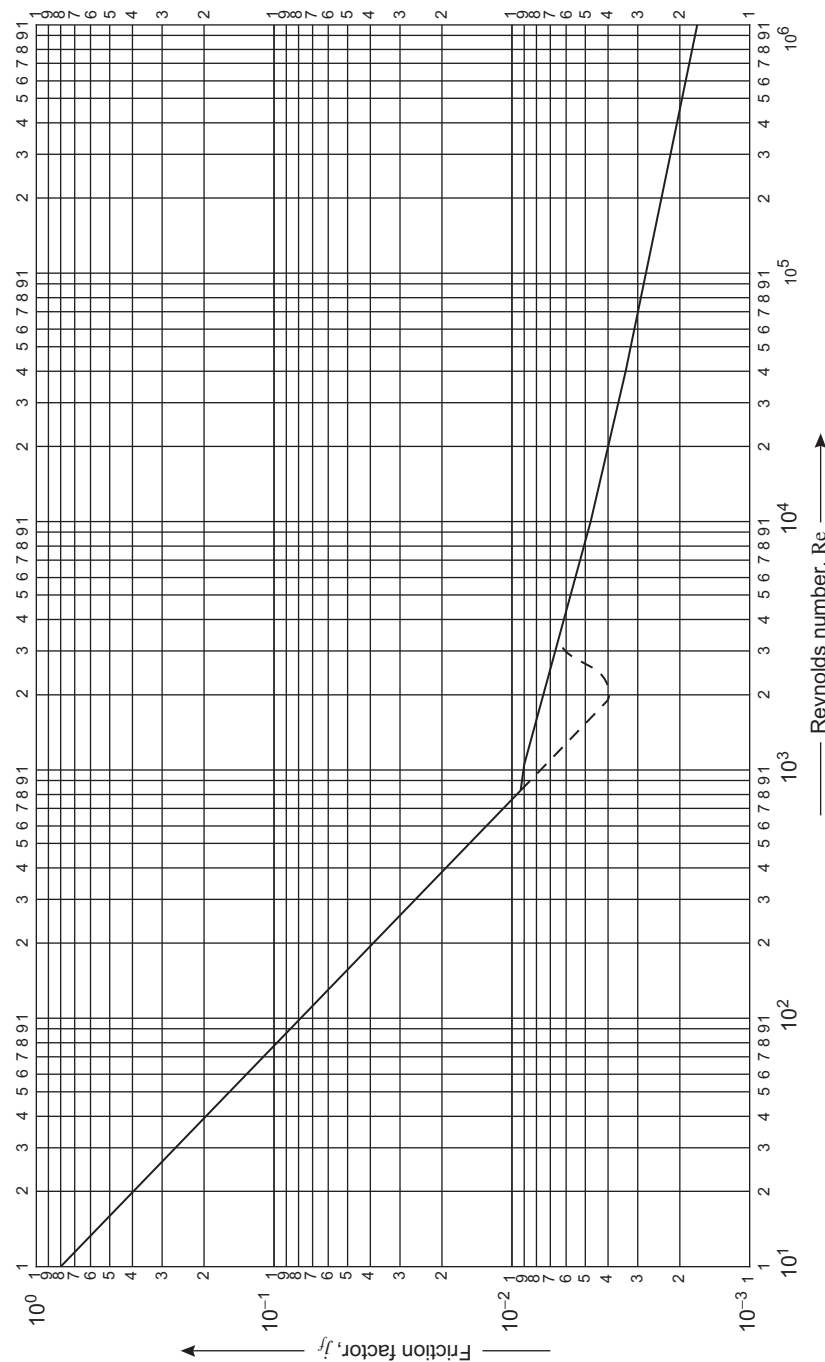


FIG. 19.24 Tube-side friction factors. *Note:* The friction factor j_f is the same as the friction factor for pipes ϕ ($= (R/\rho u^2)$).

19.9 Shell-side heat transfer and pressure drop (single phase)

19.9.1 Flow pattern

The flow pattern in the shell of a segmentally baffled heat exchanger is complex, and this makes the prediction of the shell-side heat transfer coefficient and pressure drop much more difficult than for the tube side. Though the baffles are installed to direct the flow across the tubes, the actual flow of the main stream of fluid will be a mixture of cross flow between the baffles, coupled with axial (parallel) flow in the baffle windows, as shown in Fig. 19.25. Not all the fluid flow follows the path shown in Fig. 19.25; some will leak through gaps formed by the clearances that have to be allowed for fabrication and assembly of the exchanger. These leakage and bypass streams are shown

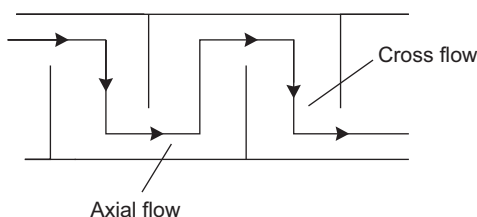


FIG. 19.25 Idealized main stream flow.

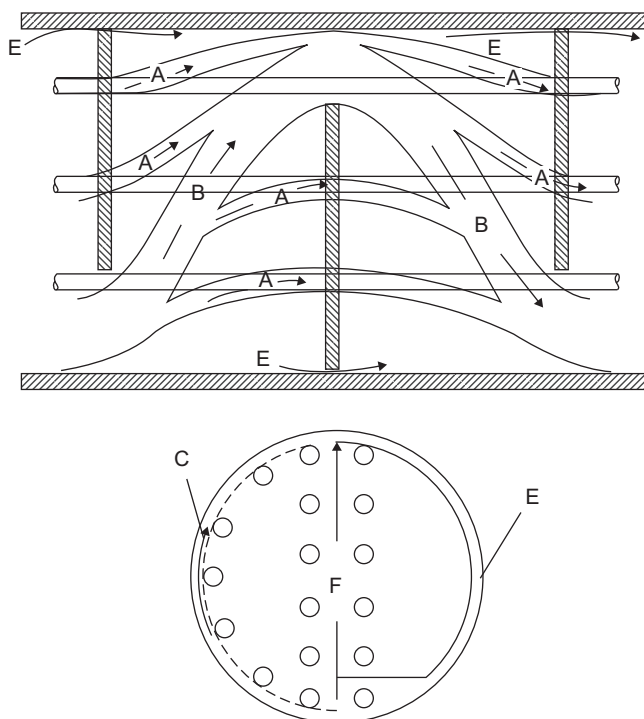


FIG. 19.26 Shell-side leakage and bypass paths.

in Fig. 19.26, which is based on the flow model proposed by Tinker (1951, 1958). In Fig. 19.26, Tinker's nomenclature is used to identify the various streams, as follows:

Stream A is the tube-to-baffle leakage stream, which is the fluid flowing through the clearance between the tube outside diameter and the tube hole in the baffle.

Stream B is the actual cross-flow stream.

Stream C is the bundle-to-shell bypass stream, which is the fluid flowing in the clearance area between the outer tubes in the bundle (bundle diameter) and the shell.

Stream E is the baffle-to-shell leakage stream, which is the fluid flowing through the clearance between the edge of a baffle and the shell wall.

Stream F is the pass-partition stream, which is the fluid flowing through the gap in the tube arrangement due to the pass partition plates. Where the gap is vertical, it will provide a low-pressure drop path for fluid flow.

Note. There is no stream D.

The fluid in streams C, E, and F bypasses the tubes, reducing the effective heat transfer area.

Stream C is the main bypass stream and is particularly significant in pull-through bundle exchangers, where the clearance between the shell and bundle is necessarily large. Stream C can be considerably reduced by using sealing strips; horizontal strips that block the gap between the bundle and the shell (Fig. 19.27). Dummy tubes are also sometimes used to block the pass-partition leakage, stream F.

The tube-to-baffle leakage stream, A, does not bypass the tubes, and its main effect is on pressure drop rather than heat transfer.

The clearances will tend to plug as the exchanger becomes fouled, and this will increase the pressure drop; see Section 19.9.6.

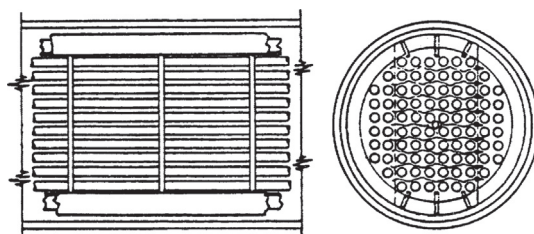


FIG. 19.27 Sealing strips.

19.9.2 Design methods

The complex flow pattern on the shell side, and the great number of variables involved, make it difficult to predict the shell-side heat transfer coefficient and pressure drop with complete assurance. In methods used for the design of exchangers before about 1960, no attempt was made to account for the leakage and bypass streams. Correlations were based on the total stream flow, and empirical methods were used to account for the performance of real exchangers compared with that for cross flow over ideal tube banks. Typical of these “bulk-flow” methods are those of Kern (1950) and Donohue (1955). Reliable predictions can only be achieved by comprehensive analysis of the contribution to heat transfer and pressure drop made by the individual streams shown in Fig. 19.26. Tinker (1951, 1958) published the first detailed stream-analysis method for predicting shell-side heat transfer coefficients and pressure drop, and the methods subsequently developed have been based on his model. Tinker’s presentation is difficult to follow, and his method is difficult and tedious to apply in manual calculations. It has been simplified by Devore (1961, 1962) using standard tolerances for commercial exchangers and only a limited number of baffle cuts. Devore gives nomographs that facilitate the application of the method in manual calculations. Mueller (1973) has further simplified Devore’s method and gives an illustrative example.

Bell (1960, 1963) developed a semi-analytical method based on work done in the cooperative research program on shell and tube exchangers at the University of Delaware. His method accounts for the major bypass and leakage streams and is suitable for a manual calculation.

The ESDU has also published a method for estimating shell-side the pressure drop and heat transfer coefficient: ESDU Design Guide 83038 (1984). The method is based on a simplification of Tinker’s work. It can be used for hand calculations, but as iterative procedures are involved, it is best programmed for use with personal computers.

Tinker’s model has been used as the basis for the proprietary computer methods developed by Heat Transfer Research Incorporated; see Palen and Taborek (1969), and by Heat Transfer and Fluid Flow Services; see Grant (1973). The HTRI method and software are available from HTRI (www.htri.net). The HTFS programs are available in process simulation programs such as Aspen Technology’s Aspen Engineering Suite and Honeywell’s UniSim Design Suite; see Chapter 4. The use of the HTFS programs is illustrated in Example 19.4.

Though Kern’s method does not take account of the bypass and leakage streams, it is simple to apply and is accurate enough for preliminary design calculations and for designs where uncertainty in other design parameters is such that the use of more elaborate methods is not justified. Kern’s method is given in Section 19.9.3 and is illustrated in Examples 19.1 and 19.3.

19.9.3 Kern’s method

This method was based on experimental work on commercial exchangers with standard tolerances and will give a reasonably satisfactory prediction of the heat transfer coefficient for standard designs. The prediction of pressure drop is less satisfactory, as pressure drop is more affected by leakage and bypassing than by heat transfer. The shell-side heat transfer and friction factors are correlated in a similar manner to those for tube-side flow by using a hypothetical shell velocity and shell diameter. As the cross-sectional area for flow varies across the shell diameter, the linear and mass velocities are based on the maximum area for cross flow: that at the shell equator. The shell equivalent diameter is calculated using the flow area between the tubes taken in the axial direction (parallel to the tubes) and the wetted perimeter of the tubes (Fig. 19.28).

Shell-side j_h and j_f factors for use in this method are given in Figs. 19.29 and 19.30 for various baffle cuts and tube arrangements. These figures are based on data given by Kern (1950) and by Coker (2015).

The procedure for calculating the shell-side heat transfer coefficient and pressure drop for a single shell pass exchanger is given next.

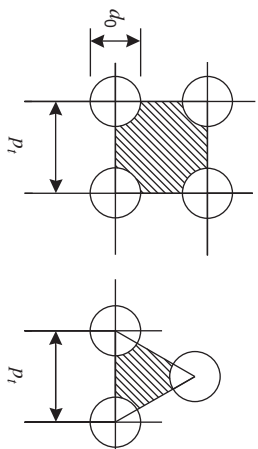


FIG. 19.28 Equivalent diameter, cross-sectional areas, and wetted perimeters.

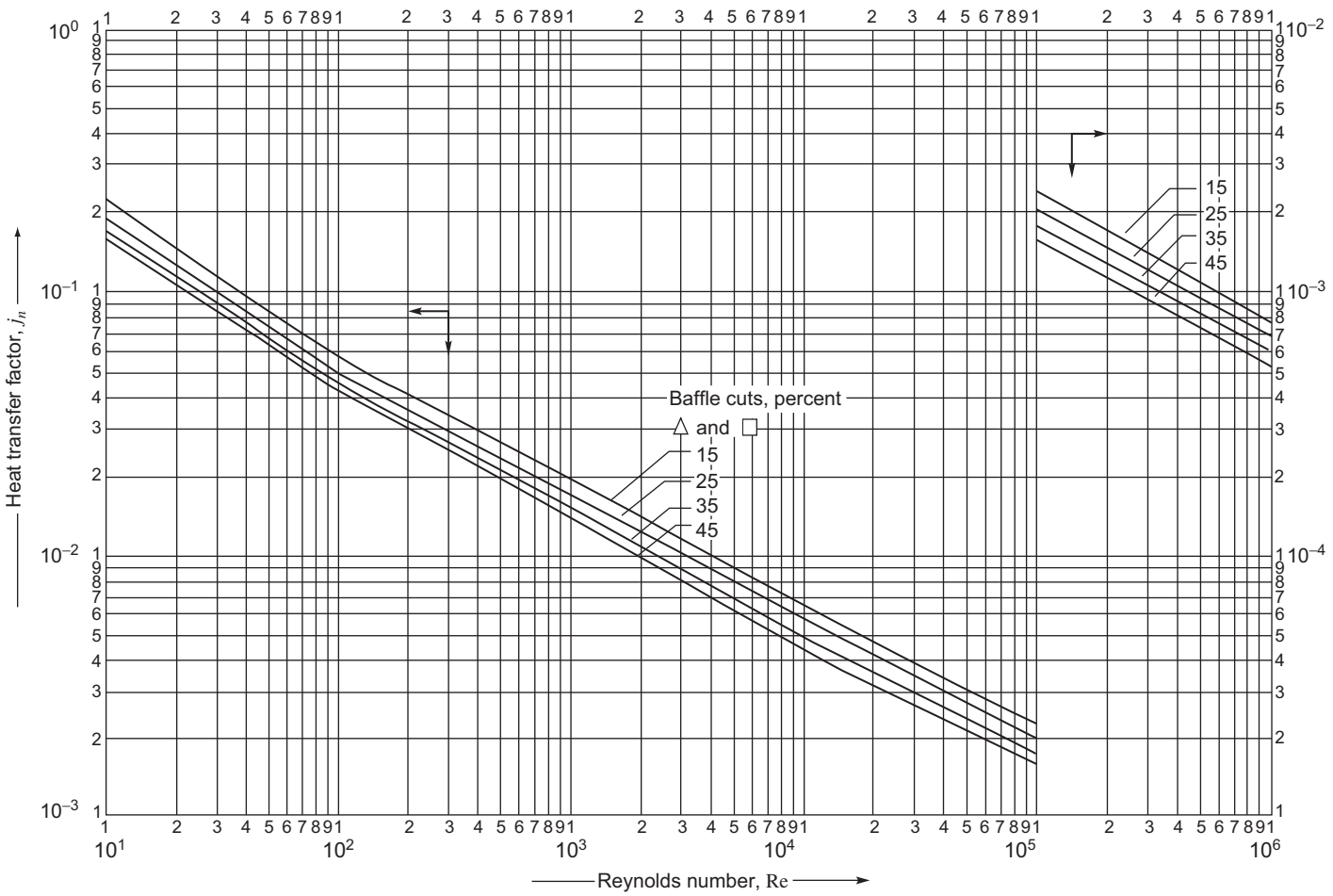


FIG. 19.29 Shell-side heat transfer factors, segmental baffles.

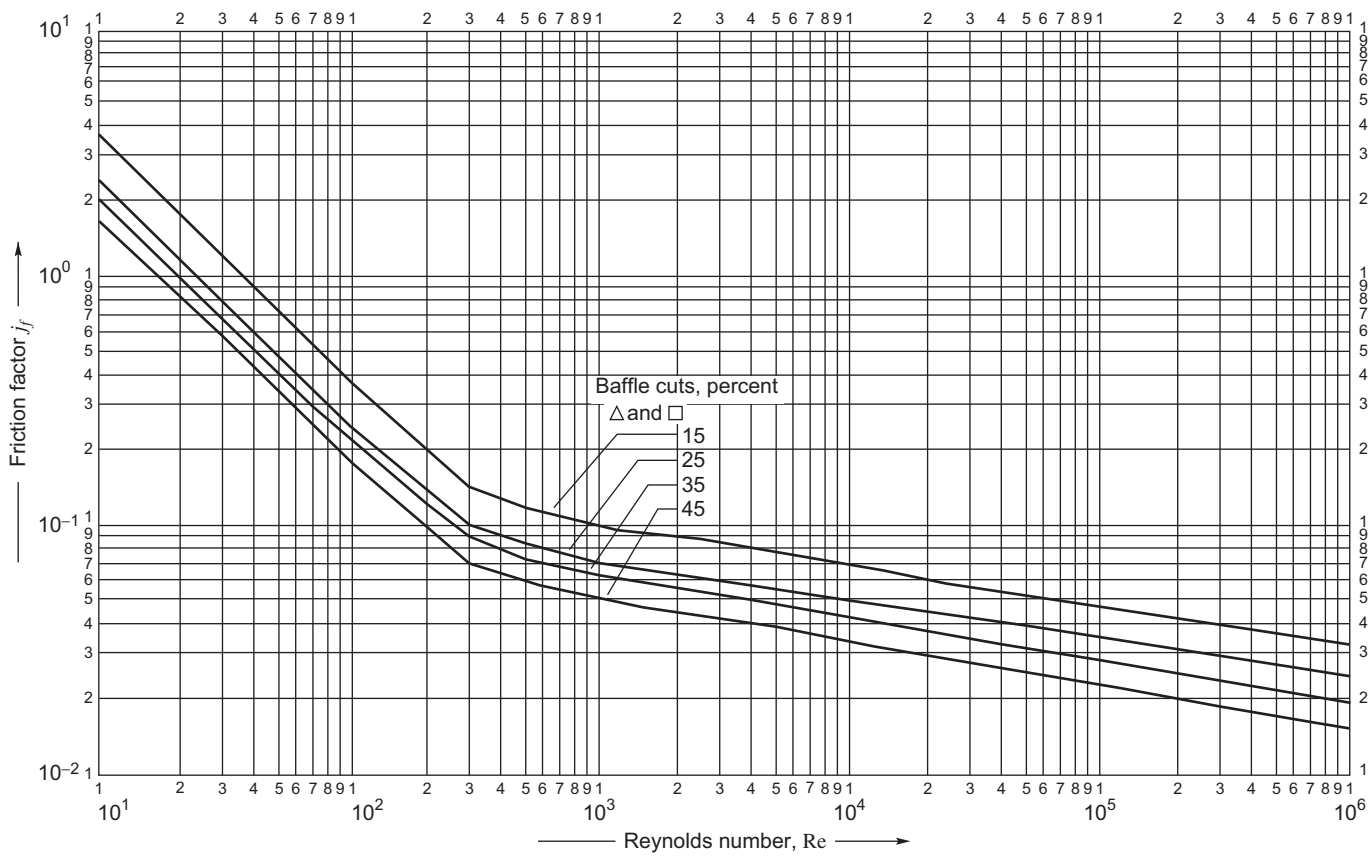


FIG. 19.30 Shell-side friction factors, segmental baffles.

Procedure

1. Calculate the area for cross-flow A_s for the hypothetical row of tubes at the shell equator, given by:

$$A_s = \frac{(p_t - d_o) D_s l_B}{p_t} \quad (19.21)$$

where p_t = tube pitch

d_o = tube outside diameter

D_s = shell inside diameter, m

l_B = baffle spacing, m

The term $(p_t - d_o)/p_t$ is the ratio of the clearance between tubes and the total distance between tube centers.

2. Calculate the shell-side mass velocity G_s and the linear velocity u_s :

$$G_s = \frac{W_s}{A_s}$$

$$u_s = \frac{G_s}{\rho}$$

where W_s = fluid flow rate on the shell side, kg/s

ρ = shell-side fluid density, kg/m³

3. Calculate the shell-side equivalent diameter (hydraulic diameter) (see Fig. 19.28). For a square pitch arrangement:
For an equilateral triangular pitch arrangement:

$$d_e = \frac{4 \left(\frac{p_t^2 - \pi d_o^2}{4} \right)}{\pi d_o} = \frac{1.27}{d_o} (p_t^2 - 0.785 d_o^2) \quad (19.22)$$

$$d_e = \frac{4 \left(\frac{p_t}{2} \times 0.87 p_t - \frac{1}{2} \pi \frac{d_o^2}{4} \right)}{\frac{\pi d_o}{2}} = \frac{1.10}{d_o} (p_t^2 - 0.917 d_o^2) \quad (19.23)$$

where d_e = equivalent diameter, m.

4. Calculate the shell-side Reynolds number, given by:

$$\text{Re} = \frac{G_s d_e}{\mu} = \frac{u_s d_e \rho}{\mu} \quad (19.24)$$

5. For the calculated Reynolds number, read the value of j_h from Fig. 19.29 for the selected baffle cut and tube arrangement and calculate the shell-side heat transfer coefficient h_s from:

$$\text{Nu} = \frac{h_s d_e}{k_f} = j_h \text{RePr}^{0.33} \left(\frac{\mu}{\mu_w} \right)^{0.14} \quad (19.25)$$

The tube wall temperature can be estimated using the method given for the tube side; see Section 19.8.1.

6. For the calculated shell-side Reynolds number, read the friction factor from Fig. 19.30 and calculate the shell-side pressure drop from:

$$\Delta P_s = 8 f_f \left(\frac{D_s}{d_e} \right) \left(\frac{L}{l_B} \right) \frac{\rho u_s^2}{2} \left(\frac{\mu}{\mu_w} \right)^{-0.14} \quad (19.26)$$

where L = tube length

l_B = baffle spacing

The term (L/l_B) is the number of times the flow crosses the tube bundle $= (N_b + 1)$, where N_b is the number of baffles.

Shell nozzle-pressure drop

The pressure loss in the shell nozzles will normally only be significant with gases. The nozzle pressure drop can be taken as equivalent to 1½ velocity heads for the inlet and ½ for the outlet, based on the nozzle area or the free area between the tubes in the row immediately adjacent to the nozzle, whichever is the lower.

Example 19.1

Design an exchanger to subcool condensate from a methanol condenser from 95 °C to 40 °C. The flow rate of methanol is 100,000 kg/h. Brackish water will be used as the coolant, with a temperature rise from 25 °C to 40 °C.

Solution

Only the thermal design will be considered. This example illustrates Kern's method. Coolant is corrosive, so assign to tube side:

$$\text{Heat capacity methanol} = 2.84 \text{ kJ/kg}^\circ\text{C}$$

$$\text{Heat load} = \frac{100,000}{3600} \times 2.84(95 - 40) = 4340 \text{ kW}$$

$$\text{Heat capacity water} = 4.2 \text{ kJ/kg}^\circ\text{C}$$

$$\text{Cooling water flow} = \frac{4340}{4.2(40 - 25)} = 68.9 \text{ kg/s}$$

$$\Delta T_{\text{lm}} = \frac{(95 - 40) - (40 - 25)}{\ln \frac{(95 - 40)}{(40 - 25)}} = 31^\circ\text{C} \quad (19.4)$$

Use one shell pass and two tube passes:

$$R = \frac{95 - 40}{40 - 25} = 3.67 \quad (19.6)$$

$$S = \frac{40 - 25}{95 - 25} = 0.21 \quad (19.7)$$

From Fig. 19.19:

$$F_t = 0.85$$

$$\Delta T_m = 0.85 \times 31 = 26^\circ\text{C}$$

From Fig. 19.1:

$$U = 600 \text{ W/m}^2\text{C}$$

Provisional area:

$$A = \frac{4340 \times 10^3}{26 \times 600} = 278 \text{ m}^2 \quad (19.1)$$

Choose 20 mm o.d., 16 mm i.d., 4.88-m-long tubes (¾ in. × 16 ft), cupro-nickel.

Allowing for tubesheet thickness, take:

$$L = 4.83 \text{ m}$$

$$\text{Area of one tube} = 4.83 \times 20 \times 10^{-3} \pi = 0.303 \text{ m}^2$$

$$\text{Number of tubes} = \frac{278}{0.303} = \underline{\underline{918}}$$

As the shell-side fluid is relatively clean, use 1.25 triangular pitch:

$$\text{Bundle diameter } D_b = 20 \left(\frac{918}{0.249} \right)^{1/2.207} = 826 \text{ mm} \quad (19.3b)$$

Use a split-ring floating head type.

From Fig. 19.10, bundle diametrical clearance = 68 mm and shell diameter, $D_s = 826 + 68 = 894$ mm. (Note: The nearest standard pipe sizes are 863.6 or 914.4 mm.) The shell size could be read from standard tube count tables.

Tube-side coefficient

$$\text{Mean water temperature} = \frac{40 + 25}{2} = 33^\circ\text{C}$$

$$\text{Tube cross-sectional area} = \frac{\pi}{4} \times 16^2 = 201 \text{ mm}^2$$

$$\text{Tubes per pass} = \frac{918}{2} = 459$$

$$\text{Total flow area} = 459 \times 201 \times 10^{-6} = 0.092 \text{ m}^2$$

$$\text{Water mass velocity} = \frac{68.9}{0.092} = 749 \text{ kg/s m}^2$$

$$\text{Density water} = 995 \text{ kg/m}^3$$

$$\text{Water linear velocity} = \frac{749}{995} = 0.75 \text{ m/s}$$

$$h_i = \frac{4200(1.35 + 0.02 \times 33)0.75^{0.8}}{16^{0.2}} = 3852 \text{ W/m}^2\text{C} \quad (19.17)$$

The coefficient can also be calculated using Equation 19.15; this is done to illustrate use of this method.

$$\frac{h_i d_i}{k_f} = j_h \text{Re} \text{Pr}^{0.33} \left(\frac{\mu}{\mu_w} \right)^{0.14}$$

$$\text{Viscosity of water} = 0.8 \text{ mNs/m}^2$$

$$\text{Thermal conductivity} = 0.59 \text{ W/m}^\circ\text{C}$$

$$\text{Re} = \frac{\rho u d_i}{\mu} = \frac{995 \times 0.75 \times 16 \times 10^{-3}}{0.8 \times 10^{-3}} = 14,925$$

$$\text{Pr} = \frac{C_p \mu}{k_f} = \frac{4.2 \times 10^3 \times 0.8 \times 10^{-3}}{0.59} = 5.7$$

$$\text{Neglect } \left(\frac{\mu}{\mu_w} \right)$$

$$\frac{L}{d_i} = \frac{4.83 \times 10^3}{16} = 302$$

From Fig. 19.23, $j_h = 3.9 \times 10^{-3}$.

$$h_i = \frac{0.59}{16 \times 10^{-3}} \times 3.9 \times 10^{-3} \times 14,925 \times 5.7^{0.33} = 3812 \text{ W/m}^2\text{C}$$

Checks reasonably well with value calculated from Equation 19.17; use lower figure.

Shell-side coefficient

$$\text{Choose baffle spacing} = \frac{D_s}{5} = \frac{894}{5} = 178 \text{ mm.}$$

$$\text{Tube pitch} = 1.25 \times 20 = 25 \text{ mm}$$

$$\text{Cross-flow area } A_s = \frac{(25 - 20)}{25} 894 \times 178 \times 10^{-6} = 0.032 \text{ m}^2 \quad (19.21)$$

$$\text{Mass velocity, } G_s = \frac{100,000}{3600} \times \frac{1}{0.032} = 868 \text{ kg/s m}^2$$

$$\text{Equivalent diameter } d_e = \frac{1.1}{20} (25^2 - 0.917 \times 20^2) = 14.4 \text{ mm} \quad (19.23)$$

$$\text{Mean shell side temperature} = \frac{95 + 40}{2} = 68^\circ\text{C}$$

$$\text{Methanol density} = 750 \text{ kg/m}^3$$

$$\text{Viscosity} = 0.34 \text{ mNs/m}^2$$

$$\text{Heat capacity} = 2.84 \text{ kJ/kg}^\circ\text{C}$$

$$\text{Thermal conductivity} = 0.19 \text{ W/m}^\circ\text{C}$$

$$\text{Re} = \frac{G_s d_e}{\mu} = \frac{868 \times 14.4 \times 10^{-3}}{0.34 \times 10^{-3}} = 36,762 \quad (19.24)$$

$$\text{Pr} = \frac{C_p \mu}{k_f} = \frac{2.84 \times 10^3 \times 0.34 \times 10^{-3}}{0.19} = 5.1$$

Choose a 25% baffle cut from Fig. 19.29:

$$j_h = 3.3 \times 10^{-3}$$

Without viscosity correction term:

$$h_s = \frac{0.19}{14.4 \times 10^{-3}} \times 3.3 \times 10^{-3} \times 36,762 \times 5.1^{1/3} = 2740 \text{ W/m}^2\text{C}$$

Estimate wall temperature:

$$\text{Mean temperature difference} = 68 - 33 = 35^\circ\text{C}$$

across all resistances

$$\text{across methanol film} = \frac{U}{h_o} \times \Delta T = \frac{600}{2740} \times 35 = 8^\circ\text{C}$$

Mean wall temperature = $68 - 8 = 60^\circ\text{C}$:

$$\mu_w = 0.37 \text{ mNs/m}^2$$

$$\left(\frac{\mu}{\mu_w}\right)^{0.14} = 0.99$$

which shows that the correction for a low-viscosity fluid is not significant.

Overall coefficient

Thermal conductivity of cupro-nickel alloys = $50 \text{ W/m}^\circ\text{C}$.

Take the fouling coefficients from Table 19.2: methanol (light organic) $5000 \text{ W/m}^2\text{C}$, brackish water (sea water), take as highest value $3000 \text{ W/m}^2\text{C}$:

$$\frac{1}{U_o} = \frac{1}{2740} + \frac{1}{5000} + \frac{20 \times 10^{-3} \ln\left(\frac{20}{16}\right)}{2 \times 50} + \frac{20}{16} \times \frac{1}{3000} + \frac{20}{16} \times \frac{1}{3812} \quad (19.2)$$

$$U_o = \underline{\underline{738 \text{ W/m}^2\text{C}}}$$

Well above the assumed value of $600 \text{ W/m}^2\text{C}$.

Pressure drop: Tube-side

From Fig. 19.24, for $\text{Re} = 14,925$:

$$j_f = 4.3 \times 10^{-3}$$

Neglecting the viscosity correction term:

$$\begin{aligned}\Delta P_t &= 2 \left(8 \times 4.3 \times 10^{-3} \left(\frac{4.83 \times 10^3}{16} \right) + 2.5 \right) \frac{995 \times 0.75^2}{2} \\ &= 7211 \text{ N/m}^2 = 7.2 \text{ kPa (1.1 psi)}\end{aligned}\quad (19.20)$$

low, could consider increasing the number of tube passes.

Pressure drop: Shell side

$$\text{Linear velocity} = \frac{G_s}{\rho} = \frac{868}{750} = 1.16 \text{ m/s}$$

From Fig. 19.30, at $Re = 36,762$:

$$j_f = 4 \times 10^{-2}$$

Neglect the viscosity correction:

$$\begin{aligned}\Delta P_s &= 8 \times 4 \times 10^{-2} \left(\frac{894}{14.4} \right) \left(\frac{4.83 \times 10^3}{178} \right) \frac{750 \times 1.16^2}{2} \\ &= 272,019 \text{ N/m}^2 \\ &= 272 \text{ kPa (39 psi) too high,}\end{aligned}\quad (19.26)$$

could be reduced by increasing the baffle pitch. Doubling the pitch halves the shell-side velocity, which reduces the pressure drop by a factor of approximately $(1/2)^2$:

$$\Delta P_s = \frac{272}{4} = 68 \text{ kPa (10 psi), acceptable}$$

This will reduce the shell-side heat transfer coefficient by a factor of $(1/2)^{0.8}$ ($h_o \propto Re^{0.8} \propto u_s^{0.8}$):

$$h_o = 2740 \times \left(\frac{1}{2} \right)^{0.8} = 1573 \text{ W/m}^2\text{°C}$$

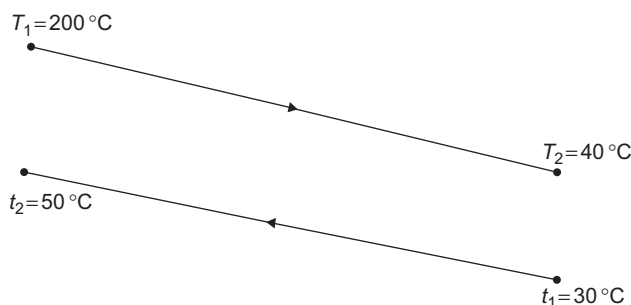
This gives an overall coefficient of $615 \text{ W/m}^2\text{°C}$, which is still above assumed value of $600 \text{ W/m}^2\text{°C}$.

Example 19.2

Gas oil at 200°C is to be cooled to 40°C . The oil flow rate is $22,500 \text{ kg/h}$. Cooling water is available at 30°C , and the temperature rise is to be limited to 20°C . The pressure drop allowance for each stream is 100 kN/m^2 . Design a suitable exchanger for this duty.

Solution

Only the thermal design will be carried out to illustrate the calculation procedure for an exchanger with a divided shell.



$$\Delta T_{lm} = \frac{(200 - 40) - (40 - 30)}{\ln \frac{(200 - 50)}{(40 - 30)}} = 51.7^\circ\text{C} \quad (19.4)$$

$$R = (200 - 50)/(50 - 30) = 8.0 \quad (19.6)$$

$$S = (50 - 30)/(200 - 30) = 0.12 \quad (19.7)$$

These values do not intercept on the figure for a single shell-pass exchanger (see Fig. 19.19) so use the figure for a two-pass shell (see Fig. 19.20), which gives:

$$F_t = 0.94, \text{ so}$$

$$\Delta T_m = 0.94 \times 51.7 = 48.6^\circ\text{C}$$

Physical properties

Water, from steam tables:

Temperature, °C	30	40	50
C_p , kJ/kg°C	4.18	4.18	4.18
k , kW/m°C	618×10^{-6}	631×10^{-6}	643×10^{-6}
μ , mNm ⁻² s	797×10^{-3}	671×10^{-3}	544×10^{-3}
ρ , kg m ⁻³	995.2	992.8	990.1

Gas oil, from Kern (1950):

Temperature, °C	200	120	40
C_p , kJ/kg°C	2.59	2.28	1.97
k , W/m°C	0.13	0.125	0.12
μ , mNm ⁻² s	0.06	0.17	0.28
ρ , kg/m ³	830	850	870

Duty:

$$\text{Oil flow rate} = 22,500/3600 = 6.25 \text{ kg/s}$$

$$Q = 6.25 \times 2.28 \times (200 - 40) = 2280 \text{ kW}$$

$$\text{Water flow rate} = \frac{2280}{4.18(50 - 30)} = 27.27 \text{ kg/h}$$

From Fig. 19.1, for cooling tower water and heavy organic liquid, take:

$$U = 500 \text{ Wm}^{-2}\text{C}^{-1}$$

$$\text{Area required} = \frac{2280 \times 10^3}{500 \times 48.6} = 94 \text{ m}^2$$

Tube-side coefficient

Select 20 mm o.d., 16 mm i.d. tubes, 4 m long, triangular pitch $1.25d_o$, carbon steel.

$$\text{Surface area of one tube} = \pi \times 20 \times 10^{-3} \times 4 = 0.251 \text{ m}^2$$

$$\text{Number of tubes required} = 94/0.251 = 375; \text{ say } 376, \text{ even number}$$

$$\text{Cross-sectional area, one tube} = \frac{\pi}{4}(16 \times 10^{-3})^2 = 2.011 \times 10^{-4} \text{ m}^2$$

$$\text{Total tube area} = 376 \times 2.011 \times 10^{-4} = 0.0756 \text{ m}^2$$

Put water through the tube for ease of cleaning.

$$\text{Tube velocity, one pass} = 27.27/(992.8 \times 0.0756) = 0.363 \text{ m/s}$$

Too low to make effective use of the allowable pressure drop; try four passes.

$$u_t = 4 \times 0.363 = 1.45 \text{ m/s}$$

A floating head will be needed due to the temperature difference. Use a pull-through type.

Tube-side heat transfer coefficient:

$$h_i = \frac{4200(1.35 + 0.02 \times 40)1.45^{0.8}}{16^{0.2}} = 6982 \text{ Wm}^{-2}\text{C}^{-1} \quad (19.17)$$

Shell-side coefficient

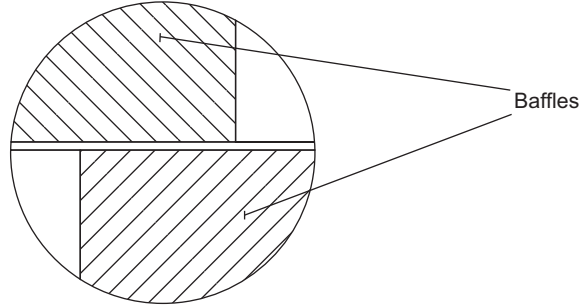
From Table 19.4 and Equation 19.3b, for four passes, $1.25d_o$ triangular pitch:

$$\text{Bundle diameter, } D_b = 20(376/0.175)^{1/2.285} = 575 \text{ mm}$$

From Fig. 19.10, for pull-through head, clearance = 92 mm:

$$\text{Shell diameter, } D_s = 575 + 92 = 667 \text{ mm (26 in. pipe)}$$

Use 25% cut baffles, with the baffle arrangement for a divided shell as shown here:



Take the baffle spacing as $\frac{1}{5}$ shell diameter = $667/5 = 133 \text{ mm}$

$$\text{Tube pitch, } p_t = 1.25 \times 20 = 25 \text{ mm}$$

Area for flow, A_s , will be half that given by Equation 19.21:

$$A_s = 0.5 \times \left(\frac{25 - 20}{25} \times 0.667 \times 0.133 \right) = 0.00887 \text{ m}^2$$

$$G_s = 6.25/0.00887 = 704.6 \text{ kg/s}$$

$$u_s = 704.6/850 = 0.83 \text{ m/s, looks reasonable} \quad (19.23)$$

$$d_e = \frac{1.10}{20} (25^2 - 0.917 \times 20^2) = 14.2 \text{ mm}$$

$$\text{Re} = \frac{0.83 \times 14.2 \times 10^{-3} \times 850}{0.17 \times 10^{-3}} = 58,930$$

From Fig. 19.29, $j_h = 2.6 \times 10^{-3}$:

$$\text{Pr} = (2.28 \times 10^3 \times 0.17 \times 10^{-3})/0.125 = 3.1$$

$$\text{Nu} = 2.6 \times 10^{-3} \times 58,930 \times 3.1^{1/3} = 223.4 \quad (19.25)$$

$$h_s = (223.4 \times 0.125)/(14.2 \times 10^{-3}) = 1967 \text{ W/m}^2\text{C}$$

Overall coefficient

Take fouling factors as 0.00025 for cooling tower water and 0.0002 for gas oil (light organic). Thermal conductivity for carbon steel tubes $45 \text{ W/m}^2\text{C}$:

$$1/U_o = 1/1967 + 0.0002 + \frac{20 \times 10^{-3} \ln(20/16)}{2 \times 45} + 20/16(1/6982 + 0.00025) = 0.00125 \quad (19.2)$$

$$U_o = 1/0.00125 = 800 \text{ Wm}^{-2}\text{C}^{-1}$$

Well above the initial estimate of $500 \text{ W/m}^2\text{C}$, so design has adequate area for the duty required.

Pressure drops: Tube side

$$\text{Re} = \frac{1.45 \times 16 \times 10^{-3} \times 992.8}{670 \times 10^{-6}} = 34,378 \quad (3.4 \times 10^{-4})$$

From Fig. 19.24, $j_f = 3.5 \times 10^{-3}$. Neglecting the viscosity correction:

$$\begin{aligned} \Delta P_t &= 4 \left[8 \times 3.5 \times 10^{-3} \times \left(\frac{4}{16 \times 10^{-3}} \right) + 2.5 \right] 992.8 \times \frac{1.45^2}{2} = 39,660 \\ &= 40 \text{ kN/m}^2 \end{aligned} \quad (19.20)$$

Well within the specification, so no need to check the nozzle pressure drop.

Pressure drops: Shell side

From Fig. 19.30, for $\text{Re} = 58,930$, $j_f = 3.8 \times 10^{-2}$

With a divided shell, the path length $= 2 \times (L/l_b)$

Neglecting the viscosity correction factor:

$$\begin{aligned} \Delta P_s &= 8 \times 3.8 \times 10^{-2} \left(\frac{662 \times 10^{-3}}{14.2 \times 10^{-3}} \right) \times \left(\frac{2 \times 4}{132 \times 10^{-3}} \right) \times 850 \times \frac{0.83^2}{2} = 251,481 \\ &= 252 \text{ kN/m}^2 \end{aligned} \quad (19.26)$$

Well within the specification, no need to check nozzle pressure drops.

So the proposed thermal design is satisfactory. As the calculated pressure drops are below that allowed, there is some scope for improving the design.

Example 19.3

Design a shell and tube exchanger for the following duty.

20,000 kg/h of kerosene (42° API) leaves the base of a kerosene side-stripping column at 200 °C and is to be cooled to 90 °C by exchange with 70,000 kg/h light crude oil (34° API) coming from storage at 40 °C. The kerosene enters the exchanger at a pressure of 5 bar and the crude oil at 6.5 bar. A pressure drop of 0.8 bar is permissible on both streams. Allowance should be made for fouling by including a fouling factor of $0.0003 \text{ (W/m}^2 \text{ °C)}^{-1}$ on the crude stream and $0.0002 \text{ (W/m}^2 \text{ °C)}^{-1}$ on the kerosene stream.

Solution

The solution to this example illustrates the iterative nature of heat exchanger design calculations. An algorithm for the design of shell and tube exchangers is shown in Fig. 19.31. The procedure set out in this figure will be followed in the solution.

Step 1: Specification

The specification is given in the problem statement:

20,000 kg/h of kerosene (42° API) at 200 °C cooled to 90 °C by exchange with 70,000 kg/h light crude oil (34° API) at 40 °C.

The kerosene pressure is 5 bar, and the crude oil pressure is 6.5 bar.

Permissible pressure drop of 0.8 bar on both streams.

Fouling factors: crude stream $0.00035 \text{ (W/m}^2 \text{ °C)}^{-1}$, kerosene stream $0.0002 \text{ (W/m}^2 \text{ °C)}^{-1}$.

To complete the specification, the duty (heat transfer rate) and the outlet temperature of the crude oil needed to be calculated.

Mean temperature of kerosene $= (200 + 90)/2 = 145 \text{ °C}$.

At this temperature, the specific heat capacity of 42° API kerosene is 2.47 kJ/kg°C (physical properties from Kern 1950).

$$\text{Duty} = \frac{20,000}{3600} \times 2.47(200 - 90) = 1509.4 \text{ kW}$$

As a first trial, take the mean temperature of the crude oil as equal to the inlet temperature, 40 °C; specific heat capacity at this temperature $= 2.01 \text{ kJ/kg°C}$.

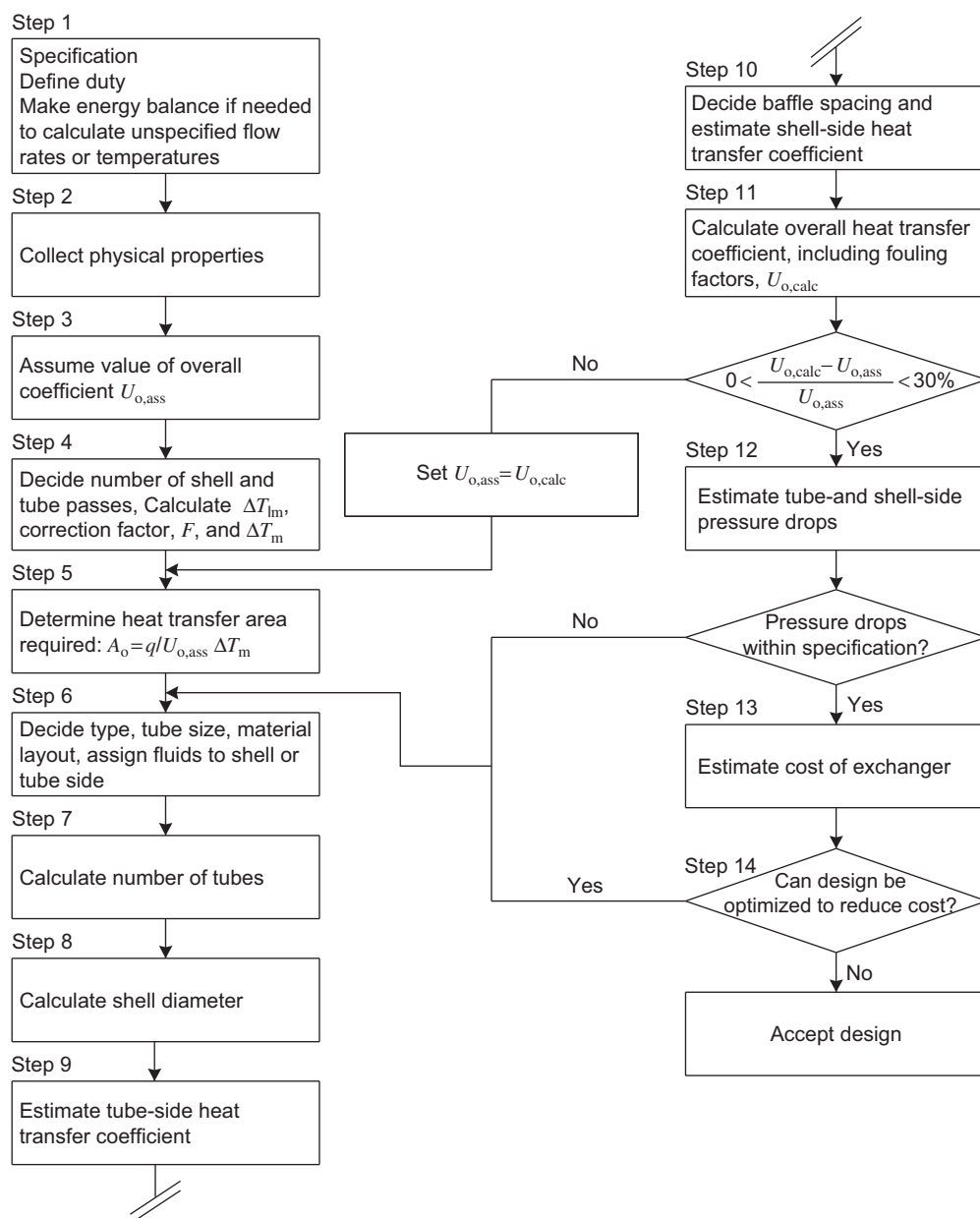


FIG. 19.31 Design procedure for shell and tube heat exchangers. Example 19.2 and Fig. 19.31 were developed by the author for the Open University Course T333 Principles and Applications of Heat Transfer. They are reproduced here by permission of the Open University.

An energy balance gives:

$$\frac{7000}{3600} \times 2.01 (t_2 - 40) = 1509.4$$

$t_2 = 78.6^\circ\text{C}$ and the stream mean temperature $= (40 + 78.6)/2 = 59.3^\circ\text{C}$.

The specific heat at this temperature is $2.05 \text{ kJ/kg}^\circ\text{C}$. A second trial calculation using this value gives $t_2 = 77.9^\circ\text{C}$ and a new mean temperature of 58.9°C . There is no significant change in the specific heat at this mean temperature from the value used, so take the crude stream outlet temperature to be 77.9°C , say 78°C .

Step 2: Physical properties

Kerosene	Inlet	Mean	Outlet	
Temperature	200	145	90	$^\circ\text{C}$
Specific heat	2.72	2.47	2.26	$\text{kJ/kg}^\circ\text{C}$
Thermal conductivity	0.130	0.132	0.135	$\text{W/m}^\circ\text{C}$
Density	690	730	770	kg/m^3
Viscosity	0.22	0.43	0.80	mN sm^{-2}

Crude oil	Outlet	Mean	Inlet	
Temperature	78	59	40	°C
Specific heat	2.09	2.05	2.01	kJ/kg°C
Thermal conductivity	0.133	0.134	0.135	W/m°C
Density	800	820	840	kg/m ³
Viscosity	2.4	3.2	4.3	mN sm ⁻²

Step 3: Overall coefficient

For an exchanger of this type, the overall coefficient will be in the range of 300 to 500 W/m²°C (see Fig. 19.1 and Table 19.1), so start with 300 W/m²°C.

Step 4: Exchanger type and dimensions

An even number of tube passes is usually the preferred arrangement, as this positions the inlet and outlet nozzles at the same end of the exchanger, which simplifies the pipework.

Start with one shell pass and two tube passes:

$$\Delta T_{lm} = \frac{(200 - 78) - (90 - 40)}{\ln \frac{(200 - 78)}{(90 - 40)}} = 80.7^\circ\text{C} \quad (19.4)$$

$$R = \frac{(200 - 90)}{(90 - 40)} = 2.9 \quad (19.6)$$

$$S = \frac{(78 - 40)}{(200 - 40)} = 0.24 \quad (19.7)$$

From Fig. 19.19, $F_t = 0.88$, which is acceptable.

So,

$$\Delta T_m = 0.88 \times 80.7 = 71.0^\circ\text{C}$$

Step 5: Heat transfer area

$$A_o = \frac{1509.4 \times 10^3}{300 \times 71.0} = 70.86 \text{ m}^2 \quad (19.1)$$

Step 6: Layout and tube size

Use a split-ring floating-head exchanger for efficiency and ease of cleaning. Neither fluid is corrosive and the operating pressure is not high, so a plain carbon steel can be used for the shell and tubes. The crude is dirtier than the kerosene, so put the crude through the tubes and the kerosene in the shell.

Use 19.05 mm (¾ inch) outside diameter, 14.83 mm inside diameter, 5 m long tubes (a popular size) on a triangular 23.81 mm pitch (pitch/dia. = 1.25).

Step 7: Number of tubes

Area of one tube (neglecting thickness of tubesheets):

$$= \pi \times 19.05 \times 10^{-3} \times 5 = 0.2992 \text{ m}^2$$

Number of tubes = $70.89/0.2992 = 237$, say 240

So, for two passes, tubes per pass = 120

Check the tube-side velocity at this stage to see if it looks reasonable:

$$\text{Tube cross-sectional area} = \frac{\pi}{4} (14.83 \times 10^{-3})^2 = 0.0001727 \text{ m}^2$$

$$\text{So area per pass} = 120 \times 0.0001727 = 0.02073 \text{ m}^2$$

$$\text{Volumetric flow} = \frac{70,000}{3600} \times \frac{1}{820} = 0.0237 \text{ m}^3/\text{s}$$

$$\text{Tube-side velocity, } u_t = \frac{0.0237}{0.02073} = 1.14 \text{ m/s}$$

The velocity is satisfactory, between 1 and 2 m/s, but may be a little low. This will show up when the pressure drop is calculated.

Step 8: Bundle and shell diameter

From Table 19.4, for two tube passes, $K_1 = 0.249$, $n_1 = 2.207$,
so,

$$D_b = 19.05 \left(\frac{240}{0.249} \right)^{1/2.207} = 428 \text{ mm (0.43 m)} \quad (19.3b)$$

For a split-ring floating-head exchanger, the typical shell clearance from Fig. 19.10 is 56 mm, so the shell inside diameter is:

$$D_s = 428 + 56 = 484 \text{ mm}$$

Step 9: Tube-side heat transfer coefficient

$$\text{Re} = \frac{\rho u d_i}{\mu} = \frac{820 \times 1.14 \times 14.83 \times 10^{-3}}{3.2 \times 10^{-3}} = 4332$$

$$\text{Pr} = \frac{C_p \mu}{k_f} = \frac{2.05 \times 10^3 \times 3.2 \times 10^{-3}}{0.134} = 48.96$$

$$\frac{L}{d_i} = \frac{5000}{14.83} = 337$$

From Fig. 19.23,

$$j_h = 3.2 \times 10^{-3}$$

$$\text{Nu} = 3.2 \times 10^{-3} (4332) (48.96)^{0.33} = 50.06 \quad (19.15)$$

$$h_i = 50.06 \left(\frac{0.134}{14.83 \times 10^{-3}} \right) = 452 \text{ W/m}^2\text{C}$$

This is clearly too low if U_o is to be 300 W/m²C. The tube-side velocity did look low, so increase the number of tube passes to four. This will halve the cross-sectional area in each pass and double the velocity.

$$\text{New } u_t = 2 \times 1.14 = 2.3 \text{ m/s}$$

$$\text{and Re} = 2 \times 4332 = 8664$$

$$j_h = 3.8 \times 10^{-3}$$

$$\begin{aligned} h_i &= \left(\frac{0.134}{14.83 \times 10^{-3}} \right) \times 3.8 \times 10^{-3} (8664) (48.96)^{0.33} \\ &= 1074 \text{ W/m}^2\text{C} \end{aligned}$$

Step 10: Shell-side heat transfer coefficient

Kern's method will be used. With four tube passes, the shell diameter will be larger than that calculated for two passes. For four passes $K_1 = 0.175$ and $n_1 = 2.285$.

$$D_b = 19.05 \left(\frac{240}{0.175} \right)^{1/2.285} = 450 \text{ mm, (0.45 m)} \quad (19.3b)$$

The bundle to shell clearance is still around 56 mm, giving:

$$D_s = 506 \text{ mm (about 20 inches)}$$

As a first trial take the baffle spacing = $D_s/5$, say 100 mm. This spacing should give good heat transfer without too high a pressure drop.

$$A_s = \frac{(23.81 - 19.05)}{23.81} 506 \times 100 = 10,116 \text{ mm}^2 = 0.01012 \text{ m}^2 \quad (19.21)$$

$$d_e = \frac{1.10}{19.05} (23.81^2 - 0.917 \times 19.05^2) = 13.52 \text{ mm} \quad (19.23)$$

$$\text{Volumetric flow – rate on shell – side} = \frac{20,000}{3600} \times \frac{1}{730} = 0.0076 \text{ m}^3/\text{s}$$

$$\text{Shell – side velocity} = \frac{0.0076}{0.01012} = 0.75 \text{ m/s}$$

$$\text{Re} = \frac{730 \times 0.75 \times 13.52 \times 10^{-3}}{0.43 \times 10^{-3}} = 17,214$$

$$\text{Pr} = \frac{2.47 \times 10^3 \times 0.43 \times 10^{-3}}{0.132} = 8.05$$

Use segmental baffles with a 25% cut. This should give a reasonable heat transfer coefficient without too large a pressure drop.

From Fig. 19.29, $j_h = 4.52 \times 10^{-3}$.

Neglecting the viscosity correction:

$$h_s = \left(\frac{0.132}{13.52} \times 10^3 \right) \times 4.52 \times 10^{-3} \times 17,214 \times 8.05^{0.33} = 1505 \text{ W/m}^2\text{°C} \quad (19.25)$$

Step 11: Overall coefficient

$$\frac{1}{U_o} = \left(\frac{1}{1074} + 0.00035 \right) \frac{19.05}{14.83} + \frac{19.05 \times 10^{-3} \text{Ln} \left(\frac{19.05}{14.83} \right)}{2 \times 55} + \frac{1}{1505} + 0.0002 \quad (19.2)$$

$$U_o = 386 \text{ W/m}^2\text{°C}$$

This is above the initial estimate of 300 W/m²°C. The number of tubes could possibly be reduced, but first check the pressure drops.

Step 12: Pressure drop: Tube side

240 tubes, four passes, tube i.d. 14.83 mm, u_t 2.3 m/s,

$\text{Re} = 8.7 \times 10^3$. From Fig. 19.24, $j_f = 5 \times 10^{-3}$.

$$\begin{aligned} \Delta P_t &= 4 \left(8 \times 5 \times 10^{-3} \left(\frac{5000}{14.83} \right) + 2.5 \right) \frac{(820 \times 2.3^2)}{2} \\ &= 4(13.5 + 2.5) \frac{(820 \times 2.3^2)}{2} \\ &= 138,810 \text{ N/m}^2, \quad 1.4 \text{ bar} \end{aligned} \quad (19.20)$$

This exceeds the specification. Return to step 6 and modify the design.

Modified design

The tube velocity needs to be reduced. This will reduce the heat transfer coefficient, so the number of tubes must be increased to compensate. There will also be a pressure drop across the inlet and outlet nozzles. Allow 0.1 bar for this, a typical figure (about 15% of the total), which leaves 0.7 bar across the tubes. Pressure drop is roughly proportional to the square of the velocity, and u_t is proportional to the number of tubes per pass. So the pressure drop calculated for 240 tubes can be used to estimate the number of tubes required.

Tubes needed = $240 / (0.6/1.4)^{0.5} = 365$

Say, 360 with four passes.

Retain four passes, as the heat transfer coefficient will be too low with two passes.

Second trial design: 360 tubes 19.05 mm o.d., 14.83 mm i.d., 5 m long, triangular pitch 23.81 mm.

$$D_b = 19.05 \left(\frac{360}{0.175} \right)^{1/2.285} = 537 \text{ mm}, \quad (0.54 \text{ m}) \quad (19.3b)$$

From Fig. 19.10, clearance with this bundle diameter = 59 mm:

$$D_s = 537 + 59 = 596 \text{ mm}$$

$$\text{Cross-sectional area per pass} = \frac{360}{4} (14.83 \times 10^{-3})^2 \frac{\pi}{4} = 0.01555 \text{ m}^2$$

$$\text{Tube velocity, } u_t = \frac{0.02337}{0.01555} = 1.524 \text{ m/s}$$

$$\text{Re} = \frac{820 \times 1.524 \times 14.83 \times 10^{-3}}{3.2 \times 10^{-3}} = 5792$$

L/d is the same as the first trial: 337, so:

$$j_h = 3.6 \times 10^{-3}$$

$$h_i = \left(\frac{0.134}{14.83} \times 10^{-3} \right) 3.6 \times 10^{-3} \times 5792 \times 48.96^{0.33} = 680 \text{ W/m}^2\text{C} \quad (19.15)$$

This looks satisfactory, but check the pressure drop before doing the shell-side calculation.

$$j_f = 5.5 \times 10^{-3}$$

$$\Delta P_t = 4 \left(8 \times 5.5 \times 10^{-3} \left(\frac{5000}{14.83} \right) + 2.5 \right) \frac{(820 \times 1.524^2)}{2} = 66,029 \text{ N/m}^2, 0.66 \text{ bar} \quad (19.20)$$

Well within specification.

Keep the same baffle cut and spacing:

$$A_s = \frac{(23.81 - 19.05)}{23.81} 596 \times 100 = 11,915 \text{ mm}^2, 0.01192 \text{ m}^2 \quad (19.21)$$

$$u_s = \frac{0.0076}{0.01193} = 0.638 \text{ m/s}$$

$d_e = 13.52 \text{ mm}$, as before

$$\text{Re} = \frac{730 \times 0.638 \times 13.52 \times 10^{-3}}{0.43 \times 10^{-3}} = 14,644$$

$$\text{Pr} = 8.05$$

$$j_h = 4.8 \times 10^{-3}, j_f = 4.6 \times 10^{-2}$$

$$h_s = \left(\frac{0.132}{13.52 \times 10^{-3}} \right) 4.8 \times 10^{-3} \times 14,644 \times (8.05)^{0.33} = 1366 \text{ W/m}^2\text{C, looks OK} \quad (19.25)$$

$$\Delta P_s = 8 \times 4.6 \times 10^{-2} \left(\frac{596}{13.52} \right) \left(\frac{5000}{100} \right) \frac{(730 \times 0.638^2)}{2} = 120,510 \text{ N/m}^2, 1.2 \text{ bar} \quad (19.26)$$

Too high; the specification only allowed 0.8 overall, including the loss over the nozzles. Check the overall coefficient to see if there is room to modify the shell-side design.

$$\frac{1}{U_o} = \left(\frac{1}{683} + 0.00035 \right) \frac{19.05}{14.83} + \frac{19.05 \times 10^{-3} \ln \left(\frac{19.05}{14.88} \right)}{2 \times 55} + \frac{1}{1366} + 0.0002$$

$$U_o = 302 \text{ W/m}^2\text{C}$$

(19.2)

$$U_o \text{ required} = \frac{Q}{(A_o \Delta T_{lm})}, \quad A_o = 360 \times 0.2992 = 107.7 \text{ m}^2,$$

$$\text{so } U_o \text{ required} = \frac{1509.4 \times 10^3}{(107.7 \times 71)} = 197 \text{ W/m}^2\text{C}$$

The estimated overall coefficient is well above that required for design, 302 compared with 192 W/m²C, which gives scope for reducing the shell-side pressure drop.

Allow a drop of 0.1 bar for the shell inlet and outlet nozzles, leaving 0.7 bar for the shell-side flow. So, to keep within the specification, the shell-side velocity will have to be reduced by around $\sqrt{1/2} = 0.707$. To achieve this, the baffle spacing will need to be increased to $100/0.707 = 141$, say 140 mm.

$$A_s = \frac{(23.81 - 19.05)}{23.81} 596 \times 140 = 6881 \text{ mm}^2, 0.167 \text{ m}^2 \quad (19.21)$$

$$u_s = \frac{0.0076}{0.0167} = 0.455 \text{ m/s},$$

Giving: $Re = 10,443$, $h_s = 1177 \text{ W/m}^2\text{°C}$, $\Delta P_s = 0.47 \text{ bar}$, and $U_o = 288 \text{ W/m}^2\text{°C}$.

The pressure drop is now well within the specification.

Step 13: Estimate cost

The cost of this design can be estimated using the methods given in [Chapter 7](#).

Step 14: Optimization

There is scope for optimizing the design by reducing the number of tubes, as the pressure drops are well within specification and the overall coefficient is well above that needed; however, the method used for estimating the coefficient and pressure drop on the shell side (Kern's method) is not accurate, so keeping to this design will give some margin of safety.

Viscosity correction factor

The viscosity correction factor $(\mu/\mu_w)^{0.14}$ was neglected when calculating the heat transfer coefficients and pressure drops. This is reasonable for the kerosene, as it has a relatively low viscosity, but it is not so obvious so for the crude oil. So, before firming up the design, the effect of this factor on the tube-side coefficient and pressure drop will be checked.

First, an estimate of the temperature at the tube wall, t_w is needed.

$$\text{The inside area of the tubes} = \pi \times 14.83 \times 10^{-3} \times 5 \times 360 = 83.86 \text{ m}^2$$

$$\text{Heat flux} = Q/A = 1509.4 \times 10^3 / 83.86 = 17,999 \text{ W/m}^2$$

As a rough approximation:

$$(t_w - t)h_i = 17,999$$

where t is the mean bulk fluid temperature = 59°C .

$$\text{So, } t_w = \frac{17,999}{680} + 59 = 86^\circ\text{C}.$$

The crude oil viscosity at this temperature = $2.1 \times 10^{-3} \text{ Ns/m}^2$.

$$\text{Giving } \left(\frac{\mu}{\mu_w}\right)^{0.14} = \left(\frac{3.2 \times 10^{-3}}{2.1 \times 10^{-3}}\right)^{0.14} = 1.06$$

Only a small factor, so the decision to neglect it was justified. Applying the correction would increase the estimated heat transfer coefficient, which is in the right direction. It would give a slight decrease in the estimated pressure drop.

Summary: The proposed design

Split ring, floating head, one shell pass, four tube passes.

360 carbon steel tubes, 5 m long, 19.05 mm o.d., 14.83 mm i.d., triangular pitch, pitch 23.18 mm.

Heat transfer area 107.7 m^2 (based on outside diameter).

Shell i.d. 597 mm (600 mm), baffle spacing 140 mm, 25% cut.

Tube-side coefficient $680 \text{ W/m}^2\text{°C}$, clean.

Shell-side coefficient $1366 \text{ W/m}^2\text{°C}$, clean.

Overall coefficient, estimated $288 \text{ W/m}^2\text{°C}$, dirty.

Overall coefficient required $197 \text{ W/m}^2\text{°C}$, dirty.

Dirt/fouling factors:

Tube-side (crude oil) $0.00035 \text{ (W/m}^2\text{°C)}^{-1}$.

Shell-side (kerosene) $0.0002 \text{ (W/m}^2\text{°C)}^{-1}$.

Pressure drops:

Tube side, estimated 0.40 bar, +0.1 for nozzles; specified 0.8 bar overall.

Shell side, estimated 0.45 bar, +0.1 for nozzles; specified 0.8 bar overall.

19.9.4 Commercial software for heat exchanger design

Computer methods for detailed design of heat exchangers are available in most of the commercial process simulation programs introduced in [Chapter 4](#); see Table 4.1. For example, Aspen Technology's Aspen Engineering Suite contains the HTFS TASC program, and Honeywell's UniSim Design Suite can be linked to Honeywell's UniSim Heat Exchanger program, which is also based on the HTFS methods. The methods developed by HTRI can be licensed from them (www.HTRI.net).

All of the commercial heat exchanger design programs allow the user to upload process data and stream properties from a process simulation. Some care is needed when uploading data for streams that undergo partial vaporization or have other effects that cause a significant variation in heat capacity or other properties across the exchanger. When there are significant changes in fluid properties between the exchanger inlet and outlet, the designer should break the exchanger into several exchangers in series in the process simulation so as to obtain several sets of property data at intermediate temperatures for input into the heat exchanger software.

The details of running the commercial heat exchanger design programs are not addressed here, as each program is slightly different from the others; consult the user manuals and online help. The heat exchanger programs have both rating and design capability. They can be configured to determine a least cost design for given desired outlet temperatures and allowable pressure drop or to calculate the outlet stream temperatures and pressures given details of the exchanger geometry and process fluid inlet conditions. All of the programs allow the user to make quick adjustments to exchanger geometry and then recalculate to see the impact on the stream outlet temperatures and pressure drops.

Example 19.4

Optimize the design of Example 19.3 using commercial heat exchanger design software.

Solution

This problem was solved using UniSim Design Heat Exchanger.

[Figs. 19.32 and 19.33](#) show the stream data and physical properties. The program was then run with the objective function set to minimum cost, giving the output shown in [Fig. 19.34](#) and the setting plan shown in [Fig. 19.35](#).

	Units	(Hot) Stream 1	(Cold) Stream 2
* Total Mass Flow	kg/h	20000	70000
* Inlet Temperature	°C	200	40
* Outlet Temperature	°C	90	
* Inlet Pressure	bar(abs)	5	6.5
* Allowable Pressure Drop	bar	0.8	0.8
Inlet Mass Quality	0 to 1		
Outlet Mass Quality	0 to 1		
Fouling Resistance	m ² K/W	0.0002	0.0003
Outlet Pressure	bar(abs)	4.2	5.7
Heat Load	kW		
Revise for Heat Balance		Default(h.load)	Default(h.load)
Inlet Specific Enthalpy	kJ/kg		
Outlet Specific Enthalpy	kJ/kg		

FIG. 19.32 Stream data for Example 19.4.

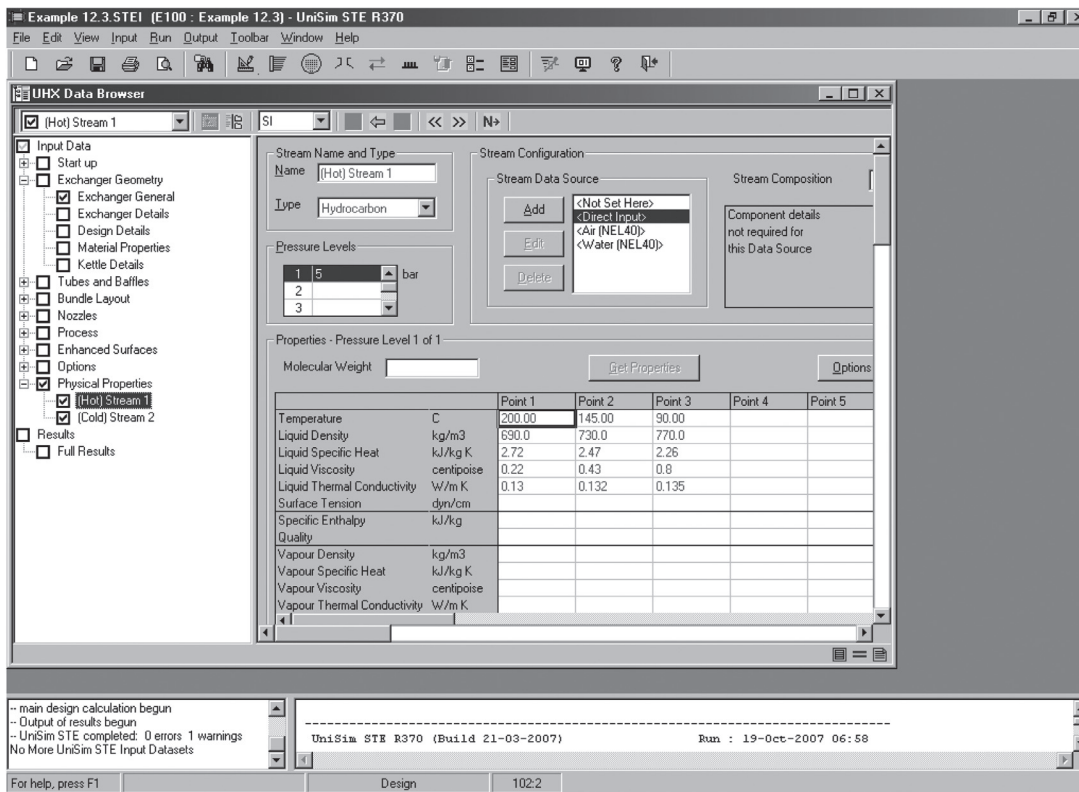


FIG. 19.33 Stream physical properties for Example 19.4.

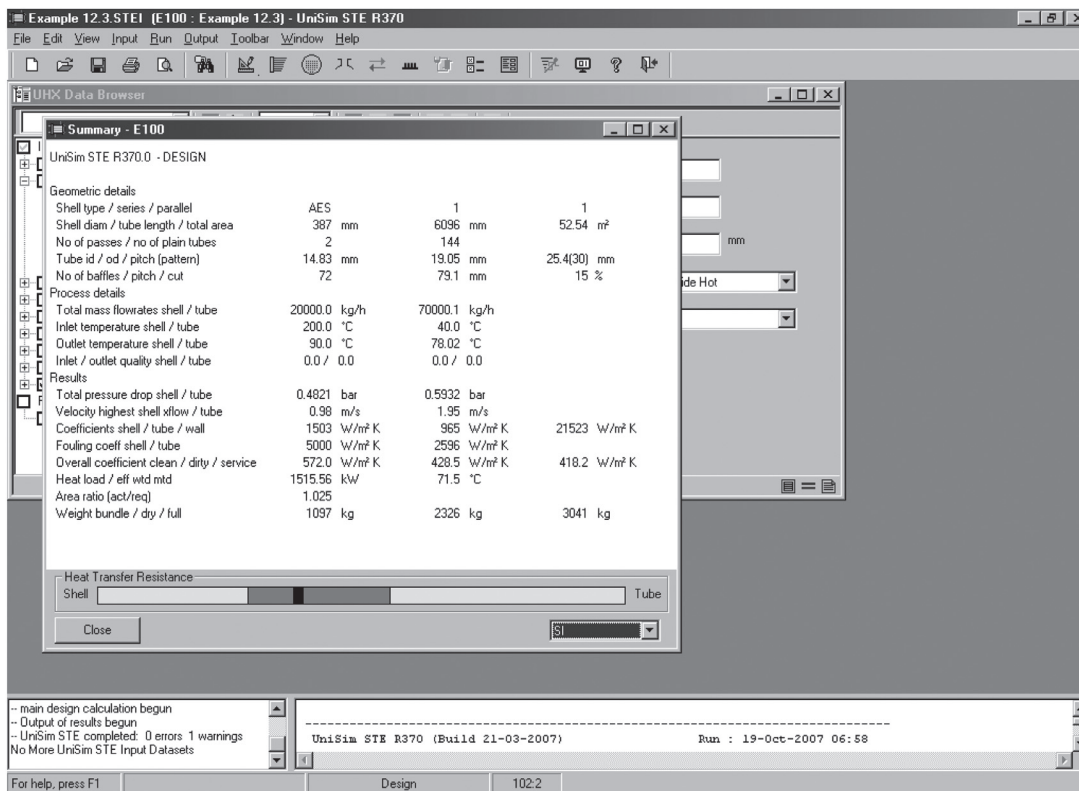


FIG. 19.34 UniSim Design STE (HTFS) program output.

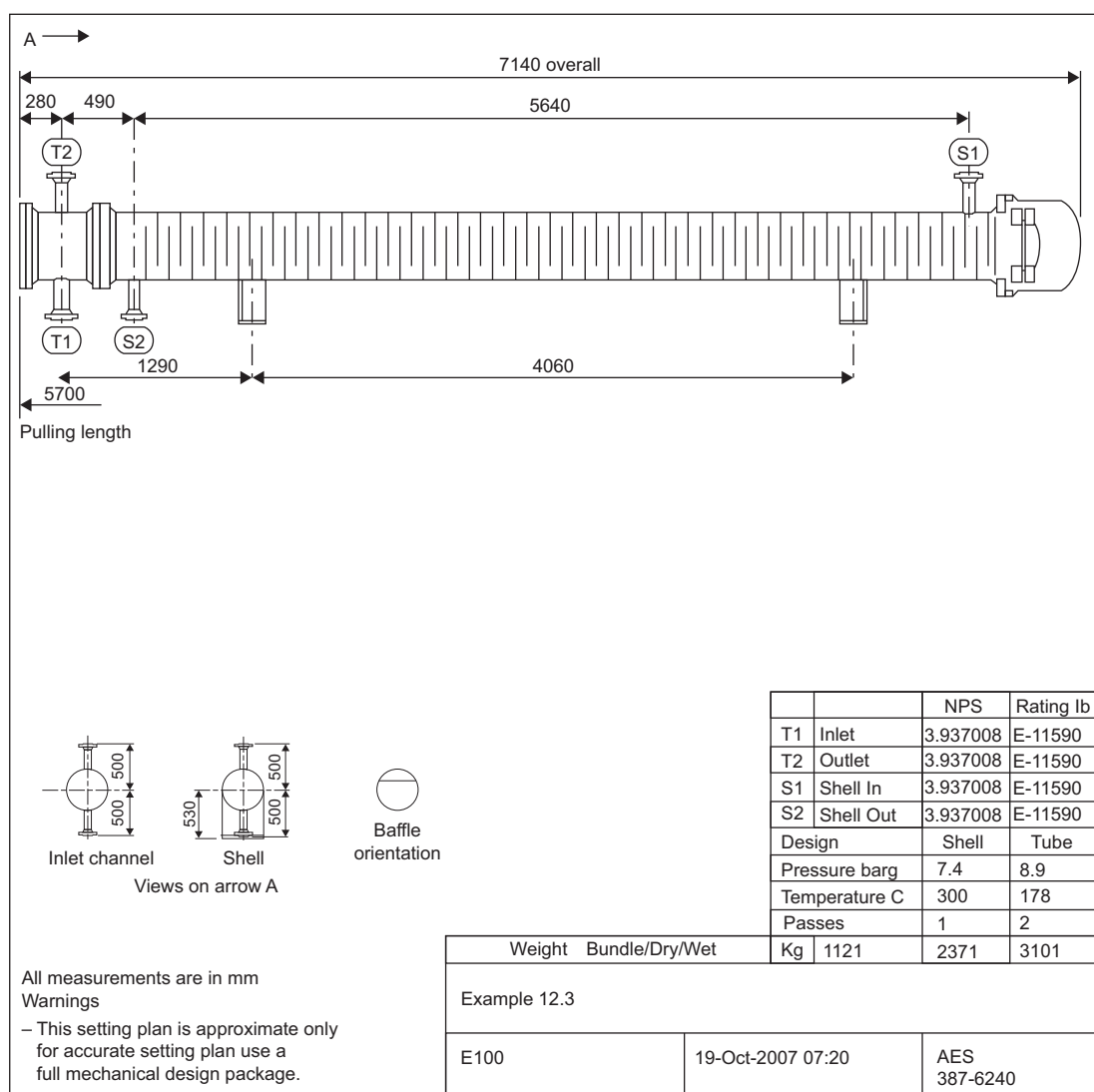


FIG. 19.35 HTFS setting plan.

The program selected two tube passes with 6096 mm (20 ft) long tubes and many baffles (72) to obtain good countercurrent flow on the shell side. This design could cause problems with plot space, or with supporting the shell, or with withdrawing the bundle for cleaning and maintenance. The program was run again with the tube length constrained to be less than 4880 mm (16 ft). This gives a more compact design with four tube passes, 12-ft-tubes, and only 28 baffles, shown in Fig. 19.36.

The program initially gives a warning: "At some point(s) the cross flow fraction in the shell side flow model was less than 30%. This is below the expected range of values and may give poor heat transfer," When the baffle-to-shell and tube-to-baffle tolerances are specified using the values given in Section 19.5.7 (1.6 mm and 0.8 mm, respectively), then the design converges with no warnings. The TEMA sheet for the resulting design is given in Fig. 19.37.

Note that the more compact design did not make as good use of the allowable pressure drop on the shell side and also has a higher bundle weight and area, corresponding to a higher capital cost than the 20-ft-long exchanger initially designed. Note also that both designs developed using the HTFS software need substantially less area than the 107.7 m² predicted using Kern's method in Example 19.3.

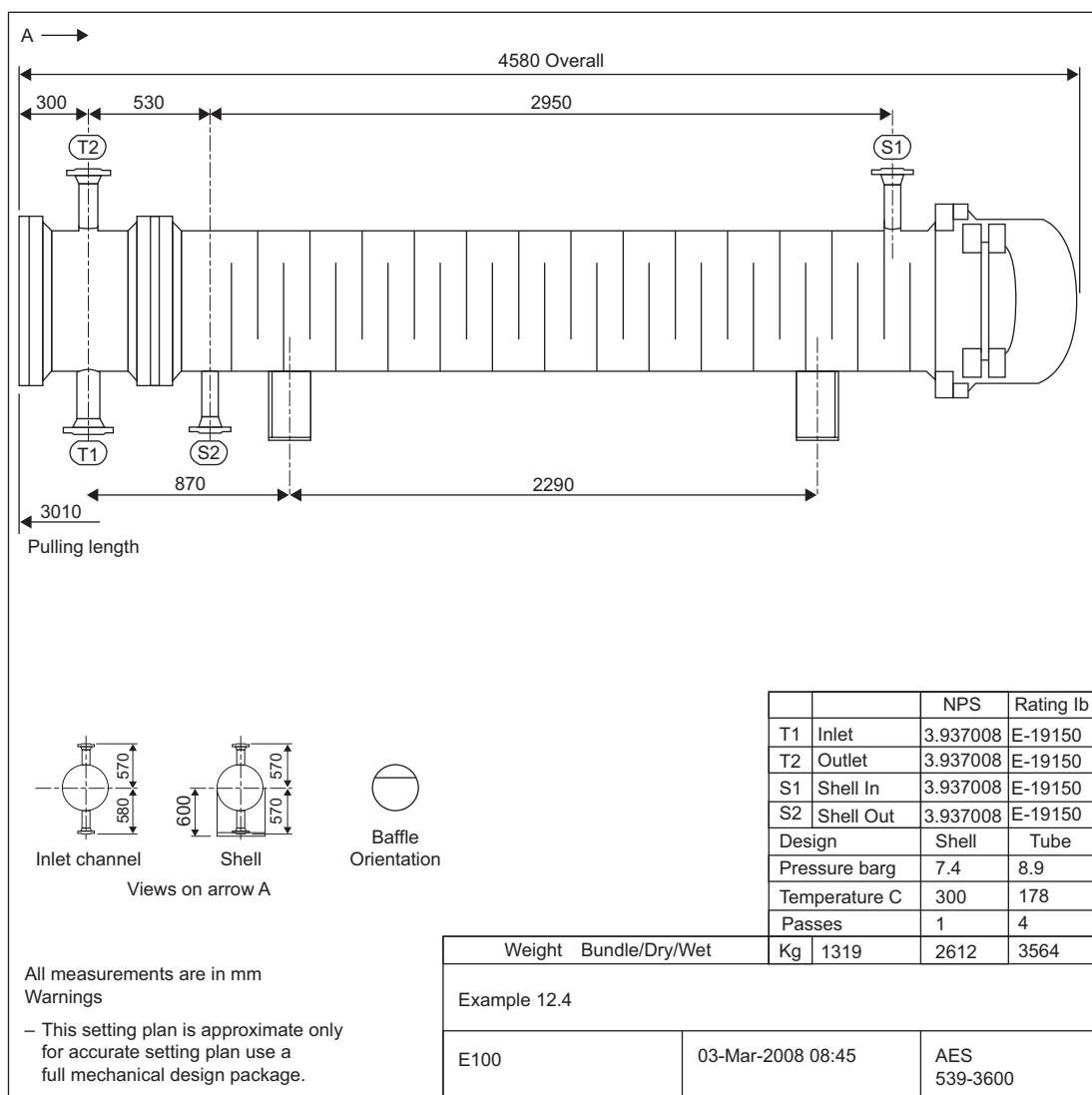


FIG. 19.36 Setting plan with length constrained.

19.10 Condensers

This section covers the design of shell and tube exchangers used as condensers. Direct contact condensers are discussed in Section 19.13.

The construction of a condenser is similar to other shell and tube exchangers, but with a wider baffle spacing, typically $l_B = D_s$.

Four condenser configurations are possible:

1. Horizontal, with condensation in the shell, and the cooling medium in the tubes.
2. Horizontal, with condensation in the tubes.
3. Vertical, with condensation in the shell.
4. Vertical, with condensation in the tubes.

Horizontal shell side and vertical tube side are the most commonly used types of condenser. A horizontal exchanger with condensation in the tubes is rarely used as a process condenser, but is the usual arrangement for heaters and vaporizers using condensing steam as the heating medium.

19.10.1 Heat transfer fundamentals

The fundamentals of condensation heat transfer are covered in [Chhabra and Shankar \(2017\)](#).

Heat exchanger specification sheet

Customer										Job No.				
Address										Reference No.				
Plant location										Proposal No.				
Service of unit Example 12.4										Date	Rev.			
Size 540-3600 Type AES Horizontal Connected 1 parallel 1 series										Item No.	E100			
Surf./Unit (Gross) 65.71 m ² Shells/Unit 1 Surface/Shell (Gross) 65.71 m ²														
Performance of one unit														
Fluid allocation				Shell side				Tube side						
Fluid name														
Fluid quantity,		Total	kg/h	20000.0				70000.2						
Vapour														
Liquid				20000.0		20000.0		70000.2		70000.2				
Steam														
Water														
Noncondensable														
Temperature (In/Out)			°C	200.0		90.0		40.0		78.0				
Density			kg/m ³	690.0		770.0		840.0		800.0				
Viscosity			centipoise	0.22		0.8		4.3		2.39925				
Molecular weight, vapour														
Molecular weight, Noncondensable														
Specific heat			kJ/kg K	2.72		2.26		2.01		2.09				
Thermal conductivity			W/m K	0.13		0.135		0.135		0.133				
Latent heat				kJ/kg										
Inlet pressure			bar(abs)	5.0				6.5						
Velocity			m/s	0.48				1.84						
Pressure drop, allow. / calc.			bar	0.8		0.14338		0.8		0.66002				
Fouling resistance (min.)			m ² K/W	0.0002				0.0003 (0.00039 referred to OD)						
Heat exchaged			1515.56	kW		MTD		71.62		°C				
Transfer rate, service			345.3	Dirty		345.4		Clean		432.9 W/m ² K				
Construction of one shell										Sketch (bundle/nozzle orientation)				
Design/test pressure		bar(g)	7.43		8.93									
Design temperature		°C	300.0		178.0									
No. passes per shell					4									
Corrosion allowance														
Connections In			62.7		90.1									
Size & Out			62.7		77.9									
Ratings Inter														
Tube no.	305	OD	19.05	mm	Thk	2.11	mm	Length	3600	mm	Pitch	25.4	mm	30deg
Tube type		plain					Material		Carbon steel					
Shell		ID	540	OD	mm		Shell cover		(Integ.)(Remov.)					
Channel or Bonnet		Carbon steel				Channel cover								
Tubesheet-stationary		Carbon steel				Tubesheet-floating								
Floating head cover						Impingement protection								
Baffles-cross		28	Type	Single segmental		% Cut	25.0	Spacing c/c	113.5	Inlet	mm			
Baffles-long						Seal type								
Supports-tube		U-Bend				Type								
Bypass seal arrangement						Tube-Tubesheet joint								
Expansion joint						Type								
ρV ² -inlet nozzle		4689.0	Bundle entrance		488.0	Bundle exit		437.3						
Gaskets		Shell side				Tube side								
Floating head														
Code requirements						Tema class		R						
Weight/shell		2612	Filled with water		3564	Bundle		1319	kg					
Remarks														

FIG. 19.37 TEMA specification sheet for Example 19.4.

The normal mechanism for heat transfer in commercial condensers is film-wise condensation. Drop-wise condensation will give higher heat transfer coefficients, but is unpredictable and is not yet considered a practical proposition for the design of condensers for general purposes.

The basic equations for film-wise condensation were derived by [Nusselt \(1916\)](#), and his equations form the basis for practical condenser design. The Nusselt equations are derived in [Chhabra and Shankar \(2017\)](#). In the Nusselt model of condensation, laminar flow is assumed in the film, and heat transfer is assumed to take place entirely by conduction through the film. In practical condensers, the Nusselt model will strictly only apply at low liquid and vapor rates and where the flowing condensate film is undisturbed. Turbulence can be induced in the liquid film at high liquid rates and by shear at high vapor rates. This will generally increase the rate of heat transfer

over that predicted using the Nusselt model. The effect of vapor shear and film turbulence are discussed in [Chhabra and Shankar \(2017\)](#); see also [Butterworth \(1978\)](#) and [Taborek \(1974\)](#).

Developments in the theory of condensation and their application in condenser design are reviewed by [Owen and Lee \(1983\)](#).

Physical properties

The physical properties of the condensate for use in the following equations are evaluated at the average condensate film temperature: the mean of the condensing temperature and the tube-wall temperature.

19.10.2 Condensation outside horizontal tubes

$$(h_c)_1 = 0.95k_L \left[\frac{\rho_L(\rho_L - \rho_v)g}{\mu_L \Gamma} \right]^{1/3} \quad (19.27)$$

where $(h_c)_1$ = mean condensation film coefficient for a single tube, $\text{W}/\text{m}^2\text{C}$

k_L = condensate thermal conductivity, $\text{W}/\text{m}^\circ\text{C}$

ρ_L = condensate density, kg/m^3

ρ_v = vapor density, kg/m^3

μ_L = condensate viscosity, Ns/m^2

g = gravitational acceleration, $9.81 \text{ m}/\text{s}^2$

Γ = the tube loading, the condensate flow per unit length of tube, $\text{kg}/\text{m s}$

In a bank of tubes, the condensate from the upper rows of tubes will add to that condensing on the lower tubes. If there are N_r tubes in a vertical row and the condensate is assumed to flow smoothly from row to row ([Fig. 19.38a](#)) and if the flow remains laminar, the mean coefficient predicted by the Nusselt model is related to that for the top tube by:

$$(h_c)_{N_r} = (h_c)_1 N_r^{-1/4} \quad (19.28)$$

In practice, the condensate will not flow smoothly from tube to tube (see [Fig. 19.38b](#)) and the factor of $(N_r)^{-1/4}$ applied to the single tube coefficient in Equation 19.28 is considered to be too conservative. Based on results from commercial exchangers, [Kern \(1950\)](#) suggests using an index of $1/6$. [Frank \(1978\)](#) suggests multiplying single tube coefficient by a factor of 0.75.

Using Kern's method, the mean coefficient for a tube bundle is given by:

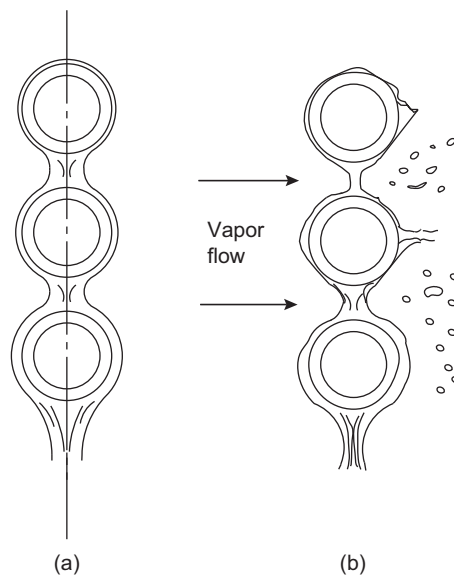


FIG. 19.38 Condensate flow over tube banks.

$$(h_c)_b = 0.95k_L \left[\frac{\rho_L(\rho_L - \rho_v)g}{\mu_L \Gamma_h} \right]^{1/3} N_r^{-1/6} \quad (19.29)$$

where $\Gamma_h = \frac{W_c}{LN_t}$

and L = tube length,

W_c = total condensate flow,

N_t = total number of tubes in the bundle,

N_r = average number of tubes in a vertical tube row.

N_r can be taken as two-thirds of the number in the central tube row.

For low-viscosity condensates, the correction for the number of tube rows is generally ignored.

A procedure for estimating the shell-side heat transfer in horizontal condensers is given in the ESDU Design Guide, ESDU 84023.

19.10.3 Condensation inside and outside vertical tubes

For condensation inside and outside vertical tubes, the Nusselt model gives:

$$(h_c)_v = 0.926k_L \left[\frac{\rho_L(\rho_L - \rho_v)g}{\mu_L \Gamma_v} \right]^{1/3} \quad (19.30)$$

where $(h_c)_v$ = mean condensation coefficient, W/m²°C

Γ_v = vertical tube loading, condensate rate per unit tube perimeter, kg/m s for a tube bundle

$$\Gamma_v = \frac{W_c}{N_t \pi d_o} \quad \text{or} \quad \frac{W_c}{N_t \pi d_i}$$

Equation 19.30 will apply up to a Reynolds number of 30; above this value, waves on the condensate film become important. The Reynolds number for the condensate film is given by:

$$\text{Re}_c = \frac{4 \Gamma_v}{\mu_L}$$

The presence of waves will increase the heat transfer coefficient, so the use of Equation 19.30 above a Reynolds number of 30 will give conservative (safe) estimates. The effect of waves on condensate film on heat transfer is discussed by Kutateladze (1963).

Above a Reynolds number of around 2000, the condensate film becomes turbulent. The effect of turbulence in the condensate film was investigated by Colburn (1934), and Colburn's results are generally used for condenser design (Fig. 19.39). Equation 19.30 is also shown in Fig. 19.39. The Prandtl number for the condensate film is given by:

$$\text{Pr}_c = \frac{C_p \mu_L}{k_L}$$

Fig. 19.39 can be used to estimate condensate film coefficients in the absence of appreciable vapor shear. Horizontal and downward vertical vapor flow will increase the rate of heat transfer, and the use of Fig. 19.39 will give conservative values for most practical condenser designs.

Boyko and Kruzhilin (1967) developed a correlation for shear-controlled condensation in tubes that is simple to use. Their correlation gives the mean coefficient between two points at which the vapor quality is known. The vapor quality x is the mass fraction of the vapor present. It is convenient to represent the Boyko–Kruzhilin correlation as:

$$(h_c)_{BK} = h'_t \left[\frac{J_1^{1/2} + J_2^{1/2}}{2} \right] \quad (19.31)$$

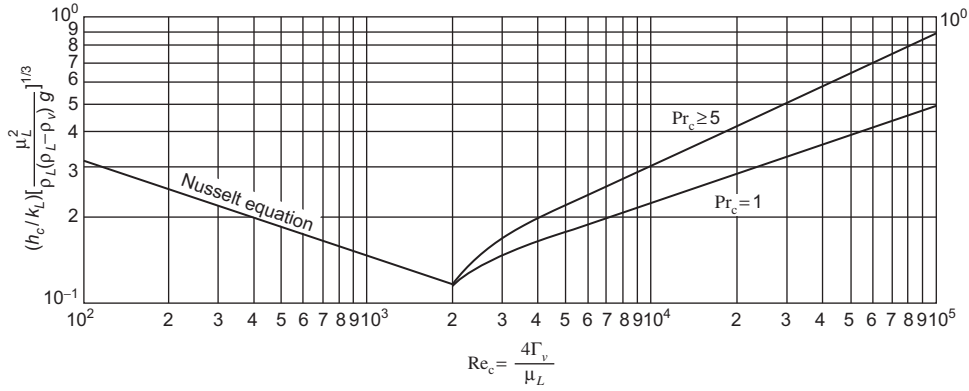


FIG. 19.39 Condensation coefficient for vertical tubes.

where

$$J = 1 + \left[\frac{\rho_L - \rho_v}{\rho_v} \right] x$$

and the suffixes 1 and 2 refer to the inlet and outlet conditions, respectively. h'_t is the tube-side coefficient evaluated for single-phase flow of the total condensate (the condensate at point 2). That is, the coefficient that would be obtained if the condensate filled the tube and was flowing alone; this can be evaluated using any suitable correlation for forced convection in tubes; see Section 19.8.

Boyko and Kruzhilin used the correlation:

$$h'_t = 0.021 \left(\frac{k_L}{d_i} \right) \text{Re}^{0.8} \text{Pr}^{0.43} \quad (19.32)$$

In a condenser, the inlet stream will normally be saturated vapor and the vapor will be totally condensed. For these conditions Equation 19.31 becomes:

$$(h_c)_{BK} = h'_t \left[\frac{1 + \sqrt{\rho_L/\rho_v}}{2} \right] \quad (19.33)$$

For the design of condensers with condensation inside the tubes and downward vapor flow, the coefficient should be evaluated using Fig. 19.39 and Equation 19.31 and the *higher* value selected.

Flooding in vertical tubes

When the vapor flows up the tube, which will be the usual arrangement for a reflux condenser, care must be taken to ensure that the tubes do not flood. Several correlations have been published for the prediction of flooding in vertical tubes; see Green and Southard (2018). One of the simplest to apply, which is suitable for use in the design of condensers handling low-viscosity condensates, is the criterion given by Hewitt and Hall-Taylor (1970); see also Butterworth (1977). Flooding should not occur if the following condition is satisfied:

$$\left[u_v^{1/2} \rho_v^{1/4} + u_L^{1/2} \rho_L^{1/4} \right] < 0.6 [g d_i (\rho_L - \rho_v)]^{1/4} \quad (19.34)$$

where u_v and u_L are the velocities of the vapor and liquid based on each phase flowing in the tube alone and d_i is in meters. The critical condition will occur at the bottom of the tube, so the vapor and liquid velocities should be evaluated at this point.

Example 19.5

Estimate the heat transfer coefficient for steam condensing on the outside and on the inside of a 25 mm o.d., 21 mm i.d. vertical tube 3.66 m long. The steam condensate rate is 0.015 kg/s per tube, and condensation takes place at 3 bar. The steam will flow down the tube.

Solution

Physical properties, from steam tables:

$$\text{Saturation temperature} = 133.5^\circ\text{C}$$

$$\rho_L = 931 \text{ kg/m}^3$$

$$\rho_v = 1.65 \text{ kg/m}^3$$

$$k_L = 0.688 \text{ W/m}^\circ\text{C}$$

$$\mu_L = 0.21 \text{ mNs/m}^2$$

$$\text{Pr}_c = 1.27$$

Condensation outside the tube

$$\Gamma_v = \frac{0.015}{\pi \times 25 \times 10^{-3}} = 0.191 \text{ kg/s m}$$

$$\text{Re}_c = \frac{4 \times 0.191}{0.21 \times 10^{-3}} = 3638$$

From Fig. 19.39:

$$\frac{h_c}{k_L} \left[\frac{\mu_L^2}{\rho_L(\rho_L - \rho_v)g} \right]^{1/3} = 1.65 \times 10^{-1}$$

$$h_c = 1.65 \times 10^{-1} \times 0.688 \left[\frac{(0.21 \times 10^{-3})^2}{931(931 - 1.65)9.81} \right]^{-1/3}$$

$$= \underline{\underline{6554 \text{ W/m}^2\text{C}}}$$

Condensation inside the tube

$$\Gamma_v = \frac{0.015}{\pi \times 21 \times 10^{-3}} = 0.227 \text{ kg/s m}$$

$$\text{Re}_c = \frac{4 \times 0.227}{0.21 \times 10^{-3}} = 4324$$

From Fig. 19.39:

$$h_c = 1.72 \times 10^{-1} \times 0.688 \left[\frac{(0.21 \times 10^{-3})^2}{931(931 - 1.65)9.81} \right]^{-1/3}$$

$$= 6832 \text{ W/m}^2\text{C}$$

Using the Boyko–Kruzhilin method:

$$\text{Cross-sectional area of tube} = (21 \times 10^{-3})^2 \frac{\pi}{4} = 3.46 \times 10^{-4} \text{ m}^2$$

Fluid velocity, total condensation:

$$u_t = \frac{0.015}{931 \times 3.46 \times 10^{-4}} = 0.047 \text{ m/s}$$

$$\text{Re} = \frac{\rho u d_i}{\mu} = \frac{931 \times 0.047 \times 21 \times 10^{-3}}{0.21 \times 10^{-3}} = 4376$$

$$h_t = 0.021 \times \frac{0.688}{21 \times 10^{-3}} (4376)^{0.8} (1.27)^{0.43} = 624 \text{ W/m}^2\text{°C} \quad (19.32)$$

$$h_c = 624 \left[\frac{1 + \sqrt{931/1.65}}{2} \right] = 7723 \text{ W/m}^2\text{°C} \quad (19.33)$$

Take higher value, $h_c = \underline{\underline{7723 \text{ W/m}^2\text{°C}}}$

Example 19.6

It is proposed to use an existing distillation column, which is fitted with a dephlegmator (reflux condenser) that has 200 vertical, 50 mm i.d. tubes for separating benzene from a mixture of chlorobenzenes. The top product will be 2500 kg/h benzene, and the column will operate with a reflux ratio of 3. Check if the tubes are likely to flood. The condenser pressure will be 1 bar.

Solution

The vapor will flow up and the liquid down the tubes. The maximum flow rates of both will occur at the base of the tube.

$$\text{Vapor flow} = (3 + 1) \times 2500 = 10,000 \text{ kg/h}$$

$$\text{Liquid flow} = 3 \times 2500 = 7,500 \text{ kg/h}$$

$$\text{Total area of tubes} = \frac{\pi}{4} (50 \times 10^{-3})^2 \times 200 = 0.39 \text{ m}^2$$

Densities at benzene boiling point:

$$\rho_L = 840 \text{ kg/m}^3, \quad \rho_v = 2.7 \text{ kg/m}^3$$

Vapor velocity (vapor flowing alone in tube):

$$u_v = \frac{10,000}{3600 \times 0.39 \times 2.7} = 2.64 \text{ m/s}$$

Liquid velocity (liquid alone):

$$u_L = \frac{7500}{3600 \times 0.39 \times 840} = 0.006 \text{ m/s}$$

From Equation 19.34 for no flooding:

$$\begin{aligned} \left[u_v^{1/2} \rho_v^{1/4} + u_L^{1/2} \rho_L^{1/4} \right] &< 0.6 [g d_i (\rho_L - \rho_v)]^{1/4} \\ \left[(2.64)^{1/2} (2.7)^{1/4} + (0.006)^{1/2} (840)^{1/4} \right] &< 0.6 [9.81 \times 50 \times 10^{-3} (840 - 2.7)]^{1/4} \\ [2.50] &< [2.70] \end{aligned}$$

Tubes should not flood, but there is little margin of safety.

19.10.4 Condensation inside horizontal tubes

Where condensation occurs in a horizontal tube, the heat transfer coefficient at any point along the tube depends on the flow pattern at that point. The various patterns that can exist in two-phase flow are shown in Fig. 19.40 and are discussed in Chhabra and Shankar (2017). In condensation, the flow will vary from a single-phase vapor at the inlet to a single-phase liquid at the outlet, with all the possible patterns of flow occurring between these points. Bell et al. (1970) give a method for following the change in flow pattern as condensation occurs on a Baker flow-regime map. Correlations for estimating the average condensation coefficient have been published by several workers, but there is no generally satisfactory method that will give accurate predictions over a wide flow range. A comparison of the published methods is given by Bell et al. (1970).

Two flow models are used to estimate the mean condensation coefficient in horizontal tubes: stratified flow (Fig. 19.41a) and annular flow (see Fig. 19.41b). The stratified-flow model represents the limiting condition at low condensate and vapor rates, and the annular model the condition at high vapor and low condensate rates. For the stratified-flow model, the condensate film coefficient can be estimated from the Nusselt equation, applying a suitable correction for the reduction in the coefficient caused by the accumulation of condensate in the bottom of the tube. The correction factor will typically be around 0.8, so the coefficient for stratified flow can be estimated from:

$$(h_c)_s = 0.76k_L \left[\frac{\rho_L(\rho_L - \rho_v)g}{\mu_L \Gamma_h} \right]^{1/3} \quad (19.35)$$

The Boyko–Kruzhilin equation, Equation 19.31, can be used to estimate the coefficient for annular flow.

For condenser design, the mean coefficient should be evaluated using the correlations for both annular and stratified flow and the *higher* value selected.

19.10.5 Condensation of steam

Steam is frequently used as a heating medium. The film coefficient for condensing steam can be calculated using the methods given in the previous sections, but as the coefficient will be high and will rarely be the limiting coefficient, it is customary to assume a typical conservative value for design purposes. For air-free steam, a coefficient of 8000 W/m²°C (1500 Btu/h ft²°F) can be used.

19.10.6 Mean temperature difference

A pure, saturated vapor will condense at a fixed temperature, at constant pressure. For an isothermal process such as this, the simple logarithmic mean temperature difference can be used in Equation 19.1; no correction factor for multiple passes is needed. The logarithmic mean temperature difference will be given by:

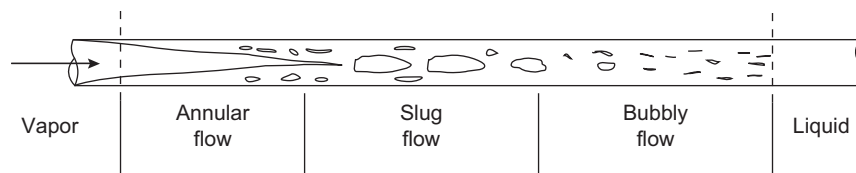


FIG. 19.40 Flow patterns, vapor condensing in a horizontal tube.

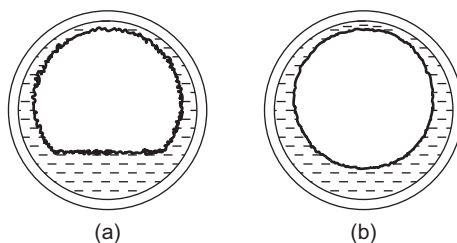


FIG. 19.41 Flow patterns in condensation. (a) Stratified flow. (b) Annular flow.

$$\Delta T_{\text{lm}} = \frac{(t_2 - t_1)}{\ln \left[\frac{T_{\text{sat}} - t_1}{T_{\text{sat}} - t_2} \right]} \quad (19.36)$$

where T_{sat} = saturation temperature of the vapor

t_1 = inlet coolant temperature

t_2 = outlet coolant temperature

When the condensation process is not exactly isothermal but the temperature change is small, such as where there is a significant change in pressure or where a narrow boiling range multicomponent mixture is being condensed, the logarithmic temperature difference can still be used, but the temperature correction factor will be needed for multi-pass condensers. The appropriate terminal temperatures should be used in the calculation.

19.10.7 Desuperheating and subcooling

When the vapor entering the condenser is superheated and the condensate leaving the condenser is cooled below its boiling point (subcooled), the temperature profile will be as shown in Fig. 19.42.

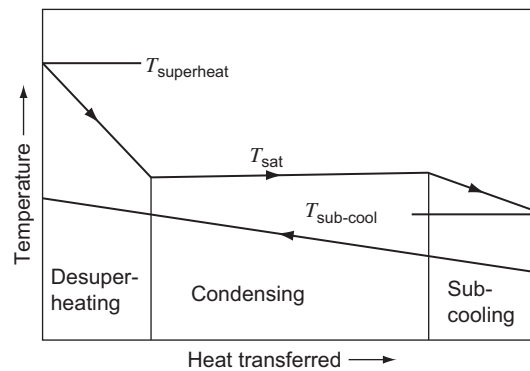


FIG. 19.42 Condensation with desuperheating and subcooling.

Desuperheating

If the degree of superheat is large, it will be necessary to divide the temperature profile into sections and determine the mean temperature difference and heat transfer coefficient separately for each section. If the tube wall temperature is below the dew point of the vapor, liquid will condense directly from the vapor onto the tubes. In these circumstances, it has been found that the heat transfer coefficient in the desuperheating section is close to the value for condensation and can be taken as the same. So, where the amount of superheat is not excessive, say less than 25% of the latent heat load, and the outlet coolant temperature is well below the vapor dew point, the sensible heat load for desuperheating can be lumped with the latent heat load. The total heat transfer area required can then be calculated using a mean temperature difference based on the saturation temperature (not the superheat temperature) and the estimated condensate film heat transfer coefficient.

Subcooling of condensate

Some subcooling of the condensate will usually be required to control the net positive suction head at the condensate pump (see Chapter 20) or to cool a product for storage. Where the amount of subcooling is large, it is more efficient to subcool in a separate exchanger. A small amount of subcooling can be obtained in a condenser by controlling the liquid level so that some part of the tube bundle is immersed in the condensate.

In a horizontal shell-side condenser, a dam baffle can be used (Fig. 19.43a). A vertical condenser can be operated with the liquid level above the bottom tubesheet (see Fig. 19.43b).

The temperature difference in the subcooled region will depend on the degree of mixing in the pool of condensate. The limiting conditions are plug flow and complete mixing. The temperature profile for plug flow is that shown in Fig. 19.42. If the pool is perfectly mixed, the condensate temperature will be constant over the subcooling region and equal to the condensate outlet temperature. Assuming perfect mixing will give a very conservative (safe) estimate of the mean temperature difference. As the liquid velocity will be low in the subcooled region, the heat transfer

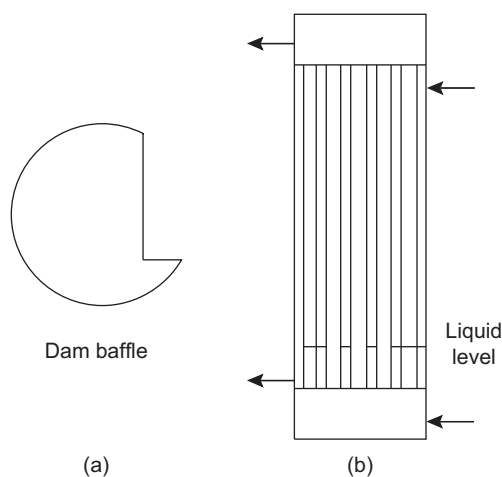


FIG. 19.43 Arrangements for subcooling.

coefficient should be estimated using correlations for natural convection (see [Chhabra & Shankar 2017](#)); a typical value would be $200 \text{ W/m}^2\text{°C}$.

19.10.8 Condensation of mixtures

The correlations given in the previous sections apply to the condensation of a single component, such as an essentially pure overhead product from a distillation column. The design of a condenser for a mixture of vapors is more difficult.

The term "*mixture of vapors*" covers three related situations of practical interest:

1. Total condensation of a multicomponent mixture, such as the overheads from a multicomponent distillation.
2. Condensation of only part of a multicomponent vapor mixture, all components of which are theoretically condensable. This situation will occur where the dew point of some of the lighter components is above the coolant temperature. The uncondensed component may be soluble in the condensed liquid, such as in the condensation of some hydrocarbon mixtures containing light "gaseous" components.
3. Condensation from a noncondensable gas, where the gas is not soluble to any extent in the liquid condensed. These exchangers are often called cooler-condensers.

The following features, common to all these situations, must be considered when developing design methods for mixed vapor condensers:

1. The condensation will not be isothermal. As the heavy component condenses out, the composition of the vapor, and therefore its dew point, changes.
2. Because the condensation is not isothermal, there will be a transfer of sensible heat from the vapor to cool the gas to the dew point. There will also be a transfer of sensible heat from the condensate, as it must be cooled from the temperature at which it condensed to the outlet temperature. The transfer of sensible heat from the vapor can be particularly significant, as the sensible heat transfer coefficient will be appreciably lower than the condensation coefficient.
3. As the composition of the vapor and liquid change throughout the condenser, their physical properties vary.
4. The heavy component must diffuse through the lighter components to reach the condensing surface. The rate of condensation will be governed by the rate of diffusion, as well as the rate of heat transfer.

Temperature profile

To evaluate the true temperature difference (driving force) in a mixed vapor condenser, a condensation curve (temperature vs. enthalpy diagram) must be calculated, showing the change in vapor temperature versus heat transferred throughout the condenser ([Fig. 19.44](#)). The temperature profile depends on the liquid-flow pattern in the condenser. There are two limiting conditions of condensate–vapor flow:

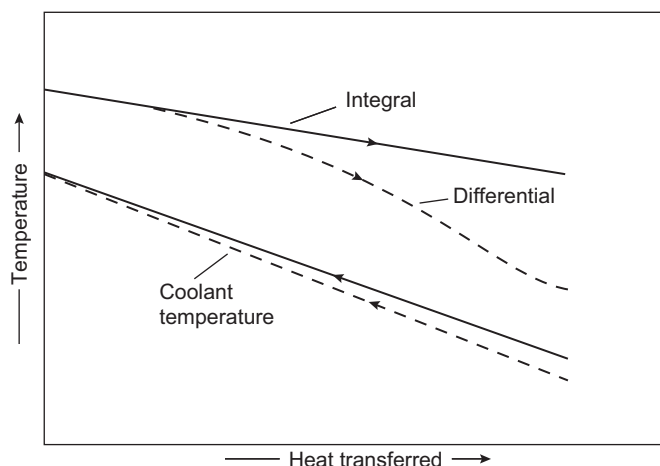


FIG. 19.44 Condensation curves.

1. Differential condensation, in which the liquid separates from the vapor from which it has condensed. This process is analogous to differential, or Rayleigh, distillation, and the condensation curve can be calculated using methods similar to those for determining the change in composition in differential distillation; see [Chhabra and Gurappa \(2019\)](#).
2. Integral condensation, in which the liquid remains in equilibrium with the uncondensed vapor. The condensation curve can be determined using procedures similar to those for multicomponent flash distillation given in [Chapter 17](#). This will be a relatively simple calculation for a binary mixture, but complex and tedious for mixtures of more than two components.

It is normal practice to assume that integral condensation occurs. The conditions for integral condensation will be approached if condensation is carried out in one pass so that the liquid and vapor follow the same path as in a vertical condenser with condensation inside or outside the tubes. In a horizontal shell-side condenser, the condensate will tend to separate from the vapor. The mean temperature difference will be lower for differential condensation, and arrangements where liquid separation is likely to occur should generally be avoided for the condensation of mixed vapors.

Where integral condensation can be considered to occur, the use of a corrected logarithmic mean temperature difference based on the terminal temperatures will generally give a conservative (safe) estimate of the mean temperature difference that can be used in preliminary design calculations.

Estimation of heat transfer coefficients

Total condensation. For the design of a multicomponent condenser in which the vapor is totally condensed, an estimate of the mean condensing coefficient can be made using the single component correlations with the liquid physical properties evaluated at the average condensate composition. It is the usual practice to apply a factor of safety to allow for the sensible heat transfer and any resistance to mass transfer. [Frank \(1978\)](#) suggests a factor of 0.65, but this is probably too pessimistic. [Kern \(1950\)](#) suggests increasing the area calculated for condensation alone by the ratio of the total heat (condensing + sensible) to the condensing load. Where a more exact estimate of the coefficient is required and justified by the data, the rigorous methods developed for partial condensation can be used.

Partial condensation. The methods developed for partial condensation and condensation from a noncondensable gas can be divided into two classes:

1. Empirical methods: Approximate methods in which the resistance to heat transfer is considered to control the rate of condensation and the mass transfer resistance is neglected. Design methods have been published by [Silver \(1947\)](#), [Bell and Ghaly \(1973\)](#), and [Ward \(1960\)](#).
2. Analytical methods: More exact procedures that are based on some model of the heat and mass transfer process and which take into account the diffusion resistance to mass transfer. The classic method is that of [Colburn and Hougren \(1934\)](#); see also [Colburn and Drew \(1937\)](#) and [Porter and Jeffreys \(1963\)](#). The analytical methods are complex, requiring iterative calculations or graphical procedures. They are suited for computer solutions using

numerical methods, and proprietary design programs are available. Examples of the application of the Colburn and Drew method are given by Kern (1950) and Jeffreys (1961). The method is discussed briefly in Chhabra and Shankar (2017).

An assessment of the methods available for the design of condensers where the condensation is from a noncondensable gas is given by McNaught (1983).

Approximate methods. The local coefficient for heat transfer can be expressed in terms of the local condensate film coefficient h'_c and the local coefficient for sensible heat transfer from the vapor (the gas film coefficient) h'_g by a relationship first proposed by Silver (1947):

$$\frac{1}{h'_{cg}} = \frac{1}{h'_c} + \frac{Z}{h'_g} \quad (19.37)$$

where h'_{cg} = the local effective cooling-condensing coefficient

$$\text{and } Z = \frac{\Delta H_s}{\Delta H_t} = x C_{pg} \frac{dT}{dH_t},$$

$(\Delta H_s/\Delta H_t)$ = the ratio of the change in sensible heat to the total enthalpy change

(dT/dH_t) = slope of the temperature–enthalpy curve

x = vapor quality, mass fraction of vapor

C_{pg} = vapor (gas) specific heat

The term dT/dH_t can be evaluated from the condensation curve, h'_c from the single component correlations, and h'_g from correlations for forced convection.

If this is done at several points along the condensation curve, the area required can be determined by graphical or numerical integration of the expression:

$$A = \int_0^{Q_t} \frac{dQ}{U(T_v - t_c)} \quad (19.38)$$

where Q_t = total heat transferred

U = overall heat transfer coefficient from Equation 19.1, using h'_{cg} ,

T_v = local vapor (gas) temperature

t_c = local cooling medium temperature

Gilmore (1963) gives an integrated form of Equation 19.37, which can be used for the approximate design of partial condensers:

$$\frac{1}{h_{cg}} = \frac{1}{h_c} + \frac{Q_g}{Q_t} \frac{1}{h_g} \quad (19.39)$$

where h_{cg} = mean effective coefficient

h_c = mean condensate film coefficient, evaluated from the single-component correlations at the average condensate composition and total condensate loading

h_g = mean gas film coefficient, evaluated using the average vapor flow rate: arithmetic mean of the inlet and outlet vapor (gas) flow rates

Q_g = total sensible heat transfer from vapor (gas)

Q_t = total heat transferred: latent heat of condensation + sensible heat for cooling the vapor (gas) and condensate

As a rough guide, the following rules of thumb suggested by Frank (1978) can be used to decide the design method to use for a partial condenser (cooler-condenser):

1. Noncondensables <0.5%: Use the methods for total condensation; ignore the presence of the uncondensed portion.
2. Noncondensables >70%: Assume the heat transfer is by forced convection only. Use the correlations for forced convection to calculate the heat transfer coefficient but include the latent heat of condensation in the total heat load transferred.
3. Between 0.5% and 70% noncondensables: Use methods that consider both mechanisms of heat transfer.

In partial condensation, it is usually better to put the condensing stream on the shell side and to select a baffle spacing that will maintain high vapor velocities, and therefore high sensible heat transfer coefficients.

Fog formation. In the condensation of a vapor from a noncondensable gas, if the bulk temperature of the gas falls below the dew point of the vapor, liquid can condense out directly as a mist or fog. This condition is undesirable, as liquid droplets may be carried out of the condenser. Fog formation in cooler-condensers is discussed by Colburn and Edison (1941) and LoPinto (1982). Steinmeyer (1972) gives criteria for the prediction of fog formation. Demisting pads can be used to separate entrained liquid droplets.

19.10.9 Pressure drop in condensers

The pressure drop on the condensing side is difficult to predict, as two phases are present and the vapor mass velocity is changing throughout the condenser.

A common practice is to calculate the pressure drop using the methods for single-phase flow and apply a factor to allow for the change in vapor velocity. For total condensation, Frank (1978) suggests taking the pressure drop as 40% of the value based on the inlet vapor conditions; Kern (1950) suggests a factor of 50%.

An alternative method, which can also be used to estimate the pressure drop in a partial condenser, is given by Gloyer (1970). The pressure drop is calculated using an average vapor flow rate in the shell (or tubes) estimated as a function of the ratio of the vapor flow rate in and out of the shell (or tubes) and the temperature profile.

$$W_s \text{ (average)} = W_s \text{ (inlet)} \times K_2 \quad (19.40)$$

K_2 is obtained from Fig. 19.45.

$\Delta T_{in}/\Delta T_{out}$ in Fig. 19.45 is the ratio of the terminal temperature differences.

These methods can be used to make a crude estimate of the likely pressure drop. A reliable prediction can be obtained by treating the problem as one of two-phase flow. For tube-side condensation, the general methods for two-phase flow in pipes can be used; see Collier and Thome (1994) and Chhabra and Shankar (2017). As the flow pattern will be changing throughout condensation, some form of step-wise procedure must be used. Two-phase flow on the shell side is discussed by Grant (1973), who gives a method for predicting the pressure drop based on Tinker's shell-side flow model. More sophisticated methods are available in the commercial heat exchanger design programs from HTFS and HTRI.

A method for estimating the pressure drop on the shell side of horizontal condensers is given in the ESDU Design Guide, ESDU 84023 (1985).

Pressure drop is only likely to be a major consideration in the design of vacuum condensers and where reflux is returned to a column by gravity flow from the condenser.

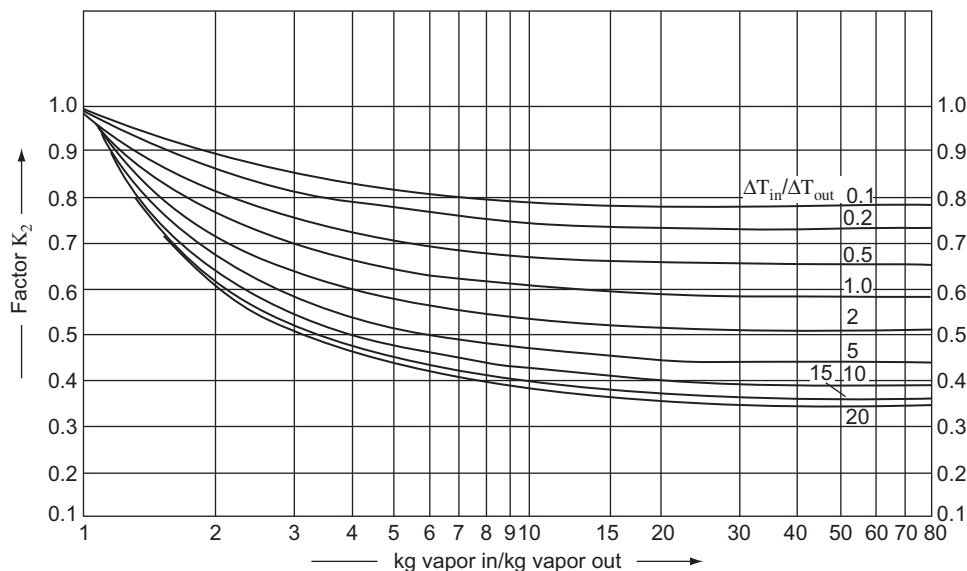


FIG. 19.45 Factor for average vapor flow rate for pressure-drop calculation.

Example 19.7

Design a condenser for the following duty: 45,000 kg/h of mixed light hydrocarbon vapors to be condensed. The condenser to operate at 10 bar. The vapor will enter the condenser saturated at 60 °C and the condensation will be complete at 45 °C. The average molecular weight of the vapors is 52. The enthalpy of the vapor is 596.5 kJ/kg and the condensate 247.0 kJ/kg. Cooling water is available at 30 °C, and the temperature rise is to be limited to 10 °C. Plant standards require tubes of 20 mm o.d., 16.8 mm i.d., 4.88 m (16 ft) long of admiralty brass. The vapors are to be totally condensed, and no sub-cooling is required.

Solution

Only the thermal design will be done. The physical properties of the mixture will be taken as the mean of those for n-propane (MW = 44) and n-butane (MW = 58) at the average temperature.

$$\text{Heat transferred from vapour} = \frac{45,000}{3600}(596.5 - 247.0) = 4368.8 \text{ kW}$$

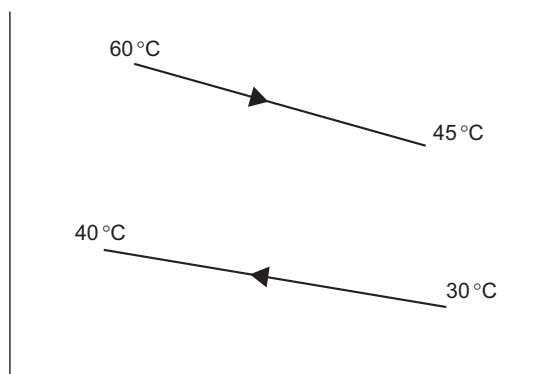
$$\text{Cooling water flow} = \frac{4368.8}{(40 - 30)4.18} = \underline{\underline{104.5 \text{ kg/s}}}$$

Assumed overall coefficient (Table 19.1) = 900 W/m²°C.

Mean temperature difference: The condensation range is small and the change in saturation temperature will be linear, so the corrected logarithmic mean temperature difference can be used.

$$R = \frac{(60 - 45)}{(40 - 30)} = 1.5 \quad (19.6)$$

$$S = \frac{(40 - 30)}{(60 - 30)} = 0.33 \quad (19.7)$$



Try a horizontal exchanger, condensation in the shell, four tube passes. For one shell pass, four tube passes, from Fig. 19.19, $F_t = 0.92$:

$$\Delta T_{lm} = \frac{(60 - 40) - (45 - 30)}{\ln \frac{(60 - 40)}{(45 - 30)}} = 17.4^\circ\text{C}$$

$$\Delta T_{lm} = 0.92 \times 17.4 = 16^\circ\text{C}$$

$$\text{Trial area} = \frac{4368.8 \times 10^3}{900 \times 16} = 303 \text{ m}^2$$

$$\text{Surface area of one tube} = 20 \times 10^{-3} \pi \times 4.88 = 0.305 \text{ m}^2 (\text{ignore tubesheet thickness})$$

$$\text{Number of tubes} = \frac{303}{0.305} = 992$$

Use square pitch, $P_t = 1.25 \times 20 \text{ mm} = 25 \text{ mm}$.

Tube bundle diameter:

$$D_b = 20 \left(\frac{992}{0.158} \right)^{1/2.263} = 954 \text{ mm} \quad (19.3b)$$

Number of tubes in center row $N_r = D_b/P_t = 954/25 = 38$

Shell-side coefficient

Estimate tube wall temperature, T_w ; assume condensing coefficient of $1500 \text{ W/m}^2\text{°C}$:

Mean temperature:

$$\text{Shell-side} = \frac{60 + 45}{2} = 52.5^\circ\text{C}$$

$$\text{Tube-side} = \frac{40 + 30}{2} = 35^\circ\text{C}$$

$$(52.5 - T_w)1500 = (52.5 - 35)900$$

$$T_w = 42.0^\circ\text{C}$$

$$\text{Mean temperature condensate} = \frac{52.5 + 42.0}{2} = 47^\circ\text{C}$$

Physical properties at 47°C :

$$\mu_L = 0.16 \text{ mNs/m}^2$$

$$\rho_L = 551 \text{ kg/m}^3$$

$$k_L = 0.13 \text{ W/m}^\circ\text{C}$$

Vapor density at mean vapor temperature:

$$\rho_v = \frac{52}{22.4} \times \frac{273}{(273 + 52.5)} \times \frac{10}{1} = 19.5 \text{ kg/m}^3$$

$$\Gamma_h = \frac{W_c}{LN_t} = \frac{45,000}{3600} \times \frac{1}{4.88 \times 992} = 2.6 \times 10^{-3} \text{ kg/s m}$$

$$N_r = \frac{2}{3} \times 38 = 25 \quad (19.29)$$

$$h_c = 0.95 \times 0.13 \left[\frac{551(551 - 19.5)9.81}{0.16 \times 10^{-3} \times 2.6 \times 10^{-3}} \right]^{1/3} \times 25^{-1/6}$$

$$= 1375 \text{ W/m}^2\text{°C}$$

Close enough to assumed value of $1500 \text{ W/m}^2\text{°C}$, so no correction to T_w needed.

Tube-side coefficient

$$\text{Tube cross-sectional area} = \frac{\pi}{4} (16.8 \times 10^{-3})^2 \times \frac{992}{4} = 0.055 \text{ m}^2$$

$$\text{Density of water, at } 35^\circ\text{C} = 993 \text{ kg/m}^3$$

$$\text{Tube velocity} = \frac{104.5}{993} \times \frac{1}{0.055} = 1.91 \text{ m/s} \quad (19.17)$$

$$h_i = \frac{4200(1.35 + 0.02 \times 35)1.91^{0.8}}{16.8^{0.2}}$$

$$= 8218 \text{ W/m}^2\text{°C}$$

Fouling factors: As neither fluid is heavily fouling, use $6000 \text{ W/m}^2\text{°C}$ for each side.

$$k_w = 50 \text{ W/m}^2\text{°C}$$

Overall coefficient

$$\frac{1}{U} = \frac{1}{1375} + \frac{1}{6000} + \frac{20 \times 10^{-3} \ln\left(\frac{20}{16.8}\right)}{2 \times 50} + \frac{20}{16.8} \times \frac{1}{6000} + \frac{20}{16.8} \times \frac{1}{8218} \quad (19.2)$$

$$U = \underline{\underline{786 \text{ W/m}^2\text{°C}}}$$

Significantly lower than the assumed value of $900 \text{ W/m}^2\text{°C}$.

Repeat calculation using new trial value of $750 \text{ W/m}^2\text{°C}$:

$$\text{Area} = \frac{4368 \times 10^3}{750 \times 16} = 364 \text{ m}^2$$

$$\text{Number of tubes} = \frac{364}{0.305} = 1194 \quad (19.3b)$$

$$D_b = 20 \left(\frac{1194}{0.158} \right)^{1/2.263} = 1035 \text{ mm}$$

$$\text{Number of tubes in center row} = \frac{1035}{25} = 41$$

$$\Gamma_h = \frac{45,000}{3600} \times \frac{1}{4.88 \times 1194} = 2.15 \times 10^{-3} \text{ kg/m s}$$

$$N_r = \frac{2}{3} \times 41 = 27$$

$$h_c = 0.95 \times 0.13 \left[\frac{551(551 - 19.5)9.81}{0.16 \times 10^{-3} \times 2.15 \times 10^{-3}} \right]^{1/3} \times 27^{-1/6} \\ = 1447 \text{ W/m}^2\text{°C} \quad (19.29)$$

$$\text{New tube velocity} = 1.91 \times \frac{992}{1194} = 1.59 \text{ m/s}$$

$$h_i = 4200(1.35 + 0.02 \times 35) \frac{1.59^{0.8}}{16.8^{0.2}} = 7097 \text{ W/m}^2\text{°C} \quad (19.17)$$

$$\frac{1}{U} = \frac{1}{1447} + \frac{1}{6000} + \frac{20 \times 10^{-3} \ln\left(\frac{20}{16.8}\right)}{2 \times 50} \\ + \frac{20}{16.8} \times \frac{1}{6000} + \frac{20}{16.8} \times \frac{1}{7097} \quad (19.2)$$

$$U = \underline{\underline{773 \text{ W/m}^2\text{°C}}}$$

Close enough to estimate, firm up design.

Shell-side pressure drop

Use pull-through floating head, no need for close clearance.

Select baffle spacing = shell diameter, 45% cut.

From Fig. 19.10, clearance = 95 mm.

Shell i.d. = $1035 + 95 = 1130 \text{ mm}$

Use Kern's method to make an approximate estimate.

$$\text{Cross-flow area } A_s = \frac{(25 - 20)}{25} 1130 \times 1130 \times 10^{-6} \\ = 0.255 \text{ m}^2 \quad (19.21)$$

Mass flow rate based on inlet conditions:

$$G_s = \frac{45,000}{3600} \times \frac{1}{0.255} = 49.02 \text{ kg/s m}^2$$

$$\text{Equivalent diameter, } d_e = \frac{1.27}{20} (25^2 - 0.785 \times 20^2)$$

$$= 19.8 \text{ mm}$$

$$\text{Vapor viscosity} = 0.008 \text{ mNs/m}^2$$

$$\text{Re} = \frac{49.02 \times 19.8 \times 10^{-3}}{0.008 \times 10^{-3}} = 121,325$$

From Fig. 19.30, $j_f = 2.2 \times 10^{-2}$:

$$u_s = \frac{G_s}{\rho_v} = \frac{49.02}{19.5} = 2.51 \text{ m/s}$$

Take pressure drop as 50% of that calculated using the inlet flow; neglect viscosity correction.

$$\Delta P_s = \frac{1}{2} \left[8 \times 2.2 \times 10^{-2} \left(\frac{1130}{19.8} \right) \left(\frac{4.88}{1.130} \right) \frac{19.5(2.51)^2}{2} \right]$$

$$= 1322 \text{ N/m}^2$$

$$= \underline{\underline{1.3 \text{ kPa}}}$$

Negligible; more sophisticated method of calculation not justified.

Tube-side pressure drop

Viscosity of water = 0.6 mN s/m²:

$$\text{Re} = \frac{u_t \rho d_i}{\mu} = \frac{1.59 \times 993 \times 16.8 \times 10^{-3}}{0.6 \times 10^{-3}} = 44,208$$

From Fig. 19.24, $j_f = 3.5 \times 10^{-3}$.

Neglect the viscosity correction.

$$\Delta P_t = 4 \left[8 \times 3.5 \times 10^{-3} \left(\frac{4.88}{16.8 \times 10^{-3}} \right) + 2.5 \right] \frac{993 \times 1.59^2}{2}$$

$$= 53,388 \text{ N/m}^2$$

$$= \underline{\underline{53 \text{ kPa}}} \text{ (7.7 psi).}$$

This pressure drop is acceptable.

19.11 Reboilers and vaporizers

The design methods given in this section can be used for reboilers and vaporizers. Reboilers are used with distillation columns to vaporize a fraction of the bottom product, whereas in a vaporizer essentially all the feed is vaporized.

Three principal types of reboiler are used:

1. Forced circulation (Fig. 19.46): The fluid is pumped through the exchanger and the vapor formed is separated in the base of the column. When used as a vaporizer, a disengagement vessel must be provided.
2. Thermosiphon, natural circulation (Fig. 19.47): Vertical exchangers with vaporization in the tubes or horizontal exchangers with vaporization in the shell. The liquid circulation through the exchanger is maintained by the

difference in density between the two-phase mixture of vapor and liquid in the exchanger and the single-phase liquid in the base of the column. As with the forced-circulation type, a disengagement vessel will be needed if this type is used as a vaporizer.

3. Kettle type (Fig. 19.48): Boiling takes place on tubes immersed in a pool of liquid; there is no circulation of liquid through the exchanger. This type is also, more correctly, called a submerged bundle reboiler. In some applications it is possible to accommodate the bundle in the base of the column (Fig. 19.49), saving the cost of the exchanger shell. This arrangement is commonly known as a “stab-in” reboiler.

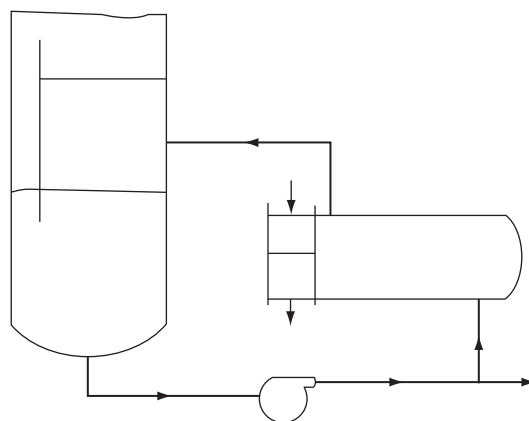


FIG. 19.46 Forced-circulation reboiler.

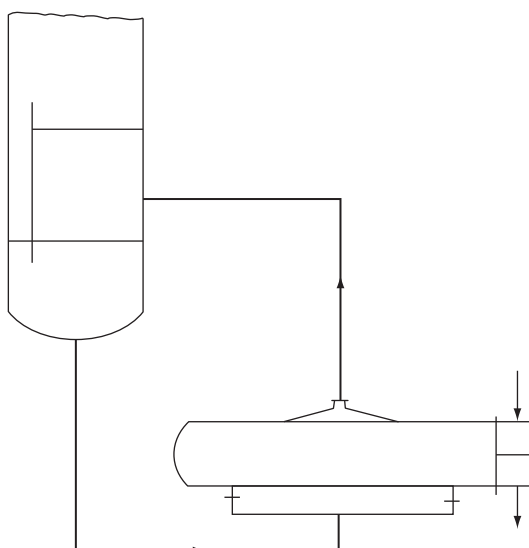


FIG. 19.47 Horizontal thermosiphon reboiler.

Choice of type

The choice of the best type of reboiler or vaporizer for a given duty will depend on the following factors:

1. The nature of the process fluid, particularly its viscosity and propensity to fouling
2. The operating pressure: vacuum or pressure
3. The equipment layout, particularly the headroom available

Forced-circulation reboilers are especially suitable for handling viscous and heavily fouling process fluids; see Chantry and Church (1958). The circulation rate is predictable, and high velocities can be used. They are also suitable for low vacuum operations and for low rates of vaporization. The major disadvantage of this type is that a pump is required and the pumping cost will be high. There is also the danger that leakage of hot fluid could occur at the pump seal; canned-rotor-type pumps can be specified to avoid the possibility of leakage.

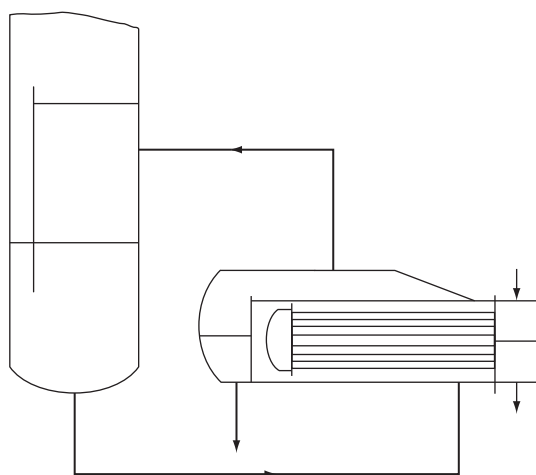


FIG. 19.48 Kettle reboiler.

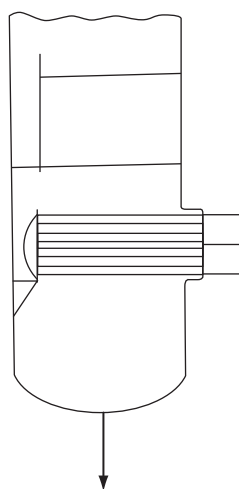


FIG. 19.49 Internal reboiler.

Thermosiphon reboilers are the most economical type for most applications, but are not suitable for high-viscosity fluids or high-vacuum operation. They would not normally be specified for pressures below 0.3 bar. A disadvantage of this type is that the column base must be elevated to provide the hydrostatic head required for the thermosiphon effect. This increases the cost of the column-supporting structure. Horizontal reboilers need less headroom than vertical but have more complex pipework. Horizontal exchangers are easier to maintain than vertical, as the tube bundle is easier to withdraw.

Kettle reboilers have lower heat transfer coefficients than the other types, as there is no liquid circulation. They are not suitable for fouling materials and have a high residence time. They will generally be more expensive than an equivalent thermosiphon type, as a larger shell is needed, but if the duty is such that the bundle can be installed in the column base, the cost will be competitive with the other types. They are often used as vaporizers, as a separate vapor–liquid disengagement vessel is not needed. They are suitable for vacuum operation and for high rates of vaporization, up to 80% of the feed. Some designs allow a liquid blowdown stream to be withdrawn to prevent accumulation of solids or nonvolatile components.

19.11.1 Boiling heat transfer fundamentals

The complex phenomena involved in heat transfer to a boiling liquid are discussed in [Chhabra and Shankar \(2017\)](#). A more detailed account is given by [Collier and Thome \(1994\)](#), [Tong and Tang \(1997\)](#), and [Hsu and Graham \(1976\)](#).

Only a brief discussion of the subject will be given in this section, sufficient for the understanding of the design methods given for reboilers and vaporizers.

The mechanism of heat transfer from a submerged surface to a pool of liquid depends on the temperature difference between the heated surface and the liquid (Fig. 19.50). At low temperature differences, when the liquid is below its boiling point, heat is transferred by natural convection. As the surface temperature is raised, incipient boiling occurs, with vapor bubbles forming and breaking loose from the surface. The agitation caused by the rising bubbles and other effects caused by bubble generation at the surface result in a large increase in the rate of heat transfer. This phenomenon is known as nucleate boiling. As the temperature is raised further, the rate of heat transfer increases until the heat flux reaches a critical value. At this point, the rate of vapor generation is such that dry patches occur spontaneously over the surface and the rate of heat transfer falls off rapidly. At higher temperature differences, the vapor rate is such that the whole surface is blanketed with vapor and the mechanism of heat transfer is by conduction through the vapor film. Conduction is augmented at very high temperature differences by radiation.

The maximum heat flux achievable with nucleate boiling is known as the critical heat flux. In a system where the surface temperature is not self-limiting, such as a nuclear reactor fuel element or a boiling tube that is heated in a fired heater, operation above the critical flux will result in a rapid increase in the surface temperature, and in the extreme situation the surface will melt. This phenomenon is known as "burn-out." The heating media used for process plants are normally self-limiting; for example, with a steam-heated reboiler, the surface temperature can never exceed the saturation temperature of the condensing steam. Care must be taken in the design of electrically heated vaporizers and directly fired vaporizers to ensure that the critical flux can never be exceeded. If the critical flux is exceeded in a directly fired vaporizer, then the tube wall temperature can approach the temperatures of the radiant zone (the bridge-wall temperature or even the flame temperature). At these temperatures damage to the tubes will occur.

The critical flux is reached at surprisingly low temperature differences: around 20 °C to 30 °C for water and 20 °C to 50 °C for light organics.

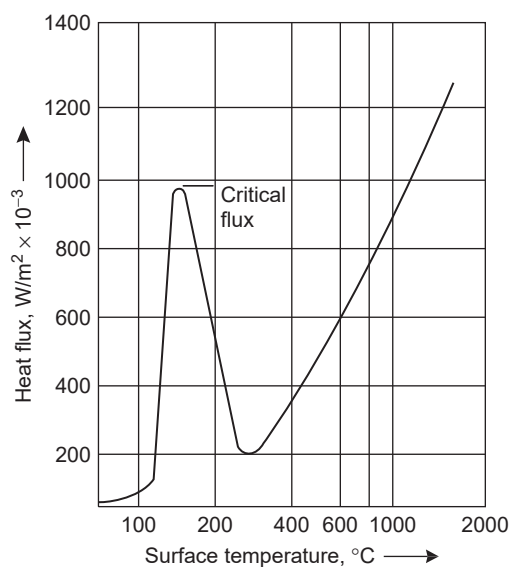


FIG. 19.50 Typical pool boiling curve (water at 1 bar).

Estimation of boiling heat transfer coefficients

In the design of vaporizers and reboilers, the designer will be concerned with two types of boiling: pool boiling and convective boiling. Pool boiling is the name given to nucleate boiling in a pool of liquid, such as in a kettle-type reboiler or a jacketed vessel. Convective boiling occurs where the vaporizing fluid is flowing over the heated surface and heat transfer takes place both by forced convection and nucleate boiling, as in forced circulation or thermosiphon reboilers.

Boiling is a complex phenomenon, and boiling heat transfer coefficients are difficult to predict with any certainty. Whenever possible, experimental values obtained for the system being considered should be used or values for a closely related system.

19.11.2 Pool boiling

In the nucleate boiling region the heat transfer coefficient is dependent on the nature and condition of the heat transfer surface, and it is not possible to present a universal correlation that will give accurate predictions for all systems. [Palen and Taborek \(1962\)](#) have reviewed the published correlations and compared their suitability for use in reboiler design.

The correlation given by [Forster and Zuber \(1955\)](#) can be used to estimate pool boiling coefficients in the absence of experimental data. Their equation can be written in the form:

$$h_{nb} = 0.00122 \left[\frac{k_L^{0.79} C_{pL}^{0.45} \rho_L^{0.49}}{\sigma^{0.5} \mu_L^{0.29} \lambda^{0.24} \rho_v^{0.24}} \right] (T_w - T_s)^{0.24} (p_w - p_s)^{0.75} \quad (19.41)$$

where h_{nb} = nucleate, pool, boiling coefficient, $W/m^2 \cdot ^\circ C$

k_L = liquid thermal conductivity, $W/m \cdot ^\circ C$

C_{pL} = liquid heat capacity, $J/kg \cdot ^\circ C$

ρ_L = liquid density, kg/m^3

μ_L = liquid viscosity, Ns/m^2

λ = latent heat, J/kg

ρ_v = vapor density, kg/m^3

T_w = wall, surface temperature, $^\circ C$

T_s = saturation temperature of boiling liquid $^\circ C$

p_w = saturation pressure corresponding to the wall temperature, T_w , N/m^2

p_s = saturation pressure corresponding to T_s , N/m^2

σ = surface tension, N/m

The reduced-pressure correlation given by [Mostinski \(1963\)](#) is simple to use and gives values that are as reliable as those given by more complex equations.

$$h_{nb} = 0.104 (P_c)^{0.69} (q)^{0.7} \left[1.8 \left(\frac{P}{P_c} \right)^{0.17} + 4 \left(\frac{P}{P_c} \right)^{1.2} + 10 \left(\frac{P}{P_c} \right)^{10} \right] \quad (19.42)$$

where P = operating pressure, bar

P_c = liquid critical pressure, bar

q = heat flux, W/m^2

Note: $q = h_{nb}(T_w - T_s)$.

Mostinski's equation is convenient to use when data on the fluid physical properties are not available.

Equations 19.41 and 19.42 are for boiling single-component fluids; for mixtures, the coefficient will generally be lower than is predicted by these equations. The equations can be used for close boiling range mixtures, say less than $5^\circ C$, and for wider boiling ranges with a suitable factor of safety (see Section 19.11.6).

Critical heat flux

It is important to check that the design and operating heat fluxes are well below the critical flux. Several correlations are available for predicting the critical flux. That given by [Zuber et al. \(1961\)](#) has been found to give satisfactory predictions for use in reboiler and vaporizer design. In SI units, Zuber's equation can be written as:

$$q_c = 0.131 \lambda [\sigma g (\rho_L - \rho_v) \rho_v^2]^{1/4} \quad (19.43)$$

where q_c = maximum, critical, heat flux, W/m^2

g = gravitational acceleration, $9.81 m/s^2$

Mostinski also gives a reduced-pressure equation for predicting the maximum critical heat flux:

$$q_c = 3.67 \times 10^4 P_c \left(\frac{P}{P_c} \right)^{0.35} \left[1 - \left(\frac{P}{P_c} \right) \right]^{0.9} \quad (19.44)$$

Film boiling

The equation given by Bromley (1950) can be used to estimate the heat transfer coefficient for film boiling on tubes. Heat transfer in the film-boiling region will be controlled by conduction through the film of vapor, and Bromley's equation is similar to the Nusselt equation for condensation, where conduction is occurring through the film of condensate.

$$h_{fb} = 0.62 \left[\frac{k_v^3 (\rho_L - \rho_v) \rho_v g \lambda}{\mu_v d_o (T_w - T_s)} \right]^{1/4} \quad (19.45)$$

where h_{fb} is the film boiling heat transfer coefficient, the suffix v refers to the vapor phase, and d_o is in meters. It must be emphasized that process reboilers and vaporizers will always be designed to operate in the nucleate boiling region. The heating medium would be selected, and its temperature controlled, to ensure that in operation the temperature difference is well below that at which the critical flux is reached. For instance, if direct heating with steam would give too high a temperature difference, the steam would be used to heat water and hot water used as the heating medium. Above temperatures where steam can be used, hot oil circuits are often used for reboilers so as to avoid direct firing of the reboiler.

Example 19.8

Estimate the heat transfer coefficient for the pool boiling of water at 2.1 bar from a surface at 125 °C. Check that the critical flux is not exceeded.

Solution

Physical properties, from steam tables:

Saturation temperature,

$$T_s = 121.8^\circ\text{C}$$

$$\rho_L = 941.6 \text{ kg/m}^3, \quad \rho_v = 1.18 \text{ kg/m}^3$$

$$C_{pL} = 4.25 \times 10^3 \text{ J/kg}^\circ\text{C}$$

$$k_L = 687 \times 10^{-3} \text{ Wm}^\circ\text{C}$$

$$\mu_L = 230 \times 10^{-6} \text{ Ns/m}^2$$

$$\lambda = 2198 \times 10^3 \text{ J/kg}$$

$$\sigma = 55 \times 10^{-3} \text{ N/m}$$

$$p_w \text{ at } 125^\circ\text{C} = 2.321 \times 10^5 \text{ N/m}^2$$

$$p_s = 2.1 \times 10^5 \text{ N/m}^2$$

Use the Foster–Zuber correlation (Equation 19.41):

$$\begin{aligned} h_{nb} &= 1.22 \times 10^{-3} \left[\frac{(687 \times 10^{-3})^{0.79} (4.25 \times 10^3)^{0.45} (941.6)^{0.49}}{(55 \times 10^{-3})^{0.5} (230 \times 10^{-6})^{0.29} (2198 \times 10^3)^{0.24} (1.18)^{0.24}} \right] (125 - 121.8)^{0.24} (2.321 \times 10^5 - 2.1 \times 10^5)^{0.75} \\ &= 3738 \text{ W/m}^2\text{C} \end{aligned}$$

Use the Zuber correlation (Equation 19.43):

$$\begin{aligned} q_c &= 1.131 \times 2198 \times 10^3 [55 \times 10^{-3} \times 9.81(941.6 - 1.18)1.18^2]^{1/4} \\ &= \underline{\underline{1.48 \times 10^6 \text{ W/m}^2}} \end{aligned}$$

$$\text{Actual flux} = (125 - 121.8) 3738 = \underline{\underline{11,962 \text{ W/m}^2}}$$

well below critical flux.

19.11.3 Convective boiling

The mechanism of heat transfer in convective boiling, where the boiling fluid is flowing through a tube or over a tube bundle, differs from that in pool boiling. It will depend on the state of the fluid at any point. Consider the situation of a liquid boiling inside a vertical tube (Fig. 19.51). The following conditions occur as the fluid flows up the tube:

1. Single-phase flow region: At the inlet, the liquid is below its boiling point (subcooled) and heat is transferred by forced convection. The equations for forced convection can be used to estimate the heat transfer coefficient in this region.
2. Subcooled boiling: In this region the liquid next to the wall has reached the boiling point, but not the bulk of the liquid. Local boiling takes place at the wall, which increases the rate of heat transfer over that given by forced convection alone.
3. Saturated boiling region: In this region bulk boiling of the liquid is occurring in a manner similar to nucleate pool boiling. The volume of vapor is increasing, and various flow patterns can form. In a long tube, the flow pattern will eventually become annular, where the liquid phase is spread over the tube wall and the vapor flows up the central core.
4. Dry wall region: Ultimately, if a large fraction of the feed is vaporized, the wall dries out and any remaining liquid is present as a mist. Heat transfer in this region is by convection and radiation to the vapor. This condition is unlikely to occur in commercial reboilers and vaporizers.

Saturated, bulk, boiling is the principal mechanism of interest in the design of reboilers and vaporizers.

A comprehensive review of the methods available for predicting convective boiling coefficients is given by Webb and Gupta (1992). The methods proposed by Chen (1966) and Shah (1976) are convenient to use in manual calculations and are accurate enough for preliminary design work. Chen's method is outlined next and illustrated in Example 19.9.

Chen's method

In forced-convective boiling, the effective heat transfer coefficient h_{cb} can be considered to be made up of convective and nucleate boiling components: h'_{fc} and h'_{nb} .

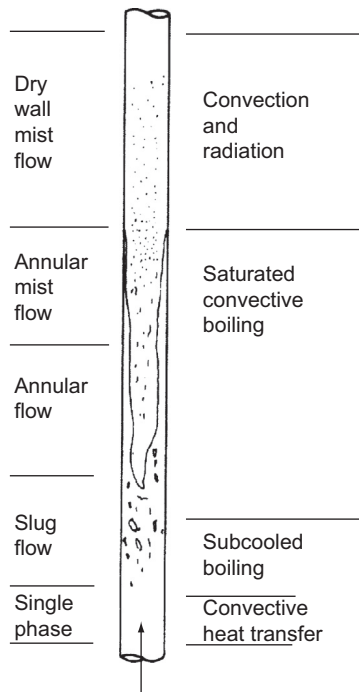


FIG. 19.51 Convective boiling in a vertical tube.

$$h_{cb} = h'_{fc} + h'_{nb} \quad (19.46)$$

The convective boiling coefficient h'_{fc} can be estimated using the equations for single-phase forced-convection heat transfer modified by a factor f_c to account for the effects of two-phase flow:

$$h'_{fc} = h_{fc} \times f_c \quad (19.47)$$

The forced-convection coefficient h_{fc} is calculated assuming that the liquid phase is flowing in the conduit alone.

The two-phase correction factor f_c is obtained from Fig. 19.52 in which the term $1/X_{tt}$ is the Lockhart–Martinelli two-phase flow parameter with turbulent flow in both phases (see Chhabra & Shankar 2017). This parameter is given by:

$$\frac{1}{X_{tt}} = \left[\frac{x}{1-x} \right]^{0.9} \left[\frac{\rho_L}{\rho_v} \right]^{0.5} \left[\frac{\mu_v}{\mu_L} \right]^{0.1} \quad (19.48)$$

where x is the vapor quality, the mass fraction of vapor.

The nucleate boiling coefficient can be calculated using correlations for nucleate pool boiling modified by a factor f_s to account for the fact that nucleate boiling is more difficult in a flowing liquid.

$$h'_{nb} = h_{nb} \times f_s \quad (19.49)$$

The suppression factor f_s is obtained from Fig. 19.53. It is a function of the liquid Reynolds number Re_L and the forced-convection correction factor f_c .

Re_L is evaluated assuming that only the liquid phase is flowing in the conduit and will be given by:

$$Re_L = \frac{(1-x) G d_e}{\mu_L} \quad (19.50)$$

where G is the total mass flow rate per unit flow area.

Chen's method was developed from experimental data on forced convective boiling in vertical tubes. It can be applied, with caution, to forced convective boiling in horizontal tubes and annular conduits (concentric pipes).

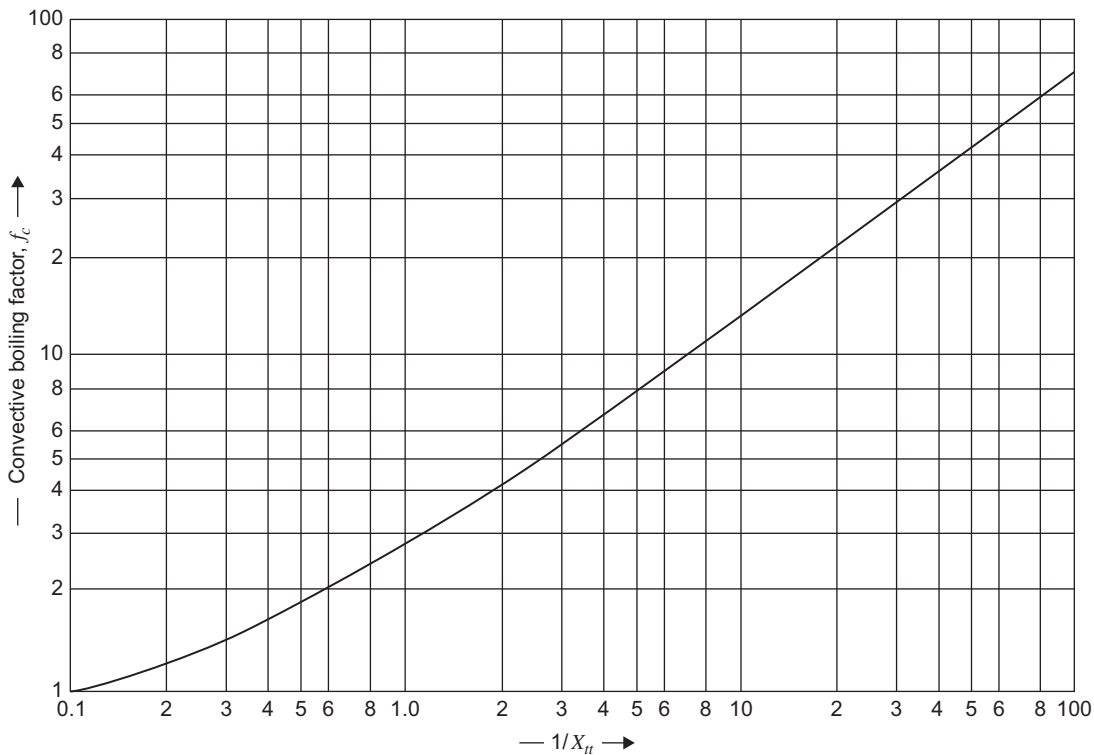


FIG. 19.52 Convective boiling enhancement factor.

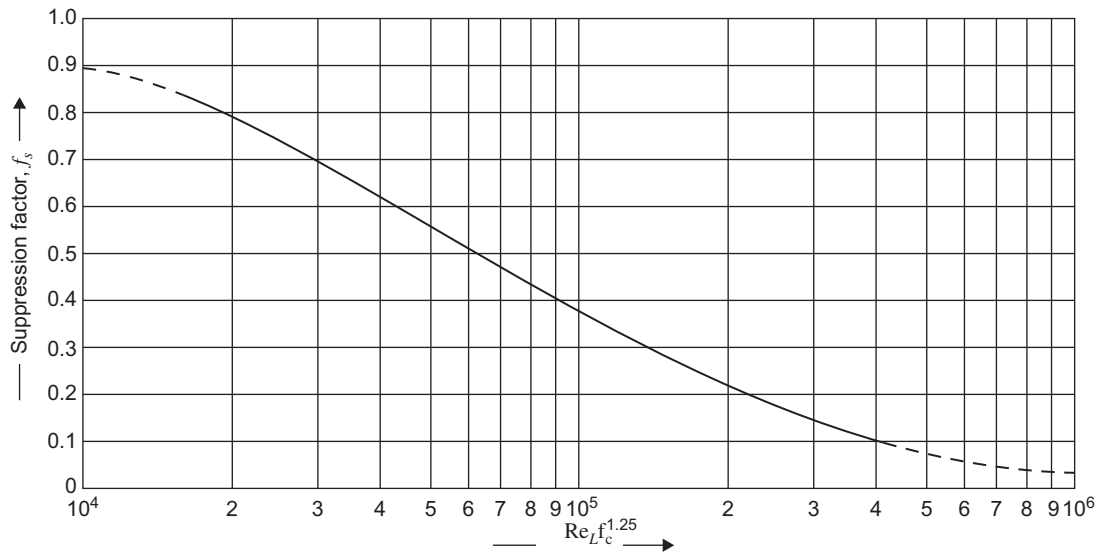


FIG. 19.53 Nucleate boiling suppression factor.

Butterworth (1977) suggests that in the absence of more reliable methods, it may be used to estimate the heat transfer coefficient for forced convective boiling in cross flow over tube bundles, using a suitable cross-flow correlation to predict the forced-convection coefficient. Shah's method was based on data for flow in horizontal and vertical tubes and annuli.

A major problem that will be encountered when applying convective boiling correlations to the design of reboilers and vaporizers is that because the vapor quality changes progressively throughout the exchanger, a step-by-step procedure will be needed. The exchanger must be divided into sections and the coefficient and heat transfer area estimated sequentially for each section.

Example 19.9

A fluid whose properties are essentially those of o-dichlorobenzene is vaporized in the tubes of a forced convection reboiler. Estimate the local heat transfer coefficient at a point where 5% of the liquid has been vaporized. The liquid velocity at the tube inlet is 2 m/s, and the operating pressure is 0.3 bar. The tube inside diameter is 16 mm, and the local wall temperature is estimated to be 120 °C.

Solution

Physical properties:

boiling point 136 °C

$$\rho_L = 1170 \text{ kg/m}^3$$

$$\mu_L = 0.45 \text{ mNs/m}^2$$

$$\mu_v = 0.01 \text{ mNs/m}^2$$

$$\rho_v = 1.31 \text{ kg/m}^3$$

$$k_L = 0.11 \text{ W/m}^\circ\text{C}$$

$$C_{pL} = 1.25 \text{ kJ/kg}^\circ\text{C}$$

$$P_c = 41 \text{ bar}$$

The forced-convective boiling coefficient will be estimated using Chen's method.

With 5% vapor, liquid velocity (for liquid flow in tube alone):

$$= 2 \times 0.95 = 1.90 \text{ m/s}$$

$$\text{Re}_L = \frac{1170 \times 1.90 \times 16 \times 10^{-3}}{0.45 \times 10^{-3}} = 79,040$$

From Fig. 19.23, $j_h = 3.3 \times 10^{-3}$:

$$\text{Pr} = \frac{1.25 \times 10^3 \times 0.45 \times 10^{-3}}{0.11} = 5.1$$

Neglect viscosity correction term:

$$h_{fc} = \frac{0.11}{16 \times 10^{-3}} \times 3.3 \times 10^{-3} (79,040)(5.1)^{0.33}$$

$$= 3070 \text{ W/m}^2\text{C} \quad (19.15)$$

$$\frac{1}{X_{tt}} = \left[\frac{0.05}{1 - 0.05} \right]^{0.9} \left[\frac{1170}{1.31} \right]^{0.5} \left[\frac{0.01 \times 10^{-3}}{0.45 \times 10^{-3}} \right]^{0.1}$$

$$= 1.44 \quad (19.48)$$

From Fig. 19.52, $f_c = 3.2$:

$$h'_{fc} = 3.2 \times 3070 = 9824 \text{ W/m}^2\text{C}$$

Using Mostinski's correlation to estimate the nucleate boiling coefficient:

$$h_{nb} = 0.104 \times 41^{0.69} [h_{nb}(136 - 120)]^{0.7} \left[1.8 \left(\frac{0.3}{41} \right)^{0.17} + 4 \left(\frac{0.3}{41} \right)^{1.2} + 10 \left(\frac{0.3}{41} \right)^{10} \right]$$

$$h_{nb} = 7.43 h_{nb}^{0.7} \quad (19.42)$$

$$h_{nb} = 800 \text{ W/m}^2\text{C}$$

$$\text{Re}_L^{1.25} = 79,040 \times 3.2^{1.25} = 338,286$$

From Fig. 19.53, $f_s = 0.13$:

$$h'_{nb} = 0.13 \times 800 = 104 \text{ W/m}^2\text{C}$$

$$h_{cb} = 9824 + 104 = \underline{\underline{9928 \text{ W/m}^2\text{C}}}$$

19.11.4 Design of forced-circulation reboilers

The normal practice in the design of forced-convection reboilers is to calculate the heat transfer coefficient, assuming that the heat is transferred by forced convection only. This will give conservative (safe) values, as any boiling that occurs will invariably increase the rate of heat transfer. In many designs, the pressure is controlled to prevent any appreciable vaporization in the exchanger. A throttle valve is installed in the exchanger outlet line, and the liquid flashes as the pressure is let down into the vapor-liquid separation vessel.

If a significant amount of vaporization does occur, the heat transfer coefficient can be evaluated using correlations for convective boiling, such as Chen's method.

Conventional shell and tube exchanger designs are used with one shell pass and two tube passes when the process fluid is on the shell side and with one shell and one tube pass when it is in the tubes. High tube velocities

are used to reduce fouling: 3 to 9 m/s. Because the circulation rate is set by the designer, forced-circulation reboilers can be designed with more certainty than natural-circulation units.

The critical flux in forced-convection boiling is difficult to predict. Kern (1950) recommends that for commercial reboiler designs the heat flux should not exceed $63,000 \text{ W/m}^2$ ($20,000 \text{ Btu/ft}^2\text{h}$) for organics and $95,000 \text{ W/m}^2$ ($30,000 \text{ Btu/ft}^2\text{h}$) for water and dilute aqueous solutions. These values are now generally considered to be too pessimistic.

19.11.5 Design of thermosiphon reboilers

The design of thermosiphon reboilers is complicated by the fact that, unlike a forced-convection reboiler, the fluid circulation rate cannot be determined explicitly. The circulation rate, heat transfer rate, and pressure drop are all interrelated, and iterative design procedures must be used. The fluid will circulate at a rate at which the pressure losses in the system are just balanced by the available hydrostatic head. The exchanger, column base, and piping act like two legs of a U-tube (Fig. 19.54). The driving force for circulation around the system is the difference in density of the liquid in the “cold” leg (the column base and inlet piping) and the two-phase fluid in the “hot” leg (the exchanger tubes and outlet piping).

To calculate the circulation rate, it is necessary to make a pressure balance around the system. A typical design procedure will include the following steps:

1. Calculate the vaporization rate required from the specified duty.
2. Estimate the exchanger area from an assumed value for the overall heat transfer coefficient. Decide the exchanger layout and piping dimensions.
3. Assume a value for the circulation rate through the exchanger.
4. Calculate the pressure drop in the inlet piping (single phase).
5. Divide the exchanger tube into sections and calculate the pressure drop section by section up the tube. Use suitable methods for the sections in which the flow is two-phase. Include the pressure loss due to the fluid acceleration as the vapor rate increases. For a horizontal reboiler, calculate the pressure drop in the shell, using a method suitable for two-phase flow.
6. Calculate the pressure drop in the outlet piping (two-phase).
7. Compare the calculated pressure drop with the available differential head, which will depend on the vapor voidage, and hence the assumed circulation rate. If a satisfactory balance has been achieved, proceed. If not, return to step 3 and repeat the calculations with a new assumed circulation rate.
8. Calculate the heat transfer coefficient and heat transfer rate section by section up the tubes. Use a suitable method for the sections in which the boiling is occurring, such as Chen’s method.

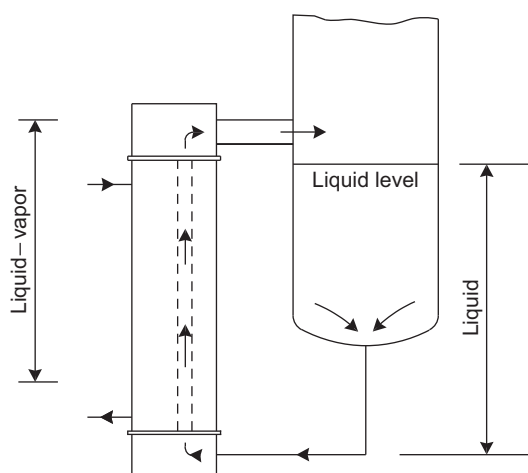


FIG. 19.54 Vertical thermosiphon reboiler, liquid and vapor flows.

9. Calculate the rate of vaporization from the total heat transfer rate, and compare with the value assumed in step 1. If the values are sufficiently close, proceed. If not, return to step 2 and repeat the calculations for a new design.
10. Check that the critical heat flux is not exceeded at any point up the tubes.
11. Repeat the complete procedure as necessary to optimize the design.

It can be seen that to design a thermosiphon reboiler using hand calculations would be tedious and time-consuming. The iterative nature of the procedure lends itself to solution by computers. [Sarma et al. \(1973\)](#) discuss the development of a computer program for vertical thermosiphon reboiler design and give algorithms and design equations.

Extensive work on the performance and design of thermosiphon reboilers has been carried out by HTFS and HTRI, and proprietary design programs are available from these organizations. The HTFS methods are available in Aspen Technology's Aspen Engineering Suite and in Honeywell's UniSim Design Suite; see Table 4.1.

In the absence of access to a computer program, the rigorous design methods given by [Fair \(1960, 1963\)](#) or [Hughmark \(1961, 1964, 1969\)](#) can be used for thermosiphon vertical reboilers. [Collins \(1976\)](#) and [Fair and Klip \(1983\)](#) give methods for the design of horizontal, shell-side thermosiphon reboilers. The design and performance of this type of reboiler is also reviewed in a paper by [Yilmaz \(1987\)](#).

Approximate methods can be used for preliminary designs. [Fair \(1960\)](#) gives a method in which the heat transfer and pressure drop in the tubes are based on the average of the inlet and outlet conditions. This simplifies step 5 in the design procedure, but trial-and-error calculations are still needed to determine the circulation rate. [Frank and Prickett \(1973\)](#) programmed Fair's rigorous design method for computer solution and used it, together with operating data on commercial exchangers, to derive a general correlation of heat transfer rate with reduced temperature for vertical thermosiphon reboilers. Their correlation, converted to SI units, is shown in [Fig. 19.55](#). The basis and limitations of the correlation are as follows:

1. Conventional designs: tube lengths 2.5 to 3.7 m (8 to 12 ft) (standard length 2.44 m), preferred diameter 25 mm (1 in.)
2. Liquid in the sump level with the top tubesheet
3. Process-side fouling coefficient 6000 W/m²°C
4. Heating medium steam, coefficient including fouling, 6000 W/m²°C
5. Simple inlet and outlet piping
6. For reduced temperatures greater than 0.8, use the limiting curve (that for aqueous solutions)
7. Minimum operating pressure 0.3 bar
8. Inlet fluid should not be appreciably subcooled
9. Extrapolation is not recommended

For heating media other than steam and process-side fouling coefficients different from 6000 W/m²°C, the design heat flux taken from [Fig. 19.55](#) may be adjusted as follows:

$$U' = \frac{q'}{\Delta T'} \quad (19.51)$$

and

$$\frac{1}{U_c} = \frac{1}{U'} - \frac{1}{6000} + \frac{1}{h_s} - \frac{1}{6000} + \frac{1}{h_{id}}$$

Where q' = flux read from [Fig. 19.55](#) at $\Delta T'$

h_s = new shell-side coefficient, W/m²°C

h_{id} = fouling coefficient on the process (tube) side W/m²°C

U_c = corrected overall coefficient

The use of Frank and Prickett's method is illustrated in Example 19.10.

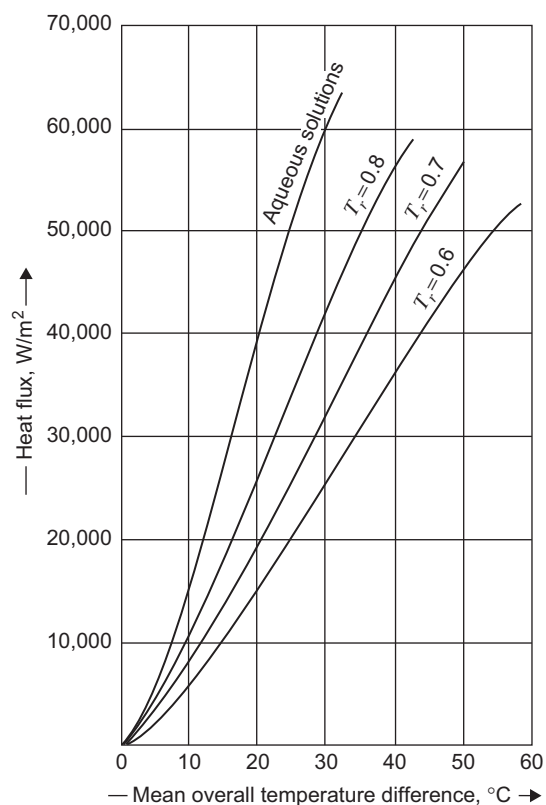


FIG. 19.55 Vertical thermosiphon design correlation.

Limitations on the use of Frank and Prickett's method

A study by [van Edmonds \(1994\)](#) using the HTFS TREB4 program found that Frank and Prickett's method gave acceptable predictions for pure components and binary mixtures with water but that the results were unreliable for other mixtures. Also, van Edmonds's results predicted higher flux values than those obtained by Prickett and Frank.

For preliminary designs for pure components, or near-pure components, Prickett and Frank's method should give a conservative estimate of the operating heat flux. It is not recommended for mixtures other than binary mixtures with water.

Approximate design method for mixtures

For mixtures, the simplified analysis used by Kern (1954) can be used to obtain an approximate estimate of the number of tubes required; see also [Aerstin and Street \(1978\)](#) and [Hewitt et al. \(1994\)](#).

This method uses simple, unsophisticated methods to estimate the two-phase pressure drop through the exchanger and piping and the convective boiling heat transfer coefficient. The calculation procedure is set out next and illustrated in Example 19.11.

Procedure

1. Determine the heat duty.
2. Estimate the heat transfer area using the maximum allowable heat flux. Take as 39,700 W/m² for vertical and 47,300 W/m² for horizontal reboilers.
3. Choose the tube diameters and length. Calculate the number of tubes required.
4. Estimate the recirculation ratio, not less than 3.
5. Calculate the vapor flow rate leaving the reboiler for the duty and liquid heat of vaporization.
6. Calculate the liquid flow rate leaving the reboiler for the vapor rate and recirculation ratio.
7. Estimate the two-phase pressure drop through the tubes due to friction. Use the homogenous model or another simple method, such as the Lochart–Martenelli equation.
8. Estimate the static head in the tubes.
9. Estimate the available head.
10. Compare the total estimated pressure drop and the available head. If the available head is greater by a sufficient amount to allow for the pressure drop through the inlet and outlet piping, proceed. If the available head is not sufficient, return to step 2 and increase the number of tubes.

11. Calculate the convective heat transfer coefficient using simple methods, such as assuming convection only, or Chen's method; see Section 19.11.3.
12. Calculate the overall heat transfer coefficient.
13. Calculate the required overall coefficient and compare with that estimated. If satisfactory, accept the design; if unsatisfactory, return to step 2 and increase the estimated area.

Maximum heat flux

Thermosiphon reboilers can suffer from flow instabilities if too high a heat flux is used. The liquid and vapor flow in the tubes is not smooth but tends to pulsate, and at high heat fluxes the pulsations can become large enough to cause vapor locking. A good practice is to install a flow restriction in the inlet line, a valve or orifice plate, so that the flow resistance can be adjusted should vapor locking occur in operation.

Kern recommends that the heat flux in thermosiphon reboilers, based on the total heat transfer area, should not exceed $37,900 \text{ W/m}^2$ ($12,000 \text{ Btu/ft}^2\text{h}$). For horizontal thermosiphon reboilers, Collins recommends a maximum flux ranging from $47,300 \text{ W/m}^2$ for 20-mm tubes to $56,800 \text{ W/m}^2$ for 25-mm tubes ($15,000$ to $18,000 \text{ Btu/ft}^2\text{h}$). These "rule of thumb" values are now thought to be too conservative; see [Shellene et al. \(1968\)](#) and [Furzer \(1990\)](#). Correlations for determining the maximum heat flux for vertical thermosiphons are given by [Lee et al. \(1956\)](#) and [Palen et al. \(1974\)](#) and for horizontal thermosiphons by [Yilmaz \(1987\)](#).

General design considerations

The tube lengths used for vertical thermosiphon reboilers vary from 1.83 m (6 ft) for vacuum service to 3.66 m (12 ft) for pressure operation. A good size for general applications is 2.44 m (8 ft) by 25 mm internal diameter. Larger tube diameters, up to 50 mm, are used for fouling systems.

The top tubesheet is normally aligned with the liquid level in the base of the column ([Fig. 19.54](#)). The outlet pipe should be as short as possible and have a cross-sectional area at least equal to the total cross-sectional area of the tubes.

Example 19.10

Make a preliminary design for a vertical thermosiphon for a column distilling crude aniline. The column will operate at atmospheric pressure, and a vaporization rate of 6000 kg/h is required. Steam is available at 22 bar (300 psig). Take the column bottom pressure as 1.2 bar.

Solution

Physical properties, taken as those of aniline:

Boiling point at 1.2 bar 190°C

Molecular weight 93.13

T_c 699 K

Latent heat $42,000 \text{ kJ/kmol}$

Steam saturation temperature 217°C

Mean overall $\Delta T = (217 - 190) = 27^\circ\text{C}$

$$\text{Reduced temperature, } T_r = \frac{(190 + 273)}{699} = 0.66$$

From [Fig. 19.55](#), design heat flux $= 25,000 \text{ W/m}^2$

$$\text{Heat load} = \frac{6000}{3600} \times \frac{42,000}{93.13} = 751 \text{ kW}$$

$$\text{Area required} = \frac{751 \times 10^3}{25,000} = 30 \text{ m}^2$$

Use 25 mm i.d., 30 mm o.d., 2.44 m long tubes.

$$\text{Area of one tube} = \pi \times 25 \times 10^{-3} \times 2.44 = 0.192 \text{ m}^2$$

$$\text{Number of tubes} = \frac{30}{0.192} = 157$$

Approximate diameter of bundle, for 1.25 square pitch:

$$D_b = 30 \left[\frac{157}{0.215} \right]^{1/2.207} = 595 \text{ mm} \quad (19.3b)$$

A fixed tubesheet will be used for a vertical thermosiphon reboiler. From Fig. 19.10, shell diametrical clearance = 14 mm:

$$\text{Shell inside dia.} = 595 + 14 = 609 \text{ mm}$$

Outlet pipe diameter; take area as equal to total tube cross-sectional area:

$$= 157(25 \times 10^{-3})^2 \frac{\pi}{4} = 0.077 \text{ m}^2$$

$$\text{Pipe diameter} = \sqrt{\frac{0.077 \times 4}{\pi}} = 0.31 \text{ m}$$

Example 19.11

Make a preliminary design for a vertical thermosiphon reboiler for a debutanizer column that has the bottoms composition given here. Take the vapor rate required to be 36 kmol/h.

Bottoms composition: C₃ 0.001, iC₄ 0.001, nC₄ 0.02, iC₅ 0.34, nC₅ 0.64, kmol.

Operating pressure 8.3 bar. Bubble point of mixture, approximately, 120 °C.

Solution

The concentrations of C₃ and iC₄ are small enough to be neglected.

Take the liquid:vapor ratio as 3:1.

Estimate the liquid and vapor compositions leaving the reboiler:

$$\text{Vapor rate, } V = 36/3600 = 0.1 \text{ kmol/s}$$

$$L/V = 3, \text{ so liquid rate, } L = 3V = 0.3 \text{ kmol/s and feed, } F = L + V = 0.4 \text{ kmol/s}$$

The vapor and liquid compositions leaving the reboiler can be estimated using the same procedure as that for a flash calculation; see Section 17.3.3.

	K_i	$A_i = K_i \times L/V$	$V_i = z_i/(1 + A_i)$	$y_i = V_i/V$	$x_i = (Fz_i - V_i)/L$
nC ₄	2.03	6.09	0.001	0.010	0.023
iC ₅	1.06	3.18	0.033	0.324	0.343
nC ₅	0.92	2.76	0.068	0.667	0.627
			—	—	—
Totals			0.102	1.001	0.993

Enthalpies of vaporization, kJ/mol (taken from Maxwell 1962):

	x_i	$\Delta H_{\text{vap},i}$	$x_i \Delta H_{\text{vap},i}$
nC ₄	0.02	16	0.32
iC ₅	0.35	17	5.95
nC ₅	0.63	19	11.97
			—
Total			18.24

Exchanger duty, feed to reboiler taken as at its boiling point:

$$= \text{vapor flow rate} \times \text{heat of vaporization}$$

$$= 0.1 \times 10^3 \times 18.24 = \underline{1824} \text{ kW}$$

(Note: We could also have estimated this using a nonadiabatic flash model in a process simulation program.)

Take the maximum flux as $37,900 \text{ W/m}^2$; see Section 19.11.5.

$$\text{Heat transfer area required} = 1,824,000/37,900 = 48.1 \text{ m}^2$$

Use 25 mm i.d., 2.5 m long tubes, a popular size for vertical thermosiphon reboilers.

$$\text{Area of one tube} = 25 \times 10^{-3} \pi \times 2.5 = 0.196 \text{ m}^2$$

$$\text{Number of tubes required} = 48.1/0.196 = 246$$

$$\text{Liquid density at base of exchanger} = 520 \text{ kg/m}^3$$

$$\text{Relative molecular mass at tube entry} = 58 \times 0.02 + 72(0.34 + 0.64) = 71.7$$

$$\text{vapor at exit} = 58 \times 0.02 + 72(0.35 + 0.63) = 71.7$$

Two-phase fluid density at tube exit:

$$\text{volume of vapor} = 0.1 \times (22.4/8.3) \times (393/273) = 0.389 \text{ m}^3$$

$$\text{volume of liquid} = (0.3 \times 71.7)/520 = 0.0413 \text{ m}^3$$

$$\text{total volume} = 0.389 + 0.0413 = 0.430 \text{ m}^3$$

$$\text{exit density} = \frac{(0.4 \times 71.7)}{0.430} \times 71.7 = 66.7 \text{ kg/m}^3$$

Friction loss

$$\text{Mass flow rate} = 0.4 \times 71.7 = 28.68 \text{ kg/s}$$

$$\text{Cross-sectional area of tube} = \frac{\pi(25 \times 10^{-3})^2}{4} = 0.00049 \text{ m}^2$$

$$\text{Total cross-sectional area of bundle} = 246 \times 0.00049 = 0.121 \text{ m}^2$$

$$\text{Mass flux, } G = \text{mass flow/area} = 28.68/0.121 = 237.0 \text{ kg m}^{-2}\text{s}^{-1}$$

At tube exit, pressure drop per unit length, using the homogeneous model:

$$\text{Homogeneous velocity} = G/\rho_m = 237/66.7 = 3.55 \text{ m/s}$$

$$\text{Viscosity, taken as that of liquid,} = 0.12 \text{ mNsm}^{-2}$$

$$\text{Re} = \frac{\rho_m u d}{\mu} = \frac{66.7 \times 3.55 \times 25 \times 10^{-3}}{0.12 \times 10^{-3}} = 49,330$$

$$\text{Friction factor, from Fig. 19.24} = 3.2 \times 10^{-3}$$

$$\Delta P_f = 8 \times 3.2 \times 10^{-3} \times \frac{1}{25 \times 10^{-3}} \times 66.7 \times \frac{3.55^2}{2} = 430 \text{ N/m}^{-2} \text{ per m} \quad (19.19)$$

At tube entry, liquid only, pressure drop per unit length:

$$\text{velocity} = G/\rho_L = 237.0/520 = 0.46 \text{ m/s}$$

$$\text{Re} = \frac{\rho_L u d}{\mu} = \frac{520 \times 0.46 \times 25 \times 10^{-3}}{0.12 \times 10^{-3}} = 49,833$$

$$\text{Friction factor, from Fig 19.24} = 3.2 \times 10^{-3}$$

$$\Delta P_f = 8 \times 3.2 \times 10^{-3} \times \frac{1}{25 \times 10^{-3}} \times 520 \times \frac{0.46^2}{2} = 56 \text{ N/m}^{-2} \text{ per m} \quad (19.19)$$

Taking the pressure drop change as linear along the tube:

$$\text{Mean pressure drop per unit length} = (430 + 56)/2 = 243 \text{ N/m}^2$$

$$\text{Pressure drop over tube } 243 \times 2.5 = 608 \text{ N/m}^2$$

The viscosity correction factor is neglected in this rough calculation.

Static pressure in tubes

Making the simplifying assumption that the variation in density in the tubes is linear from bottom to top, the static pressure will be given by:

$$\Delta P_s = g \int_0^L \frac{dx}{v_i + x(v_0 - v_i)/L} = \frac{gL}{(v_0 - v_i)} \times \text{Ln}(v_0/v_i)$$

where v_i and v_0 are the inlet and outlet specific volumes.

$$v_i = 1/520 = 0.00192 \text{ and } v_0 = 1/66.7 = 0.0150 \text{ m}^3/\text{kg}$$

$$\Delta P_s = \frac{9.8 \times 2.5}{(0.0150 - 0.00192)} \times \text{Ln}(0.0150/0.00192) = 3850 \text{ N/m}^2$$

$$\text{Total pressure drop over tubes} = 608 + 3850 = \underline{\underline{4460}} \text{ N/m}^2$$

Available head (driving force)

$$\Delta P_s = \rho_L g L = 520 \times 9.8 \times 2.5 = \underline{\underline{12,740}} \text{ N/m}^2$$

This is adequate to maintain a circulation ratio of 3:1, including allowances for the pressure drop across the piping.

Heat transfer

The convective boiling coefficient will be calculated using Chen's method; see [Section 19.13.3](#). As the heat flux is known and only a rough estimate of the coefficient is required, use Mostinski's equation to estimate the nucleate boiling coefficient; see [Section 19.11.2](#).

Take the critical pressure as that for n-pentane, 33.7 bar.

$$\begin{aligned} h_{nb} &= 0.104(33.7)^{0.69}(37,900)^{0.7} \left[1.8(8.3/33.7)^{0.17} + 4(8.3/33.7)^{1.2} + 10(8.3/33.7)^{10} \right] \\ &= 1888.6(1.418 + 0.744 + 0.000) = 4083 \text{ Wm}^{-2}\text{C}^{-1} \end{aligned} \quad (19.42)$$

$$\text{Vapor quality, } x = \text{mass vapor/total mass flow} = 0.1/0.4 = 0.25$$

$$\text{Viscosity of vapor} = 0.0084 \text{ mNm}^{-2}\text{s}$$

$$\text{Vapor density at tube exit} = (0.1 \times 71.7)/0.389 = 18.43 \text{ kg/m}^3$$

$$1/X_{tt} = [0.25/(1 - 0.25)]^{0.9} [520/18.43]^{0.5} [0.0084/0.12]^{0.1} = 1.51$$

$$\text{Specific heat of liquid} = 2.78 \text{ kJkg}^{-1}\text{C}^{-1}, \text{ thermal conductivity of liquid} = 0.12 \text{ W/m}^\circ\text{C} \quad (19.46)$$

$$\text{Pr}_L = (2.78 \times 10^3 \times 0.12 \times 10^{-3})/0.12 = 2.78$$

$$\text{Mass flux, liquid phase only flowing in tubes} = (0.3 \times 71.7)/0.121 = 177.8 \text{ kg m}^{-2}\text{s}^{-1}$$

$$\text{Velocity} = 177.8/520 = 0.34 \text{ m/s}$$

$$\text{Re}_L = \frac{520 \times 0.34 \times 25 \times 10^{-3}}{0.12 \times 10^{-3}} = 36,833$$

From Fig. 19.23 $j_h = 3.3 \times 10^{-3}$:

$$\begin{aligned} \text{Nu} &= 3.3 \times 10^{-3} \times 36,833 \times 2.78^{0.33} = 170.3 \\ h_i &= 170.3 \times (0.12/25 \times 10^{-3}) = 817 \text{ W/m}^2\text{C} \end{aligned} \quad (19.15)$$

again, neglecting the viscosity correction factor.

From Fig. 19.52, the convective boiling factor, $f_c = 3.6$:

$$\text{Re}_L \times f_c^{1.25} = 36,833 \times 3.6^{1.25} = 182,896 (1.8 \times 10^{-5})$$

From Fig. 19.53 the nucleate boiling suppression factor, $f_s = 0.23$.

$$\text{So, } h_{cb} = 3.6 \times 817 + 0.23 \times 4083 = 3880 \text{ W/m}^2\text{C}$$

This value has been calculated at the outlet conditions.

Assuming that the coefficient changes linearly from the inlet to outlet, then the average coefficient will be given by:

$$[\text{inlet coefficient (all liquid)} + \text{outlet coefficient (liquid + vapor)}] / 2$$

$$\text{Re}_L \text{ at inlet} = 36,833 \times 0.4/0.3 = 49,111 (4.9 \times 10^{-4})$$

From Fig. 19.23, $j_h = 3.2 \times 10^{-3}$

$$\begin{aligned} \text{Nu} &= 3.2 \times 10^{-3} \times 49,111 \times 2.78^{0.33} = 220.2 \\ h_i &= 220.2 \times (0.12/25 \times 10^{-3}) = 1057 \text{ W/m}^2\text{C} \end{aligned}$$

$$\text{Mean coefficient} = (1057 + 3880) / 2 = \underline{2467} \text{ Wm}^{-2}\text{C}^{-1}$$

The overall coefficient, U , neglecting the resistance of the tube wall and taking the steam coefficient as $8000 \text{ W/m}^2\text{C}$, is given by:

$$\begin{aligned} 1/U &= 1/8000 + 1/2467 = 5.30 \times 10^{-4} \\ U &= \underline{1886} \text{ Wm}^{-2}\text{C}^{-1} \end{aligned}$$

The overall coefficient required for the design = duty / ΔT_{LM}

$\Delta T_{LM} = 158.8 - 120 = 38.8^\circ\text{C}$, taking both streams as isothermal

So, U required = $37,900 / 38.3 = \underline{990} \text{ Wm}^{-2}\text{C}^{-1}$

So the area available in the proposed design is more than adequate and will take care of any fouling.

The analysis could be improved by dividing the tube length into sections, calculating the heat transfer coefficient and pressure drop over each section, and totaling. More accurate, but more complex, methods could be used to predict the two-phase pressure drop and heat transfer coefficients. The pressure drop over the inlet and outlet pipes could also be estimated, taking into account the bends and expansions and contractions. An allowance could also be included for the energy (pressure drop) required to accelerate the liquid–vapor mixture as the liquid is vaporized. This can be taken as two velocity head, based on the mean density.

19.11.6 Design of kettle reboilers

Kettle reboilers and other submerged bundle equipment are essentially pool-boiling devices, and their design is based on data for nucleate boiling.

In a tube bundle, the vapor rising from the lower rows of tubes passes over the upper rows. This has two opposing effects: there will be a tendency for the rising vapor to blanket the upper tubes, particularly if the tube spacing is close, which will reduce the heat transfer rate, but this is offset by the increased turbulence caused by the rising vapor bubbles. Palen and Small (1964) give a detailed procedure for kettle reboiler design in which the heat transfer coefficient calculated using equations for boiling on a single tube is reduced by an empirically derived tube bundle factor to account for the effects of vapor blanketing. Later work by HTRI reported by Palen et al. (1972), showed that the coefficient for bundles was usually greater than that estimated for a single tube. On balance, it seems reasonable to use the correlations for single tubes to estimate the coefficient for tube bundles without applying any correction (Equations 19.41 and 19.42).

The maximum heat flux for stable nucleate boiling will, however, be less for a tube bundle than for a single tube. Palen and Small (1964) suggest modifying the Zuber equation for single tubes (Equation 19.43) with a tube density factor. This approach was supported by Palen et al. (1972).

The modified Zuber equation can be written as:

$$q_{cb} = K_b \left(\frac{p_t}{d_o} \right) \left(\frac{\lambda}{\sqrt{N_t}} \right) [\sigma g (\rho_L - \rho_v) \rho_v^2]^{0.25} \quad (19.52)$$

where q_{cb} = maximum (critical) heat flux for the tube bundle, W/m^2

K_b = 0.44 for square pitch arrangements

= 0.41 for equilateral triangular pitch arrangements

p_t = tube pitch

d_o = tube outside diameter

N_t = total number of tubes in the bundle

Note. For U-tubes, N_t will be equal to twice the number of actual U-tubes.

Palen and Small suggest that a factor of safety of 0.7 be applied to the maximum flux estimated from Equation 19.52. This will still give values that are well above those that have traditionally been used for the design of commercial kettle reboilers, such as that of $37,900 \text{ W/m}^2$ ($12,000 \text{ Btu/ft}^2\text{h}$) recommended by Kern (1950). This has had important implications in the application of submerged bundle reboilers, as the high heat flux allows a smaller bundle to be used, which can then often be installed in the base of the column, saving the cost of shell and piping.

General design considerations

A typical layout is shown in Fig. 19.8. The tube arrangement, triangular or square pitch, will not have a significant effect on the heat transfer coefficient. A tube pitch of between 1.5 and 2.0 times the tube outside diameter should be used to avoid vapor blanketing. Long, thin bundles will be more efficient than short, fat bundles.

The shell should be sized to give adequate space for the disengagement of the vapor and liquid. The shell diameter required will depend on the heat flux. The following values can be used as a guide:

Heat flux W/m^2	Shell dia./bundle dia.
25,000	1.2–1.5
25,000–40,000	1.4–1.8
40,000	1.7–2.0

The freeboard between the liquid level and shell should be at least 0.25 m. To avoid excessive entrainment, the maximum vapor velocity $\hat{u}_v$ (m/s) at the liquid surface should be less than that given by the expression:

$$\hat{u}_v < 0.2 \left[\frac{\rho_L - \rho_v}{\rho_v} \right]^{1/2} \quad (19.53)$$

When only a low rate of vaporization is required, a vertical cylindrical vessel with a heating jacket or coils should be considered. The boiling coefficients for internal submerged coils can be estimated using the equations for nucleate pool boiling.

Mean temperature differences

When the fluid being vaporized is a single component and the heating medium is steam (or another condensing vapor), both shell- and tube-side processes are isothermal and the mean temperature difference will be simply the difference between the saturation temperatures. If one side is not isothermal, the logarithmic mean temperature difference should be used. If the temperature varies on both sides, the logarithmic temperature difference must be corrected for departures from true cross-current or countercurrent flow (see Section 19.6).

If the feed is subcooled, the mean temperature difference should still be based on the boiling point of the liquid, as the feed will rapidly mix with the boiling pool of liquid; the quantity of heat required to bring the feed to its boiling point must be included in the total duty.

Mixtures

The equations for estimating nucleate boiling coefficients given in Section 19.11.1 can be used for close boiling mixtures, say less than 5 °C, but will overestimate the coefficient if used for mixtures with a wide boiling range. [Palen and Small \(1964\)](#) give an empirical correction factor for mixtures that can be used to estimate the heat transfer coefficient in the absence of experimental data:

$$(h_{nb})_{\text{mixture}} = f_m (h_{nb})_{\text{single component}} \quad (19.54)$$

where $f_m = \exp[-0.0083(T_{bo} - T_{bi})]$

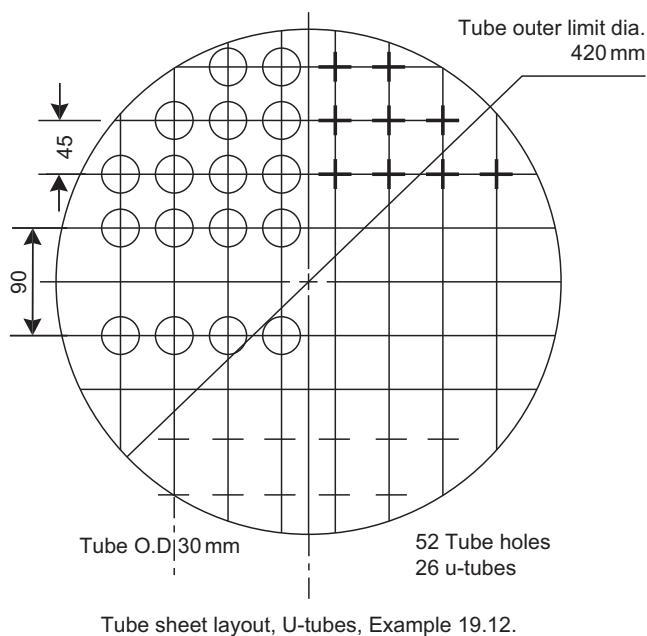
and T_{bo} = temperature of the vapor mixture leaving the reboiler °C

T_{bi} = temperature of the liquid entering the reboiler °C

The inlet temperature is the saturation temperature of the liquid at the base of the column, and the vapor temperature is the saturation temperature of the vapor returned to the column. The composition of these streams will be fixed by the distillation column design specification.

Example 19.12

Design a vaporizer to vaporize 5000 kg/h n-butane at 5.84 bar. The minimum temperature of the feed (winter conditions) will be 0 °C. Steam is available at 1.70 bar (10 psig).



Solution

Only the thermal design and general layout will be done. Select kettle type.

Physical properties of n-butane at 5.84 bar:

boiling point = 56.1 °C

latent heat = 326 kJ/kg

mean specific heat, liquid = 2.51 kJ/kg°C

critical pressure, P_c = 38 bar

Heat loads:

$$\text{sensible heat(maximum)} = (56.1 - 0)2.51 = 140.8 \text{ kJ/kg}$$

$$\text{total heat load} = (140.8 + 326) \times \frac{5000}{3600} = 648.3 \text{ kW}$$

add 5 % for heat losses

$$\begin{aligned}\text{maximum heat load(duty)} &= 1.05 \times 648.3 \\ &= 681 \text{ kW}\end{aligned}$$

From Fig. 19.1, assume $U = 1000 \text{ W/m}^2\text{C}$.

Mean temperature difference; both sides isothermal, steam saturation temperature at 1.7 bar = 115.2 °C

$$\Delta T_m = 115.2 - 56.1 = 59.1^\circ\text{C}$$

$$\text{Area (outside) required} = \frac{681 \times 10^3}{1000 \times 59.1} = 11.5 \text{ m}^2$$

Select 25 mm i.d., 30 mm o.d. plain U-tubes:

Nominal length 4.8 m(one U – tube)

$$\text{Number of U tubes} = \frac{11.5}{(30 \times 10^{-3})\pi 4.8} = 25$$

Use square pitch arrangement, pitch = 1.5 × tube o.d.

$$= 1.5 \times 30 = 45 \text{ mm}$$

Draw a tube layout diagram; take minimum bend radius:

$$1.5 \times \text{tube o.d.} = 45 \text{ mm}$$

Proposed layout gives 26 U-tubes, tube outer limit diameter 420 mm.

Boiling coefficient

Use Mostinski's equation. Heat flux, based on estimated area:

$$\begin{aligned}q &= \frac{681}{11.5} = 59.2 \text{ kW/m}^2 \\ h_{nb} &= 0.104(38)^{0.69}(59.2 \times 10^3)^{0.7} \left[1.8 \left(\frac{5.84}{38} \right)^{0.17} + 4 \left(\frac{5.84}{38} \right)^{1.2} + 10 \left(\frac{5.84}{38} \right)^{10} \right] \\ &= 4855 \text{ W/m}^2\text{C}\end{aligned} \tag{19.42}$$

Take steam condensing coefficient as 8000 W/m²°C, fouling coefficient 5000 W/m²°C; butane fouling coefficient, essentially clean, 10,000 W/m²°C. Tube material will be plain carbon steel, $k_w = 55 \text{ W/m}^2\text{C}$.

$$\begin{aligned}\frac{1}{U_o} &= \frac{1}{4855} + \frac{1}{10,000} + \frac{30 \times 10^{-3} \ln \frac{30}{25}}{2 \times 55} + \frac{30}{25} \left(\frac{1}{5000} + \frac{1}{8000} \right) \\ U_o &= \underline{1341} \text{ W/m}^2\text{C}\end{aligned} \tag{19.2}$$

Close enough to original estimate of 1000 W/m²°C for the design to stand.

Myers and Katz (1953) give some data on the boiling of n-butane on banks of tubes. To compare the value estimated with their values, an estimate of the boiling film temperature difference is required:

$$= \frac{1341}{4855} \times 59.1 = 16.3^\circ\text{C} \text{ (29}^\circ\text{F)}$$

Myers data, extrapolated, gives a coefficient of around 3000 Btu/h ft² °F at a 29 °F temperature difference = 17,100 W/m²°C, so the estimated value of 4855 is certainly on the safe side.

Check the maximum allowable heat flux. Use the modified Zuber equation.

$$\text{Surface tension (estimated)} = 9.7 \times 10^{-3} \text{ N/m}$$

$$\rho_L = 550 \text{ kg/m}^3$$

$$\rho_v = \frac{58}{22.4} \times \frac{273}{(273 + 56)} \times 5.84 = 12.6 \text{ kg/m}^3$$

$$N_t = 52$$

For square arrangement $K_b = 0.44$:

$$\begin{aligned} q_c &= 0.44 \times 1.5 \times \frac{326 \times 10^3}{\sqrt{52}} [9.7 \times 10^{-3} \times 9.81(550 - 12.6)12.6^2]^{0.25} \\ &= 283,224 \text{ W/m}^2 \\ &= 280 \text{ kW/m}^2 \end{aligned} \quad (19.52)$$

Applying a factor of 0.7, maximum flux should not exceed $280 \times 0.7 = 196 \text{ kW/m}^2$. Actual flux of 59.2 kW/m^2 is well below maximum allowable.

Layout

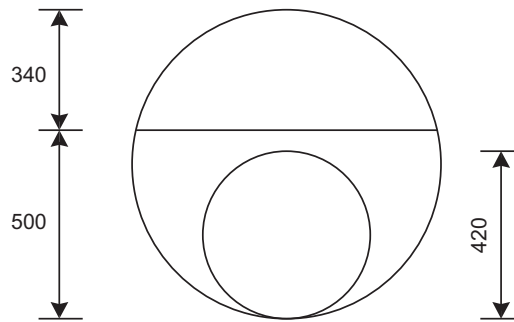
From the tubesheet layout, $D_b = 420 \text{ mm}$.

Take shell diameter as twice bundle diameter:

$$D_s = 2 \times 420 = 840 \text{ mm}$$

Take liquid level as 500 mm from base:

$$\text{freeboard} = 840 - 500 = 340 \text{ mm, satisfactory.}$$



From sketch, width at liquid level = 0.8 m.

$$\text{Surface area of liquid} = 0.8 \times 2.4 = 1.9 \text{ m}^2.$$

$$\text{Vapor velocity at surface} = \frac{5000}{3600} \times \frac{1}{12.6} \times \frac{1}{1.9} = \underline{\underline{0.06 \text{ m/s}}}$$

Maximum allowable velocity:

$$\hat{u}_v = 0.2 \left[\frac{550 - 12.6}{12.6} \right]^{1/2} = \underline{\underline{1.3 \text{ m/s}}} \quad (19.53)$$

So actual velocity is well below maximum allowable velocity. A smaller shell diameter could be considered.

19.12 Plate heat exchangers

19.12.1 Gasketed plate heat exchangers

A gasketed plate heat exchanger consists of a stack of closely spaced thin plates clamped together in a frame. A thin gasket seals the plates around their edges. The plates are normally between 0.5 and 3 mm thick, and the gap between them is 1.5 to 5 mm. Plate surface areas range from 0.03 to 1.5 m², with a plate width:length ratio from 2.0 to 3.0. The size of plate heat exchangers can vary from very small, 0.03 m², to very large, 1500 m². The maximum flow rate of fluid is limited to around 2500 m³/h.

The basic layout and flow arrangement for a gasketed plate heat exchanger are shown in Fig. 19.56. Corner ports in the plates direct the flow from plate to plate. The plates are embossed with a pattern of ridges, which increase the rigidity of the plate and improve the heat transfer performance.

Plates are available in a wide range of metals and alloys, including stainless steel, aluminum, and titanium. A variety of gasket materials can be used (Table 19.7).

Selection

The advantages and disadvantages of gasketed plate heat exchangers compared with conventional shell and tube exchangers are listed next:

Advantages:

1. Plates are attractive when material costs are high.
2. Plate heat exchangers are easier to maintain.
3. Low approach temperatures can be used, as low as 1 °C, compared with 5 °C to 10 °C for shell and tube exchangers.
4. Plate heat exchangers are more flexible; it is easy to add extra plates.
5. Plate heat exchangers are more suitable for highly viscous materials.
6. The temperature correction factor, F_t , will normally be higher with plate heat exchangers, as the flow is closer to true countercurrent flow.
7. Fouling tends to be significantly less in plate heat exchangers (Table 19.8).

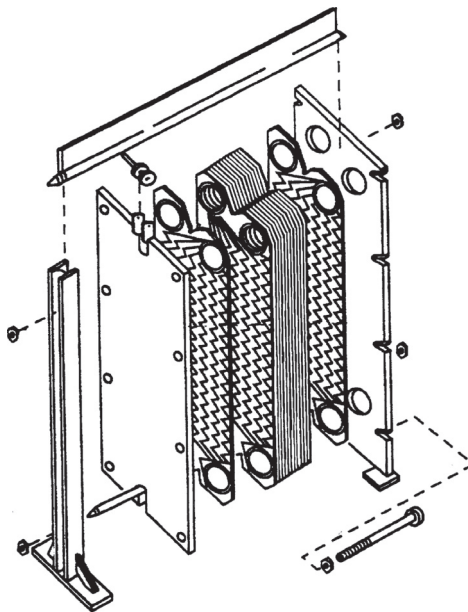


FIG. 19.56 Gasketed plate heat exchanger.

Disadvantages:

1. A plate is not a good shape to resist pressure, and plate heat exchangers are not suitable for pressures greater than about 30 bar or for high differential pressures between the two streams transferring heat.
2. The selection of a suitable gasket is critical; see [Table 19.7](#).
3. The maximum operating temperature is limited to about 250 °C due to the performance of the available gasket materials.

Plate heat exchangers are used extensively in the food and beverage industries, as they can be readily taken apart for cleaning and inspection. Their use in the chemical industry depends on the relative cost for the particular application compared with a conventional shell and tube exchanger; see [Parker \(1964\)](#) and [Trom \(1990\)](#).

TABLE 19.7 Typical gasket materials for plated heat exchangers

Material	Approximate temperature limit, °C	Fluids
Styrene–butane rubber	85	Aqueous systems
Acrylonitrile–butane rubber	140	Aqueous system, fats, aliphatic hydrocarbons
Ethylene–propylene rubber	150	Wide range of chemicals
Fluorocarbon rubber	175	Oils
Compressed asbestos	250	General resistance to organic chemicals

TABLE 19.8 Fouling factors (coefficients), typical values for plate heat exchangers

Fluid	Coefficient (W/m ² °C)	Factor (m ² °C/W)
Process water	30,000	0.00003
Towns water (soft)	15,000	0.00007
Towns water (hard)	6000	0.00017
Cooling water (treated)	8000	0.00012
Sea water	6000	0.00017
Lubricating oil	6000	0.00017
Light organics	10,000	0.0001
Process fluids	5000–20,000	0.0002–0.00005

Plate heat exchanger design

It is not possible to give exact design methods for plate heat exchangers. They are proprietary designs and are normally specified in consultation with the manufacturers. Information on the performance of the various patterns of plate used is not generally available. [Emerson \(1967\)](#) gives performance data for some proprietary designs, and [Kumar \(1984\)](#) and [Bond \(1981\)](#) have published design data for APV chevron patterned plates.

The approximate method given next can be used to size an exchanger for comparison with a shell and tube exchanger and to check performance of an existing exchanger for new duties. More detailed design methods are given by [Hewitt et al. \(1994\)](#) and [Cooper and Usher \(1983\)](#).

Procedure:

The design procedure is similar to that for shell and tube exchangers.

1. Calculate duty, the rate of heat transfer required.
2. If the specification is incomplete, determine the unknown fluid temperature or fluid flow rate from a heat balance.
3. Calculate the log mean temperature difference, ΔT_{lm} .
4. Determine the log mean temperature correction factor, F_t ; see the method given later.

5. Calculate the corrected mean temperature difference $\Delta T_m = F_t \times \Delta T_{lm}$.
6. Estimate the overall heat transfer coefficient; see Table 19.1.
7. Calculate the surface area required; see Equation 19.1.
8. Determine the number of plates required = total surface area/area of one plate.
9. Decide the flow arrangement and number of passes.
10. Calculate the film heat transfer coefficients for each stream; see the method given later.
11. Calculate the overall coefficient, allowing for fouling factors.
12. Compare the calculated with the assumed overall coefficient. If satisfactory, say -0% to $+10\%$ error, proceed. If unsatisfactory, return to step 8 and increase or decrease the number of plates.
13. Check the pressure drop for each stream; see method given later.

This design procedure is illustrated in Example 19.13.

Flow arrangements

The stream flows can be arranged in series or parallel, or a combination of series and parallel, (Fig. 19.57). Each stream can be subdivided into a number of passes, analogous to the passes used in shell and tube exchangers.

Estimation of the temperature correction factor

For plate heat exchangers, it is convenient to express the log mean temperature difference correction factor, F_t , as a function of the number of transfer units, NTU , and the flow arrangement (number of passes) (Fig. 19.58). The correction will normally be higher for a plate heat exchanger than for a shell and tube exchanger operating with the same temperatures. For rough sizing purposes, the factor can be taken as 0.95 for series flow.

The number of transfer units is given by:

$$NTU = (t_o - t_i) / \Delta T_{lm} \quad (19.55)$$

where t_i = stream inlet temperature, $^{\circ}\text{C}$

t_o = stream outlet temperature, $^{\circ}\text{C}$

ΔT_{lm} = log mean temperature difference, $^{\circ}\text{C}$

Typically, the NTU will range from 0.5 to 4.0, and for most applications will lie between 2.0 and 3.0.

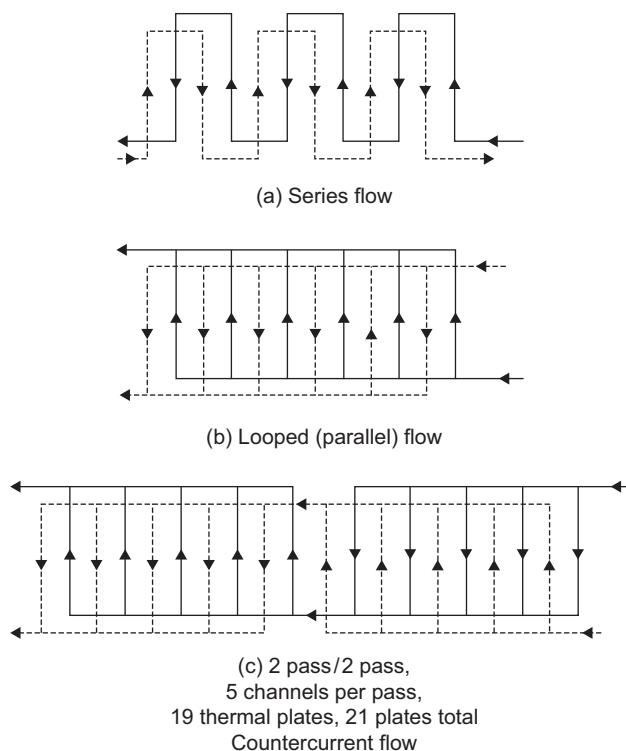


FIG. 19.57 Plate heat-exchanger flow arrangements.

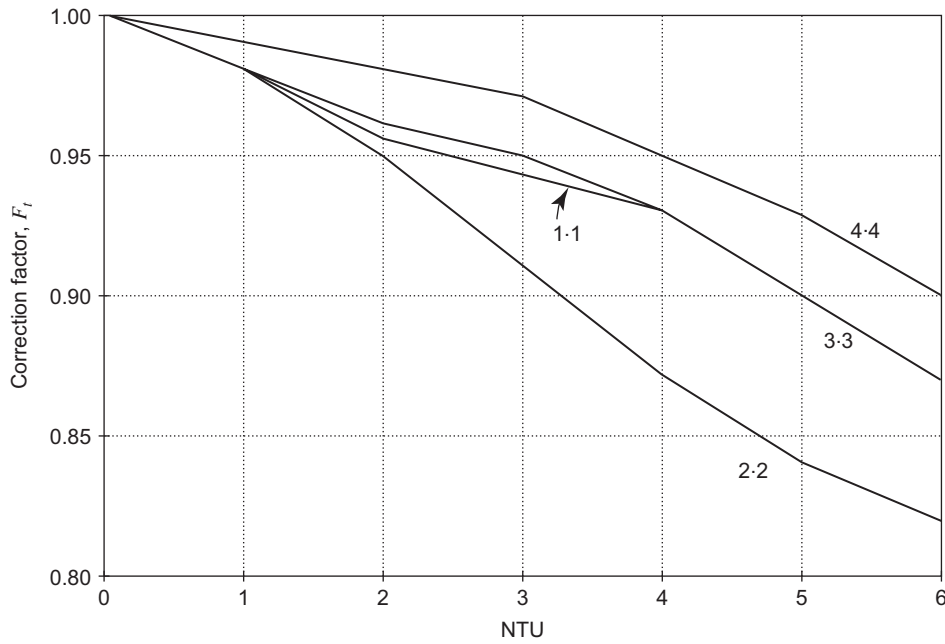


FIG. 19.58 Log mean temperature correction factor for plate heat exchangers. Adapted from Raju & Chand 1980.

Heat transfer coefficient

The equation for forced-convective heat transfer in conduits can be used for plate heat exchangers; see Equation 19.10.

The values for the constant C and the indices a , b , and c will depend on the particular type of plate being used. Typical values for turbulent flow are given in the following equation, which can be used to make a preliminary estimate of the area required.

$$\frac{h_p d_e}{k_f} = 0.26 \text{Re}^{0.65} \text{Pr}^{0.4} \left(\frac{\mu}{\mu_w} \right)^{0.14} \quad (19.56)$$

where h_p = plate film coefficient

$\text{Re} = \text{Reynolds number} = \frac{G_p d_e}{\mu} = \frac{\rho u_p d_e}{\mu}$

$G_p = \text{mass flow rate per unit cross-sectional area} = w/A_f, \text{kgm}^{-2} \text{s}^{-1}$

$w = \text{mass flow rate per channel, kg/s}$

$A_f = \text{cross-sectional area for flow, m}^2$

$u_p = \text{channel velocity, m/s}$

$d_e = \text{equivalent (hydraulic) diameter, taken as twice the gap between the plates, m}$

The corrugations on the plates will increase the projected plate area and reduce the effective gap between the plates. For rough sizing, where the actual plate design is not known, this increase can be neglected. The channel width equals the plate pitch minus the plate thickness.

There is no heat transfer across the end plates, so the number of effective plates will be the total number of plates less two.

Pressure drop

The plate pressure drop can be estimated using a form of the equation for flow in a conduit, Equation 19.18.

$$\Delta P_p = 8j_f (L_p / d_e) \frac{\rho u_p^2}{2} \quad (19.57)$$

where $L_p = \text{the path length and } u_p = G_p / \rho$.

The value of the friction factor, j_f , depends on the design of plate used. For preliminary calculations, the following relationship can be used for turbulent flow:

$$j_f = 0.6\text{Re}^{-0.3}$$

The transition from laminar to turbulent flow will normally occur at a Reynolds number of 100 to 400, depending on the plate design. With some designs, turbulence can be achieved at very low Reynolds numbers, which makes plate heat exchangers quite suitable for use with viscous fluids.

The pressure drop due to the contraction and expansion losses through the ports in the plates must be added to the friction loss. Kumar (1984) suggests adding 1.3 velocity heads per pass, based on the velocity through the ports.

$$\Delta P_{pt} = 1.3 \frac{(\rho u_{pt}^2)}{2} N_p \quad (19.58)$$

where u_{pt} = the velocity through the ports $w/\rho A_p$, m/s

w = mass flow through the ports, kg/s

A_p = area of the port = $(\pi d_{pt}^2)/4$, m²

d_{pt} = port diameter, m

N_p = number of passes

Example 19.13

Investigate the use of a gasketed plate heat exchanger for the duty set out in Example 19.1: cooling methanol using brackish water as the coolant. Titanium plates are to be specified to resist corrosion by the saline water.

Summary of Example 19.1

Cool 100,000 kg/h of methanol from 95 °C to 40 °C, duty 4340 kW. Cooling water inlet temperature 25 °C and outlet temperature 40 °C. Flow rates: methanol 27.8 kg/s, water 68.9 kg/s.

Physical properties:	Methanol	Water
Density, kg/m ³	750	995
Viscosity, mN m ⁻² s	3.4	0.8
Prandtl number	5.1	5.7

Logarithmic mean temperature difference: 31 °C.

Solution

NTU , based on the maximum temperature difference:

$$= \frac{95 - 40}{31} = 1.8$$

Try a 1:1 pass arrangement.

From Fig. 19.58, $F_t = 0.96$

From Table 19.2, take the overall coefficient, light organic–water, to be 2000 W/m²°C.

Then, area required

$$= \frac{4340 \times 10^3}{2000 \times 0.96 \times 31} = 72.92 \text{ m}^2$$

Select an effective plate area of 0.75 m², effective length 1.5 m, and width 0.5 m; these are typical plate dimensions. The actual plate size will be larger to accommodate the gasket area and ports.

$$\text{Number of plates} = \text{total heat transfer area} / \text{effective area of one plate} = 72.92/0.75 = 97$$

No need to adjust this, 97 will give an even number of channels per pass, allowing for an end plate.

$$\text{Number of channels per pass} = (97 - 1)/2 = 48$$

Take plate spacing as 3 mm, a typical value, then :

$$\text{Channel cross-sectional area} = 3 \times 10^{-3} \times 0.5 = 0.0015 \text{ m}^2$$

$$\text{Hydraulic mean diameter} = 2 \times 3 \times 10^{-3} = 6 \times 10^{-3} \text{ m}$$

Methanol

$$\text{Channel velocity} = \frac{27.8}{750} \times \frac{1}{0.0015} \times \frac{1}{48} = 0.51 \text{ m/s}$$

$$\text{Re} = \frac{\rho u_p d_e}{\mu} = \frac{750 \times 0.51 \times 6 \times 10^{-3}}{0.34 \times 10^{-3}} = 6750$$

$$\text{Nu} = 0.26(6750)^{0.65} \times 5.1^{0.4} = 153.8 \quad (19.56)$$

$$h_p = 153.8 \left(\frac{0.19}{6 \times 10^{-3}} \right) = 4870 \text{ Wm}^{-2}\text{C}$$

Brackish water

$$\text{Channel velocity} = \frac{68.9}{995} \times \frac{1}{0.0015} \times \frac{1}{48} = 0.96 \text{ m/s}$$

$$\text{Re} = \frac{995 \times 0.96 \times 6 \times 10^{-3}}{0.8 \times 10^{-3}} = 7164$$

$$\text{Nu} = 0.26(7164)^{0.65} \times 5.7^{0.4} = 167.2 \quad (19.56)$$

$$h_p = 167.2 \left(\frac{0.59}{6 \times 10^{-3}} \right) = 16,439 \text{ Wm}^{-2}\text{C}$$

Overall coefficient

From Table 19.2, take the fouling factors (coefficients) as brackish water (seawater) $6000 \text{ W/m}^2\text{C}$ and methanol (light organic) $10,000 \text{ W/m}^2\text{C}$.

Take the plate thickness as 0.75 mm. Thermal conductivity of titanium $21 \text{ W/m}^\circ\text{C}$.

$$\frac{1}{U} = \frac{1}{4870} + \frac{1}{10,000} + \frac{0.75 \times 10^{-3}}{21} + \frac{1}{16,439} + \frac{1}{6000}$$

$$U = 1759 \text{ W/m}^2\text{C, too low}$$

Increase the number of channels per pass to 60, giving $(2 \times 60) + 1 = 121$ plates.

Then, methanol channel velocity $= 0.51 \times (48/60) = 0.41 \text{ m/s}$ and $\text{Re} = 5400$.

Cooling water channel velocity $= 0.96 \times (48/60) = 0.77 \text{ m/s}$ and $\text{Re} = 5746$.

Giving $h_p = 4215 \text{ W/m}^2\text{C}$ for methanol and $14,244 \text{ W/m}^2\text{C}$ for water.

Which gives an overall coefficient of $1640 \text{ W/m}^2\text{C}$.

Overall coefficient required $2000 \times 48/60 = \underline{1600} \text{ Wm}^{-2}\text{C}^{-1}$, so 60 plates per pass should be satisfactory.

Pressure drops*Methanol:*

$$j_f = 0.60(5400)^{-0.3} = 0.046$$

Path length = plate length $\times$ number of passes = $1.5 \times 1 = 1.5$ m.

$$\Delta P_p = 8 \times 0.046 \left(\frac{1.5}{6 \times 10^{-3}} \right) \times 750 \times \frac{0.41^2}{2} = 5799 \text{ N/m}^2 \quad (19.57)$$

Port pressure loss, take port diameter as 100 mm, area = 0.00785 m^2 .

Velocity through port = $(27.8/750)/0.00785 = 4.72 \text{ m/s}$.

$$\Delta P_{pt} = 1.3 \times \frac{750 \times 4.72^2}{2} = 10,860 \text{ N/m}^2 \quad (19.58)$$

Total pressure drop = $5799 + 10,860 = \underline{\underline{16,659 \text{ N/m}^2}}$, 0.16 bar.

Water:

$$j_f = 0.6(5501)^{-0.3} = 0.045$$

Path length = plate length $\times$ number of passes = $1.5 \times 1 = 1.5$ m.

$$\Delta P_p = 8 \times 0.045 \times \left(\frac{1.5}{6 \times 10^{-3}} \right) \times 995 \times \frac{0.77^2}{2} = 26,547 \text{ N/m}^2 \quad (19.57)$$

Velocity through port = $(68.9/995)/0.0078 = 8.88 \text{ m/s}$ (rather high)

$$\Delta P_{pt} = 1.3 \times \frac{995 \times 8.88^2}{2} = 50,999 \text{ N/m}^2$$

Total pressure drop = $26,547 + 50,999 = \underline{\underline{77,546 \text{ N/m}^2}}$, 0.78 bar

Could increase the port diameter to reduce the pressure drop.

The trial design should be satisfactory, so a plate heat exchanger could be considered for this duty.

19.12.2 Welded plate exchangers

Welded plate heat exchangers use plates similar to those in gasketed plate exchangers, but the plate edges are sealed by welding. This increases the pressure and temperature rating to up to 80 bar and temperatures in excess of 500°C . They retain the advantages of plate heat exchangers (compact size and good rates of heat transfer) while giving security against leakage. An obvious disadvantage is that the exchangers cannot be dismantled for cleaning, so their use is restricted to applications where fouling is not a problem. The plates are fabricated in a variety of materials.

A welded plate exchanger can be specified for a high-pressure service if it is contained inside a pressure vessel and the space between the exchanger and the vessel is pressurized to the same pressure as the fluid inside the exchanger. The differential pressure between the two sides of the exchanger must still be small.

A combination of gasketed and welded plate construction can also be used, with an aggressive process fluid flowing between welded plates and a benign process stream, or service stream, between gasketed plates.

19.12.3 Plate-fin exchangers

Plate-fin exchangers consist essentially of plates separated by corrugated sheets, which form the fins. They are made up in a block and are often referred to as matrix exchangers (Fig. 19.59). They are usually constructed of aluminum and joined and sealed by brazing. The main application of plate-fin exchangers has been in the cryogenics industries, such as air separation plants, where large heat transfer surface areas are needed. They are now finding wider applications in the chemical process industries, in applications where large surface area, compact, exchangers are required. Their compact size and low weight have led to some use in offshore applications. The brazed

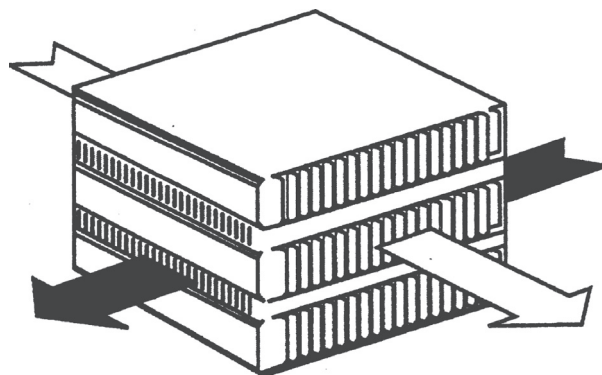


FIG. 19.59 Plate-fin exchanger.

aluminum construction is limited to pressures up to around 60 bar and temperatures up to 150 °C. The units cannot be mechanically cleaned, so their use is restricted to clean process and service steams. The construction and design of plate-fin exchangers and their applications are discussed by [Saunders \(1988\)](#) and [Burley \(1991\)](#) and their use in cryogenic service by [Lowe \(1987\)](#).

19.12.4 Spiral heat exchangers

A spiral heat exchanger can be considered as a plate heat exchanger in which the plates are formed into a spiral. The fluids flow through the channels formed between the plates. The exchanger is made up of long sheets, between 150 and 1800 mm wide, formed into a pair of concentric spiral channels. The channels are closed by gasketed end plates bolted to an outer case. Inlet and outlet nozzles are fitted to the case and connect to the channels ([Fig. 19.60](#)). The gap between the sheets varies between 4 and 20 mm; depending on the size of the exchanger and the application. They can be fabricated in any material that can be cold-worked and welded.

Spiral heat exchangers are compact units: a unit with around 250 m² area occupying a volume of approximately 10 m³. The maximum operating pressure is limited to 20 bar and the temperature to 400 °C.

For a given duty, the pressure drop over a spiral heat exchanger will usually be lower than that for the equivalent shell and tube exchanger. Spiral heat exchangers give true countercurrent flow and can be used where the temperature correction factor F_t for a shell and tube exchanger would be too low; see [Section 19.6](#). Because they are easily cleaned and the turbulence in the channels is high, spiral heat exchangers can be used for very dirty process fluids and slurries.

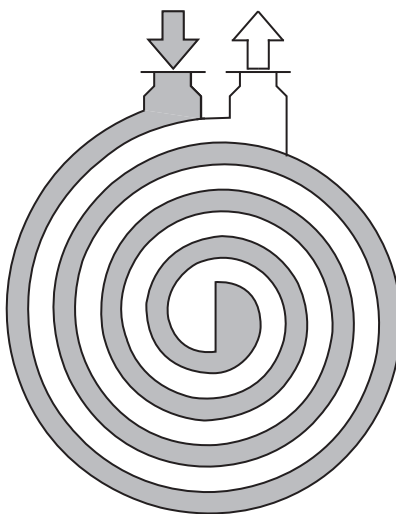


FIG. 19.60 Spiral heat exchanger.

The correlations for flow in conduits can be used to estimate the heat transfer coefficient and pressure drop in the channels, using the hydraulic mean diameter as the characteristic dimension.

The design of spiral heat exchangers is discussed by [Minton \(1970\)](#).

19.13 Direct-contact heat exchangers

In direct-contact heat exchange the hot and cold streams are brought into contact without any separating wall, and high rates of heat transfer are achieved. Applications include reactor off-gas quenching, vacuum condensers, cooler-condensers, desuperheating, and humidification. Water cooling towers are a particular example of direct-contact heat exchange; see [Section 3.2.5](#). In direct-contact cooler-condensers, the condensed liquid is frequently used as the coolant ([Fig. 19.61](#)).

Direct-contact heat exchangers should be considered whenever the process stream and coolant are compatible. The equipment used is simple and cheap and is suitable for use with heavily fouling fluids and with liquids containing solids; spray chambers, spray columns, and plate and packed columns are used.

There is no general design method for direct-contact exchangers. Most applications involve the transfer of latent heat as well as sensible heat, and the process is one of simultaneous heat and mass transfer. When the approach to thermal equilibrium is rapid, as it will be in many applications, the size of the contacting vessel is not critical and the design can be based on experience with similar processes. For other situations, the designer must work from first principles, setting up the differential equations for mass and heat transfer and using judgment in making the simplifications necessary to achieve a solution. The design procedures used are analogous to those for gas absorption and distillation. The rates of heat transfer will be high, with coefficients for packed columns typically in the range of 2000 to 20,000 W/m³°C (i.e., per cubic meter of packing).

The design and application of direct-contact heat exchangers is discussed by Fair (1961, 1972*a*, 1972*b*) and [Chen-Chia and Fair \(1989\)](#). They give practical design methods and data for a range of applications.

The design of water cooling towers and humidification is discussed by [Chhabra and Shankar \(2017\)](#). The same principles apply to the design of other direct-contact exchangers.

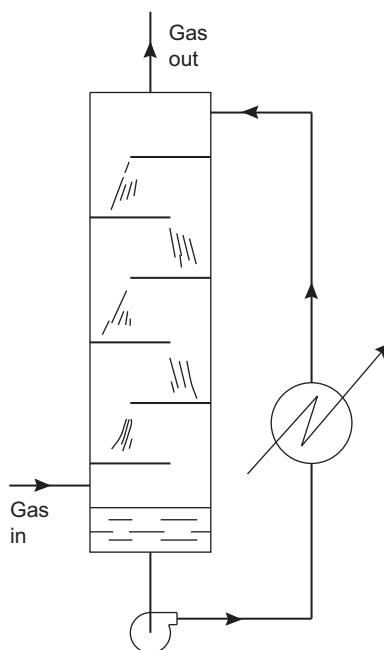


FIG. 19.61 Typical direct-contact cooler (baffle plates).

19.14 Finned tubes

Fins are used to increase the effective surface area of heat exchanger tubing. Many different types of fins have been developed, but the plain transverse fin shown in [Fig. 19.62](#) is the most commonly used type for process heat

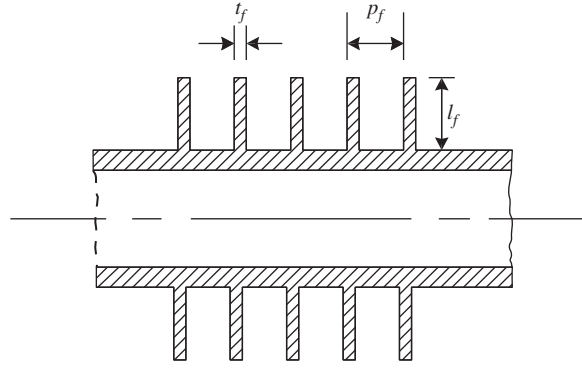


FIG. 19.62 Finned tube.

exchangers. Typical fin dimensions are pitch 2.0 to 4.0 mm, height 12 to 16 mm, and ratio of fin area to bare tube area 15:1 to 20:1.

Finned tubes are used when the heat transfer coefficient on the outside of the tube is appreciably lower than that on the inside, as in heat transfer from a liquid to a gas, such as in air-cooled heat exchangers.

The fin surface area will not be as effective as the bare tube surface, as the heat has to be conducted along the fin. This is allowed for in design by the use of a fin effectiveness, or fin efficiency, factor. The equations describing heat transfer from a fin are derived in [Chhabra and Shankar \(2017\)](#); see also [Kern \(1950\)](#). The fin effectiveness is a function of the fin dimensions and the thermal conductivity of the fin material. Fins are therefore usually made from metals with a high thermal conductivity; for copper and aluminum, the effectiveness is typically between 0.9 and 0.95.

When using finned tubes, the coefficients for the outside of the tube in Equation 19.2 are replaced by a term involving fin area and effectiveness:

$$\frac{1}{h_o} + \frac{1}{h_{od}} = \frac{1}{E_f} \left(\frac{1}{h_f} + \frac{1}{h_{df}} \right) \frac{A_o}{A_f} \quad (19.59)$$

where h_f = heat transfer coefficient based on the fin area

h_{df} = fouling coefficient based on the fin area

A_o = outside area of the bare tube

A_f = fin area

E_f = fin effectiveness

It is not possible to give a general correlation for the coefficient h_f covering all types of fins and fin dimensions. Design data should be obtained from the tube manufacturers for the particular type of tube to be used. For banks of tubes in cross flow with plain transverse fins, the correlation given by [Briggs and Young \(1963\)](#) can be used to make an approximate estimate of the fin coefficient.

$$\text{Nu} = 0.134 \text{Re}^{0.681} \text{Pr}^{0.33} \left(\frac{p_f - t_f}{l_f} \right)^{0.2} \left(\frac{p_f}{t_f} \right)^{0.1134} \quad (19.60)$$

where p_f = fin pitch

l_f = fin height

t_f = fin thickness

The Reynolds number is evaluated for the bare tube (i.e., assuming that no fins exist).

[Kern and Kraus \(1972\)](#) give full details of the use of finned tubes in process heat exchanger design and design methods.

Low fin tubes

Tubes with low transverse fins, about 1 mm high, can be used with advantage as replacements for plain tubes in many applications. The fins are formed by rolling, and the tube outside diameters are the same as those for standard plain tubes. Details are given in the manufacturer's data books, [Wolverine \(1984\)](#), and an electronic version of their design manual, www.wlv.com; see also [Webber \(1960\)](#).

19.15 Double-pipe heat exchangers

One of the simplest and cheapest types of heat exchanger is the concentric pipe arrangement shown in Fig. 19.63. These can be made up from standard pipe fittings and are useful when only a small heat transfer area is required. Several units can be connected in series to extend their capacity.

The correlation for forced convective heat transfer in conduits (Equation 19.10) can be used to predict the heat transfer coefficient in the annulus, using the appropriate equivalent diameter:

$$d_e = \frac{4 \times \text{cross-sectional area}}{\text{wetted perimeter}} = \frac{4 \left(\frac{\pi}{4} (d_2^2 - d_1^2) \right)}{\pi (d_2 + d_1)} = d_2 - d_1$$

where d_2 is the inside diameter of the outer pipe and d_1 the outside diameter of the inner pipe. Some designs of double-pipe exchanger use inner tubes fitted with longitudinal fins.

A variant of the double-pipe heat exchanger is the hairpin exchanger shown in Fig. 19.64. A hairpin exchanger is formed by inserting one or more U-tubes into two pipe sections welded to a large flanged end, which is then closed using a removable bonnet. Each straight section of the U-tubes acts as a double-pipe exchanger. True countercurrent flow is obtained if a single U-tube is used.

Hairpin exchangers are cheaper than shell and tube exchangers at very small sizes and can be specified for areas from 7 m² to 150 m².

19.16 Air-cooled exchangers

Air-cooled exchangers consist of banks of finned tubes over which air is blown or drawn by fans mounted below or above the tubes (forced or induced draft). Typical designs are shown in Fig. 19.65.

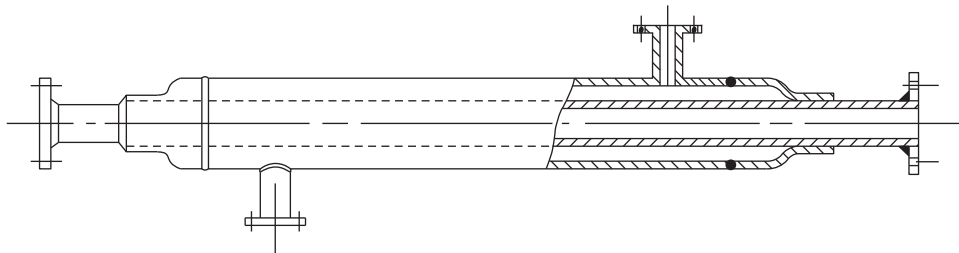


FIG. 19.63 Double-pipe exchanger (constructed for weld fittings).

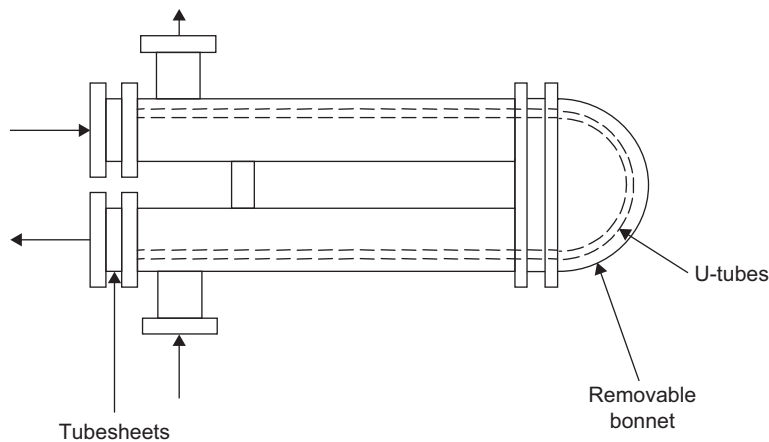


FIG. 19.64 Hairpin exchanger.

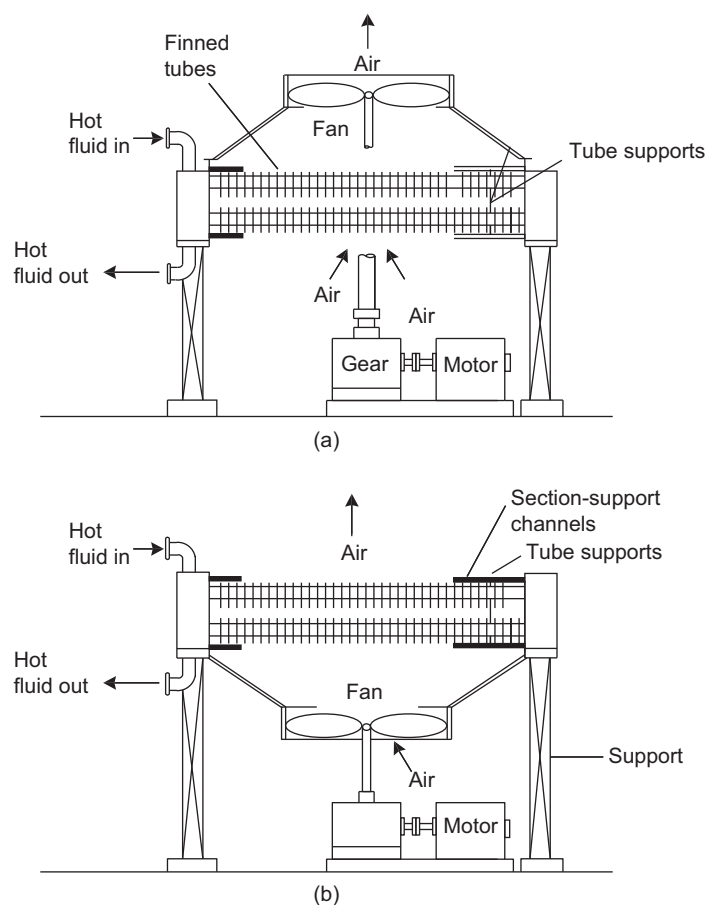


FIG. 19.65 Air-cooled exchangers. (a) Induced draft. (b) Forced draft.

Air-cooled exchangers should be considered when cooling water is in short supply or expensive. They can be competitive with water-cooled units even when water is plentiful. Frank (1978) suggests that in moderate climates air cooling will usually be the best choice for minimum process temperatures above 65 °C and water cooling for minimum processes temperatures below 50 °C. Between these temperatures a detailed economic analysis must be carried out to decide the best coolant. Air-cooled exchangers are used for cooling and condensing.

Cooling water circuits require a humidity driving force to achieve cooling of the water; see Section 3.2.5 and Chhabra and Shankar (2017). In climates that often experience a combination of high temperature and high humidity, air cooling will usually be cheaper than water cooling. Air coolers are also often specified for revamps or additions to existing plants so as to avoid increasing the cooling tower load and obviate investments in the site utility system.

The design and application of air-cooled exchangers is discussed by Rubin (1960), Lerner (1972), Brown (1978), and Mukherjee (1997). Design procedures are also given in the books by Kern (1950), Kern and Kraus (1972), and Kroger (2004). Lerner and Brown give typical values for the overall coefficient for a range of applications and provide methods for the preliminary sizing of air-cooled heat exchangers.

Details of the construction features of air-cooled exchangers are given by Coker (2015). The construction features of air-cooled heat exchangers are covered by the American Petroleum Institute Standard, API 661, which has been adopted as the recognized international standard for air coolers, ISO 13706-1:2011.

Air-cooled exchangers are packaged units and are normally selected and specified in consultation with a manufacturer. Some typical overall coefficients are given in Table 19.1. These can be used to make an approximate estimate of the area required for a given duty. The commercial heat exchanger design programs developed by HTRI and HTFS include programs for air cooler design; see Section 19.1.

19.16.1 Air coolers: Construction details

Air coolers can be designed with the fan mounted either above or below the tube rack, as illustrated in Fig. 19.65.

In an induced draft cooler (Fig. 19.65a), the fan is mounted above the bank of tubes and air is pulled up over the tubes. The housing around the fan provides for some chimney effect, which can give better cooling when the fan is not running. This design generally has better air distribution across the tube rack and lower chance of air recirculation, but the location of the fan makes maintenance access difficult, and the tubes can become damaged during fan maintenance.

Fig. 19.65(b) shows a forced draft cooler in which the fan is mounted below the bank of tubes and air is blown up across the tube bank. The natural draft capability of this design is lower, but the fan is easier to access for maintenance. The fan draws cooler air in a forced draft design, which reduces the power requirement. Forced draft designs can also recirculate air in winter to offset the effect of lower ambient temperature. Care must be taken when designing the cooler and plant layout to ensure that unwanted air recirculation does not occur during normal operation. Forced draft coolers are usually less expensive than induced draft coolers.

The tubes used in air coolers are usually finned to provide additional surface area to compensate for the poor heat transfer coefficient on the air side. Because of the use of finned tubes, air coolers are sometimes referred to as “fin-fan coolers.” The ratio of finned area to bare tube area is typically about 20:1.

The tubes are usually welded into a header at each end of the exchanger, and the tube-side flow is usually two-pass so that the inlet and outlet are at the same (front) end of the cooler. The opposite (floating) header is mounted to allow for thermal expansion of the tubes. Multiple banks of tubes may be used as long as the air-side pressure drop and fan power are not excessive. Standard tube lengths are typically used, but longer tubes up to 20 m (60 ft) can be accommodated by arranging multiple fans in parallel. The height of the tube bank above the ground must be large enough to give an inlet velocity equal to the face velocity over the tube bundle. A typical height is half of the tube length per fan.

19.16.2 Heat transfer in air coolers

The air-side heat transfer coefficient in air coolers is usually very low and dominates the overall heat transfer coefficient. If the detailed design of the finned tubes is known, then the air-side coefficient can be calculated using Equation 19.60 or a similar expression provided by the tube manufacturer. When the finned tube design is not known, the fin coefficient can be estimated using the expression given by Lohrisch (1966):

$$\text{Nu} = 0.28 \text{ Re}^{0.6} \text{Pr}^{0.33} \quad (19.61)$$

Using typical fin dimensions of fin pitch 2.3 mm (12 fins per inch), fin length 15.9 mm, and fin thickness 0.48 mm, Equation 19.60 reduces to:

$$\text{Nu} = 0.104 \text{ Re}^{0.681} \text{Pr}^{0.33} \quad (19.62)$$

Air coolers are typically designed for the highest temperature that will be exceeded for 40 hours per year less 4 °C (API 661). This temperature is usually about 40 °C (~104 °F). At this temperature, dry air has the properties $\text{Pr} \sim 0.7$, $\rho \sim 1.13 \text{ kg/m}^3$, $\mu \sim 1.9 \times 10^{-5} \text{ Nm}^{-2}\text{s}$ and $k \sim 0.0272 \text{ W/m}^\circ\text{C}$. Using a typical tube diameter of 25.4 mm and face velocity of 2.5 m/s (500 ft/min), we obtain:

From Equation 19.61: $\text{Nu} \sim 34.8$, $h_f \sim 37.2$

From Equation 19.62: $\text{Nu} \sim 25.2$, $h_f \sim 27.0$

The fin efficiency is typically about 0.9 at these values of h_f (Lohrisch, 1966), and the ratio of finned area to bare tube area is typically 20:1. Fouling factors for air are usually quite high, and the fouling coefficient can be taken as 5000 $\text{W/m}^2\text{ }^\circ\text{C}$ (from Table 19.2). Taking the average of the values of h_f calculated earlier, equation 19.59 gives:

$$\frac{1}{h_o} + \frac{1}{h_{od}} = \frac{1}{0.9} \left(\frac{1}{32} + \frac{1}{5000} \right) \frac{1}{20} \quad (19.59)$$

$$\frac{1}{h_o} + \frac{1}{h_{od}} \approx \frac{1}{600} \quad (19.63)$$

where h_o is based on the bare tube area.

Equation 19.63 can be used with Equation 19.2 to make an initial estimate of the overall heat transfer coefficient. More accurate correlations are available in the air-cooled exchanger design programs developed by HTRI (www.HTRI.net) and HTFS. The HTFS programs are part of Aspen Technology's Aspen Engineering Suite and Honeywell's UniSim Design Suite; see Table 4.1. Equations for heat transfer to tube banks in natural convection are given by Chu (2005).

19.16.3 Air cooler design

Air-cooled exchangers are designed to give satisfactory performance under most expected climate conditions. The API 661/ISO 13706 standard recommends that for critical applications, the ambient temperature shall be the highest air temperature that is exceeded for 40 hours per year. For noncritical applications, the air temperature can be taken as the higher of:

- The highest temperature that is exceeded for 400 h/y
- The highest temperature that is exceeded for 40 h/y minus 4 °C

A small allowance on ambient temperature is usually made to provide a safety margin to compensate for recirculation, typically 2 °C to 3 °C.

The procedure for designing an air cooler is then as follows:

1. Estimate the cooling duty, including heat of condensation, if phase change occurs.
2. Collect physical property data.
3. Estimate the tube inside heat transfer coefficient and fouling coefficient. Hence calculate U using Equations 19.2 and 19.63.
4. Determine the ambient air temperature for design. Ambient temperature statistics can be found from government meteorological offices and sources such as www.weather.com. Add an allowance for recirculation.
5. Estimate ΔT_{lm} , assuming that the air-side temperature is approximately constant. Hence, estimate ΔT_m , assuming F_t is 0.9. The air-side temperature is obviously not constant and typically increases by a few °C. The low change in temperature on the air side gives a high value of R and a low value of S unless the process-side outlet temperature is very close to the air temperature. Consequently, F_t is close to 1.0 for air coolers. Assuming $F_t = 0.9$ is a conservative approximation that compensates for the error introduced by using a constant air-side temperature in calculating ΔT_{lm} .
6. Estimate the bare tube area using Equation 19.1.
7. Choose a tube diameter (typically 25.4 mm) and length (typically a multiple of 6 m or 20 ft) and determine the number of tubes required. Select the tube material of construction.
8. Decide the bundle layout. Determine the number of tubes per bank and number of banks of tubes (generally fewer than 10 banks per bundle).
9. Determine the bundle area. The tube spacing is typically 50.8 mm or 63.5 mm (2.0 in. or 2.5 in.) on a triangular pitch, but wider spacing may be used if the fin length is large. The bundle area, A_b , is given by:

$$A_b = L' p_t N_{bk} \quad (19.64)$$

where $L' =$ effective tube length

$p_t =$ tube pitch

$N_{bk} =$ number of tubes per bank

10. Estimate the air-side flow rate. A typical face velocity is 2.5 m/s (500 ft/min). The air flow rate is the product of face velocity and bundle area.
11. Estimate the fan power consumption. The fan power, W_f , can be estimated as:

$$W_f = \frac{u_f A_b \Delta P_b}{\eta_f} \quad (19.65)$$

where $u_f =$ face velocity

$\Delta P_b =$ pressure drop across the bundle

$\eta_f =$ fan efficiency, typically about 0.7

In customary units, Equation 19.65 is usually written as:

$$\text{Fan power (hp)} = \left(\frac{\text{ACFM } \Delta P_b}{6837 \eta_f} \right) \quad (19.65b)$$

where ACFM = air flow rate in actual cubic feet per minute and ΔP_b is in inches of water. The pressure drop across the bundle is usually very low, and a value of 150 N/m² (0.6 in. of water) can be used as an initial estimate.

12. Estimate the overall power consumption by allowing an additional 5% to 10% for motor efficiency.
13. Carry out a detailed simulation of the specified geometry to confirm that the heat transfer coefficients and pressure drops that have been assumed are realistic. This step is usually carried out using commercial design programs. Modify the design if necessary.
14. Determine the air cooler capital and operating costs. Modify the design as necessary to optimize the total annualized cost.

This procedure is illustrated in Example 19.14.

19.16.4 Air cooler operation and control

An intrinsic problem of air coolers is that they must be designed to perform under the warmest climate conditions and so are oversized for the current ambient temperature most of the time. This can be particularly problematic in winter, when ambient temperature may be low enough to cause precipitation of components from the process fluid or even freeze the process fluid.

Temperature control of air coolers can be accomplished by varying the air flow or bypassing some hot fluid around the exchanger, as discussed in Chapter 5 and illustrated in Fig. 5.12. Several methods are used to vary air flow:

1. Variable-speed drives can be used on the fans. These have a higher cost than single-speed fans, but give the best energy efficiency and temperature control.
2. Variable-pitch fans can be specified. Changing the fan pitch varies the air flow, and power consumption is reduced at low pitch.
3. Louvers can be used to provide a restriction in the air flow path and reduce the air rate. The louvers may be manually or automatically adjusted. The use of louvers does not reduce power consumption, and louver operation can be impeded by snow or ice, but this is the cheapest method.

In regions that experience cold winters, air flow control may not be adequate for winterization. Annex C of API 661/ISO 13706 provides guidelines on when winterization is needed and describes other winterization approaches such as internal or external air recirculation.

Example 19.14

Investigate the use of an air-cooled exchanger for the duty set out in Example 19.1. Determine the fan power consumption.

Solution

Summary of Example 19.1

Cool 100,000 kg/h of methanol from 95 °C to 40 °C, duty 4340 kW. Methanol flow rate 27.8 kg/s, density 750 kg/m³, viscosity 0.34 mNm⁻²s, Prandtl number 5.1, thermal conductivity 0.19 W/m°C.

Step 3: Heat transfer coefficient

Assuming a typical tube-side velocity of 1.5 m/s and selecting 25.4 mm o.d. (22.1 mm i.d.) tubes:

$$\text{Re} = \frac{\rho u d_i}{\mu} = \frac{750 \times 1.5 \times 22.1 \times 10^{-3}}{0.34 \times 10^{-3}} = 73.1 \times 10^3$$

From Equation 19.11, neglecting the viscosity correction:

$$\begin{aligned}\text{Nu} &= 0.023 \text{Re}^{0.8} \text{Pr}^{0.33} \\ &= 0.023 (73.1 \times 10^3)^{0.8} (5.1)^{0.33} = 306.5\end{aligned}$$

$$\begin{aligned}\text{Hence, } h_i &= 306.5 \times (0.19 / (22.1 \times 10^{-3})) \\ &= 2635 \text{ W/m}^2 \text{ } ^\circ\text{C}\end{aligned}$$

Because the exchanger no longer uses brine as coolant, a cheaper material of construction such as plain carbon steel can be selected. For carbon steel, the thermal conductivity, k_w , is 55 W/m $^\circ\text{C}$ over the range of temperature of interest.

Using the same fouling coefficient of 5000 W/m 2 $^\circ\text{C}$ for methanol and substituting Equation 19.63 in Equation 19.2 gives:

$$\begin{aligned}\frac{1}{U_o} &= \frac{1}{600} + \frac{25.4 \times 10^{-3} \ln\left(\frac{25.4}{22.1}\right)}{2 \times 55} + \frac{25.4}{22.1} \times \frac{1}{2635} + \frac{25.4}{22.1} \times \frac{1}{5000} \\ \frac{1}{U_o} &= \frac{1}{600} + \frac{1}{31120} + \frac{1}{2292} + \frac{1}{4350}\end{aligned}\quad (19.2)$$

From which it is clear that the air-side contribution governs the overall heat transfer coefficient, and:

$$U_o = 423 \text{ W/m}^2 \text{ } ^\circ\text{C}$$

Step 4: Ambient temperature

If the cooler is located in the U.S. Midwest, then a reasonable initial estimate of the design air temperature is 32 $^\circ\text{C}$ (90 $^\circ\text{F}$). The ambient temperature is obviously highly site-specific, and for many regions a product temperature specification of 40 $^\circ\text{C}$ would immediately rule out the use of air cooling as impractical. Adding 2 $^\circ\text{C}$ to allow for some recirculation gives a design temperature of 34 $^\circ\text{C}$.

Step 5: Estimate ΔT_m

$$\Delta T_{lm} = \frac{(95 - 34) - (40 - 34)}{\ln\left(\frac{95 - 34}{40 - 34}\right)} = 23.7^\circ\text{C}$$

Assuming $F_t = 0.9$,

$$\Delta T_m = 0.9 \times 23.7 = 21.3^\circ\text{C}$$

Step 6: Estimate bare tube area

$$Q = UA\Delta T_m \quad (19.1)$$

hence:

$$A = \frac{4.34 \times 10^6}{423 \times 21.3} = 481.7 \text{ m}^2$$

Steps 7 and 8: Pick tubes and determine bundle layout

The area of a 6 m long, 25.4 mm o.d. (20 ft, 1 in. o.d.) tube is $\pi \times 25.4 \times 10^{-3} \times 6 = 0.479 \text{ m}^2$, so we require $481.7 / 0.479 = 1006$ tubes. If we increase tube length to 18 m (60 ft), then we need 335 tubes, which can be arranged as five banks of 67 tubes each.

Step 9: Determine bundle area

If the tube pitch is 76.2 mm (3 in.), then the bundle area is:

$$A_b = 18 \times 76.2 \times 10^{-3} \times 67 = 91.9 \text{ m}^2$$

Step 10: Estimate air flow rate

Assuming a face velocity of 2.5 m/s, the air flow rate is $2.5 \times 91.9 = 229.7$ actual m 3 /s.

Steps 11 and 12: Estimate fan power and total power

From Equation 19.65, assuming 150 N/m² pressure drop and 70% fan efficiency, the fan power is:

$$W_f = \frac{u_f A_b \Delta P_b}{\eta_f} = \frac{2.5 \times 91.9 \times 150}{0.7} = 49.2 \text{ kW} \quad (19.65)$$

Allowing for motor efficiency of 95%, the total power consumed = 49.2/0.95 = 51.8 kW.

Note that the fan power consumed is much less than the cooling duty.

Step 13: Confirm heat transfer coefficients and pressure drop

Check tube-side heat transfer:

With 335 tubes, we have 167 tubes on one pass and 168 on the other pass. Using 168 tubes, methanol flow rate = 27.8 kg/s = 27.8/750 m³/s = 0.037 m³/s.

Tube inside area = $\pi (22.1 \times 10^{-3})^2/4 = 3.836 \times 10^{-4} \text{ m}^2$ per tube.

So, tube-side velocity = $0.037/(168 \times 3.836 \times 10^{-4}) = 0.574 \text{ m/s}$.

This is less than 40% of the assumed 1.5 m/s, so we need to correct the inside heat transfer coefficient. The corrected value of h_i will be:

$$\begin{aligned} h_i &= 2635 \times \left(\frac{0.574}{1.5} \right)^{0.8} \\ &= 1222 \text{ W/m}^2\text{°C} \end{aligned}$$

The corrected value of U_o is:

$$\begin{aligned} \frac{1}{U_o} &= \frac{1}{600} + \frac{1}{31120} + \frac{25.4}{22.1} \times \frac{1}{1222} + \frac{1}{4350} \\ U_o &= 349 \text{ W/m}^2\text{°C} \end{aligned}$$

The area required would now be $481.7 \times (423/349) = 584 \text{ m}^2$, i.e., 1220 tubes

Check air-side temperature change and ΔT_m :

Air flow rate = 229.7 m³/s

At 34 °C, air has density 1.15 kg/m³ and $C_p \sim 1 \text{ kJ/kg°C}$.

So,

$$\text{air side temperature change} = \frac{4.34 \times 10^3}{229.7 \times 1.15 \times 1} = 16.4 \text{ °C}$$

Such a large change in temperature on the air side is clearly not acceptable with a temperature approach of 6 °C at the cold end of the exchanger. To reach an acceptable air-side temperature change, we would need at least three times the air flow calculated, which would cause a factor of three increase in the fan duty. At this point, most experienced design engineers would conclude that because of the low outlet temperature specified for the cooler, an air cooler would not be economically attractive compared with a water cooler in this service.

If water cooling was not possible, then we could consider three air coolers in parallel, each with a single bank of 150 tubes, each 18 m long. Keeping the tube pitch at 76.2 mm, the bundle area is:

$$A_b = 18 \times 76.2 \times 10^{-3} \times 150 = 205.7 \text{ m}^2 (\text{roughly } 18 \text{ m} \times 12 \text{ m})$$

Assuming a face velocity of 2.5 m/s, the air flow rate in each cooler is $2.5 \times 205.7 = 514.3 \text{ actual m}^3/\text{s}$, so the total air flow is 1543 actual m³/s, roughly three times the initial design.

$$\text{air side temperature change} = \frac{4.34 \times 10^3}{514.3 \times 3 \times 1.15 \times 1} = 2.45 \text{ °C}$$

This gives

$$\begin{aligned} R &= \frac{95 - 40}{2.45} = 22.4 \\ S &= \frac{2.45}{95 - 34} = 0.040 \end{aligned}$$

From which $F_t \sim 0.95$,

$$\Delta T_{lm} = \frac{(95 - 36.5) - (40 - 34)}{\ln\left(\frac{95 - 36.5}{40 - 34}\right)} = 23.1^\circ\text{C}$$

and

$$\Delta T_m = 0.95 \times 23.1 = 21.9^\circ\text{C}.$$

With 3×150 tubes, we have $3 \times 75 = 225$ tubes per pass, so tube-side velocity $= 0.037 / (225 \times 3.836 \times 10^{-4}) = 0.429 \text{ m/s}$.

$$\begin{aligned} h_i &= 2635 \times \left(\frac{0.429}{1.5}\right)^{0.8} \\ &= 967 \text{ W/m}^2\text{C} \end{aligned}$$

The corrected value of U_o is:

$$\begin{aligned} \frac{1}{U_o} &= \frac{1}{600} + \frac{1}{31120} + \frac{25.4}{22.1} \times \frac{1}{967} + \frac{1}{4350} \\ U_o &= 321 \text{ W/m}^2\text{C} \end{aligned}$$

hence

$$A = \frac{4.34 \times 10^6}{321 \times 21.9} = 617.6 \text{ m}^2$$

The area of a 18 m long, 25.4 mm o.d. (20 ft, 1 in. o.d.) tube is 1.437 m^2 , so we require $617.6/1.437 = 430$ tubes. This is less than the $3 \times 150 = 450$ tubes assumed in the design, so we now have more than the required area. We could reduce the tube count slightly and iterate towards a converged solution, but the design is probably good enough at this point. The air-side heat transfer coefficient and pressure drop should now be confirmed and the design finalized using commercial air-cooler design software.

The new fan duty for each bundle is:

$$W_f = \frac{u_f A_b \Delta P_b}{\eta_f} = \frac{2.5 \times 205.7 \times 150}{0.7} = 110.2 \text{ kW} \quad (19.65)$$

Allowing for motor efficiency of 95%, the total power consumed for all three coolers $= 3 \times 110.2/0.95 = 348 \text{ kW}$.

The design generated in this example is feasible, but has high plot-space requirements, as well as a higher capital cost than the water cooler. The operating cost of running these air coolers at $0.06 \text{ \$/kWh}$ would be $348 \times 0.06 = \$20.9/\text{h}$. The cost of providing cooling water to the exchanger of Example 19.1 would be roughly $\$6.50/\text{h}$ (based on cooling water cost of $\$0.1/1000 \text{ gal}$), so the operating cost of the air cooler is also higher. An air cooler is clearly less economically attractive than a water cooler in this service and would only be considered if water costs were excessive or if there was flexibility to cool the methanol to a higher outlet temperature.

19.17 Fired heaters (furnaces and boilers)

When high temperatures and high flow rates are required, fired heaters are used. Fired heaters are directly heated by the products of combustion of a fuel. The highest temperature at which steam is used for process heating is typically about 250°C . Circulating heating oils are used up to about 330°C , but hot oil loops themselves require a fired heater as the primary heat source. Small vertical cylindrical fired heaters are used for duties up to 45 MW , and larger cabin furnaces are used for higher duties.

Typical applications of fired heaters are:

1. Process feed-stream heaters, such as the feed heaters for high-temperature reactors and refinery crude columns (pipe still furnaces) in which up to 60% of the feed may be vaporized.

2. Reboilers for columns, using relatively small size direct-fired units.
3. Direct-fired reactors, for example, the pyrolysis of dichloroethane to form vinyl chloride.
4. Interstage reheaters for high-temperature endothermic reactions.
5. Reformers for hydrogen production, giving outlet temperatures of 800 to 900 °C.
6. Steam boilers.
7. Heaters for hot oil circuits.

19.17.1 Basic construction

Many different designs and layouts are used, depending on the application; see Berman (1979a) and [Trinks et al. \(2004\)](#).

The basic construction consists of a rectangular or cylindrical steel chamber, lined with refractory bricks. Tubes are arranged around the wall in either horizontal or vertical banks. The fluid to be heated flows through the tubes. Typical layouts are shown in [Fig. 19.66](#) (a, b, and c). A more detailed diagram of a pyrolysis furnace is given in [Fig. 19.67](#).

Heat transfer to the tubes on the furnace walls is predominantly by radiation. In modern designs, this radiant section is surmounted by a smaller section in which the combustion gases flow over banks of tubes and transfer heat by convection. Extended surface tubes, with fins or pins, can be used in the convection section to improve the heat transfer from the combustion gases. Plain tubes known as *shock tubes* are used in the bottom rows of the convection section to act as a heat shield from the hot gases in the radiant section. Heat transfer to the shock tubes will be by both radiation and convection. The tube sizes used will normally be between 75 and 150 mm diameter. The tube size and number of passes used depend on the application and the process-fluid flow rate. Typical tube velocities are from 1 to 2 m/s for heaters, with lower rates used for reactors. Carbon steel is used for low-temperature duties; stainless steel and special alloy steels are used for elevated temperatures. For high temperatures, a material that resists creep must be used. Special metallurgies are also needed if the process fluid undergoes coking or can cause metal dusting corrosion, or if it is a mixed oxidizing-sulfiding gas that attacks the metal surface; see [Chapter 6](#).

The burners are positioned at the base or along the sides of the radiant section. The combustion air may be pre-heated in tubes in the convection section.

The fuel is typically natural gas, fuel oil, or off-gases from the process. When process off-gases are used, they are often blended with natural gas to allow for start-up and to dampen out any variations in the fuel quality. The selection of fuel is determined by cost and environmental constraints. It will sometimes be necessary to use a more expensive fuel to meet the requirements of environmental emissions permits.

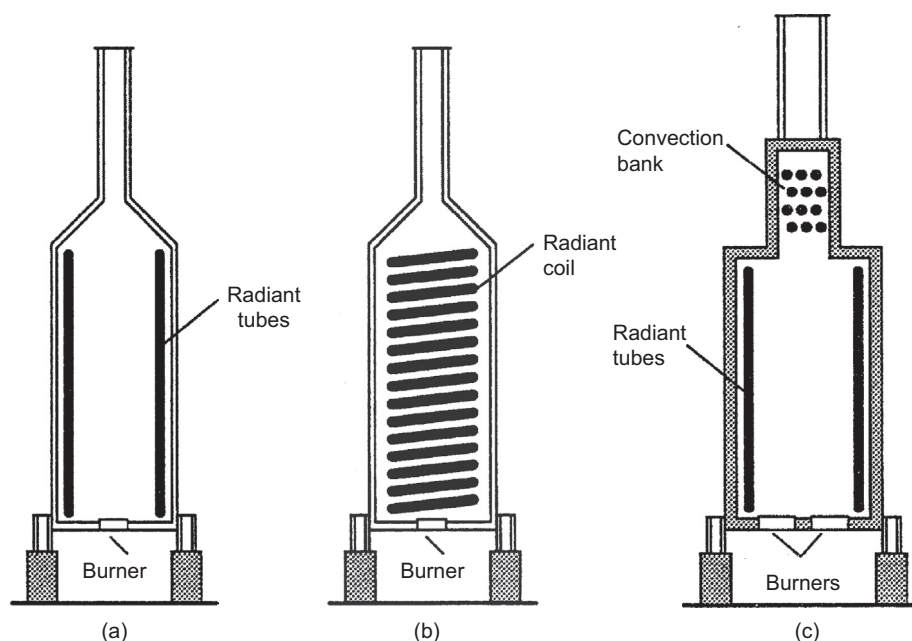


FIG. 19.66 Fired heaters. (a) Vertical-cylindrical, all radiant. (b) Vertical-cylindrical, helical coil. (c) Vertical-cylindrical with convection section.

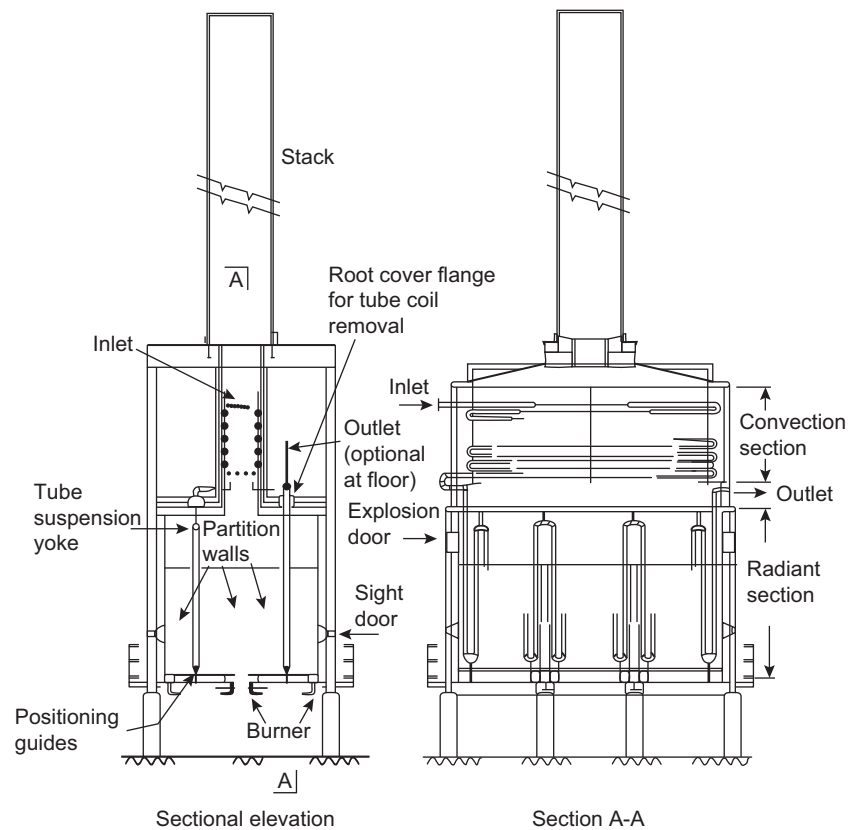


FIG. 19.67 (Foster Wheeler) Multizoned pyrolysis furnace.

19.17.2 Design of fired heaters

Commercial computer programs for the design of fired heaters are available from HTFS and HTRI; see [Section 19.1](#). Manual calculation methods, suitable for the preliminary design of fired heaters, are given by [Kern \(1950\)](#), [Wimpress \(1978\)](#), and [Evans \(1980\)](#). A good overview of fired heater design and optimization is given by [Zhu \(2014\)](#). A brief review of the factors to be considered is given in the following sections.

19.17.3 Heat transfer in fired heaters

Radiant section

Between 50% and 70% of the total heat is transferred in the radiant section. The gas temperature depends on the fuel used and the amount of excess air. Around 20% excess air is normally used for gaseous fuels and 25% for liquid fuels. Excess air is used to prevent the formation of soot and carbon monoxide. The effect of increasing excess air is to reduce the adiabatic flame temperature and increase the stack gas flow rate, hence shifting heat availability from the radiant section to the convective section.

Radiant heat transfer from a surface is governed by the Stefan–Boltzmann equation:

$$q_r = \sigma T^4 \quad (19.66)$$

where q_r = radiant heat flux, W/m^2

σ = Stefan–Boltzmann constant, $5.67 \times 10^{-8} \text{ W}/\text{m}^2 \text{ K}^4$

T = temperature of the surface, K

For the exchange of heat between the combustion gases and the hot tubes, the equation can be written as:

$$Q_r = \sigma(\alpha A_{cp})F(T_g^4 - T_t^4) \quad (19.67)$$

where Q_r = radiant heat transfer rate, W

A_{cp} = the "cold-plane" area of the tubes

= number of tubes $\times$ the exposed length $\times$ tube pitch

α = the absorption efficiency factor

F = the radiation exchange factor

T_g = temperature of the hot gases, K

T_t = tube surface temperature, K

Part of the radiation from the hot combustion gases will strike the tubes and be absorbed, and part will pass through the spaces between the tubes and be radiated back into the furnace. If the tubes are in front of the wall, some of the radiation from the wall will also be absorbed by the tubes. This complex situation is allowed for by calculating what is known as the cold plane area of the tubes, A_{cp} , and then applying the absorption efficiency factor α to allow for the fact that the tube area will not be as effective as a plane area. The absorption efficiency factor is a function of the tube arrangement and will vary from around 0.4 for widely spaced tubes to 1.0 for the theoretical situation when the tubes are touching. It will be around 0.7 to 0.8 when the pitch equals the tube diameter. Values for α are available in handbooks for different tube arrangements; see [Green and Southard \(2018\)](#) and [Wimpress \(1978\)](#).

The radiation exchange factor, F , depends on the arrangement of the surfaces and their emissivity and absorptivity. Combustion gases are poor radiators because only the carbon dioxide and water vapor, about 20% to 25% of the total, will emit radiation in the thermal spectrum. For a fired heater, the exchange factor depends on the partial pressure and emissivity of these gases and the layout of the heater. The partial pressure is dependent on the kind of fuel used, liquid or gas, and the amount of excess air. The gas emissivity is a function of temperature. Methods for estimating the exchange factor for typical furnace designs are given in the handbooks; see [Green and Southard \(2018\)](#) and [Wimpress \(1978\)](#).

The heat flux to the tubes in the radiant section will be between 20 and 40 kW/m² for most applications. A value of 30 kW/m² can be used to make a rough estimate of the tube area needed in this section.

A small amount of heat will be transferred to the tubes by convection in the radiant section, but as the superficial velocity of the gases will be low, the heat transfer coefficient will be low, around 10 W/m²°C.

The temperature of the flue gas leaving the radiant section is known as the bridgewall temperature. This temperature can be estimated by assuming that roughly 60% of the heat released by combustion is transferred to the process fluid in the radiant section. The bridgewall temperature is needed for detailed design of the convective section.

Convection section

The combustion gases flow across the tube banks in the convection section, and the correlations for cross-flow in tube banks can be used to estimate the heat transfer coefficient. The gas-side coefficient will be low, and where extended surfaces are used, an allowance must be made for the fin efficiency. Procedures are given in the tube vendors' literature and in handbooks; see Section 19.14 and [Berman \(1978b\)](#).

The overall coefficient will depend on the gas velocity and temperature and the tube size. Typical values range from 20 to 50 W/m²°C.

The shock tubes in the lower rows of the convection section receive heat by radiation from the radiant section. This can be allowed for by including the area of the lower row of tubes with the tubes in the radiant section.

19.17.4 Pressure drop

Most of the pressure drop on the flue gas side occurs in the convection section. The procedures for estimating the pressure drop across banks of tubes can be used to estimate the pressure drop in this section; see Section 19.9.3.

The pressure drop in the radiant section will be small compared with that across the convection section and can usually be neglected.

19.17.5 Process-side heat transfer and pressure drop

The tube-inside heat transfer coefficients and pressure drop can be calculated using the conventional methods for flow inside tubes; see Section 19.8. If the unit is being used as a vaporizer, the existence of two-phase flow in some of the tubes must be taken into account. [Berman \(1978b\)](#) gives a quick method for estimating two-phase pressure drop in the tubes of fired heaters.

Typical approach temperatures, flue gas to inlet process fluid, are around 100 °C.

19.17.6 Stack design

Most fired heaters operate with natural draft, and the stack height must be sufficient to achieve the flow of combustion air required and to remove the combustion products.

It is normal practice to operate with a slight vacuum throughout the heater so that air will leak in through sight-boxes and dampers, rather than combustion products leak out. Typically, the aim would be to maintain a vacuum of around 2 mm water gauge just below the convection section.

The stack height required depends on the temperature of the combustion gases leaving the convection section and the elevation of the site above sea level. The draft arises from the difference in density of the hot gases and the surrounding air.

The draft in millimeters of water (mm H₂O) can be estimated using the equation:

$$P_d = 0.35(L_s)(p') \left[\frac{1}{T_a} - \frac{1}{T_{ga}} \right] \quad (19.68)$$

where L_s = stack height, m

p' = atmospheric pressure, millibar ($\text{N/m}^2 \times 10^{-2}$)

T_a = ambient temperature, K

T_{ga} = average flue gas temperature, K

Because of heat losses, the temperature at the top of the stack will be around 80 °C below the inlet temperature.

The frictional pressure loss in the stack must be added to the loss in the heater when estimating the stack draft required. This can be calculated using the usual methods for pressure loss in circular conduits; see Section 19.8. The mass velocity in the stack will be around 1.5 to 2 kg/m². These values can be used to determine the cross-section needed.

An approximate estimate of the pressure losses in the convection section can be made by multiplying the velocity head ($u^2/2g$) by factors for each restriction; typical values are given here:

0.2 to 0.5 for each row of plain tubes

1.0 to 2.0 for each row of finned tubes

0.5 for the stack entrance

1.0 for the stack exit

1.5 for the stack damper

19.17.7 Thermal efficiency

Modern fired heaters operate at thermal efficiencies of between 80% and 90%, depending on the fuel and the excess air requirement. In some applications, additional excess air may be used to reduce the flame temperature to avoid overheating of the tubes. If only the radiant section is used, the efficiency is in the range 60% to 65%.

Where the inlet temperature of the process fluid is such that the outlet temperature from the convection section would be excessive, giving low thermal efficiency, this excess heat can be used to pre-heat the air to the furnace. Tubes would be installed above the process fluid section in the convection section. Forced-draft operation would be needed to drive the air flow through the pre-heat section.

Heat losses from the heater casing are normally between 1.5% and 2.5% of the heat input.

The largest contribution to heat losses is the sensible heat of the flue gas leaving the stack. There are several practical limits on how much heat can be recovered from the flue gas:

1. For process control reasons, the process heating duty is often carried out in the radiant section only; see [Chapter 5](#).
2. There may not be sufficient process heat requirement in the lower temperature range of the convection section. In this case, the convection section can be used to raise steam or pre-heat boiler feed water if the heat recovered justifies the cost of the additional piping required.
3. The stack gas should not be cooled to the dew point; otherwise, condensation can occur, leading to corrosion of the stack. The presence of carbon dioxide and sulfur oxides from the combustion reactions in the flue gas raises the dew point and causes the condensate to be acidic.
4. A large temperature difference is usually specified between the flue gas and the process fluid so as to decrease the tube cost in the convection section. This leads to high stack temperatures.

5. Many companies avoid cooling the flue gas close to the dew point to prevent the formation of a visible plume from the stack. If the flue gas leaving the stack is near the dew point, then when it mixes with cold ambient air, a mist will form. This gives the stack the appearance of smoking. If the flue gas is hotter, the gas can disperse before condensation occurs and the plume is eliminated. The general public prefers not to see plumes coming from chemical plants, so public relations often triumphs over energy efficiency.

19.17.8 Fired heater emissions

Fired heaters are a major source of atmospheric emissions and are tightly regulated. Permits are usually required for operation of process heaters, and modifications to the heater or burners often require approval from environmental agencies or reissuance of the permit.

Fired heater emissions concerns include:

1. Carbon monoxide, unburned hydrocarbons, and soot can be formed if combustion is not complete. These emissions are usually minimized by operating with at least 20% excess air.
2. Sulfur oxides and metals can be emitted if sulfur or metals were present in the fuel. These emissions occur mainly when burning heavy fuel oils. Emissions of sulfur oxides can be reduced by switching to a fuel with lower sulfur content, such as natural gas.
3. Nitrogen oxides, NO_x , are formed during combustion. Unfortunately, use of excess air tends to make NO_x formation worse. Formation of NO_x is controlled by using special burner designs such as staged-air or staged-fuel burners, by using steam injection or flue gas recirculation to reduce the flame temperature, or by catalytic decomposition of the NO_x in the flue gas.
4. Carbon dioxide is formed from combustion of hydrocarbon fuels. The penalties for CO_2 emissions are not yet large enough to drive companies to recover CO_2 from flue gas, but CO_2 can be recovered by scrubbing if necessary. Carbon dioxide capture can be made easier by using novel furnace designs in which the fuel is burnt in oxygen and recirculating carbon dioxide—a system known as “oxyfuel combustion.”

19.18 Heat transfer to vessels

The simplest way to transfer heat to a process or storage vessel is to fit an external jacket or an internal coil. If these methods cannot provide sufficient heat transfer area, a stream is withdrawn from the vessel, pumped through a heat exchanger, and returned to the vessel.

19.18.1 Jacketed vessels

Conventional jackets

The most commonly used type of jacket is that shown in Fig. 19.68. It consists of an outer cylinder that surrounds part of the vessel. The heating or cooling medium circulates in the annular space between the jacket and vessel walls, and the heat is transferred through the wall of the vessel. Circulation baffles are usually installed in the annular space to increase the velocity of the liquid flowing through the jacket and improve the heat transfer coefficient

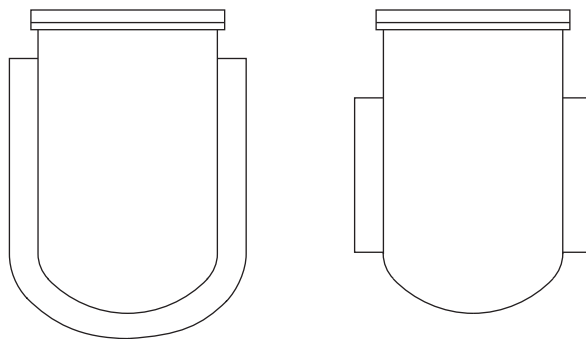


FIG. 19.68 Jacketed vessel.

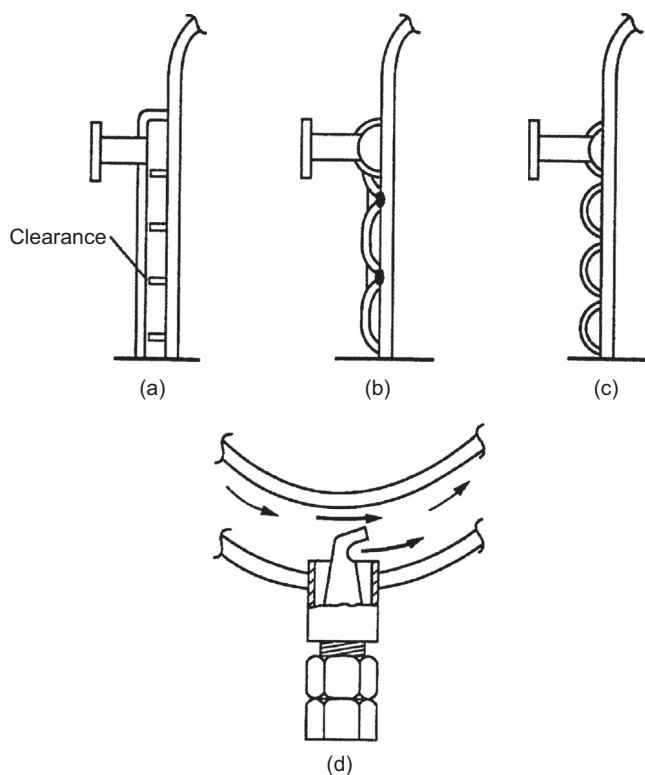


FIG. 19.69 Jacketed vessels. (a) Spirally baffled jacket. (b) Dimple jacket. (c) Half-pipe jacket. (d) Agitation nozzle.

(Fig. 19.69a). The same effect can be obtained by introducing the fluid through a series of nozzles spaced down the jacket. The momentum of the jets issuing from the nozzles sets up a swirling motion in the jacket liquid (see Fig. 19.69d). The spacing between the jacket and vessel wall will depend on the size of the vessel, but will typically range from 50 mm for small vessels to 300 mm for large vessels.

Half-pipe jackets

Half-pipe jackets are formed by welding sections of pipe, cut in half along the longitudinal axis, to the vessel wall. The pipe is usually wound around the vessel in a helix (Fig. 19.69c).

The pitch of the coils and the area covered can be selected to provide the heat transfer area required. Standard pipe sizes are used, ranging from 60 to 120 mm outside diameter. The half-pipe construction makes a strong jacket capable of withstanding pressure better than the conventional jacket design.

Dimpled jackets

Dimpled jackets are similar to the conventional jackets, but are constructed of thinner plates. The jacket is strengthened by a regular pattern of hemispherical dimples pressed into the plate and welded to the vessel wall (see Fig. 19.69b).

Jacket selection

Factors to consider when selecting the type of jacket to use are:

1. Cost: In terms of cost, the designs can be ranked, from cheapest to most expensive, as:
 - simple, no baffles
 - agitation nozzles
 - spiral baffle
 - dimple jacket
 - half-pipe jacket
2. Heat transfer rate required: Select a spirally baffled or half-pipe jacket if high rates are required.
3. Pressure: As a rough guide, the pressure rating of the designs can be taken as:

jackets, up to 10 bar
 dimpled jackets, up to 20 bar
 half-pipe, up to 70 bar.

So, half-pipe jackets would be used for high pressure.

Jacket heat transfer and pressure drop

The heat transfer coefficient to the vessel wall can be estimated using the correlations for forced convection in conduits, such as Equation 19.11. The fluid velocity and the path length can be calculated from the geometry of the jacket arrangement. The hydraulic mean diameter (equivalent diameter, d_e) of the channel or half-pipe should be used as the characteristic dimension in the Reynolds and Nusselt numbers; see Section 19.8.1.

In dimpled jackets, a velocity of 0.6 m/s can be used to estimate the heat transfer coefficient. A method for calculating the heat transfer coefficient for dimpled jackets is given by Makovitz (1971).

The coefficients for jackets using agitation nozzles are similar to those given by using baffles. A method for calculating the heat transfer coefficient using agitation nozzles is given by Bolliger (1982).

To increase heat transfer rates, the velocity through a jacket can be increased by recirculating the cooling or heating liquid.

For simple jackets without baffles, heat transfer will be mainly by natural convection, and the heat transfer coefficient will range from 200 to 400 W/m²°C.

When steam is used in a jacket, the heat transfer coefficient is in the range of 4000 to 5000 W/m²°C.

19.18.2 Internal coils

The simplest and cheapest form of heat transfer surface for installation inside a vessel is a helical coil (Fig. 19.70). The pitch and diameter of the coil can be made to suit the application and the area required. The diameter of the pipe used for the coil is typically equal to $D_v/30$, where D_v is the vessel diameter. The coil pitch is usually around twice the pipe diameter. Small coils can be self-supporting, but for large coils some form of supporting structure will be necessary. Single- or multiple-turn coils are used.

Coil heat transfer and pressure drop

The heat transfer coefficient at the inside wall and pressure drop through the coil can be estimated using the correlations for flow through pipes; see Section 19.8. Correlations for forced convection in coiled pipes are also given in the ESDU Design Guide, ESDU 78031 (2001).

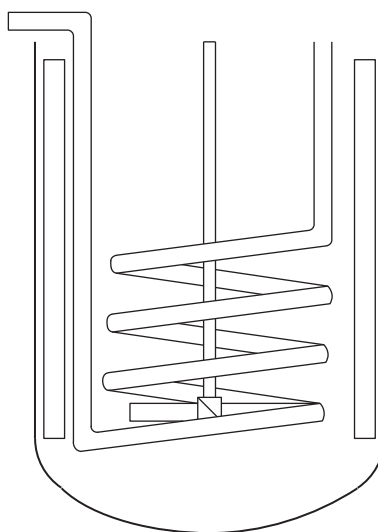


FIG. 19.70 Internal coils.

19.18.3 Agitated vessels

Unless only small rates of heat transfer are required, as when maintaining the temperature of liquids in storage vessels, some form of agitation will be needed. The various types of agitator used for mixing and blending described in Chapter 15, Section 15.5, are also used to promote heat transfer in vessels. The correlations used to estimate the heat transfer coefficient to the vessel wall, or to the surface of coils, have the same form as those used for forced convection in conduits; see Equation 19.10. The fluid velocity is replaced by a function of the agitator diameter and rotational speed, $D \times N$, and the characteristic dimension is the agitator diameter.

$$\text{Nu} = C \text{Re}^a \text{Pr}^b \left(\frac{\mu}{\mu_w} \right)^c \quad (19.10)$$

For agitated vessels:

$$\frac{h_v D}{k_f} = C \left(\frac{ND^2 \rho}{\mu} \right)^a \left(\frac{C_p \mu}{k_f} \right)^b \left(\frac{\mu}{\mu_w} \right)^c \quad (19.69)$$

where h_v = heat transfer coefficient to vessel wall or coil, $\text{W/m}^2\text{°C}$

D = agitator diameter, m

N = agitator, speed, rps (revolutions per second)

ρ = liquid density, kg/m^3

k_f = liquid thermal conductivity, $\text{W/m}^{\circ}\text{C}$

C_p = liquid-specific heat capacity, $\text{J/kg}^{\circ}\text{C}$

μ = liquid viscosity, Nm^{-2}s

The values of the constant C and the indices a , b , and c depend on the type of agitator, the use of baffles, and whether the transfer is to the vessel wall or to coils. Some typical correlations are given next.

Baffles will normally be used in most applications.

1. Flat blade paddle, baffled or unbaffled vessel, transfer to vessel wall, $Re < 4000$:

$$\text{Nu} = 0.36 \text{Re}^{0.67} \text{Pr}^{0.33} \left(\frac{\mu}{\mu_w} \right)^{0.14} \quad (19.70a)$$

2. Flat blade disc turbine, baffled or unbaffled vessel, transfer to vessel wall, $Re < 400$:

$$\text{Nu} = 0.54 \text{Re}^{0.67} \text{Pr}^{0.33} \left(\frac{\mu}{\mu_w} \right)^{0.14} \quad (19.70b)$$

3. Flat blade disc turbine, baffled vessel, transfer to vessel wall, $Re > 400$:

$$\text{Nu} = 0.74 \text{Re}^{0.67} \text{Pr}^{0.33} \left(\frac{\mu}{\mu_w} \right)^{0.14} \quad (19.70c)$$

4. Propeller, three blades, transfer to vessel wall, $Re > 5000$:

$$\text{Nu} = 0.64 \text{Re}^{0.67} \text{Pr}^{0.33} \left(\frac{\mu}{\mu_w} \right)^{0.14} \quad (19.70d)$$

5. Turbine, flat blades, transfer to coil, baffled, Re , 2000 to 700,000:

$$\text{Nu} = 1.10 \text{Re}^{0.62} \text{Pr}^{0.33} \left(\frac{\mu}{\mu_w} \right)^{0.14} \quad (19.70e)$$

6. Paddle, flat blades, transfer to coil, baffled:

$$\text{Nu} = 0.87 \text{Re}^{0.62} \text{Pr}^{0.33} \left(\frac{\mu}{\mu_w} \right)^{0.14} \quad (19.70f)$$

More comprehensive design data are given by [Uhl and Gray \(1967\)](#), [Wilkinson and Edwards \(1972\)](#), [Penny \(1983\)](#), and [Fletcher \(1987\)](#).

Example 19.15

A jacketed, agitated reactor consists of a vertical cylinder 1.5 m diameter with a hemispherical base and a flat, flanged top. The jacket is fitted to the cylindrical section only and extends to a height of 1 m. The spacing between the jacket and vessel walls is 75 mm. The jacket is fitted with a spiral baffle. The pitch between the spirals is 200 mm.

The jacket is used to cool the reactor contents. The coolant used is chilled water at 10 °C; flow rate 32,500 kg/h, exit temperature 20 °C.

Estimate the heat transfer coefficient at the outside wall of the reactor and the pressure drop through the jacket.

Solution

The baffle forms a continuous spiral channel, [section 75 mm](#) $\times$ 200 mm.

$$\text{Number of spirals} = \text{height of jacket/pitch} = 1/(200 \times 10^{-3}) = 5$$

$$\text{Length of channel} = 5 \times \pi \times 1.5 = 23.6 \text{ m}$$

$$\text{Cross-sectional area of channel} = (75 \times 200) \times 10^{-6} = 15 \times 10^{-3} \text{ m}^2$$

$$\begin{aligned} \text{Hydraulic mean diameter, } d_e &= \frac{4 \times \text{cross-sectional area}}{\text{wetted perimeter}} \\ &= \frac{4 \times (75 \times 200)}{2(75 + 200)} = 109 \text{ mm} \end{aligned}$$

Physical properties at mean temperature of 15 °C, from steam tables: $\rho = 999 \text{ kg/m}^3$, $\mu = 1.136 \text{ mNm}^{-2}\text{s}$, $\text{Pr} = 7.99$, $k_f = 595 \times 10^{-3} \text{ W/m}^\circ\text{C}$.

$$\text{Velocity through channel, } u = \frac{32,000}{3600} \times \frac{1}{999} \times \frac{1}{15 \times 10^{-3}} = 0.602 \text{ m/s}$$

$$\text{Re} = \frac{999 \times 0.602 \times 109 \times 10^{-3}}{1.136 \times 10^{-3}} = 57,705$$

Chilled water is not viscous, so use Equation 19.11 with $C = 0.023$ and neglect the viscosity correction term.

$$\text{Nu} = 0.023 \text{Re}^{0.8} \text{Pr}^{0.33}$$

$$\frac{h_j \times 109 \times 10^{-3}}{595 \times 10^{-3}} = 0.023 (57705)^{0.8} (7.99)^{0.33} \quad (19.11)$$

$$h_j = \underline{1606} \text{ W/m}^2\text{ }^\circ\text{C}$$

Use Equation 19.18 for estimating the pressure drop, taking the friction factor from [Fig. 19.24](#). As the hydraulic mean diameter will be large compared with the roughness of the jacket surface, the relative roughness will be comparable with that for heat exchanger tubes. The relative roughness of pipes and channels and the effect on the friction factor are covered in [Chapter 20](#).

From [Fig. 19.24](#), for $\text{Re} = 5.8 \times 10^4$, $j_f = 3.2 \times 10^{-3}$:

$$\begin{aligned} \Delta P &= 8j_f \left(\frac{L}{d_e} \right) \rho \frac{u^2}{2} \\ \Delta P &= 8 \times 3.2 \times 10^{-3} \left(\frac{23.6}{109} \times 10^{-3} \right) 999 \times \frac{0.602^2}{2} \\ &= \underline{1003} \text{ N/m}^2 \end{aligned} \quad (19.18)$$

Example 19.16

The reactor described in Example 19.15 is fitted with a flat blade disc turbine agitator 0.6 m diameter, running at 120 rpm. The vessel is baffled and is constructed of stainless steel plate 10 mm thick.

The physical properties of the reactor contents are $\rho = 850 \text{ kg/m}^3$, $\mu = 80 \text{ mNm}^{-2}\text{s}$, $k_f = 400 \times 10^{-3} \text{ W/m}^\circ\text{C}$, $C_p = 2.65 \text{ kJ/kg}^\circ\text{C}$.

Estimate the heat transfer coefficient at the vessel wall and the overall coefficient in the clean condition.

Solution

Agitator speed (revs per sec) $= 1200/60 = 2 \text{ s}^{-1}$

$$\text{Re} = \frac{\rho ND^2}{\mu} = \frac{850 \times 2 \times 0.6^2}{80 \times 10^{-3}} = 7650$$

$$\text{Pr} = \frac{C_p \mu}{k_f} = \frac{2.65 \times 10^3 \times 80 \times 10^{-3}}{400 \times 10^{-3}} = 530$$

For a flat blade turbine use Equation 19.70(c):

$$\text{Nu} = 0.74 \text{Re}^{0.67} \text{Pr}^{0.33} \left(\frac{\mu}{\mu_w} \right)^{0.14}$$

Neglect the viscosity correction term:

$$\frac{h_v \times 0.6}{400 \times 10^{-3}} = 0.74 (7650)^{0.67} (530)^{0.33}$$

$$h_v = 1564 \text{ W/m}^2^\circ\text{C}$$

Taking the thermal conductivity of stainless steel as $16 \text{ W/m}^\circ\text{C}$ and the jacket coefficient from Example 19.15:

$$\frac{1}{U} = \frac{1}{1606} + \frac{10 \times 10^{-3}}{16} + \frac{1}{1564}$$

$$U = \underline{\underline{530}} \text{ W/m}^2^\circ\text{C}$$

19.19 Capital cost of heat transfer equipment

The capital cost of most heat transfer devices is dominated by the cost of the tubes or plates that provide the heat transfer area. Cost correlations for heat transfer devices are therefore usually expressed as a function of the heat transfer area. Cost correlations for shell and tube, double-pipe, plate-and-frame exchangers, and kettle and thermosiphon reboilers are given in Table 7.2.

For fired heaters, the heat flux through the tubes is very high. The cost of surface area is still significant, but other costs from site fabrication, refractory installation, and burners are also important. Fired heater costs are usually better correlated against heater duty than tube wall area. Costs for fired heaters are also given in Table 7.2.

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19.21 Nomenclature

		Dimensions in $MLT\theta$
A	Heat transfer area	L^2
A_b	Bundle cross-sectional area of an air-cooled exchanger	L^2
A_{cp}	Cold-plane area of tubes	L^2
A_o	Outside area of bare tube	L^2
A_f	Fin area in Equation 19.59	L^2
A_f	Cross-sectional area for flow in Equation 19.56	L^2
A_o	Outside area of bare tube	L^2
A_p	Area of a port plate heat exchanger	L^2
A_s	Cross-flow area between tubes	L^2
a	Index in Equation 19.10	—
b	Index in Equation 19.10	—
C	Constant in Equation 19.10	—
C_p	Heat capacity at constant pressure	$L^2T^{-2}\theta^{-1}$
C_{pg}	Heat capacity of gas	$L^2T^{-2}\theta^{-1}$
C_{pL}	Heat capacity of liquid phase	$L^2T^{-2}\theta^{-1}$
c	Index in Equation 19.10	—
D	Agitator diameter	L
D_b	Bundle diameter	L
D_s	Shell diameter	L
D_v	Vessel diameter	L
d	Diameter	L
d_e	Equivalent (hydraulic mean) diameter	L
d_i	Tube inside diameter	L
d_{pt}	Diameter of the ports in the plates of a plate heat exchanger	L
d_o	Tube outside diameter	L
d_1	Outside diameter of inner of concentric tubes	L
d_2	Inside diameter of outer of concentric tubes	L
E	Term in Equation 19.12	—
E_f	Fin efficiency	—
F	Radiation exchange factor in Equation 19.67	—

Dimensions in $MLT\theta$

F	Feed molar flow rate (Example 19.11)	MT^{-1}
F_t	Log mean temperature difference correction factor	—
f_c	Two-phase flow factor	—
f_m	Temperature correction factor for mixtures	—
f_s	Nucleate boiling suppression factor	—
G	Total mass flow rate per unit area	$ML^{-2}T^{-1}$
G_p	Mass flow rate per unit cross-sectional area between plates	$ML^{-2}T^{-1}$
G_s	Shell-side mass flow rate per unit area	$ML^{-2}T^{-1}$
G_t	Tube-side mass flow rate per unit area	$ML^{-2}T^{-1}$
g	Gravitational acceleration	LT^{-2}
H_s	Sensible heat of stream	ML^2T^{-3}
H_t	Total heat of stream (sensible + latent)	ML^2T^{-3}
h_c	Mean heat transfer coefficient in condensation	$MT^{-3}\theta^{-1}$
$(h_c)_1$	Mean condensation heat transfer coefficient for a single tube	$MT^{-3}\theta^{-1}$
$(h_c)_b$	Heat transfer coefficient for condensation on a horizontal tube bundle	$MT^{-3}\theta^{-1}$
$(h_c)_{N_r}$	Mean condensation heat transfer coefficient for a tube in a row of tubes	$MT^{-3}\theta^{-1}$
$(h_c)_v$	Heat transfer coefficient for condensation on a vertical tube	$MT^{-3}\theta^{-1}$
$(h_c)_{BK}$	Condensation coefficient from Boyko–Kruzhilin correlation	$MT^{-3}\theta^{-1}$
$(h_c)_s$	Condensation heat transfer coefficient for stratified flow in tubes	$MT^{-3}\theta^{-1}$
h'_c	Local condensing film coefficient, partial condenser	$MT^{-3}\theta^{-1}$
h_{cb}	Convective boiling heat transfer coefficient	$MT^{-3}\theta^{-1}$
h_{cg}	Mean effective cooling-condensing heat transfer coefficient, partial condenser	$MT^{-3}\theta^{-1}$
h'_{cg}	Local effective cooling-condensing heat transfer coefficient, partial condenser	$MT^{-3}\theta^{-1}$
h_{df}	Fouling coefficient based on fin area	$MT^{-3}\theta^{-1}$
h_f	Heat transfer coefficient based on fin area	$MT^{-3}\theta^{-1}$
h_{fb}	Film boiling heat transfer coefficient	$MT^{-3}\theta^{-1}$
h_{fc}	Forced-convection coefficient calculated assuming liquid is flowing alone (Equation 19.47)	$MT^{-3}\theta^{-1}$
h'_{fc}	Forced-convection heat transfer coefficient in Equation 19.46	$MT^{-3}\theta^{-1}$
h_g	Mean gas film heat transfer coefficient	$MT^{-3}\theta^{-1}$
h'_g	Local sensible-heat transfer coefficient, partial condenser	$MT^{-3}\theta^{-1}$
h_i	Film heat transfer coefficient inside a tube	$MT^{-3}\theta^{-1}$
h_{id}	Fouling coefficient on inside of tube	$MT^{-3}\theta^{-1}$
h_{nb}	Nucleate boiling heat transfer coefficient	$MT^{-3}\theta^{-1}$
h'_{nb}	Nucleate boiling coefficient in Equation 19.46	$MT^{-3}\theta^{-1}$
h_o	Heat transfer coefficient outside a tube	$MT^{-3}\theta^{-1}$
h_{od}	Fouling coefficient on outside of tube	$MT^{-3}\theta^{-1}$
h_p	Heat transfer coefficient in a plate heat exchanger	$MT^{-3}\theta^{-1}$

		Dimensions in $MLT\theta$
h_s	Shell-side heat transfer coefficient	$MT^{-3}\theta^{-1}$
h'_t	Inside film coefficient in Boyko–Kruzhilin correlation	$MT^{-3}\theta^{-1}$
h_v	Heat transfer coefficient to vessel wall or coil	$MT^{-3}\theta^{-1}$
J	Term in Boyko–Kruzhilin correlation, Equation 19.31	—
j_h	Heat transfer factor defined by Equation 19.14	—
j_H	Heat transfer factor defined by Equation 19.15	—
j_f	Friction factor	—
K_1	Constant in Equation 19.3, from Table 19.4	—
K_2	Constant in Equation 19.40	—
K_b	Constant in Equation 19.52	—
K_i	Phase equilibrium constant for component i	—
k_f	Thermal conductivity of fluid	$MLT^{-3}\theta^{-1}$
k_L	Thermal conductivity of liquid	$MLT^{-3}\theta^{-1}$
k_v	Thermal conductivity of vapor	$MLT^{-3}\theta^{-1}$
k_w	Thermal conductivity of tube wall material	$MLT^{-3}\theta^{-1}$
L	Tube length	L
L	Liquid molar flow rate in Example 19.11	MT^{-1}
L'	Effective tube length	L
L_P	Path length in a plate heat exchanger	L
L_s	Stack height	L
l_B	Baffle spacing (pitch)	L
l_f	Fin height	L
m	Index in Equation 19.19	—
N	Rotational speed	T^{-1}
N_b	Number of baffles	—
N_{bk}	Number of tubes per bank	—
N_p	Number of passes	—
N_r	Number of tubes in a vertical row	—
N_t	Number of tubes in a tube bundle	—
NTU	Number of transfer units	—
n_1	Index in Equation 19.3 from Table 19.4	—
P	Total pressure	$ML^{-1}T^{-2}$
P_c	Critical pressure	$ML^{-1}T^{-2}$
P_d	Stack draft	L
ΔP	Pressure drop	$ML^{-1}T^{-2}$
ΔP_b	Pressure drop across air-cooled exchanger bundle	$ML^{-1}T^{-2}$
ΔP_f	Pressure drop due to friction	$ML^{-1}T^{-2}$
ΔP_p	Pressure drop in a plate heat exchanger	$ML^{-1}T^{-2}$
ΔP_{pt}	Pressure loss through the ports in a plate heat exchanger	$ML^{-1}T^{-2}$

Dimensions in MLT θ

ΔP_s	Shell-side pressure drop	$ML^{-1}T^{-2}$
ΔP_s	Static pressure in thermosiphon tubes	$ML^{-1}T^{-2}$
ΔP_t	Tube-side pressure drop	$ML^{-1}T^{-2}$
p'	Atmospheric pressure	$ML^{-1}T^{-2}$
p_f	Fin pitch	L
p_s	Saturation vapor pressure	$ML^{-1}T^{-2}$
p_t	Tube pitch	L
p_w	Saturation vapor pressure corresponding to wall temperature	$ML^{-1}T^{-2}$
Q	Heat transferred in unit time	ML^2T^{-3}
Q_g	Sensible heat transfer rate from gas phase	ML^2T^{-3}
Q_r	Radiant heat transfer rate	ML^2T^{-3}
Q_t	Total heat transfer rate from gas phase	ML^2T^{-3}
q	Heat flux (heat transfer rate per unit area)	MT^{-3}
q'	Uncorrected value of flux from Fig. 19.55	MT^{-3}
q_c	Maximum (critical) flux for a single tube	MT^{-3}
q_{cb}	Maximum flux for a tube bundle	MT^{-3}
q_r	Radiant heat flux	MT^{-3}
R	Dimensionless temperature ratio defined by Equation 19.6	—
S	Dimensionless temperature ratio defined by Equation 19.7	—
T	Shell-side temperature	θ
T	Temperature of surface	θ
T_a	Ambient temperature	θ
T_{bi}	Temperature of liquid entering reboiler	θ
T_{bo}	Temperature of vapor leaving reboiler	θ
T_c	Critical temperature	θ
T_g	Temperature of combustion gases	θ
T_{ga}	Average flue-gas temperature	θ
T_r	Reduced temperature	—
T_s	Saturation temperature	θ
T_{sat}	Saturation temperature	θ
T_t	Tube surface temperature	θ
T_v	Vapor (gas) temperature	θ
T_w	Wall (surface) temperature	θ
T_1	Hot-side inlet temperature	θ
T_2	Hot-side exit temperature	θ
ΔT	Temperature difference	θ
ΔT_{lm}	Logarithmic mean temperature difference	θ
ΔT_m	Mean temperature difference in Equation 19.1	θ
ΔT_s	Temperature change in vapor (gas) stream	θ

Dimensions in $MLT\theta$

t	Tube-side temperature	θ
t	Water temperature in Equation 19.17	θ
t_c	Local coolant temperature	θ
t_f	Fin thickness	L
t_i	Stream inlet temperature	θ
t_o	Stream outlet temperature	θ
t_w	Estimated wall temperature	θ
t_1	Cold-side inlet temperature	θ
t_2	Cold-side exit temperature	θ
U	Overall heat transfer coefficient	$MT^{-3}\theta^{-1}$
U'	Uncorrected overall coefficient, Equation 19.51	$MT^{-3}\theta^{-1}$
U_c	Corrected overall coefficient, Equation 19.52	$MT^{-3}\theta^{-1}$
U_o	Overall heat transfer coefficient based on tube outside area	$MT^{-3}\theta^{-1}$
U_1, U_2	Overall heat transfer coefficients evaluated at the ends of the exchanger	$MT^{-3}\theta^{-1}$
u	Fluid velocity	LT^{-1}
u_L	Liquid velocity, Equation 19.34	LT^{-1}
u_p	Fluid velocity in a plate heat exchanger	LT^{-1}
u_{pt}	Velocity through the ports of a plate heat exchanger	LT^{-1}
u_p	Velocity through channels of a plate heat exchanger	LT^{-1}
u_s	Shell-side fluid velocity	LT^{-1}
u_t	Tube-side fluid velocity	LT^{-1}
u_v	Vapor velocity, Equation 19.34	LT^{-1}
$\hat{u}_v$	Maximum vapor velocity in kettle reboiler	LT^{-1}
V	Vapor molar flow rate	MT^{-1}
v_i	Specific volume at inlet	L^3M^{-1}
v_o	Specific volume at outlet	L^3M^{-1}
W	Mass flow rate of fluid	MT^{-1}
W_c	Total condensate mass flow rate	MT^{-1}
W_s	Shell-side fluid mass flow rate	MT^{-1}
w	Mass flow through the channels and ports in a plate heat exchanger	MT^{-1}
w_f	Fan power	ML^2T^{-3}
X_{tt}	Lockhart–Martinelli two-phase flow parameter	—
x	Mass fraction of vapor	—
x_i	Mole fraction of component i in liquid phase	—
y_i	Mole fraction of component i in vapor phase	—
Z	Ratio of change in sensible heat to change in total heat of gas stream (sensible + latent)	—
z_i	Mole fraction of component i in feed	—
a	Absorption efficiency factor	—

		Dimensions in $MLT\theta$
Γ	Tube loading	$ML^{-1}T^{-1}$
Γ_h	Condensate loading on a horizontal tube	$ML^{-1}T^{-1}$
Γ_v	Condensate loading on a vertical tube	$ML^{-1}T^{-1}$
η_f	Fan efficiency	—
λ	Latent heat	L^2T^{-2}
μ	Viscosity at bulk fluid temperature	$ML^{-1}T^{-1}$
μ_L	Liquid viscosity	$ML^{-1}T^{-1}$
μ_v	Vapor viscosity	$ML^{-1}T^{-1}$
μ_w	Viscosity at wall temperature	$ML^{-1}T^{-1}$
ρ	Fluid density	ML^{-3}
ρ_L	Liquid density	ML^{-3}
ρ_m	Mean density	ML^{-3}
ρ_v	Vapor density	ML^{-3}
σ	Stephan–Boltzmann constant in Equation 19.66	$MT^{-3}\theta^{-4}$
σ	Surface tension	MT^{-3}
Dimensionless numbers		
Nu	Nusselt number	—
Pr	Prandtl number	—
Pr_c	Prandtl number for condensate film	—
Re	Reynolds number	—
Re_c	Reynolds number for condensate film	—
Re_L	Reynolds number for liquid phase	—
St	Stanton number	—

19.22 Problems

- 19.1.** A solution of sodium hydroxide leaves a dissolver at 80 °C and is to be cooled to 40 °C using cooling water. The maximum flow rate of the solution will be 8000 kg/h. The maximum inlet temperature of the cooling water will be 20 °C, and the temperature rise is limited to 20 °C.

Design a double-pipe exchanger for this duty, using standard carbon steel pipe and fittings. Use pipe of 50 mm inside diameter, 55 mm outside diameter for the inner pipe, and 75 mm inside diameter pipe for the outer. Make each [section 5](#) m long. The physical properties of the caustic solution are:

Temperature, °C	40	80
specific heat, $\text{kJkg}^{-1}\text{°C}^{-1}$	3.84	3.85
density, kg/m^3	992.2	971.8
thermal conductivity, $\text{W/m}^\circ\text{C}$	0.63	0.67
viscosity, $\text{mN m}^{-2} \text{ s}$	1.40	0.43

- 19.2.** A double-pipe heat exchanger is to be used to heat 6000 kg/h of 22 mol% hydrochloric acid. The exchanger will be constructed from karbate (impervious carbon) and steel tubing. The acid will flow through the inner karbate tube, and saturated steam at 100 °C will be used for heating. The tube dimensions will be

karbate tube inside diameter 50 mm, outside diameter 60 mm; steel tube inside diameter 100 mm. The exchanger will be constructed in sections, with an effective length of 3 m each.

How many sections will be needed to heat the acid from 15 °C to 65 °C?

Physical properties of 22% HCl at 40 °C: specific heat $4.93 \text{ kJ kg}^{-1} \text{ °C}^{-1}$, thermal conductivity 0.39 W/m °C , density 866 kg/m^3 .

Viscosity:	temperature	20	30	40	50	60	70 °C
	$\text{mN m}^{-2}\text{s}$	0.68	0.55	0.44	0.36	0.33	0.30

Karbate thermal conductivity 480 W/m °C .

- 19.3. In a food processing plant there is a requirement to heat 50,000 kg/h of town water from 10 °C to 70 °C. Steam at 2.7 bar is available for heating the water. An existing heat exchanger is available, with the following specifications:

Shell inside diameter 337 mm, E type

Baffles 25% cut, set at a spacing of 106 mm

Tubes 15 mm inside diameter, 19 mm outside diameter, 4094 mm long

Tube pitch 24 mm, triangular

Number of tubes 124, arranged in a single pass

Would this exchanger be suitable for the specified duty?

- 19.4. Design a shell and tube exchanger to heat 50,000 kg/h of liquid ethanol from 20 °C to 80 °C. Steam at 1.5 bar is available for heating. Assign the ethanol to the tube side. The total pressure drop must not exceed 0.7 bar for the alcohol stream. Plant practice requires the use of carbon steel tubes, 25 mm inside diameter, 29 mm outside diameter, 4 m long.

Set out your design on a data sheet and make a rough sketch of the heat exchanger. The physical properties of ethanol can be readily found in the literature.

- 19.5. 4500 kg/h of ammonia vapor at 6.7 bara pressure is to be cooled from 120 °C to 40 °C using cooling water. The maximum supply temperature of the cooling water available is 30 °C, and the outlet temperature is to be restricted to 40 °C. The pressure drops over the exchanger must not exceed 0.5 bar for the ammonia stream and 1.5 bar for the cooling water.

A contractor has proposed using a shell and tube exchanger with the following specifications for this duty:

Shell: E-type, inside diameter 590 mm

Baffles: 25% cut, 300 mm spacing

Tubes: carbon steel, 15 mm inside diameter, 19 mm outside diameter, 2400 mm long, number 360

Tube arrangement: 8 passes, triangular tube pitch, pitch 23.75 mm

Nozzles: shell 150 mm inside diameter, tube headers 75 mm inside diameter

It is proposed to put the cooling water through the tubes.

Is the proposed design suitable for the duty?

Physical properties of ammonia at the mean temperature of 80 °C: specific heat $2.418 \text{ kJ kg}^{-1} \text{ °C}^{-1}$, thermal conductivity 0.0317 W/m °C , density 4.03 kg/m^3 , viscosity $1.21 \times 10^{-5} \text{ N m}^{-2}\text{s}$.

- 19.6. A vaporizer is required to evaporate 10,000 kg/h of a process fluid at 6 bar. The liquid is fed to the vaporizer at 20 °C.

The plant has a spare kettle reboiler available with the following specifications:

U-tube bundle, 50 tubes, mean length 4.8 m, end to end

Carbon steel tubes, inside diameter 25 mm, outside diameter 30 mm, square pitch 45 mm

Steam at 1.7 bara will be used for heating

Check if this reboiler would be suitable for the duty specified. Only check the thermal design. You may take it that the shell will handle the vapor rate.

Take the physical properties of the process fluid as liquid: density 535 kg/m^3 , specific heat $2.6 \text{ kJ kg}^{-1} \text{ °C}^{-1}$, thermal conductivity 0.094 W/m °C , viscosity $0.12 \text{ mN m}^{-2}\text{s}$, surface tension 0.85 N/m , heat of vaporization 322 kJ/kg .

Vapor density 14.4 kg/m^3 .

Vapor pressure:

temperature °C	50	60	70	80	90	100	110	120
pressure bar	5.0	6.4	8.1	10.1	12.5	15.3	18.5	20.1

- 19.7.** A condenser is required to condense n-propanol vapor leaving the top of a distillation column. The n-propanol is essentially pure and is a saturated vapor at a pressure of 2.1 bara. The condensate needs to be subcooled to 45 °C.
Design a horizontal shell and tube condenser capable of handling a vapor rate of 30,000 kg/h. Cooling water is available at 30 °C, and the temperature rise is to be limited to 30 °C. The pressure drop on the vapor stream is to be less than 50 kN/m² and on the water stream less than 70 kN/m². The preferred tube size is 16 mm inside diameter, 19 mm outside diameter, and 2.5 m long.
Take the saturation temperature of n-propanol at 2.1 bar as 118 °C. The other physical properties required can be found in the literature or estimated.
- 19.8.** Design a vertical shell and tube condenser for the duty given in Question 19.7. Use the same preferred tube size.
- 19.9.** In the manufacture of methyl ethyl ketone (MEK) from 2-butanol, the reactor products are precooled and then partially condensed in a shell and tube exchanger. A typical analysis of the stream entering the condenser is mol fractions: MEK 0.47, unreacted alcohol 0.06, hydrogen 0.47. Only 85% of the MEK and alcohol are condensed. The hydrogen is noncondensable.
The vapors enter the condenser at 125 °C and the condensate and uncondensed material leave at 27 °C. The condenser pressure is maintained at 1.1 bara. Make a preliminary design of this condenser for a feed rate of 1500 kg/h. Chilled water will be used as the coolant at an inlet temperature of 10 °C and allowable temperature rise of 30 °C.
Any of the physical properties of the components not available in Appendix C or the general literature should be estimated. Appendix C is available in the online material at www.elsevier.com/books-and-journals/book-companion/9780128211793.
- 19.10.** A vertical thermosiphon reboiler is required for a column. The liquid at the base of the column is essentially pure n-butane. A vapor rate of 5 kg/s is required.
The pressure at the base of the column is 20.9 bar. Saturated steam at 5 bar will be used for heating. Estimate the number of 25 mm outside diameter, 22 mm inside diameter, 4 m long tubes needed.
At 20.9 bar the saturation temperature of n-butane is 117 °C and the heat of vaporization is 828 kJ/kg.
- 19.11.** An immersed bundle vaporizer is to be used to supply chlorine vapor to a chlorination reactor at a rate of 10,000 kg/h. The chlorine vapor is required at 5 bar pressure. The minimum temperature of the chlorine feed will be 10 °C. Hot water at 50 °C is available for heating. The pressure drop on the water side must not exceed 0.8 bar.
Design a vaporizer for this duty. Use stainless steel U-tubes 6 m long, 21 mm inside diameter, 25 mm outside diameter, on a square pitch of 40 mm.
The physical properties of chlorine at 5 bar are saturation temperature 10 °C, heat of vaporization 260 kJ/kg, specific heat 0.99 kJ/kg⁻¹°C⁻¹, thermal conductivity 0.13 W/m°C, density 1440 kg/m³, viscosity 0.3 mN m⁻²s, surface tension 0.013 N/m, vapor density 16.3 kg/m³.
The vapor pressure can be estimated from the equation:
- $$\ln(P) = 9.34 - 1978/(T + 246); P \text{ bar}, T^\circ\text{C}$$
- 19.12.** There is a requirement to cool 200,000 kg/h of a dilute solution of potassium carbonate from 70 °C to 30 °C. Cooling water will be used for cooling, with inlet and outlet temperatures of 20 °C and 60 °C.
A gasketed-plate heat exchanger is available with the following specifications:
Number of plates 329
Effective plate dimensions: length 1.5 m, width 0.5 m, thickness 0.75 mm
Channel width 3 mm
Flow arrangement two pass: two pass
Port diameters 150 mm

Check if this exchanger is likely to be suitable for the thermal duty required, and estimate the pressure drop for each stream.

Take the physical properties of the dilute potassium carbonate solution to be the same as those for water.

- 19.13.** Design an air cooler to cool 30,000 kg/h of diesel oil from 120 °C to 50 °C. The highest ambient temperature that is exceeded for 40 h/yr on average is 40 °C.
Physical properties of diesel oil over the temperature range of interest can be taken as specific heat capacity 2.1 kJ/kg°C, thermal conductivity 0.135 W/m°C, density 800 kg/m³, viscosity 1.2 mN m⁻²s.
- 19.14.** The heat duty of the jacketed vessel of Examples 19.15 and 19.16 can be estimated from the coolant flow rate and temperatures. What is the minimum temperature at which the reactor can operate using the jacket as designed? Does the choice of coolant make sense at this temperature? Propose a better design.
- 19.15.** A stirred tank fermentation reactor has height 2 m and diameter 1.5 m and is filled with a fermentation broth that can be assumed to have the physical properties of water. The fermenter must be maintained at a temperature less than 42 °C to prevent damage to the cell culture. After allowing for sensible heat losses to the cold feed, the heat that must be removed from the fermenter is 80 kW. Cooling water is available at 20 °C and can be returned at any temperature up to 35 °C. Recommend a design for cooling the contents of the fermenter.

Transport and storage of fluids

KEY LEARNING OBJECTIVES

- How to store gases and liquids
- How to calculate line pressure drop
- How valves, controllers, compressors, and pumps work
- How to select pumps and compressors and size piping systems and control valves

20.1 Introduction

This chapter describes the methods used for storing and transporting fluids within a process. Methods for storing and conveying solids are discussed in [Chapter 18](#).

The design of piping systems and rotating equipment such as pumps and compressors is usually done by specialist design groups, and a detailed discussion of piping design and plant hydraulics is beyond the scope of this book. Only general guide rules are given. The piping handbook edited by [Nayyar \(2000\)](#) is particularly recommended for guidance on the detailed design of piping systems. The references cited in the text and listed at the end of the chapter should also be consulted.

Piping and instrumentation diagrams (P&IDs) are introduced in [Chapter 5](#). A P&ID usually contains information on pipe sizes and ancillary valves and instruments, as well as the main process control valves. The sizing and selection of control valves must be carried out in concert with pipe sizing and pump or compressor design and selection. Accurate design of control valves requires a good understanding of process hydraulics and equipment pressure drop and so is best carried out after the major plant items have been specified.

20.2 Storage of fluids

All processes require storage of some inventory of feeds, products, solvents, or intermediate compounds. Inventories are maintained to ensure continuity of supply of feeds and products, to smooth out process operations, to allow scheduling of shipments, and to ensure the availability of solvents and consumables that may be needed by the process. Even plants that operate without local inventory usually require maintenance of inventory elsewhere in the supply chain. For example, gas processing plants often have no local inventory but rely on storage facilities located at various points in the gas distribution pipeline system. This section discusses methods used to store gases and liquids.

20.2.1 Storage of gases

Gases can be stored at low pressure in gas holders similar to those used for town gas. The liquid-sealed type is most commonly used. These consist of a number of telescopic sections (lifts) that rise and fall as gas is added to or withdrawn from the holder. The dry-sealed type is used where the gas must be kept dry. In this type, the gas is contained by a piston moving in a large vertical cylindrical vessel. Water seal holders are intrinsically safer for use with flammable gases than the dry seal type, as any leakage through the piston seal may form an explosive mixture in the closed space between the piston and the vessel roof. Details of the construction of gas holders can be found in textbooks on gas engineering, such as [Meade \(1921\)](#) and [Smith \(1945\)](#).

Gases are stored at high pressures where this is a process requirement and to reduce the storage volume. For some gases, the volume can be further reduced by liquefying the gas by pressure or refrigeration. Cylindrical and spherical vessels (Horton spheres) are used. High-pressure gas storage vessels are designed as pressure vessels, as discussed in [Chapter 14](#).

When a process consumes gas at a low flow rate, road or rail tankers are sometimes used to store the gas before transfer to a small process storage vessel.

Underground storage in depleted oil or gas reservoirs is used to store large quantities of natural gas to meet seasonal variations in demand. Underground storage tanks are sometimes used to maintain local inventories of natural gas for retail supply.

Industrial gases such as oxygen, nitrogen, and hydrogen that are consumed in large quantities are usually supplied to chemical plants by pipeline and are not stored on-site. The industrial gas companies contract to supply these gases “across the fence” and maintain sufficient storage and production capacity at their own facilities to ensure continuity of supply. When a chemical plant is located far from existing pipelines, the industrial gas companies will sometimes build a local plant to supply these gases.

20.2.2 Storage of liquids

Liquids are usually stored in bulk in vertical cylindrical steel tanks. Fixed- and floating-roof tanks are used. In a floating-roof tank a movable piston floats on the surface of the liquid and is sealed at the tank walls. Floating-roof tanks are used to eliminate evaporation losses and, for flammable liquids, to obviate the use of inert gas blanketing to prevent an explosive mixture forming above the liquid, as would be the situation with a fixed-roof tank.

Horizontal cylindrical tanks and rectangular tanks are also used for storing liquids, usually for relatively small quantities such as surge tanks for process intermediates.

Gas blanketing is used for fixed-roof tanks and for pressure vessels that are used to store liquids. The use of a gas bubble in the top of the vessel allows measurement of the level in the tank and prevents hydrostatic overpressure or discharge of liquid through vapor vents. Nitrogen is most commonly used for gas-blanketed storage, but natural gas, hydrogen, or fuel gas can be used if the designer does not want inert nitrogen to enter the process. Liquids that are not volatile, hazardous, or flammable can be stored in tanks that are vented to atmosphere, but care must be taken that dissolved oxygen does not introduce problems in downstream processing.

The mechanical design of fixed-roof, vertical tanks is discussed in [Chapter 14, Section 14.15](#).

Liquid storage tanks are usually grouped together in a “tank farm” located away from the main plant area. The tank farm should preferably be downwind of the plant area so that in the event of a tank failure or leakage, flammable vapors will not be carried towards ignition sources in the plant. The tank farm must have good access to road, rail, or port facilities, depending on the method used to ship feeds and products.

Raw water can be stored in retaining ponds or underground aquifers when seasonal variations create a need for water storage. Treated water (demineralized water, deionized water, boiler feed water, distilled water, and reverse osmosis water) is usually stored in tanks to prevent recontamination. Cooling tower water is not typically stored, as the cooling water system provides an adequate inventory during operation and the system can be flushed to wastewater treatment during major maintenance.

20.3 Transport of gases and liquids

20.3.1 Gases

The type of equipment best suited for the pumping of gases in pipelines depends on the flow rate, the differential pressure required, and the operating pressure.

Reciprocating, centrifugal, and axial flow compressors are the principal types used in the chemical process industries, and the range of application of each type is shown in Fig. 20.1, which has been adapted from a similar diagram by Dimoplon (1978). A more comprehensive selection guide is given in Table 20.1. In general, fans are used where the pressure drop is small, less than 35 cm H₂O (0.03 bar); axial flow compressors for high flow rates and moderate differential pressures; centrifugal compressors for high flow rates; and, by staging, high differential pressures. Reciprocating compressors can be used over a wide range of pressures and capacities, but are normally only specified in preference to centrifugal compressors where high pressures are required at relatively low flow rates. Diagrammatic sketches of the compressors listed are given in Fig. 20.2. Gas compression is discussed in more detail in Section 20.6.

Vacuum production

The production of vacuum (subatmospheric pressure) is required for many chemical engineering processes, for example, vacuum distillation, drying, and filtration. The type of vacuum pump needed will depend on the degree of vacuum required, the capacity of the system, and the rate of air in-leakage.

Reciprocating and rotary positive displacement pumps are commonly used where moderately low vacuum is required, about 10 mmHg (0.013 bar), at moderate to high flow rates, such as in vacuum filtration.

Steam-jet ejectors are versatile and economic vacuum pumps and are frequently used, particularly in vacuum distillation. They can handle high vapor flow rates and, by using several ejectors in series, can produce low pressures, down to about 0.1 mmHg (0.13 mbar).

The operating principle of steam-jet ejectors is explained in Chhabra and Shankar (2017). Their specification, sizing, and operation are covered in a comprehensive series of papers by Power (1964). Diffusion pumps are used where very low pressures are required (hard vacuum) for processes such as molecular distillation.

For general references on the design and application of vacuum systems, see Ryan and Roper (1986) or Jousten and Nakhosteen (2016).

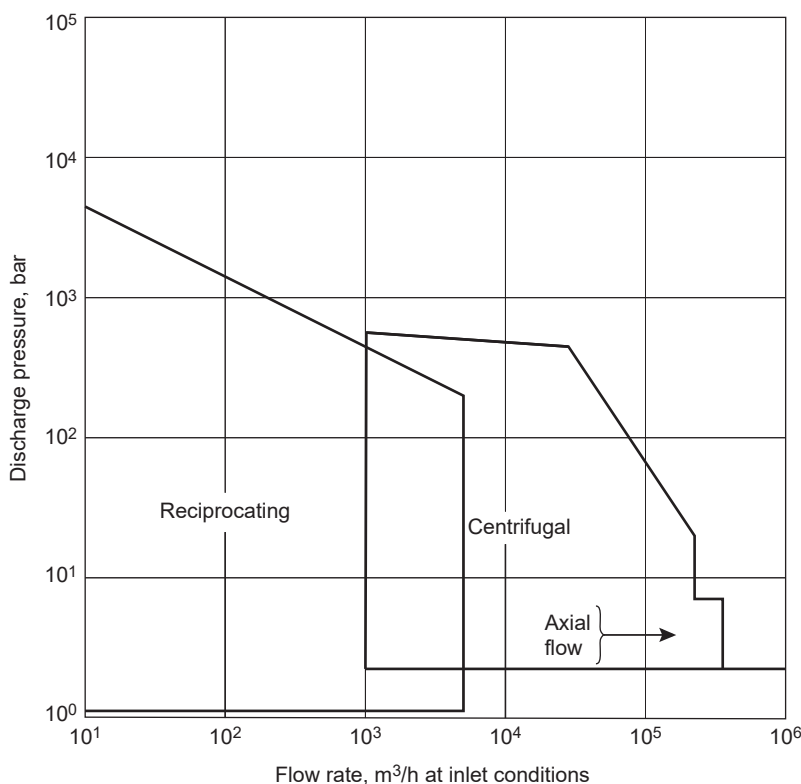


FIG. 20.1 Compressor operating ranges.

TABLE 20.1 Operating range of compressors and blowers

Type of compressor	Normal Maximum Speed (rpm)	Normal Maximum Capacity (m ³ /h)	Normal Maximum Pressure (Differential) (bar)	
			Single stage	Multiple stage
Displacement				
1. Reciprocating	300	85,000	3.5	5000
2. Sliding vane	300	3400	3.5	8
3. Liquid ring	200	2550	0.7	1.7
4. Rootes	250	4250	0.35	1.7
5. Screw	10,000	12,750	3.5	17
Dynamic				
6. Centrifugal fan	1000	170,000		0.2
7. Turbo blower	3000	8500	0.35	1.7
8. Turbo compressor	10,000	136,000	3.5	100
9. Axial flow fan	1000	170,000	0.35	2.0
10. Axial flow blower	3000	170,000	3.5	10

From Begg 1966.

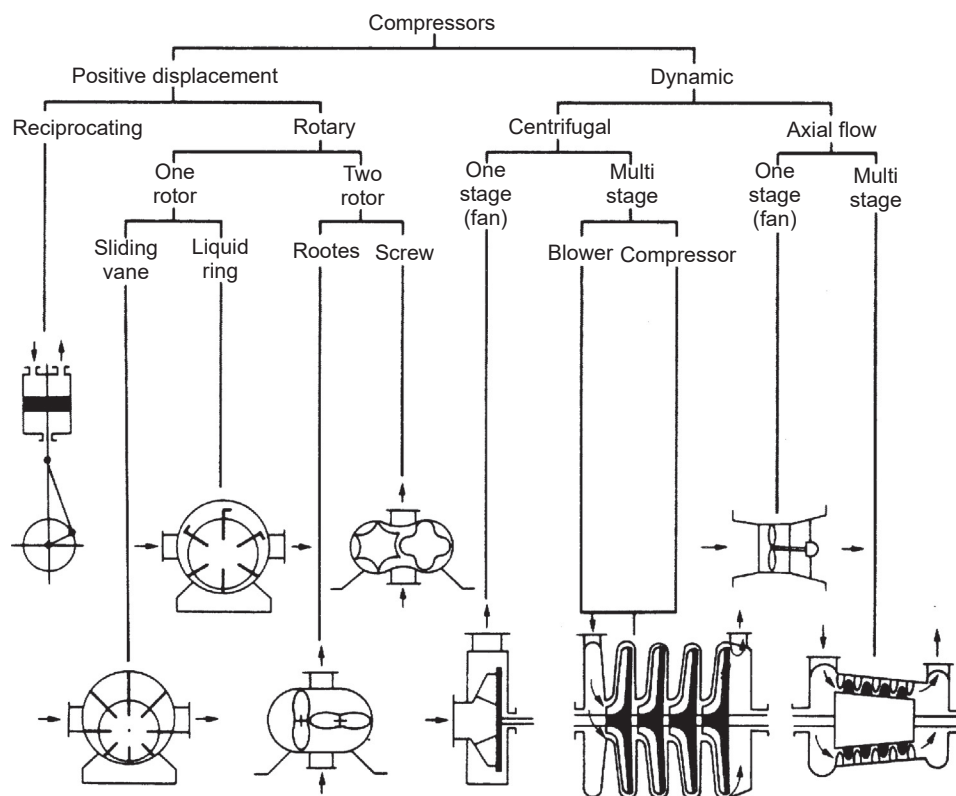


FIG. 20.2 Type of compressor. From Begg, 1966.

20.3.2 Liquids

Liquids and dilute slurries are almost always pumped through pipe systems. Pumps can be classified into two general types:

1. Dynamic pumps, such as centrifugal pumps
2. Positive displacement pumps, such as reciprocating and diaphragm pumps

The single-stage, horizontal, overhung centrifugal pump is by far the most common type in the chemical process industries. Other types are used where a high head or other special process considerations are specified. For example, positive displacement metering pumps are often used when small flow rates of additives must be fed to a process. Positive displacement pumps are also intrinsically easier to use with variable-speed drivers when power consumption efficiency is important; see [Hall \(2010\)](#). Descriptions of the types of pumps used in the chemical process industries are given in textbooks and handbooks on this subject: [Garay \(1997\)](#), [Karassik et al. \(2007\)](#), and [Parmley \(2000\)](#). A general guide to the selection, installation, and operation of pumps for the process industries is given by [Davidson and von Bertele \(2005\)](#) and [Jandiel \(2000\)](#).

Pump selection is made on the flow rate and head required, together with other process considerations, such as corrosion or the presence of solids in the fluid. The principal types used and their operating pressures and capacity ranges are summarized in [Table 20.2](#). The chart shown in [Fig. 20.3](#) can be used to determine the type of pump required for a particular head and flow rate. This figure is based on one published by [Doolin \(1977\)](#). For a detailed discussion of the factors governing the selection of the best centrifugal pump for a given duty, the reader should refer to the articles by [De Santis \(1976\)](#), [Neerkin \(1974\)](#), [Jacobs \(1965\)](#), and [Couper et al. \(2012\)](#). Special factors to consider when selecting a pump for slurry service are described by [Nwaokocha \(2019\)](#). The design and selection of centrifugal pumps are discussed in more detail in [Section 20.7](#) and in the article by [Kelly \(2010\)](#).

[Fig. 20.4](#) shows operating ranges for different types of positive displacement pumps. Centrifugal pumps will normally be the first choice for pumping process fluids, with the other types only being used for special applications, such as the use of reciprocating and gear pumps for metering. Positive displacement pumps are normally used where a high head is required at a low flow rate. Positive displacement pumps are also used for pumping highly viscous fluids. [Holland and Chapman \(1966\)](#) review the various types of positive displacement pumps available and discuss their applications. [Hall \(2010\)](#) discusses the use of positive displacement pumps with variable-speed drivers.

TABLE 20.2 Normal operating range of pumps

Type	Capacity range (m ³ /h)	Typical head (m of water)
Centrifugal	0.25–103	10–50 300 (multistage)
Reciprocating	0.5–500	50–200
Diaphragm	0.05–50	5–60
Rotary gear and similar	0.05–500	60–200
Rotary sliding vane or similar	0.25–500	7–70

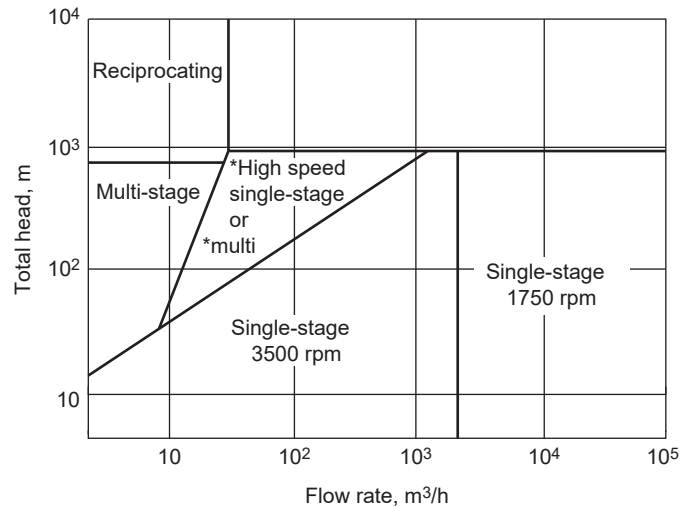


FIG. 20.3 Centrifugal pump selection guide. *Single-stage >1750 rpm, multistage 1750 rpm.

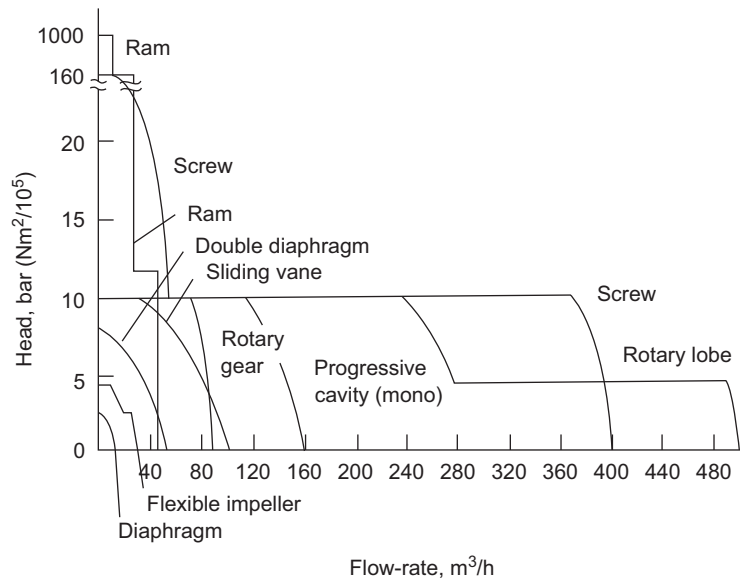


FIG. 20.4 Selection of positive displacement pumps. Adapted from Marshall, 1985.

20.4 Pressure drop in pipelines

Pumps, compressors, and control valves cannot be designed independently of the piping system that connects them. The pressure head that the pump or compressor must deliver is determined by the pressure drop in the process equipment and piping, the change in elevation, and the pressure drop that is required for satisfactory operation of the control valve.

20.4.1 Pressure drop in pipes

The pressure drop in a pipe due to friction is a function of the fluid flow rate, fluid density and viscosity, pipe diameter, pipe surface roughness, and length of the pipe. It can be calculated using the following equation:

$$\Delta P_f = 8f(L/d_i) \frac{\rho u^2}{2} \quad (20.1)$$

where ΔP_f = pressure drop, N/m²

f = friction factor
 L = pipe length, m
 d_i = pipe inside diameter, m
 ρ = fluid density, kg/m³
 u = fluid velocity, m/s

The friction factor is dependent on the Reynolds number and pipe roughness. The friction factor for use in Equation 20.1 can be found from Fig. 20.5.

$$\text{The Reynolds number is given by: } Re = (\rho \times u \times d_i) / \mu \quad (20.2)$$

where μ is the fluid viscosity. Values for the absolute surface roughness of commonly used pipes are given in Table 20.3. The parameter to use with Fig. 20.5 is the relative roughness, given by:

$$\text{relative roughness, } e = \text{absolute roughness} / \text{pipe inside diameter}$$

Note: The friction factor used in Equation 20.1 is related to the shear stress at the pipe wall, R , by the equation $f = (R / \rho u^2)$. Other workers use different relationships. Their charts for friction factor will give values that are multiples of those given by Fig. 20.5, so it is important to make sure that the pressure drop equation used matches the friction factor chart. One of the most commonly used is that of Fanning, which defines the coefficient of friction as $C_f = (2R / \rho u^2)$, that is, $C_f = 2f$, in which case Equation 20.1 becomes:

$$\Delta P_f = 4 C_f (L / d_i) \frac{\rho u^2}{2} \quad (20.1b)$$

Non-Newtonian fluids

In Equation 20.1, and when calculating the Reynolds number for use with Fig. 20.5, the fluid viscosity and density are taken to be constant. This will be true for Newtonian liquids but not for non-Newtonian liquids, where the apparent viscosity is a function of the shear stress.

More complex methods are needed to determine the pressure drop of non-Newtonian fluids in pipelines. Suitable methods are given in Chhabra and Gurappa (2019) and in Chhabra and Richardson (2008); see also Darby (2001).

Gases

When a gas flows through a pipe, the gas density is a function of the pressure and so is determined by the pressure drop. If the pressure of the gas does not change by more than about 20%, the gas can be treated as an incompressible fluid with density equal to the density at the mean pressure. Equation 20.1 and Fig. 20.5 can then be used to estimate the pressure drop, but it may be necessary to divide the pipeline into short sections and sum the results.

For longer pipelines or low-pressure systems, the gas flow must be treated as flow of a compressible fluid. Because the density of a compressible fluid is a function of both temperature and pressure, the analysis of pipe flow becomes much more complex and sophisticated models are needed. Furthermore, it can be shown that if the pressure ratio across a restriction, valve, or pipe opening reaches a critical value, a maximum flow rate is obtained, corresponding to sonic flow at the restriction. When sonic flow occurs, further decreases in downstream pressure have no effect on the flow rate. The critical pressure ratio for an ideal gas under isothermal conditions is only 0.607, so sonic flow can easily occur. Compressible flow is discussed in Chhabra and Shankar (2017) and Teng et al. (2014).

Two-phase mixtures

For vapor–liquid mixtures, the pressure drop in horizontal pipes can be found using the correlation of Lockhart and Martinelli (1949), which relates the two-phase pressure drop to the pressure drop that would be calculated if each phase were flowing separately in the pipe. Details of the correlation and methods for two-phase flow in vertical pipes are given in Green and Southard (2018) and Chhabra and Shankar (2017).

Liquid–solid mixtures that do not settle out rapidly are usually treated as non-Newtonian fluids. This will usually be the case if the solid particle size is less than about 200 microns (0.2 mm). Larger particle sizes form settling

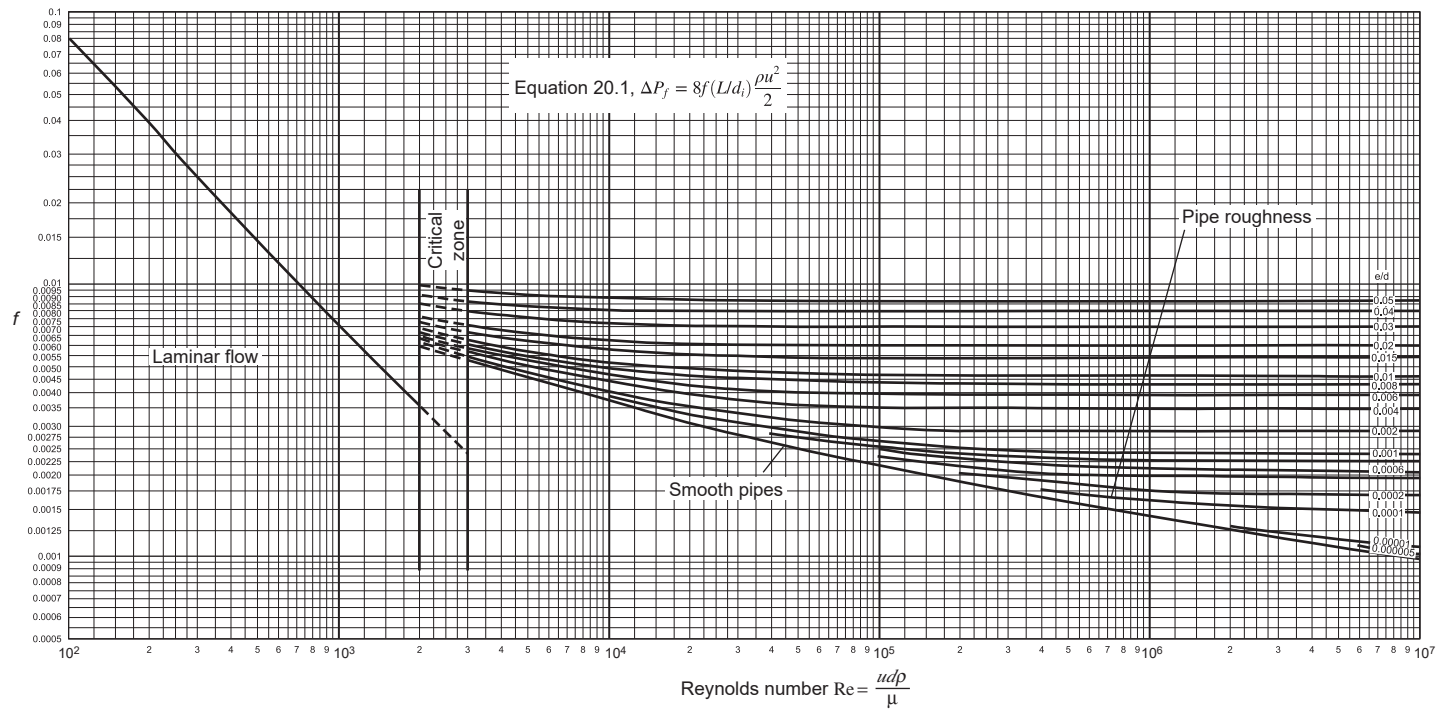


FIG. 20.5 Pipe friction versus Reynolds number and relative roughness.

TABLE 20.3 Pipe roughness

Material	Absolute roughness, mm
Drawn tubing	0.0015
Commercial steel pipe	0.046
Cast iron pipe	0.26
Concrete pipe	0.3–3.0

slurries and require a critical velocity to maintain the solids in suspension. Correlations for critical velocity and pressure drop are given in [Green and Southard \(2018\)](#) and [Chhabra and Shankar \(2017\)](#).

Gas–solid mixtures are commonly encountered in pneumatic conveying. This is discussed in [Chhabra and Shankar \(2017\)](#) and by [Mills \(2015\)](#) and [Mills et al. \(2004\)](#).

20.4.2 Pressure drop in pipe fittings

Any obstruction to flow will generate turbulence and cause a pressure drop, so pipe fittings such as bends, elbows, reducing or enlargement sections, and tee junctions will increase the pressure drop in a pipeline. There will also be a pressure drop due to the valves used to isolate equipment and control the fluid flow, and there can be pressure losses due to flow measurement equipment; see [Livelli \(2013\)](#). The pressure drop due to these miscellaneous losses can be estimated using either of two methods:

1. As the number of velocity heads, K , lost at each fitting or valve. A velocity head is $u^2/2g$, meters of the fluid, equivalent to $(\rho u^2/2)$, N/m², where the fluid velocity, u , is calculated at the pipe diameter. The total number of velocity heads lost due to all the fittings and valves is added to the pressure drop due to pipe friction.
2. As a length of pipe that would cause the same pressure loss as the fitting or valve. Because this will be a function of the pipe diameter, it is expressed as the number of equivalent pipe diameters. The length of pipe to add to the actual pipe length is found by multiplying the total number of equivalent pipe diameters by the diameter of the pipe being used.

The number of velocity heads lost, or equivalent pipe diameter, is a characteristic of the particular fitting or type of valve used. Values can be found in handbooks and manufacturers' literature. The values for a selected number of fittings and valves are given in [Table 20.4](#).

The two methods used to estimate the miscellaneous losses are illustrated in Example 20.1.

Pipe fittings are discussed in Section 20.9.4; see also [Green and Southard \(2018\)](#). Valve types and applications are discussed in [Section 20.5](#).

Example 20.1

A pipeline connecting two tanks contains four standard elbows, a globe valve that is fully open, and a gate valve that is half-open. The line is commercial steel pipe, 25 mm internal diameter, length 120 m.

The properties of the fluid are viscosity 0.99 mNm⁻² s and density 998 kg/m³. Calculate the total pressure drop due to friction when the flow rate is 3500 kg/h.

Solution

$$\text{Cross-sectional area of pipe} = \frac{\pi}{4} (25 \times 10^{-3})^2 = 0.491 \times 10^{-3} \text{ m}^2$$

$$\text{Fluid velocity, } u = \frac{3500}{3600} \times \frac{1}{0.491 \times 10^{-3}} \times \frac{1}{998} = 1.98 \text{ m/s}$$

$$\begin{aligned} \text{Reynolds number, } Re &= (998 \times 1.98 \times 25 \times 10^{-3}) / 0.99 \times 10^{-3} \\ &= 49,900 = 5 \times 10^4 \end{aligned} \quad (20.2)$$

Absolute roughness commercial steel pipe, [Table 20.3](#) = 0.046 mm

TABLE 20.4 Pressure loss in pipe fittings and valves (for turbulent flow)

Fitting or valve	K, number of velocity heads	number of equivalent pipe diameters
45-degree standard elbow	0.35	15
45-degree long radius elbow	0.2	10
90-degree standard radius elbow	0.6–0.8	30–40
90-degree standard long elbow	0.45	23
90-degree square elbow	1.5	75
Tee entry from leg	1.2	60
Tee entry into leg	1.8	90
Union and coupling	0.04	2
Sharp reduction (tank outlet)	0.5	25
Sudden expansion (tank inlet)	1.0	50
Gate valve		
fully open	0.15	7.5
quarter open	16	800
half open	4	200
three-fourths open	1	40
Globe valve, bevel seat-		
fully open	6	300
half open	8.5	450
Globe valve, plug-disk		
fully open	9	450
half open	36	1800
quarter open	112	5600
Plug valve – open	0.4	18

Relative roughness = $0.046/(25 \times 10^{-3}) = 0.0018$, round to 0.002

From friction factor chart, Fig. 20.5, $f = 0.0032$

Miscellaneous losses

Fitting/valve	Number of velocity heads, K	Equivalent pipe diameters
Entry	0.5	25
Elbows	(0.8×4)	(40×4)
Globe valve, open	6.0	300
Gate valve, half-open	4.0	200
Exit	1.0	50
Total	14.7	735

Method 1: Velocity heads

A velocity head = $u^2/2g = 1.98^2/(2 \times 9.8) = 0.20$ m of liquid.

Head loss = $0.20 \times 14.7 = 2.94$ m

as pressure = $2.94 \times 998 \times 9.8 = 28,754$ N/m²

$$\begin{aligned} \text{Friction loss in pipe, } \Delta P_f &= 8 \times 0.0032 \frac{(120)}{(25 \times 10^{-3})} 998 \times \frac{1.98^2}{2} \\ &= 240,388 \text{ N/m}^2 \end{aligned} \quad (20.1)$$

$$\text{Total pressure} = 28,754 + 240,388 = 269,142 \text{ N/m}^2 = \underline{\underline{270 \text{ kN/m}^2}}$$

Method 2: Equivalent pipe diameters

Extra length of pipe to allow for miscellaneous losses:

$$= 735 \times 25 \times 10^{-3} = 18.4 \text{ m}$$

So, total length for ΔP calculation = $120 + 18.4 = 138.4$ m

$$\begin{aligned} \Delta P_f &= 8 \times 0.0032 \frac{(138.4)}{(25 \times 10^{-3})} 998 \times \frac{1.98^2}{2} = 277,247 \text{ N/m}^2 \\ &= \underline{\underline{277 \text{ kN/m}^2}} \end{aligned} \quad (20.1)$$

Note: The two methods will not give exactly the same result. The method using velocity heads is the more fundamentally correct approach, but the use of equivalent diameters is easier to apply and sufficiently accurate for use in preliminary design calculations.

20.5 Valves

The valves used for chemical process plant can be divided into two broad classes, depending on their primary function:

1. Shut-off valves (block valves or isolation valves), whose purpose is to close off the flow
2. Control valves, both manual and automatic, used to regulate flow

The main types of valves used are:

Gate	Fig. 20.6(a)
Plug	Fig. 20.6(b)
Ball	Fig. 20.6(c)
Globe	Fig. 20.6(d)
Diaphragm	Fig. 20.6(e)
Butterfly	Fig. 20.6(f)

A valve selected for shut-off purposes should give a positive seal in the closed position and minimum resistance to flow when open. Gate, plug, and ball valves are most frequently used for this purpose. Gate valves are available in

the widest range of sizes and can be operated manually or by a motor. They have a straight-through flow channel and low pressure drop when fully open. Several turns of the valve handle are usually required to close the valve, so they are best used when operated infrequently. Gate valves should not be operated partially open, as the valve seals can become deformed, causing the valve not to seal properly. Plug valves and ball valves have the advantage in that they only require a quarter-turn to open or close. These valves are often actuated by solenoids and are used where quick on–off switching is needed. The selection of valves is discussed by [Merrick \(1986, 1990\)](#), [Smith and Vivian \(1995\)](#), and [Smith and Zappe \(2003\)](#).

If flow control is required, the valve should be capable of giving smooth control over the full range of flow, from fully open to closed. Globe valves are normally used, though diaphragm valves are also common. Butterfly valves are often used for the control of gas and vapor flows. Automatic control valves are usually globe valves with special trim designs (see [Chhabra & Rohani, 2017](#)).

The careful selection and design of control valves are important; good flow control must be achieved without introducing excessive pressure drop. Control valve sizing is discussed in more detail in [Section 20.11](#).

Nonreturn valves are used to prevent back flow of fluid in a process line. They do not normally give an absolute shut-off of the reverse flow. A typical design is shown in [Fig. 20.6\(g\)](#). Because swing-type check valves depend on gravity to close the valve, care must be taken to orient the valve properly when locating and installing it.

Standards for valves are set by the ASME B16 Standards Committee and can be ordered from the American Society of Mechanical Engineers. General standards are described in ASME B16.34-2017, while valve dimensions are given in ASME B16.10-2017. Valve design is covered by [Pearson \(1978\)](#).

20.6 Compression and expansion of gases

The equipment used to compress a gas through a process piping system is different from that used for liquids. At low pressure drops, a simple fan may be adequate. At higher pressure drops, multistage compressors are commonly used. The different types of compressors were introduced in [Section 20.3.1](#). This section discusses the sizing of compressors and estimation of the work required to compress a gas or vapor. Several textbooks are available on compressor design, selection, and operation: [Bloch et al. \(1982\)](#), [Brown \(2005\)](#), and [Aungier \(1999, 2003\)](#). An introduction to centrifugal compressor selection is given in the article by [Sorokes \(2013\)](#).

20.6.1 Compression of gases

The work done in compressing or expanding a gas or vapor is given by:

$$-W = \int_{v_1}^{v_2} P \, dv \quad (20.3)$$

To calculate the work done, a relationship between pressure and volume during the expansion is needed.

If the compression or expansion is isothermal (at constant temperature), then for unit mass of an ideal gas:

$$Pv = \text{constant} \quad (20.4)$$

and the work done,

$$-W = P_1 v_1 \ln \frac{P_2}{P_1} = \frac{RT_1}{M_w} \ln \frac{P_2}{P_1} \quad (20.5)$$

where P_1 = initial pressure

P_2 = final pressure

v_1 = initial volume

R = universal gas constant

M_w = molecular mass (weight) of gas

In industrial compressors or expanders, the compression or expansion path will be “polytropic,” approximated by the expression:

$$Pv^n = \text{constant} \quad (20.6)$$

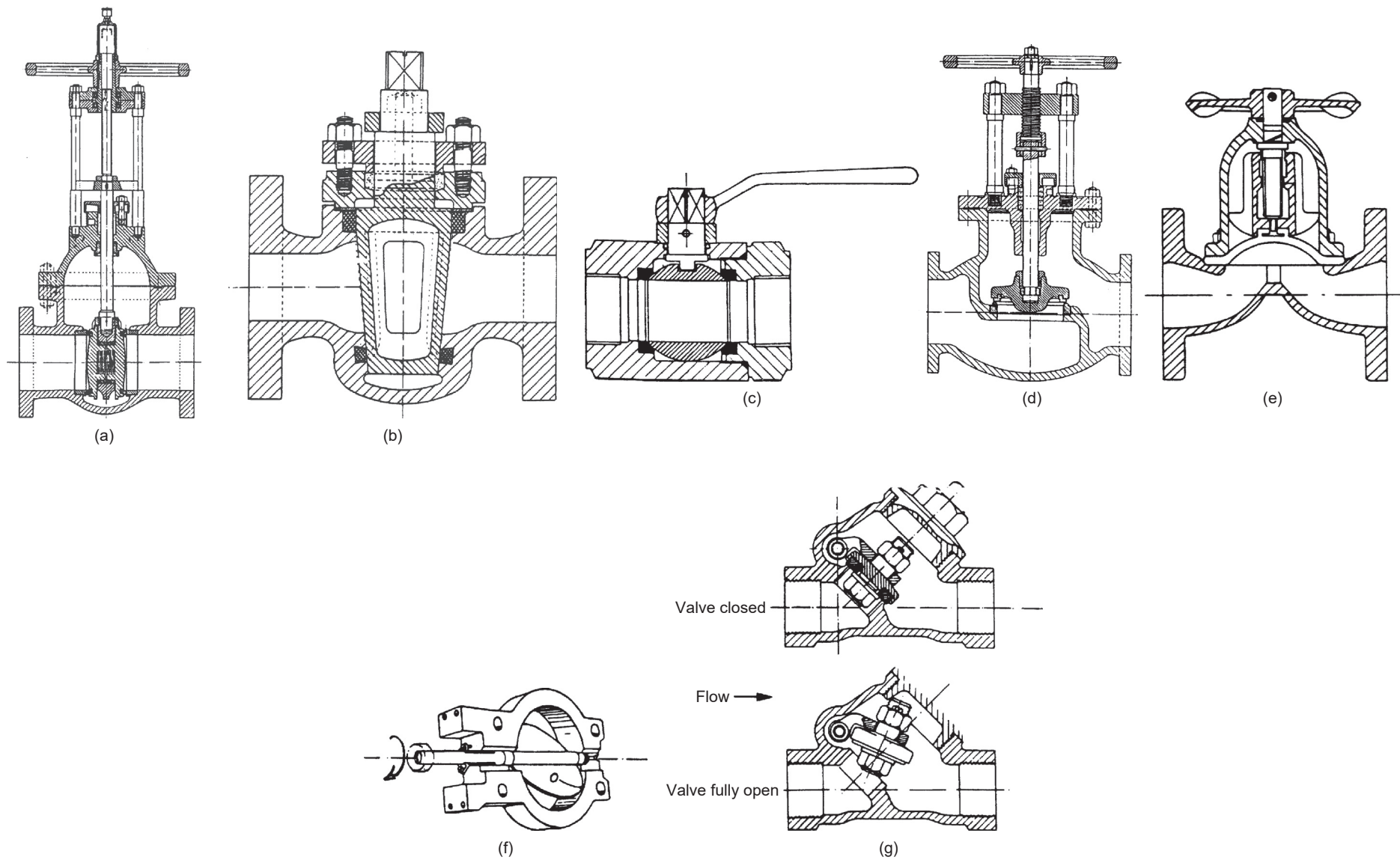


FIG. 20.6 (a) Gate valve (slide valve). (b) Plug valve. (c) Ball valve. (d) Globe valve. (e) Diaphragm valve. (f) Butterfly valve. (g) Nonreturn valve, check valve, hinged disc type.

where n is the polytropic expansion coefficient. The work produced (or required) is given by the general expression (see Chhabra & Shankar, 2017):

$$-W = P_1 v_1 \frac{n}{n-1} \left[\left(\frac{P_2}{P_1} \right)^{(n-1)/n} - 1 \right] = Z \frac{RT_1}{M_w} \frac{n}{n-1} \left[\left(\frac{P_2}{P_1} \right)^{(n-1)/n} - 1 \right] \quad (20.7)$$

where Z = compressibility factor (1 for an ideal gas)

R = universal gas constant, $8.314 \text{ J K}^{-1} \text{ mol}^{-1}$

T_1 = inlet temperature, K

W = work done, kJ/kg

The value of n will depend on the design and operation of the machine.

The energy required to compress a gas, or the energy obtained from expansion, can be estimated by calculating the ideal work and applying a suitable efficiency value. For reciprocating compressors, the isentropic work is normally used ($n = \gamma$, where γ is the ratio of specific heats) (see Fig. 20.9), and for centrifugal or axial machines the polytropic work (see Fig. 20.8 and Section 20.6.3).

20.6.2 Mollier diagrams

If a Mollier diagram (enthalpy–pressure–temperature–entropy chart) is available for the working fluid, the isentropic work can be easily calculated.

$$W = H_1 - H_2 \quad (20.8)$$

where H_1 = the specific enthalpy at the pressure and temperature corresponding to point 1, the initial gas conditions

H_2 = the specific enthalpy corresponding to point 2, the final gas condition

Point 2 is found from point 1 by tracing a path (line) of constant entropy on the diagram.

The method is illustrated in Example 20.2.

Example 20.2

Methane is compressed from 1 bar and 290 K to 10 bar. If the isentropic efficiency is 0.85, calculate the energy required to compress 10,000 kg/h. Estimate the exit gas temperature.

Solution

From the Mollier diagram, shown diagrammatically in Fig. 20.7:

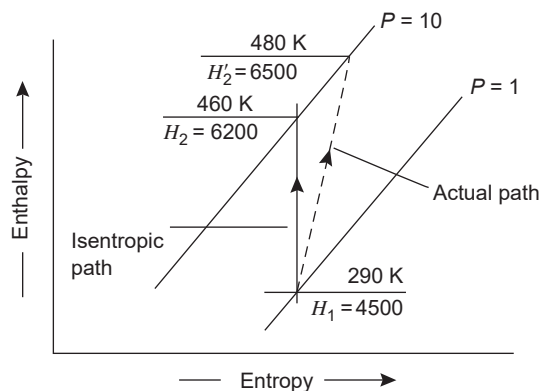


FIG. 20.7 Mollier diagram, methane.

$$H_1 = 4500 \text{ cal/mol}$$

$$H_2 = 6200 \text{ cal/mol (isentropic path)}$$

$$\begin{aligned} \text{Isentropic work} &= 6200 - 4500 \\ &= \underline{\underline{1700 \text{ cal/mol}}} \end{aligned}$$

For an isentropic efficiency of 0.85:

$$\text{Actual work done on gas} = \frac{1700}{0.85} = 2000 \text{ cal/mol}$$

So, actual final enthalpy:

$$H'_2 = H_1 + 2000 = \underline{\underline{6500 \text{ cal/mol}}}$$

From the Mollier diagram, if all the extra work is taken as irreversible work done on the gas, the exit gas temperature = 480 K

Molecular weight of methane = 16

$$\text{Energy required} = (\text{moles per hour}) \times (\text{specific enthalpy change})$$

$$= \frac{10,000}{16} \times 2000 \times 10^3$$

$$= 1.25 \times 10^9 \text{ cal/h}$$

$$= 1.25 \times 10^9 \times 4.187 \text{ J/h}$$

$$= 5.23 \times 10^9 \text{ J/h}$$

$$\text{Power} = \frac{5.23 \times 10^9}{3600} = \underline{\underline{1.45 \text{ MW}}}$$

20.6.3 Polytropic compression and expansion

If no Mollier diagram is available, it is more difficult to estimate the ideal work in compression or expansion processes.

Equation 20.7 can be used if the compressibility Z and polytropic coefficient n are known. Compressibility can be plotted against reduced temperature and pressure, as shown in Fig. 20.10.

At conditions away from the critical point:

$$n = \frac{1}{1 - m} \quad (20.9)$$

where:

$$m = \frac{(\gamma - 1)}{\gamma E_p} \text{ for compression} \quad (20.10)$$

$$m = \frac{(\gamma - 1)E_p}{\gamma} \text{ for expansion} \quad (20.11)$$

and E_p is the polytropic efficiency, defined by:

$$\text{for compression } E_p = \frac{\text{polytropic work}}{\text{actual work required}}$$

$$\text{for expansion } E_p = \frac{\text{actual work obtained}}{\text{polytropic work}}$$

An estimate of E_p can be obtained from Fig. 20.8.

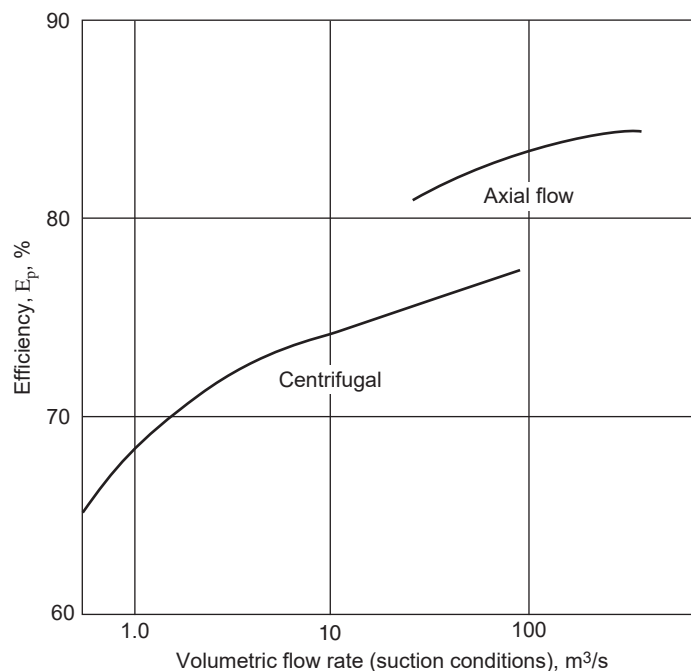


FIG. 20.8 Approximate polytropic efficiencies of centrifugal and axial-flow compressors.

The outlet temperature can be estimated from:

$$T_2 = T_1 \left(\frac{P_2}{P_1} \right)^m \quad (20.12)$$

Close to the critical conditions these equations should not be used. The procedure for calculation of polytropic work of compression or expansion close to the critical point is more complex (Schultz, 1962), and it is easiest to make such calculations using process simulation programs.

Example 20.3

Estimate the power required to compress 5000 kmol/h of HCl at 5 bar, 15 °C, to 15 bar.

Solution

For HCl, critical conditions are $P_c = 82$ bar, $T_c = 324.6$ K.

Estimate T_2 from Equation 20.12. For diatomic gases $\gamma \approx 1.4$.

Note: γ could be estimated from the relationship:

$$\gamma = \frac{C_p}{C_v}$$

From Fig. 20.8, $E_p = 0.73$.

From Equation 20.10:

$$m = \frac{1.4 - 1}{1.4 \times 0.73} = 0.391$$

$$T_2 = 288 \left(\frac{15}{5} \right)^{0.39} = 442 \text{ K}$$

$$T_{r(\text{mean})} = \frac{442 + 288}{2 \times 324.6} = 1.12$$

$$P_{r(\text{mean})} = \frac{5 + 15}{2 \times 82} = 0.12$$

From Fig. 20.10, at mean conditions:

$$Z = 0.98$$

From Equation 20.9:

$$n = \frac{1}{1 - 0.391} = 1.64$$

From Equation 20.7:

$$\begin{aligned} W_{\text{polytropic}} &= 0.98 \times 288 \times 8.314 \times \frac{1.64}{1.64 - 1} \left(\left(\frac{15}{5} \right)^{(1.64-1)/1.64} - 1 \right) \\ &= \underline{\underline{3219 \text{ kJ/kmol}}} \end{aligned}$$

$$\text{Actual work required} = \frac{\text{polytropic work}}{E_p} = \frac{3219}{0.73} = \underline{\underline{4409 \text{ kJ/kmol}}}$$

$$\text{Power} = \frac{4409 \times 5000}{3600} = 6124 \text{ kW, say } \underline{\underline{6.1 \text{ MW}}}$$

Example 20.4

Consider the extraction of energy from the tail gases from a nitric acid adsorption tower.

Gas composition, kmol/h:

O ₂	371.5
N ₂	10,014.7
NO	21.9
NO ₂	Trace
H ₂ O	Saturated at 25 °C

If the gases leave the tower at 6 atm, 25 °C, and are expanded to, say, 1.5 atm, calculate the turbine exit gas temperatures, without pre-heat, and if the gases are pre-heated to 400 °C with the reactor off-gas. Also, estimate the power recovered from the pre-heated gases.

Solution

For the purposes of this calculation, it will be sufficient to consider the tail gas as all nitrogen, flow 10,410 kmol/h.

$$P_c = 33.5 \text{ atm}, T_c = 126.2 \text{ K}$$

Fig. 20.8 can be used to estimate the turbine efficiency:

$$\text{Exit gas volumetric flow rate} = \frac{10,410}{3600} \times 22.4 \times \frac{1}{1.5} \approx 43 \text{ m}^3/\text{s}$$

from Fig. 20.8 $E_p = 0.75$:

$$P_{r \text{ inlet}} = \frac{6}{33.5} = 0.18$$

$$T_{r \text{ inlet}} = \frac{298}{126.2} = 2.4$$

Using Equations 20.9 and 20.11, for N_2 , $\gamma = 1.4$:

$$m = \frac{1.4 - 1}{1.4} \times 0.75 = 0.21$$

$$n = \frac{1}{1 - m} = \frac{1}{1 - 0.21} = 1.27$$

$$\begin{aligned} \text{Without pre-heat } T_2 &= 298 \left(\frac{1.5}{6.0} \right)^{0.21} = 223 \text{ K} \\ &= \underline{\underline{-50^\circ \text{C}}} \end{aligned}$$

This temperature would be problematic. Acidic water would condense out, probably damaging the turbine.

$$\begin{aligned} \text{With pre-heat, } T_2 &= 673 \left(\frac{1.5}{6.0} \right)^{0.21} = 503 \text{ K} \\ &= \underline{\underline{230^\circ \text{C}}} \end{aligned}$$

From Equation 20.7, the work done by the gas as a result of polytropic expansion is:

$$\begin{aligned} &= -1 \times 673 \times 8.314 \times \frac{1.27}{1.27 - 1} \left\{ \left(\frac{1.5}{6.0} \right)^{(1.27-1)/1.27} - 1 \right\} \\ &= 6718 \text{ kJ/kmol} \end{aligned}$$

Actual work = polytropic work $\times E_p$

$$= 6718 \times 0.75 = \underline{\underline{5039 \text{ kJ/kmol}}}$$

$$\begin{aligned} \text{Power output} &= \text{work/kmol} \times \text{kmol/s} = 5039 \times \frac{10,410}{3600} \\ &= 14,571 \text{ kJ/s} = \underline{\underline{14.6 \text{ MW}}} \end{aligned}$$

This is a significant amount of power and will probably justify the cost of the expansion turbine.

20.6.4 Multistage compressors

Single-stage compressors can only be used for low pressure ratios. At high pressure ratios, the temperature rise is too high for efficient operation.

To cope with the need for high pressure generation, the compression is split into a number of separate stages, with intercoolers between each stage. The interstage pressures are normally selected to give equal work in each stage. For an n -stage compressor, this occurs when the compression ratio across each stage is the n th root of the overall compression ratio.

For a two-stage compressor, the interstage pressure is given by:

$$P_i = \sqrt{(P_1 \times P_2)} \quad (20.13)$$

where P_i is the intermediate-stage pressure.

Example 20.5

Estimate the power required to compress 1000 m³/h air from ambient conditions to 700 kN/m² gauge, using a two-stage reciprocating compressor with an intercooler.

Solution

Take the inlet pressure, P_1 , as 1 atmosphere = 101.33 kN/m², absolute.

Outlet pressure, P_2 , = 700 + 101.33 = 801.33 kN/m², absolute.

For equal work in each stage, the intermediate pressure, P_i ,

$$= \sqrt{(1.0133 \times 10^5 \times 8.0133 \times 10^5)} = \underline{\underline{2.8495 \times 10^5 \text{ N/m}^2}}$$

For air, take ratio of the specific heats, γ , to be 1.4.

For equal work in each stage, the total work will be twice that in the first stage.

Take the inlet temperature to be 20 °C, At that temperature, the specific volume is given by:

$$v_1 = \frac{29}{22.4} \times \frac{293}{273} = 1.39 \text{ m}^3/\text{kg}$$

$$\begin{aligned} \text{Isentropic work done, } -W &= 2 \times 1.0133 \times 10^5 \times 1.39 \times \frac{1.4}{1.4 - 1} \left(\left(\frac{2.8495}{1.0133} \right)^{(1.4-1)/1.4} - 1 \right) \\ &= 338,844 \text{ J/kg} = 339 \text{ kJ/kg} \end{aligned}$$

From Fig. 20.9, for a compression ratio of 2.85, the efficiency is approximately 84%. So work required:

$$= 339/0.84 = \underline{404 \text{ kJ/kg}}$$

$$\text{Mass flow rate} = \frac{1000}{1.39 \times 3600} = 0.2 \text{ kg/s}$$

$$\text{Power required} = 404 \times 0.2 = \underline{80 \text{ kW}}$$

20.6.5 Compressor performance curves

Multistage centrifugal compressors and axial compressors can be operated over a range of conditions by varying the compressor speed. A typical set of performance curves is shown in Fig. 20.11. The shape of the curves depends on the number of stages in the compressor. As more stages are added to the compressor, the curves become longer and flatter, approaching constant pressure head delivered for a given speed.

At low-volume flow rates, compressor operation can become unstable. If the flow rate is low, the pressure in the discharge pipe can become temporarily greater than the pressure delivered by the compressor, leading to momentary flow reversal. When the flow reverses, the outlet pressure drops and the compressor begins to deliver gas again. This causes a pulsation in outlet pressure and flow known as *surge* or *pumping*, which can cause damage to the compressor. Compressors are usually instrumented to prevent surge from occurring.

Some compressors are equipped with adjustable inlet guide vanes to further extend the range of operation by allowing the inlet flow to be throttled.

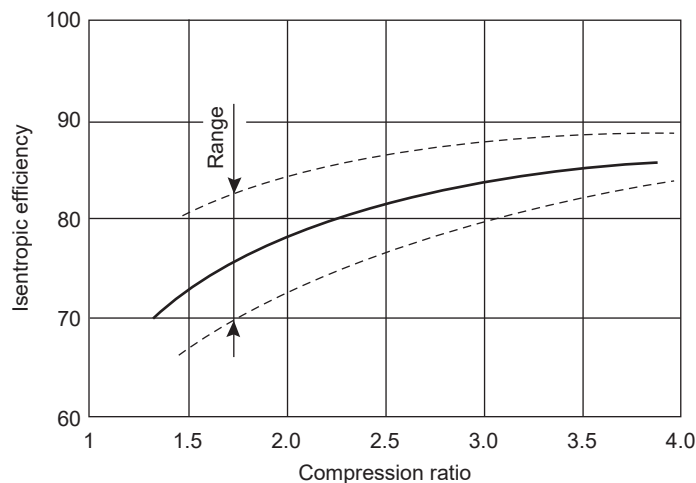


FIG. 20.9 Typical efficiencies for reciprocating compressors.

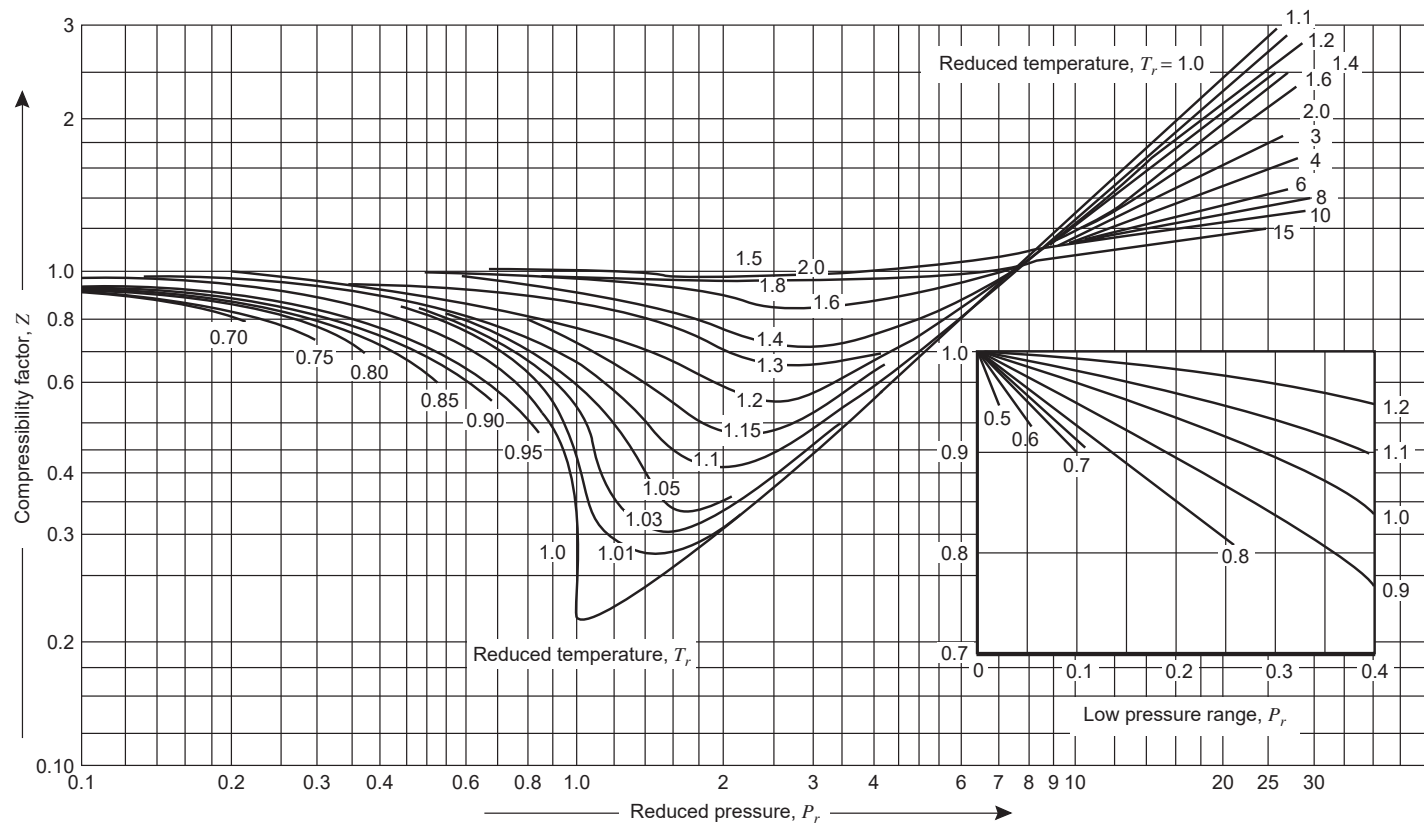


FIG. 20.10 Compressibility factors of gases and vapors.

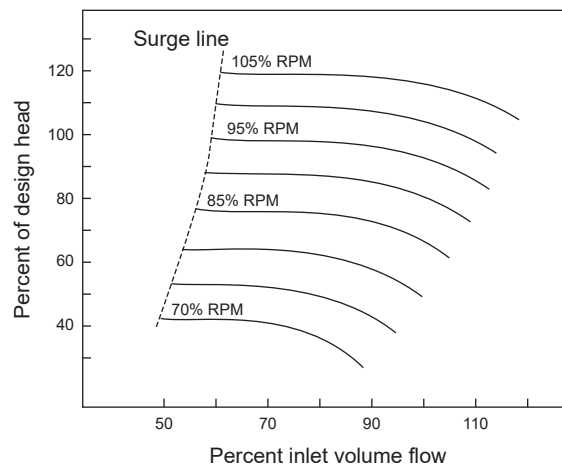


FIG. 20.11 Typical centrifugal compressor performance curves.

20.7 Pumping of liquids

The different types of pumps used for transporting liquids are introduced in [Section 20.3.2](#). This section describes the design and selection of centrifugal pumps in more detail. Centrifugal pumps are by far the most widely used type, as they are compact, inexpensive, robust, and available in a wide range of materials.

20.7.1 Centrifugal pump design

A centrifugal pump consists of a shaped impeller with curved radial vanes, which is housed inside a casing ([Fig. 20.12](#)). A shaft connects the impeller to a power source such as an electric motor or steam turbine, which causes the impeller to rotate at high speed. Fluid enters the casing in the axial direction, towards the center of the impeller, and is pushed out towards the edge of the casing by centrifugal force. The high speed of the impeller gives the fluid a high kinetic energy, which is then converted into higher pressure as the liquid decelerates to the tangential discharge pipe.

Different types and sizes of impellers can be used with the same pump. Specialized impellers are available for pumping slurries and suspensions; see [Nwaokocha \(2019\)](#). The pump casing is usually designed as a volute casing, which is a spiral that increases uniformly in cross-sectional area towards the discharge. Some pumps also incorporate fixed guide vanes or diffusers between the impeller and casing, which gives a more gradual change in direction to the fluid and leads to improved efficiency.

Centrifugal pumps are characterized by their specific speed. In the dimensionless form, specific speed is given by:

$$N_s = \frac{NQ^{1/2}}{(gh)^{3/4}} \quad (20.14)$$

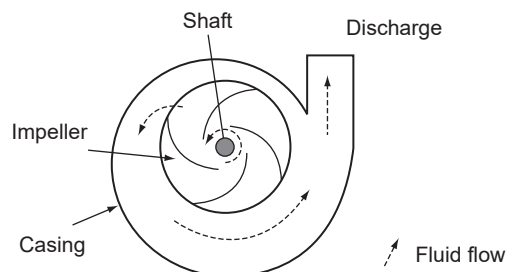


FIG. 20.12 Schematic diagram of a centrifugal pump.

where N = revolutions per second

Q = flow, m^3/s

h = head, m

g = gravitational acceleration m/s^2

Pump manufacturers do not generally use the dimensionless specific speed, but define the impeller specific speed by the equation:

$$N'_S = \frac{N'Q^{1/2}}{h^{3/4}} \quad (20.15)$$

where N' = revolutions per minute (rpm)

Q = flow, U.S. gal/min

h = head, ft

Values of the nondimensional specific speed, as defined by Equation 20.14, can be converted to the form defined by Equation 20.15 by multiplying by $1.72 \times 10^4 \text{ gal}^{0.5} \text{ min}^{-1.5} \text{ ft}^{-0.75}$.

The impeller specific speed for centrifugal pumps (Equation 20.15) usually lies between 400 and 20,000, depending on the type of impeller. Generally, pump impellers are classified as centrifugal or radial for specific speeds between 400 and 4000, mixed flow between 4000 and 9000, and axial above 9000 (Heald, 2002). Doolin (1977) states that below a specific speed of 1000, the efficiency of single-stage centrifugal pumps is low and multistage pumps should be considered.

20.7.2 Power requirements for pumping liquids

To transport a liquid from one vessel to another through a pipeline, energy has to be supplied to:

1. Overcome the friction losses in the pipes
2. Overcome the miscellaneous losses in the pipe fittings (e.g., bends), valves, instruments, etc.
3. Overcome the losses in process equipment (e.g., heat exchangers, packed beds)
4. Overcome any difference in elevation from end to end of the pipe
5. Overcome any difference in pressure between the vessels at each end of the pipeline

The total energy required can be calculated from the energy equation:

$$g\Delta z + \Delta P/\rho - \Delta P_f/\rho - W = 0 \quad (20.16)$$

where W = work done by the fluid, J/kg

Δz = difference in elevations ($z_1 - z_2$), m (Fig. 20.13)

ΔP = difference in system pressures ($P_1 - P_2$), N/m^2

ΔP_f = pressure drop due to friction, including miscellaneous losses and equipment losses (see Section 20.4), N/m^2

ρ = liquid density, kg/m^3

g = acceleration due to gravity, m/s^2

If W is negative, a pump is required; if it is positive, a turbine could be installed to extract energy from the system.

$$\text{The head required from the pump} = \Delta P_f/\rho g - \Delta P/\rho g - \Delta z \quad (20.17)$$

Pump shaft power

The power required is given by:

$$\text{Power} = (W \times m) \times 100/\eta_p, \text{ for a pump} \quad (20.18)$$

where m = mass flow rate, kg/s

η_p = pump efficiency, percent

For an incompressible fluid, the power required can also be written as:

$$\text{Power} = \frac{\Delta P Q}{\eta_p} \times 100 \quad (20.19)$$

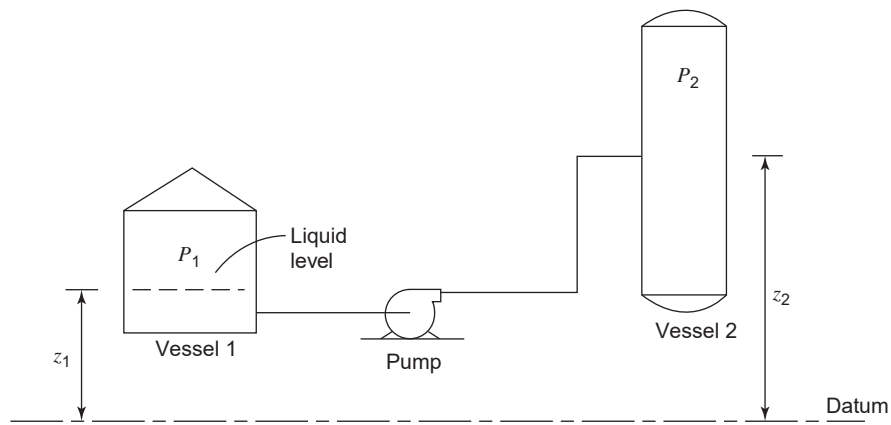


FIG. 20.13 Piping system.

where ΔP = pressure differential across the pump, N/m^2
 Q = flow rate, m^3/s

The efficiency of centrifugal pumps will depend on the size and type of pump used and the operating conditions. The values given in Fig. 20.14 can be used to estimate the power and energy requirements for preliminary design purposes. The efficiency of reciprocating pumps is usually around 90%.

When a pump is used as a hydraulic power recovery turbine, the work delivered is given by:

$$\text{Power delivered} = (W \times m) \times \eta_t \quad (20.20)$$

where η_t = turbine efficiency = power out/power in.

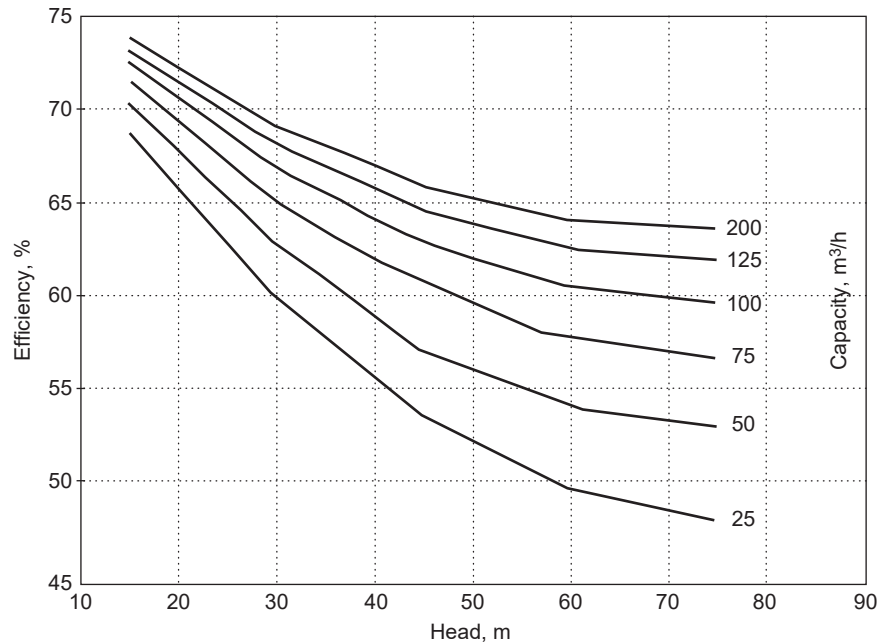


FIG. 20.14 Centrifugal pump efficiency.

Example 20.6

A tanker carrying toluene is unloaded, using the ship's pumps, to an on-shore storage tank. The pipeline is 225 mm internal diameter and 900 m long. Miscellaneous losses due to fittings, valves, etc., amount to 600 equivalent pipe diameters. The maximum liquid level in the storage tank is 30 m above the lowest level in the ship's tanks. The ship's tanks are nitrogen blanketed and maintained at a pressure of 1.05 bar. The storage tank has a floating roof, which exerts a pressure of 1.1 bar on the liquid.

The ship must unload 1000 metric tons within 5 hours to avoid demurrage charges. Estimate the power required by the pump. Take the pump efficiency as 70%.

Physical properties of toluene: density 874 kg/m^3 , viscosity $0.62 \text{ mNm}^{-2} \text{ s}$.

Solution

$$\text{Cross-sectional area of pipe} = \frac{\pi}{4}(225 \times 10^{-3})^2 = 0.0398 \text{ m}^2$$

$$\text{Minimum fluid velocity} = \frac{1000 \times 10^3}{5 \times 3600} \times \frac{1}{0.0398} \times \frac{1}{874} = 1.6 \text{ m/s}$$

$$\begin{aligned} \text{Reynolds number} &= (874 \times 1.6 \times 225 \times 10^{-3}) / 0.62 \times 10^{-3} \\ &= 507,484 = 5.1 \times 10^5 \end{aligned} \quad (20.2)$$

Absolute roughness commercial steel pipe, Table 20.3 = 0.046 mm

Relative roughness = $0.046/225 = 0.0002$

Friction factor from Fig. 20.5, $f = 0.0019$

Total length of pipeline, including miscellaneous losses:

$$= 900 + 600 \times 225 \times 10^{-3} = 1035 \text{ m}$$

Friction loss in pipeline,

$$\begin{aligned} \Delta P_f &= 8 \times 0.0019 \times \left(\frac{1035}{225 \times 10^{-3}} \right) \times 874 \times \frac{1.6^2}{2} \\ &= 78,221 \text{ N/m}^2 \end{aligned}$$

Maximum difference in elevation, $(z_1 - z_2) = (0 - 30) = \underline{-30 \text{ m}}$

Pressure difference, $(P_1 - P_2) = (1.05 - 1.1)10^5 = \underline{-5 \times 10^3 \text{ N/m}^2}$ (20.1)

Energy balance:

$$\begin{aligned} 9.8(-30) + (-5 \times 103)/874 - (78,221)/874 - W &= 0 \\ W &= \underline{-389.2 \text{ J/kg}} \end{aligned} \quad (20.16)$$

$$\text{Power} = (389.2 \times 55.56) / 0.7 = 30,981 \text{ W, say } \underline{31 \text{ kW}}. \quad (20.18)$$

Note that this is the maximum power required by the pump at the end of the unloading when the ship's tank is nearly empty and the storage tank is nearly full. Initially, the difference in elevation is lower and the power required is reduced. For design purposes, the maximum power case would be the governing case and would be used to size the pump and motor.

20.7.3 Characteristic curves for centrifugal pumps

The performance of a centrifugal pump is characterized by plotting the head developed against the flow rate. The pump efficiency can be shown on the same curve. A typical plot is shown in Fig. 20.15. The head developed by the pump falls as the flow rate is increased. The efficiency rises to a maximum and then falls.

For a given type and design of pump, the performance will depend on the impeller diameter, the pump speed, and the number of stages. Pump manufacturers publish families of operating curves for the range of pumps they sell, corresponding to different combinations of casing and impeller sizes. These can be used to select the best pump for a given duty. A typical set of curves is shown in Fig. 20.16.

Pump characteristic curves are usually plotted in terms of the head delivered by the pump instead of pressure difference. A centrifugal pump will deliver the same head for any liquid of viscosity less than about 50 mNs/m² (50 cP). The pressure rise delivered by the pump depends on the liquid density. The design engineer should check with the pump vendor when specifying a pump for a high-viscosity liquid, a non-Newtonian liquid, or a suspension. A good guide to reading pump curves is given by Kelly (2010).

20.7.4 Cavitation and net positive suction head (NPSH)

The pressure at the inlet to a pump must be high enough to prevent cavitation from occurring in the pump. Cavitation occurs when bubbles of vapor, or gas, form in the pump casing. Vapor bubbles will form if the pressure falls below the vapor pressure of the liquid. These bubbles subsequently collapse, creating localized shock waves that cause noise and can lead to damage to the pump.

The net positive suction head available ($NPSH_{avail}$) is the pressure at the pump suction, above the vapor pressure of the liquid, expressed as head of liquid.

The net positive head required ($NPSH_{reqd}$) is a function of the design parameters of the pump and will be specified by the pump manufacturer. As a general guide, the NPSH should be above 3 m for pump capacities up to 100 m³/h and 6 m above this capacity. Special impeller designs can be used to overcome problems of low suction head; see Doolin (1977). The NPSH required varies with flow rate and is sometimes indicated on pump characteristic curves.

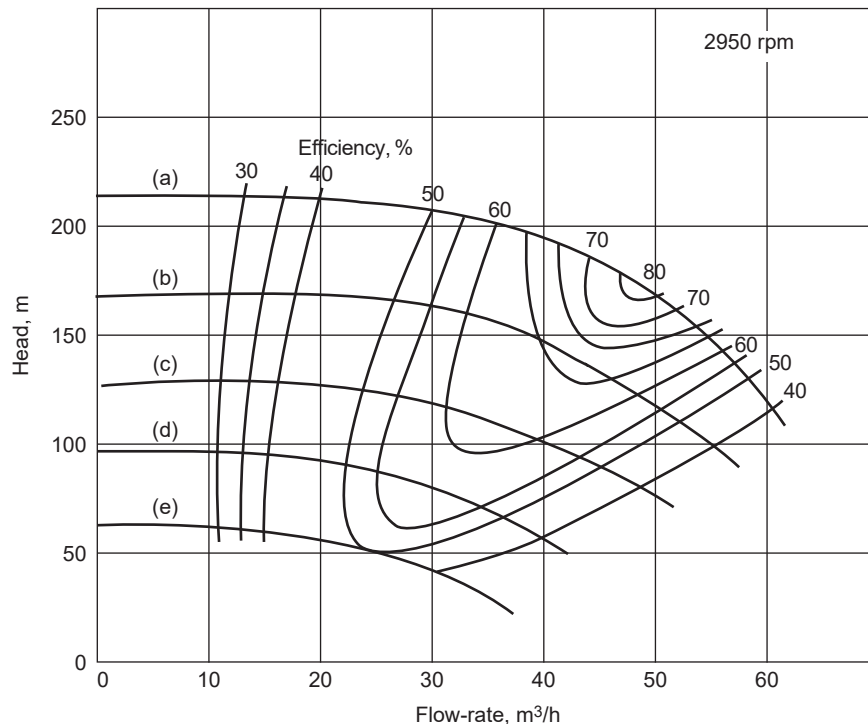


FIG. 20.15 Pump characteristics for a range of impeller sizes. (a) 250 mm. (b) 225 mm. (c) 200 mm. (d) 175 mm. (e) 150 mm.

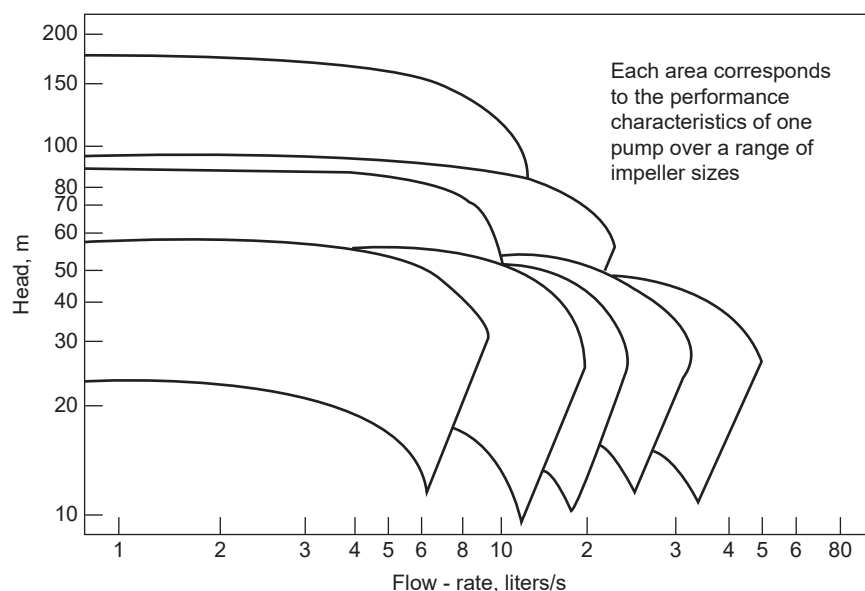


FIG. 20.16 Family of pump curves.

The net positive head available is given by the following equation:

$$NPSH_{avail} = P/\rho \cdot g + H - P_f/\rho \cdot g - P_v/\rho \cdot g \quad (20.21)$$

where $NPSH_{avail}$ = net positive suction head available at the pump suction, m

P = the pressure above the liquid in the feed vessel, N/m^2

H = the height of liquid above the pump suction, m

P_f = the pressure loss in the suction piping, N/m^2

P_v = the vapor pressure of the liquid at the pump suction, N/m^2

ρ = the density of the liquid at the pump suction temperature, kg/m^3

g = the acceleration due to gravity, m/s^2

The inlet piping arrangement must be designed to ensure that $NPSH_{avail}$ exceeds $NPSH_{reqd}$ under all operating conditions.

The calculation of $NPSH_{avail}$ is illustrated in Example 20.7.

Example 20.7

Liquid chlorine is unloaded from rail tankers into a storage vessel. To provide the necessary NPSH, the transfer pump is placed in a pit below ground level. Given the following information, calculate the NPSH available at the inlet to the pump at a maximum flow rate of 16,000 kg/h.

The total length of the pipeline from the rail tanker outlet to the pump inlet is 50 m. The vertical distance from the tank outlet to the pump inlet is 10 m. Commercial steel piping, 50 mm internal diameter, is used.

Miscellaneous friction losses due to the tanker outlet constriction and the pipe fittings in the inlet piping are equal to 1000 equivalent pipe diameters. The vapor pressure of chlorine at the maximum temperature reached at the pump is 685 kN/m² and its density and viscosity, 1286 kg/m³ and 0.364 mNm⁻² s. The pressure in the tanker is 7 bara.

Solution**Friction losses**

$$\text{Miscellaneous losses} = 1000 \times 50 \times 10^{-3} = 50 \text{ m of pipe}$$

$$\text{Total length of inlet piping} = 50 + 50 = 100 \text{ m}$$

$$\text{Relative roughness, } e/d = 0.046/50 = 0.001$$

$$\text{Pipe cross-sectional area} = \frac{\pi}{4}(50 \times 10^{-3})^2 = 1.96 \times 10^{-3} \text{ m}^2 \quad (20.2)$$

$$\text{Velocity, } u = \frac{16,000}{3600} \times \frac{1}{1.96 \times 10^{-3}} \times \frac{1}{1286} = 1.76 \text{ m/s}$$

$$\text{Reynolds number} = \frac{1286 \times 1.76 \times 50 \times 10^{-3}}{0.364 \times 10^{-3}} = 3.1 \times 10^5$$

Friction factor from Fig. 20.5, $f = 0.00225$:

$$\Delta P_f = 8 \times 0.00225 \frac{(100)}{(50 \times 10^{-3})} \times 1286 \times \frac{1.76^2}{2} = 71,703 \text{ N/m}^2 \quad (20.1)$$

$$\begin{aligned} NPSH &= \frac{7 \times 10^5}{1286 \times 9.8} + 10 - \frac{71.703}{1286 \times 9.8} - \frac{685 \times 10^3}{1286 \times 9.8} \\ &= 55.5 + 10 - 5.7 - 54.4 = \underline{\underline{5.4 \text{ m}}} \end{aligned} \quad (20.21)$$

20.7.5 System curve (operating line)

There are two components to the pressure head that have to be supplied by the pump in a piping system:

1. The static pressure, to overcome the differences in head (height) and pressure
2. The dynamic loss due to friction in the pipe, the miscellaneous losses, and the pressure loss through equipment

The static pressure difference will be independent of the fluid flow rate. The dynamic loss will increase as the flow rate is increased. It will be roughly proportional to the flow rate squared; see Equation 20.1. The system curve, or operating line, is a plot of the total pressure head versus the liquid flow rate. The operating point of a centrifugal pump can be found by plotting the system curve on the pump's characteristic curve; see Example 20.8. The operating point is the point where the system curve and pump curve intersect.

When selecting a centrifugal pump for a given duty, it is important to match the pump characteristic with the system curve. The operating point should be as close as is practical to the point of maximum pump efficiency, allowing for the range of flow rate over which the pump may be required to operate. This requires a good understanding of the pressure drop across the control valve if a valve is used in the line downstream of the pump. The control valve pressure drop will be proportional to the velocity squared and can range from as low as 6 to over 100 velocity heads over the range of operation of the valve, depending on the type of valve chosen (see Table 20.4). The system curve should be plotted for the case when the valve is fully open and the case when the valve is quarter-open (or at the minimum fraction open recommended by the valve manufacturer) to determine the range over which flow can be controlled with a given combination of valve and pump. Details of valve pressure drop can be obtained from manufacturers.

Most centrifugal pumps are controlled by throttling the flow with a valve on the pump discharge. This varies the dynamic pressure loss, and thus the position of the operating point on the pump characteristic curve. Throttling the flow causes an energy loss. This energy loss is acceptable in most applications; however, when the flow rates are large, the use of variable-speed control on the pump drive should be considered as a means of conserving energy.

Methods for calculating the pressure drop through pipes and fittings are given in Section 20.4. It is important that a proper analysis is made of the system, and the use of a calculation form (worksheet) to standardize pump-head calculations is recommended. A standard calculation form ensures that a systematic method of calculation is used and provides a checklist to ensure that all the usual factors have been considered. It is also a permanent record of the calculation. A template for a standard pump and line calculation is given in Appendix G and can be downloaded in Microsoft Excel format from www.elsevier.com/books-and-journals/book-companion/9780128211793. Example 20.12 has been set out using this calculation form. The calculation should include a check on the NPSH available; see Section 20.7.4.

A more detailed discussion of the operating characteristics of centrifugal and other types of pumps is given by Couper et al. (2012) and Karassik et al. (2007).

Example 20.8

A process liquid is pumped from a storage tank to a distillation column using a centrifugal pump. The pipeline is 80 mm internal diameter commercial steel pipe, 100 m long. Miscellaneous losses are equivalent to 600 pipe diameters. The storage tank operates at atmospheric pressure and the column at 1.7 bara. The lowest liquid level in the tank will be 1.5 m above the pump inlet, and the feed point to the column is 3 m above the pump inlet.

Plot the system curve on the pump characteristic given in Fig. 20.17 and determine the operating point and pump efficiency.

Properties of the fluid: density 900 kg/m^3 , viscosity $1.36 \text{ mN m}^{-2}\text{s}$.

Solution

Static head

$$\text{Difference in elevation, } \Delta z = 3.0 - 1.5 = 1.5 \text{ m}$$

$$\text{Difference in pressure, } \Delta P = (1.7 - 1.013)10^5 = 0.7 \times 10^5 \text{ N/m}^2$$

$$\text{as head of liquid} = (0.7 \times 10^5) / (900 \times 9.8) = 7.9 \text{ m}$$

$$\text{Total static head} = 1.5 + 7.9 = \underline{9.4 \text{ m}}$$

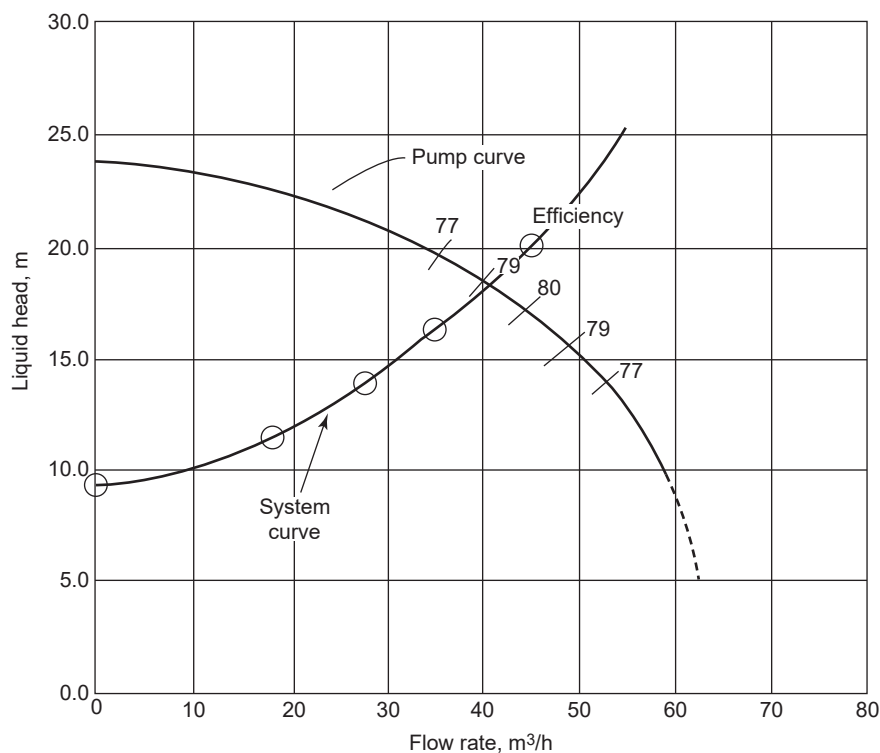


FIG. 20.17 Example 20.8.

Dynamic head

As an initial value, take the fluid velocity as 1 m/s, a reasonable value.

$$\text{Cross-sectional area of pipe} = \frac{\pi}{4}(80 \times 10^{-3})^2 = 5.03 \times 10^{-3} \text{ m}^2$$

$$\text{Volumetric flow rate} = 1 \times 5.03 \times 10^{-3} \times 3600 = 18.1 \text{ m}^3/\text{h} \quad (20.2)$$

$$\text{Reynolds number} = \frac{900 \times 1 \times 80 \times 10^{-3}}{1.36 \times 10^{-3}} = 5.3 \times 10^4$$

$$\text{Relative roughness} = 0.046/80 = 0.0006$$

$$\text{Friction factor from Fig. 20.5, } f = 0.0027$$

$$\text{Length including miscellaneous losses} = 100 + (600 \times 80 \times 10^{-3}) = 148 \text{ m}$$

$$\begin{aligned} \text{Pressure drop, } \Delta P_f &= 8 \times 0.0027 \frac{(148)}{(80 \times 10^{-3})} \times 900 \times \frac{1^2}{2} = \underline{\underline{17,982 \text{ N/m}^2}} \\ &= 17,982/(900 \times 9.8) = \underline{\underline{2.03 \text{ m liquid}}} \end{aligned} \quad (20.1)$$

$$\text{Total head} = 9.4 + 2.03 = 11.4 \text{ m}$$

To find the system curve, the calculations were repeated for the velocities shown in the following table:

Velocity m/s	Flow rate m ³ /h	Static head m	Dynamic head m	Total head m
1	18.1	9.4	2.0	11.4
1.5	27.2	9.4	4.3	14.0
2.0	36.2	9.4	6.8	16.2
2.5	45.3	9.4	10.7	20.1
3.0	54.3	9.4	15.2	24.6

Plotting these values on the pump characteristic gives the operating point as 18.5 m at 41 m³/h and the pump efficiency as 79%.

20.7.6 Pump and other shaft seals

A seal must be made where a rotating shaft passes through the casing of a pump or the wall of a vessel. The seal must serve several functions:

1. To keep the liquid contained
2. To prevent ingress of incompatible fluids, such as air
3. To prevent escape of flammable or toxic materials

Packed glands

The simplest, and oldest, form of seal is the packed gland, or stuffing box (Fig. 20.18). Its applications range from sealing the stems of the water faucets in every home to providing the seal on industrial pump, agitator, and valve shafts.

The shaft runs through a housing (gland), and the space between the shaft and the wall of the housing is filled with rings of packing. A gland follower is used to apply pressure to the packing to ensure that the seal is tight. Proprietary packing materials are used. A summary of the factors to be considered in the selection of packing materials for packed glands is given by Hoyle (1978). To make a completely tight seal, the pressure on the packing must be two to three times the system pressure. This can lead to excessive wear on rotating shafts, and lower pressures are used, allowing some leakage, which lubricates the packing. Packed glands should therefore only be specified for fluids that are not toxic, corrosive, or flammable.

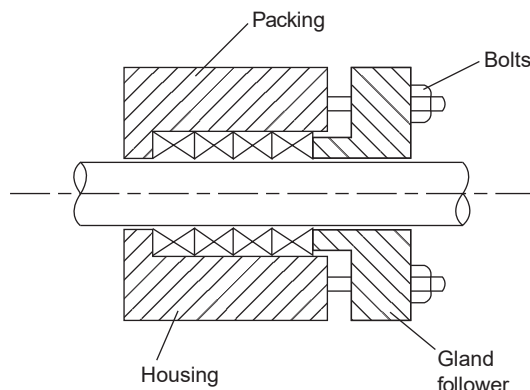


FIG. 20.18 Packed gland.

To provide positive lubrication, a lantern ring is often incorporated in the packing and lubricant forced through the ring into the packing (Fig. 20.19). With a pump seal, a flush is often taken from the pump discharge and returned to the seal, through the lantern ring, to lubricate and cool the packing. If any leakage to the environment must be avoided, a separate flush liquid can be used. A liquid must be selected that is compatible with the process fluid and the environment; water is often used.

Mechanical seals

In the process industries the conditions at the pump seal are often harsh, and more complex seals are needed. Mechanical face seals are used (Fig. 20.20). They are generally referred to simply as mechanical seals and are used only on rotating shafts.

The seal is formed between two flat faces set perpendicular to the shaft. One face rotates with the shaft; the other is stationary. The seal is made, and the faces lubricated, by a very thin film of liquid, about $0.0001\ \mu\text{m}$ thick. A particular advantage of this type of seal is that it can provide a very effective seal without causing any wear on the shaft. The wear is transferred to the special seal faces. Some leakage will occur, but it is small, normally only a few drops per hour. Unlike a packed gland, a mechanical seal, when correctly installed and maintained, can be considered leak-tight.

A great variety of mechanical seal designs are available, and seals can be found to suit virtually all applications. Only the basic mechanical seal is described here. Full details and specifications of the range of seals available and their applications can be obtained from manufacturers' catalogs.

The basic mechanical seal

The components of a mechanical seal (see Fig. 20.20) are:

1. A stationary sealing ring (mating ring)
2. A seal for the stationary ring, usually O-rings or gaskets
3. A rotating seal ring (primary ring), mounted so that it can slide along the shaft to take up wear in the seal faces
4. A secondary seal for the rotating ring mount, usually O-rings or chevron seals
5. A spring to maintain contact pressure between the seal faces to push the faces together
6. A thrust support for the spring, either a collar keyed to the shaft or a step in the shaft

The assembled seal is fitted into a gland housing (stuffing box) and held in place by a retaining ring (gland plate).

Mechanical seals are classified as inside or outside, depending on whether the primary (rotating ring) is located inside the housing, running in the fluid, or outside. Outside seals are easier to maintain, but inside seals are more commonly used in the process industries, as it is easier to lubricate and flush this type. Fig. 20.20 illustrates an inside seal.

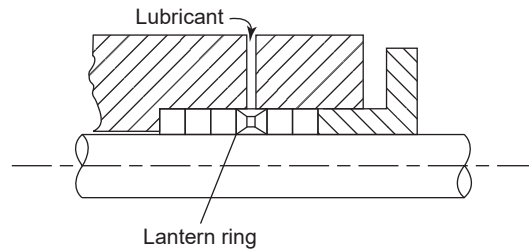


FIG. 20.19 Packed gland with lantern ring.

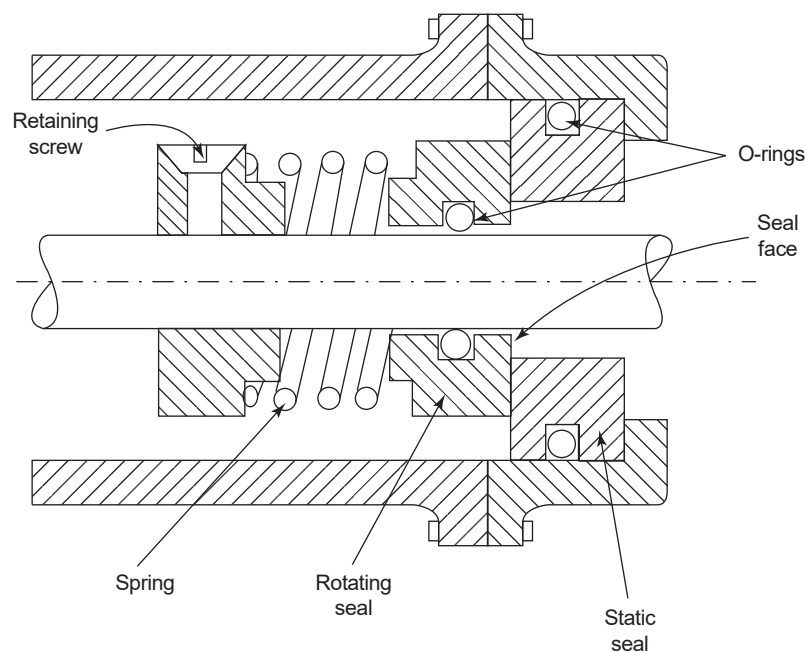


FIG. 20.20 Basic mechanical seal.

Double seals

Where it is necessary to prevent any leakage of fluid to the atmosphere, a double mechanical seal is used. The space between the two seals is flushed with a harmless fluid, compatible with the process fluid, and provides a buffer between the two seals.

Seal-less pumps (canned pumps)

Pumps that have no seal on the shaft between the pump and the drive motor are available. They are used for severe duties, where it is essential that there is no leakage into the process fluid or the environment.

The drive motor and pump are enclosed in a single casing, and the stator windings and armature are protected by metal cans; they are usually referred to as canned pumps. The motor runs in the process fluid. The use of canned pumps to control environmental pollution is discussed by Webster (1979).

20.8 Selection of drivers for rotating equipment

A mechanical driver is required to provide the shaft work to operate a pump or compressor. Electric motors are used for most small pumps, fans, and compressors, as well as most positive displacement pumps. Large compressors and pumps are sometimes powered by steam turbines. Large reciprocating compressors are occasionally powered by stand-alone diesel engines. Some axial compressors are powered by integral shaft gas turbine engines.

20.8.1 Electric motors as drivers

Alternating current electric motors are the most widely used drivers for pumps, compressors, agitators, conveyors, and most other process equipment that requires a mechanical drive. Constant-speed motors are most common, but variable-speed drivers are gaining in popularity, as they are more energy efficient. Electric motors are commercially available up to 10 MW, but special electrical infrastructure is needed for the larger sizes.

The electric power required to drive a compressor or pump can be calculated using the motor efficiency:

$$\text{Power} = \frac{-W \times m}{E_e} \quad (20.22)$$

where $-W$ = work of compression per unit mass

m = mass flow rate

E_e = electric motor efficiency

The efficiency of the drive motor will depend on the type, speed, and size. The values given in Table 20.5 can be used to make a rough estimate of the power required.

20.8.2 Steam turbines as drivers

Steam turbines are used as drivers when the process equipment requires a moderately high shaft work input, between about 1 MW and 20 MW, and when sufficient process steam is available. The use of steam turbines allows conversion of process waste heat to mechanical work, which reduces electricity costs and demands on the site electrical infrastructure.

Three main types of steam turbines are used:

- **Condensing turbines:** The downstream pressure is set by a condenser that condenses all of the steam, giving a high pressure ratio and high power per unit flow.
- **Noncondensing or back-pressure turbines:** The outlet pressure is controlled at a level above atmospheric pressure, usually equal to the pressure of one of the levels of the steam system so that the discharge steam can be used for process heating.
- **Extraction turbines:** Part of the steam is extracted at intermediate pressure after one or more stages of expansion, while the remainder is expanded to a condenser.

Extraction turbines allow some flexibility in meeting variations in process power and heat requirements.

The steam consumption of a steam turbine driver can be calculated by energy balance between the steam inlet and outlet conditions. Specific enthalpies of steam and condensate are given in steam tables. This calculation was illustrated in Example 3.2.

Steam turbine efficiency varies from about 60% for smaller turbines to about 85% for the larger machines.

TABLE 20.5 Approximate efficiencies of electric motors

Size (kW)	Efficiency (%)
5	80
15	85
75	90
200	92
750	95
>4000	97

20.9 Mechanical design of piping systems

20.9.1 Piping system design codes

The most widely used international design codes for pressure piping are those set by the ASME B31 Committee. Different standards are required for different services, as shown in [Table 20.6](#). Most chemical plant and oil refinery piping is designed in accordance with ASME B31.3, which will be used as a reference in the following sections. The ASME B31.3 code applies to piping for raw, intermediate, and finished chemicals; petroleum products; gas, steam, air, and water; fluidized solids; refrigerants; and cryogenic fluids. It does not apply to:

1. Piping systems designed for pressures less than 15 psi gauge that handle fluids that are nonflammable, nontoxic, and not damaging to human tissues and are at temperatures between -29°C (-20°F) and 186°C (366°F).
2. Power boiler piping that conforms to ASME B31.1 and power boilers that conform to ASME Boiler and Pressure Vessel Code Section I.
3. Fired heater tubes, tube headers, and manifolds that are internal to the heater enclosure.
4. Internal piping and external pipe connections of pressure vessels, heat exchangers, pumps, compressors, and other fluid-handling or process equipment.

Note, though, that different design standards apply for refrigeration plants, fuel gas piping, hydrogen pipelines, power plant, and slurry-handling systems.

20.9.2 Wall thickness: Pipe schedule

The pipe wall thickness is selected to resist the internal pressure, with allowances for corrosion, erosion, and other mechanical allowances for pipe threads, etc. Process pipes can normally be considered thin cylinders; only high-pressure pipes, such as high-pressure steam lines, are likely to be classified as thick cylinders and must be given special consideration (see [Chapter 14](#)).

The ASME B31.3 code gives the following formula for pipe thickness:

$$t_m = t_p + c \quad (20.23)$$

$$t_p = \frac{P d}{2(SE + P\gamma_T)}$$

where t_m = minimum required thickness

TABLE 20.6 ASME pipe codes

Code no.	Scope	Latest revision
B31.1	Power piping	2018
B31.2	Fuel gas piping (withdrawn)	1968
B31.3	Process piping	2018
B31.4	Pipeline transportation systems for liquid hydrocarbons and other liquids	2019
B31.5	Refrigeration piping and heat transfer components	2016
B31.8	Gas transmission and distribution piping systems	2018
B31.9	Building services piping	2017
B31.11	Slurry transportation piping systems (superseded by ASME B31.4)	2002
B31.12	Hydrogen piping and pipelines	2014

t_p = pressure design thickness

c = sum of mechanical allowances (thread depth) plus corrosion and erosion allowances

P = internal design gauge pressure, lb/in² (or N/mm²)

d = pipe outside diameter

S = basic allowable stress for pipe material, lb/in² (or N/mm²)

E = casting quality factor

γ_T = temperature coefficient

Allowable stresses and values of the coefficients for different materials are given in Appendix A of the design code. Standard dimensions for stainless steel pipe are given in ASME B36.19M and for wrought steel and wrought iron pipe in ASME B36.10M. Standard pipe dimensions are also summarized by [Green and Southard \(2018\)](#).

Pipes are often specified by a schedule number (based on the thin cylinder formula). The schedule number is defined by:

$$\text{Schedule number} = \frac{P_s \times 1000}{\sigma_s} \quad (20.24)$$

P_s = safe working pressure, lb/in² (or N/mm²)

σ_s = safe working stress, lb/in² (or N/mm²)

Schedule 40 pipe is commonly used for general-purpose applications at low pressures (below 50 bar).

Example 20.9

Estimate the safe working pressure for a 4 in. (100 mm) dia., schedule 40 pipe, SA53 carbon steel, butt welded, working temperature 100 °C. The maximum allowable stress for butt welded steel pipe up to 120 °C is 11,700 lb/in² (79.6 N/mm²).

Solution

$$P_s = \frac{(\text{schedule no.}) \times \sigma_s}{1000} = \frac{40 \times 11700}{1000} = \underline{\underline{468 \text{ lb/in}^2}} = \underline{\underline{3180 \text{ kN/m}^2}}$$

20.9.3 Pipe supports

Over long runs, between buildings and equipment, pipes are usually carried on pipe racks. These carry the main process and service pipes and are laid out to allow easy access to the equipment.

Various designs of pipe hangers and supports are used to support individual pipes. Details of typical supports can be found in the books by [Green and Southard \(2018\)](#) and [Nayyar \(2000\)](#). Pipe supports should incorporate provision for thermal expansion.

20.9.4 Pipe fittings

Pipe runs are normally made up of lengths of pipe incorporating standard fittings for joints, bends, and tees. Joints are usually welded, but small sizes may be screwed. Flanged joints are used where this is a more convenient method of assembly or if the joint will have to be frequently broken for maintenance. Flanged joints are normally used for the final connection to the process equipment, valves, and ancillary equipment.

Details of the standard pipe fittings, welded, screwed, and flanged, can be found in manufacturer's catalogues and in the appropriate national standards. Standards for pipe fittings are set by the ASME B16 committee. The standards for metal pipes and fittings are discussed by [Masek \(1968\)](#). Flanges and flange standards are discussed in [Section 14.10](#).

20.9.5 Pipe stressing

Piping systems must be designed so as not to impose unacceptable stresses on the equipment to which they are connected.

Loads arise from:

1. Thermal expansion of the pipes and equipment
2. The weight of the pipes, their contents, insulation, and any ancillary equipment
3. The reaction to the fluid pressure drop
4. Loads imposed by the operation of ancillary equipment, such as relief valves
5. Vibration

Thermal expansion is a major factor to be considered in the design of piping systems. The reaction load due to pressure drop will normally be negligible. The dead-weight loads can be carried by properly designed supports.

Flexibility is incorporated into piping systems to absorb the thermal expansion. A piping system will have a certain amount of flexibility due to the bends and loops required by the layout. If necessary, expansion loops, bellows, and other special expansion devices can be used to take up expansion.

A discussion of the methods used for the calculation of piping flexibility and stress analysis is beyond the scope of this book. Manual calculation techniques, and the application of computers in piping stress analysis, are discussed in the handbook edited by [Nayyar \(2000\)](#).

20.9.6 Layout and design

An extensive discussion of the techniques used for piping system design and specification is beyond the scope of this book. The subject is covered thoroughly in the books by [Kentish \(1982a, 1982b\)](#), [Lamit \(1981\)](#), [Parisher \(2011\)](#), [Whistance and Sherwood \(2009\)](#), [Silowash \(2009\)](#), and [Beale and Bowers \(2017\)](#). The piping layout should generally attempt to make connections between vessels using the shortest length of pipe with the fewest bends possible, subject to the constraints of supporting the pipe, allowing for thermal expansion and providing open access for operators and maintenance.

20.10 Pipe size selection

If the motive power to drive the fluid through the pipe is available free, for instance when pressure is let down from one vessel to another or if there is sufficient head for gravity flow, the smallest pipe diameter that gives the required flow rate would normally be used. If the fluid has to be pumped through the pipe, the size should be selected to give the smallest total annualized cost.

Typical pipe velocities and allowable pressure drops that can be used to estimate pipe sizes are given here:

	Velocity m/s	ΔP kPa/m
Liquids, pumped (not viscous)	1–3	0.5
Liquids, gravity flow	—	0.05
Gases and vapors	15–30	0.02% of line pressure
High-pressure steam, >8 bar	30–60	—

[Rase \(1953\)](#) gives expressions for design velocities in terms of the pipe diameter. His expressions, converted to SI units, are:

Pump discharge	$0.06d_i + 0.4$ m/s
Pump suction	$0.02d_i + 0.1$ m/s
Steam or vapor	$0.2d_i$ m/s

where d_i is the internal diameter in mm.

[Simpson \(1968\)](#) gives values for the optimum velocity in terms of the fluid density. His values, converted to SI units and rounded, are:

Fluid density kg/m ³	Velocity m/s
1600	2.4
800	3.0
160	4.9
16	9.4
0.16	18.0
0.016	34.0

The maximum velocity should be kept below that at which erosion is likely to occur. For gases and vapors, the velocity cannot exceed the critical velocity (sonic velocity) (see [Section 10.9.3](#) and [Chhabra & Shankar \(2017\)](#)) and would normally be limited to 30% of the critical velocity. [Kern \(1975\)](#) discusses the practical design of pump suction piping in a series of articles on the practical aspects of piping system design published in the journal *Chemical Engineering* from December 1973 to November 1975. A detailed presentation of pipe-sizing techniques is also given by [Simpson \(1968\)](#), who covers liquid, gas, and two-phase systems.

Economic pipe diameter

The capital cost of a pipe run increases with diameter, whereas the pumping costs decrease with increasing diameter. The most economic pipe diameter will be the one that gives the lowest total annualized cost.

A rule of thumb for the economic pipe diameter that is widely used in oil refining is:

$$\text{Economic diameter in inches} = (\text{flow rate in gpm})^{0.5}$$

In metric units this converts to:

$$d_i, \text{ optimum} = 0.33(m/\rho)^{0.5}$$

where m = mass flow rate, kg/s

ρ = density, kg/m³

d_i = pipe inside diameter, m

The formulae developed in this section are presented as an illustration of a simple optimization problem in design and to provide an estimate of economic pipe diameter in SI units. The method used is essentially that first published by [Genereaux \(1937\)](#).

The cost equations can be developed by considering a 1-meter length of pipe.

The purchase cost will be roughly proportional to the diameter raised to some power.

$$\text{Purchase cost} = Bd^n \$/\text{m}$$

The value of the constant B and the index n depend on the pipe material and schedule.

The installed cost can be calculated by using the factorial method of costing discussed in [Chapter 7](#).

$$\text{Installed cost} = Bd^n(1+F)$$

where the factor F includes the cost of valves, fittings, and erection for a typical run of the pipe.

The capital cost can be included in the operating cost as an annual capital charge. There will also be an annual charge for maintenance, based on the capital cost.

$$C_C = Bd^n(1+F)(a+b) \quad (20.25)$$

where C_C = annualized capital cost of the piping, \$/m.yr

a = capital annualization factor, yr⁻¹

b = maintenance costs as fraction of installed capital, yr⁻¹

The power required for pumping is given by:

$$\text{Power} = \text{volumetric flow rate} \times \text{pressure drop}$$

Only the friction pressure drop need be considered, as any static head is not a function of the pipe diameter.

To calculate the pressure drop, the pipe friction factor must be known. This is a function of Reynolds number, which is in turn a function of the pipe diameter. Several expressions have been proposed for relating friction factor to Reynolds number. For simplicity, the relationship proposed by [Genereaux \(1937\)](#) for turbulent flow in clean commercial steel pipes will be used.

$$C_f = 0.04 \text{Re}^{-0.16}$$

where C_f is the Fanning friction factor $= 2(R/\rho u^2)$.

Substituting this into the Fanning pressure drop equation gives:

$$\Delta P = 0.125 m^{1.84} \mu^{0.16} \rho^{-1} d_i^{-4.84} \quad (20.26)$$

where ΔP = pressure drop, N/m² (Pa)

μ = viscosity, Nm⁻² s

The annual pumping costs will be given by:

$$C_w = \frac{Ap}{1000\eta} \Delta P \frac{m}{\rho} \quad (20.27)$$

where A = plant attainment, hours/year

p = cost of power, \$/kWh

η = pump efficiency

Substituting from [Equation 20.26](#):

$$C_w = \frac{Ap}{\eta} 1.25 \times 10^{-4} m^{2.84} \mu^{0.16} \rho^{-2} d_i^{-4.84} \quad (20.28)$$

The total annual operating cost $C_t = C_C + C_w$

Adding [Equations 20.25 and 20.28](#), differentiating, and equating to zero to find the pipe diameter to give the minimum cost gives:

$$d_{i, \text{ optimum}} = \left(\frac{6.05 \times 10^{-4} A p m^{2.84} \mu^{0.16} \rho^{-2}}{\eta n B (1 + F)(a + b)} \right)^{1/(4.84 + n)} \quad (20.29)$$

[Equation 20.29](#) is a general equation and can be used to estimate the economic pipe diameter for any particular situation. It can be set up on a spreadsheet and the effect of the various factors investigated.

The equation can be simplified by substituting typical values for the constants.

- A The normal attainment for a chemical process plant will be between 90% and 95%, so take the operating hours per year as 8000.
- η Pump and compressor efficiencies will be between 50% and 70%, so take 0.6.
- p A typical wholesale cost of electric power for a large user is 0.06 \$/kWh (mid-2011).
- F This is the most difficult factor to estimate. Other authors have used values ranging from 1.5 ([Peters & Timmerhaus, 1968](#)) to 6.75 ([Nolte, 1978](#)). It is best taken as a function of the pipe diameter, as has been done to derive the simplified equations given later.
- B, n Can be estimated from the current cost of piping.
- a Will depend on the current cost of capital and could range from 0.1 to 0.25, but is typically around 0.16. See [Chapter 9](#) for a detailed discussion.
- b A typical figure for a process plant will be 5%; see [Chapter 8](#).

F , B , and n are best established from a recent correlation of piping costs and should include the costs of fittings, paint or insulation, and installation. The following correlations were derived in January 2006 and can be updated by multiplying by the Chemical Engineering piping cost index, which is published monthly in *Chemical Engineering*. The value of the Chemical Engineering piping index in January 2006 was 655.9, and in January 2010 was 794.5, so the correlations can be updated to a January 2010 basis by multiplying by $794.5/655.9 = 1.21$. Note that the cost index is specific to the costs of piping, valves, and fittings and is not the same as the Chemical Engineering plant cost index.

A106 Carbon steel

$$1 \text{ to } 8 \text{ inches, } \$/\text{ft} = 17.4 (d_i \text{ in inches})^{0.74}$$

$$10 \text{ to } 24 \text{ inches, } \$/\text{ft} = 1.03 (d_i \text{ in inches})^{1.73}$$

304 Stainless steel

$$1 \text{ to } 8 \text{ inches, } \$/\text{ft} = 24.5 (d_i \text{ in inches})^{0.9}$$

$$10 \text{ to } 24 \text{ inches, } \$/\text{ft} = 2.74 (d_i \text{ in inches})^{1.7}$$

In metric units (d_i in meters), these become:

A106 Carbon steel

$$25 \text{ to } 200 \text{ mm, } \$/\text{m} = 880 d_i^{0.74}$$

$$250 \text{ to } 600\text{mm, } \$/\text{m} = 1900 d_i^{1.73}$$

304 Stainless steel

$$25 \text{ to } 200 \text{ mm, } \$/\text{m} = 2200 d_i^{0.94}$$

$$250 \text{ to } 600\text{mm, } \$/\text{m} = 4700 d_i^{1.7}$$

For small-diameter carbon steel pipes, substitution in Equation 20.29 gives:

$$d_i, \text{ optimum} = 0.830 m^{0.51} \mu^{0.03} \rho^{-0.36}$$

Because the exponent of the viscosity term is small, its value will change very little over a wide range of viscosity

$$\mu = 10^{-5} \text{ Nm}^{-2}\text{s (0.01 cp)}, \mu^{0.03} = 0.71$$

at

$$\mu = 10^{-2} \text{ Nm}^{-2}\text{s (10 cp)}, \mu^{0.03} = 0.88$$

Taking a mean value of 0.8 gives the following equations for the optimum diameter for turbulent flow:

A106 Carbon steel pipe:

$$25 \text{ to } 200 \text{ mm, } d_i, \text{ optimum} = 0.664 m^{0.51} \rho^{-0.36} \quad (20.30)$$

$$250 \text{ to } 600 \text{ mm, } d_i, \text{ optimum} = 0.534 m^{0.43} \rho^{-0.30}$$

304 Stainless steel pipe:

$$25 \text{ to } 200 \text{ mm, } d_i, \text{ optimum} = 0.550 m^{0.49} \rho^{-0.35} \quad (20.31)$$

$$250 \text{ to } 600 \text{ mm, } d_i, \text{ optimum} = 0.465 m^{0.43} \rho^{-0.31}$$

Note that the optimum diameter for stainless steel is smaller than for carbon steel, as would be expected given the higher materials cost of the pipe. Note also that Equations 20.30 and 20.31 predict optimum pipe diameters that are roughly double those given by the rule of thumb at the start of this section. This most likely reflects a change in the relative values of capital and energy since the period when the rule of thumb was deduced.

Equations 20.30 and 20.31 can be used to make an approximate estimate of the economic pipe diameter for normal pipe runs. For a more accurate estimate, or if the fluid or pipe run is unusual, the method used to develop Equation 20.29 can be used, taking into account the special features of the particular pipe run.

For very long pipe systems, such as transportation pipelines, the capital costs of the required pumps should also be included. For gases, the capital cost of compression is much more significant and should always be included in the analysis.

Equations for the optimum pipe diameter with laminar flow can be developed by using a suitable equation for pressure drop in the equation for pumping costs.

The approximate equations should not be used for steam, as the quality of steam depends on its pressure, and hence the pressure drop.

Nolte (1978) gives detailed methods for the selection of economic pipe diameters, taking into account all the factors involved. He gives equations for liquids, gases, steam, and two-phase systems. He includes in his method an allowance for the pressure drop due to fittings and valves, which was neglected in the development of Equation 20.26 and by most other authors.

The use of Equations 20.30 and 20.31 are illustrated in Examples 20.10 and 20.11 and the results compared with those obtained by other authors. The older correlations give lower values for the economic pipe diameters, probably due to changes in the relative values of capital and energy. Note, however, that the refining rule of thumb given at the

start of this section gives optimum pipe diameters roughly twice those predicted from Equations 20.30 and 20.31, perhaps reflecting more conservative pipe sizing in that industry.

Example 20.10

Estimate the optimum pipe diameter for a water flow rate of 10 kg/s, at 20 °C. Carbon steel pipe will be used. Density of water 1000 kg/m³.

Solution

$$d_{i, \text{optimum}} = 0.664 \times (10)^{0.51} 1000^{-0.36} = \underline{177 \text{ mm}} \quad (20.30)$$

177 mm is 6.97 inches, which is not a standard pipe size. We can choose either 6-inch or 8-inch pipe, so try 6-inch pipe, sch 40, inside diameter 6.065 inches (154 mm).

Viscosity of water at 20 °C = 1.1×10^{-3} Ns/m²,

$$\text{Re} = \frac{4m}{\pi \mu d} = \frac{4 \times 10}{\pi \times 1.1 \times 10^{-3} \times 154 \times 10^{-3}} = 7.51 \times 10^4$$

>4000, so flow is turbulent.

Comparison of methods:

	Economic diameter
Equation 20.30	180 mm
Peters and Timmerhaus (1991)	4 in. (100 mm)
Nolte (1978)	80 mm

Example 20.11

Estimate the optimum pipe diameter for a flow of HCl of 7000 kg/h at 5 bar, 15 °C, stainless steel pipe. Molar volume 22.4 m³/kmol, at 1 bar, 0 °C.

Solution

Molecular weight HCl = 36.5.

$$\text{Density at operating conditions} = \frac{36.5}{22.4} \times \frac{5}{1} \times \frac{273}{288} = \underline{7.72 \text{ kg/m}^3}$$

$$\text{Optimum diameter} = 0.465 \left(\frac{7000}{3600} \right)^{0.43} \times 7.72^{-0.31} = \underline{328.4 \text{ mm}} \quad (20.31)$$

which is 12.9 inches, so we can use 14-inch pipe, sch 40, with inside diameter 13.124 inch (333 mm).

Viscosity of HCl 0.013 m Ns/m²

$$\text{Re} = \frac{4}{\pi} \times \frac{7000}{3600} \times \frac{1}{0.013 \times 10^{-3} \times 333 \times 10^{-3}} = \underline{5.71 \times 10^5}, \text{ turbulent}$$

Comparison of methods:

	Economic diameter
Equation 20.31	14 in. (333 mm)
Peters and Timmerhaus (1991)	9 in. (220 mm) carbon steel
Nolte (1978)	7 in. (180 mm) carbon steel

Example 20.12

Calculate the line size and specify the pump required for the line shown in Fig. 20.21: material ortho-dichlorobenzene (ODCB), flow rate 10,000 kg/h, temperature 20 °C, pipe material carbon steel.

Solution

ODCB density at 20 °C = 1306 kg/m³.

Viscosity: 0.9 mNs/m² (0.9 cp).

Estimation of pipe diameter required

typical velocity for liquid 1 m/s

$$\text{mass flow} = \frac{10^4}{3600} = 2.78 \text{ kg/s}$$

$$\text{volumetric flow} = \frac{2.78}{1306} = 2.13 \times 10^{-3} \text{ m}^3/\text{s}$$

$$\text{area of pipe} = \frac{\text{volumetric flow}}{\text{velocity}} = \frac{2.13 \times 10^{-3}}{1} = 2.13 \times 10^{-3} \text{ m}^2$$

$$\text{diameter of pipe} = \sqrt{\left(2.13 \times 10^{-3} \times \frac{4}{\pi}\right)} = 0.052 \text{ m} = 52 \text{ mm}$$

Or use the economic pipe diameter formula:

$$\begin{aligned} d, \text{ optimum} &= 0.664 \times 2.78^{0.51} \times 1306^{-0.36} \\ &= 78.7 \text{ mm} \end{aligned} \quad (20.30)$$

Take diameter as 77.9 mm (3-in., sch 40 pipe):

$$\text{cross-sectional area} = \frac{\pi}{4} (77.9 \times 10^{-3})^2 = 4.77 \times 10^{-3} \text{ m}^2$$

Pressure drop calculation

$$\text{fluid velocity} = \frac{2.13 \times 10^{-3}}{4.77 \times 10^{-3}} = 0.45 \text{ m/s}$$

Friction loss per unit length, Δf_1 :

$$\text{Re} = \frac{1306 \times 0.45 \times 77.9 \times 10^{-3}}{0.9 \times 10^{-3}} = 5.09 \times 10^4 \quad (20.2)$$

Absolute roughness commercial steel pipe, Table 20.3 = 0.46 mm

Relative roughness, $e/d = 0.046/80 = 0.0005$

Friction factor from Fig. 20.5, $f = 0.0025$

$$\Delta f_1 = 8 \times 0.0025 \times \frac{1}{77.9 \times 10^{-3}} \times 1306 \times \frac{0.45^2}{2} = 33.95 \text{ N/m}^2 \quad (20.1)$$

Design for a maximum flow rate of 20% above the average flow:

$$\text{Friction loss} = 0.0339 \times 1.2^2 = 0.0489 \text{ kPa/m}$$

Miscellaneous losses

Take as equivalent pipe diameters. All bends will be taken as 90-degree standard radius elbow.

Line to pump suction:

$$\text{length} = 1.5 \text{ m}$$

$$\text{bend, } 1 \times 30 \times 80 \times 10^{-3} = 2.4 \text{ m}$$

$$\text{valve, } 1 \times 18 \times 80 \times 10^{-3} = 1.4 \text{ m}$$

$$\text{total } 5.3 \text{ m}$$

$$\text{entry loss} = \frac{\rho u^2}{2} \text{ (see Section 20.4)}$$

$$\text{at maximum design velocity} = \frac{1306 (0.45 \times 1.2)^2}{2 \times 10^3} = 0.19 \text{ kPa}$$

Control valve pressure drop, allow normal 140 kPa

($\times 1.2^2$) maximum 200 kPa

Heat exchanger, allow normal 70 kPa

($\times 1.2^2$) maximum 100 kPa

Orifice, allow normal 15 kPa

($\times 1.2^2$) maximum 22 kPa

Line from pump discharge:

$$\text{length} = 4 \times 5.5 + 20 + 5 + 0.5 + 1 + 6.5 + 2 = 44.5 \text{ m}$$

$$\text{bends, } 6 \times 30 \times 80 \times 10^{-3} = 14.4 = 14.4 \text{ m}$$

$$\text{valves, } 3 \times 18 \times 80 \times 10^{-3} = 4.4 = 4.4 \text{ m}$$

$$63.4 \text{ m}$$

The line pressure-drop calculation is set out on the calculation sheet shown in Table 20.7. A blank version of this calculation sheet can be found online in Appendix G and is also available in Microsoft Excel format at www.elsevier.com/books-and-journals/book-companion/9780128211793.

Pump selection:

$$\text{flow rate} = 2.13 \times 10^{-3} \times 3600 = 7.7 \text{ m}^3/\text{h}$$

differential head, maximum, 38 m

select single-stage centrifugal (Fig. 20.3)

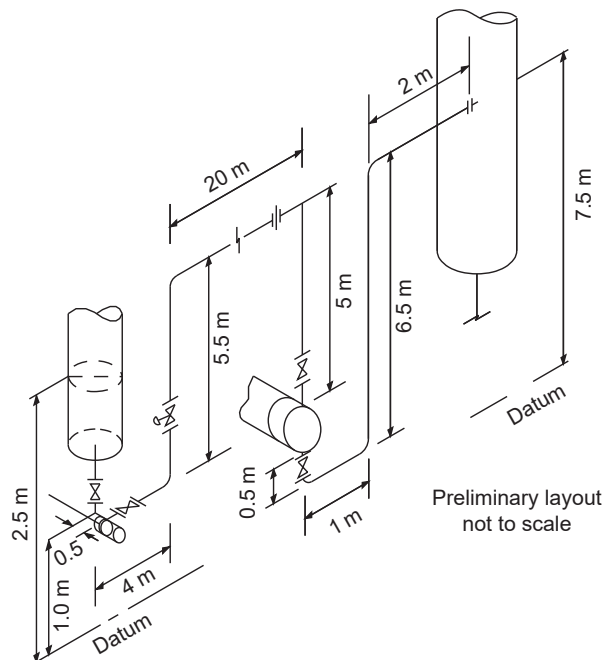


FIG. 20.21 Piping isometric drawing (Example 20.12).

TABLE 20.7 Line calculation form (Example 20.12)

Company Name Address Pump and Line Calculation Sheet				Project Name Project Number				Sheet 1 of 1																																																																																																																																																																																																											
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Equipment label P101 Plant section Process service Fluid ODCB Operating temperature Normal 20 °C Min 15 °C Max 30 °C				Equipment name Stripper bottoms pump Density 1306 kg/m ³ Viscosity 0.9 N.s/m ² Normal flow rate 2.78 kg/s Design flow rate 3.34 kg/s																																																																																																																																																																																																															
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u_2	Velocity	0.4	0.5	m/s																																																																																																																																																																																																															
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L_2	Line length	63.4	63.4	m																																																																																																																																																																																																															
$\Delta f_2 L_2$	Line loss	2.15	3.09	kPa																																																																																																																																																																																																															
	Orifice / Flow meter	15	22	kPa																																																																																																																																																																																																															
	Control valve	140	200	kPa																																																																																																																																																																																																															
	Equipment																																																																																																																																																																																																																		
S&THX	H 205	70	100	kPa																																																																																																																																																																																																															
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Total	(6) Dynamic loss	227	325	kPa																																																																																																																																																																																																															
z_2	Static head	6.5	6.5	m																																																																																																																																																																																																															
$\rho g z_2$		83.3	83.3	kPa																																																																																																																																																																																																															
	Equip. press (max)	200	200	kPa																																																																																																																																																																																																															
	Contingency	0	0	kPa																																																																																																																																																																																																															
	(7) Sub-total	283.3	283.3	kPa																																																																																																																																																																																																															
(7) + (6)	Discharge pressure	510.4	608.4	kPa																																																																																																																																																																																																															
(3)	Suction pressure	118.9	111.8	kPa																																																																																																																																																																																																															
	(8) Differential pressure	391.5	496.6	kPa																																																																																																																																																																																																															
(8)/ ρg	Pump head	30.6	38.8	m																																																																																																																																																																																																															
	Control valve																																																																																																																																																																																																																		
Valve(6)	% Dyn. loss		62%																																																																																																																																																																																																																
PUMP DATA																																																																																																																																																																																																																			
Pump manufacturer Catalog No. Pump flow rate normal 7.7 m ³ /h max. 9.2 m ³ /h Differential pressure 391.5 kPa 30.6 m 39.9 m water m NPSH required Pump type No. of stages single Impeller type closed Mounting horizontal				Driver type Electric 3-phase Power supply 440 V Seal type Mechanical, external flush Hydraulic power 0.833 kW Rated power kW Efficiency % Suction specific speed Casing design pressure 610 kPa Casing design temperature 30 °C Casing type Casing material																																																																																																																																																																																																															
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1. Process data completed, remaining information to be filled in after equipment selection 2. 3. 4. 5.																																																																																																																																																																																																																			

20.11 Sizing of control valves

The pressure drop required across a control valve will be a function of the valve, pump, and piping design and will set the size of valve that is selected. Sufficient pressure drop must be allowed for when sizing the pump, valve, and piping system to ensure that the control valve operates satisfactorily over the full range of flow required. If possible, the control valve and pump should be sized together, as a unit, to ensure that the optimum size is selected for both.

All control valves operate by manipulating a resistance to flow and hence controlling a flow rate, regardless of the function of the control loop. The signal from the valve actuator causes a variation in the size of a restriction in the fluid flow path, leading to an increase or decrease in fluid flow.

Most automatic control valves are globe valves, although butterfly valves are sometimes used for gases and vapors, and diaphragm valves and ball valves are occasionally used. Illustrations of the different types of valves are given in [Section 20.5](#). A plug-disk-type globe valve has several contributions to fluid pressure drop. As the fluid enters the valve, it makes a sharp 90-degree turn, enters an area of expanded cross-section below the valve seat, passes through the variable annular space between the disk and the seat of the valve into the space above the disk, and then makes another sharp 90-degree turn to leave the valve. As the valve opens, the annular space increases and the pressure drop across the valve falls, but even in the fully open position there is still a substantial pressure drop across the valve. The pressure drop across the valve varies between about 9 velocity heads in the fully open position and about 112 velocity heads when one-quarter open, depending on the valve design (see [Table 20.4](#)). The system pressure drop curves for both the fully open and 25% open cases (or minimum percent open recommended by the valve manufacturer) can be plotted against the pump curve to determine the controllable range of flow rate for the proposed pump and piping system, as described in [Section 20.7.5](#).

As a rough guide, if the characteristics are not specified, the control valve pressure drop should be taken as at least 30% of the total dynamic pressure drop through the system, with a minimum value of 100 kPa (14 psi). A good rule of thumb in the early stages of process design is to allow 140 kPa (20 psi) pressure drop for each control valve if the pressure drop in the process equipment is not yet known. Some of the pressure drop across the valve will be recovered downstream, with the amount depending on the type of valve used.

Detailed methods for control valve sizing are given in the ISA S75.01 standard and have been implemented in computer design tools that are used by instrument engineers and valve manufacturers. The detailed methods can account for choked flow, compressible flow, laminar flow, flashing liquids, and cavitation.

For nonchoked, turbulent flow of an incompressible fluid, ISA 75.01.01-2007 gives the following equation for valve sizing:

$$C_v = \frac{Q}{N} \sqrt{\frac{\rho_1/\rho_0}{\Delta P}} \quad (20.32)$$

where C_v = valve flow coefficient

Q = volumetric flow

ρ_1 = fluid density at flow conditions

ρ_0 = density of water at 15 °C, i.e., ρ_1/ρ_0 = fluid specific gravity

N = a constant, depending on the units used (see later)

ΔP = pressure drop = $P_1 - P_2$, where:

P_1 = pressure measured two nominal pipe diameters upstream of the valve

P_2 = pressure measured six nominal pipe diameters downstream of the valve

The value of the constant N depends on the units used. If the volumetric flow rate is given in gallons per minute and the pressure drop in pounds per square inch, then $N = 1$, so the valve coefficient can be interpreted as the flow rate of water in gpm that will flow through the valve with 1 psi pressure drop at 15 °C (59 °F). If the flow rate is given in m³/h and the pressure drop in bar, then $N = 0.865$.

Different formulae are given for C_v for compressible fluids, choked flow, laminar flow, and other conditions in the ISA S75.01 standard.

Control valves should always be sized for a flow rate larger than the normal flow condition. Different companies use different methods for determining how much design margin to incorporate in control valve sizing. A simple method is to calculate C_v based on the normal flow rate and then select a valve that has twice this value of C_v so that in normal operation the valve will be roughly 50% open. Another popular method is to calculate C_v based

on a maximum flow rate 30% above the normal stream flow rate, then divide the result by 0.7 and pick the valve with the next largest C_v . This gives a valve with C_v equal to 1.85 times the value calculated for normal flow, which in normal operation will be about 54% open.

The ratio between the maximum and minimum controllable flow rates is known as the *rangeability* of the valve. The installed rangeability is influenced by the actuator and positioner, as well as the valve design. The ratio of the normal maximum flow to the minimum controllable flow is called the *turndown* of the valve. For most control valves, the turndown is about 70% of the rangeability. Rangeability typically varies between about 20 and 50. The designer should consider the desired turndown and rangeability when selecting valves, particularly in level control applications where the process may call for a high range of operable flow rates. When high rangeability is required, two control valves of different sizes with slightly overlapping operating ranges can be used in parallel. This is known as a *split-range* arrangement. Control valve characteristics and rangeability are discussed in more detail in the ISA S75.11 standard.

Example 20.13

The feed media to a fermentation reactor consists of glucose solution in water with added nutrients and has specific gravity 1.03 at 40 °C. Determine the size of control valve to use for a normal design flow rate of 0.2 m³/h if the valve is sized with a pressure drop of 1.25 bar.

Solution

For metric units, $N = 0.865$ in Equation 20.32, so for normal flow conditions:

$$C_v = \frac{0.2}{0.865} \sqrt{\frac{1.03}{1.25}} = 0.210 \quad (20.32)$$

Select a valve that has $C_v \approx 0.42$ so that valve normally operates about 50% open.

20.12 References

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- ASME B16.34-2017. Valves, flanged, threaded and welding end.
- ASME B31.1-2018. Power piping.
- ASME B31.2-1968 (withdrawn). Fuel gas piping.
- ASME B31.3-2018. Process piping.
- ASME B31.4-2019. *Pipeline transportation systems for liquid hydrocarbons and other liquids*.
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International standards

- ISO 5209-2019. General purpose industrial valves – marking (2nd ed.).

20.13 Nomenclature

		Dimensions in MLT\$
A	Plant attainment (hours operated per year)	—
B	Purchased cost factor, pipes	$\$L^{-1}$
a	Capital charges factor, piping	T^{-1}
b	Maintenance cost factor, piping	T^{-1}
C	Sum of mechanical, corrosion and erosion allowances	L
C_f	Fanning friction factor	—
C_C	Annual capital charge, piping	$\$L^{-1}T^{-1}$
C_p	Specific heat at constant pressure	$L^2T^{-2}\theta^{-1}$
C_t	Total annual cost, piping	$\$L^{-1}T^{-1}$
C_v	Specific heat at constant volume	$L^2T^{-2}\theta^{-1}$
C_v	Control valve sizing parameter (Equation 20.32)	—
C_w	Annual pumping cost, piping	$\$L^{-1}T^{-1}$
d	Pipe diameter	L
d_i	Pipe inside diameter	L
E	Pipe casting quality factor	—
E_e	Efficiency, electric motors	—
E_p	Polytropic efficiency, compressors, and turbines	—
e	Relative roughness	—
F	Installed cost factor, piping	—
f	Friction factor	—
g	Gravitational acceleration	LT^{-2}
H	Enthalpy (Equation 20.8)	ML^2T^{-2}
H	Height of liquid above the pump suction	L
h	Pump head	L
K	Number of velocity heads	—
L	Pipe length	L
M_w	Molecular mass (weight)	—
m	Polytropic temperature exponent	—
$\dot{m}$	Mass flow rate	MT^{-1}
N	Pump speed, revolutions per unit time	T^{-1}
N	Control valve constant (Equation 20.32)	—
N_s	Pump specific speed	—
n	Index relating pipe cost to diameter	—
P	Pressure	$ML^{-1}T^{-2}$
P_c	Critical pressure	$ML^{-1}T^{-2}$
P_F	Pressure loss in suction piping	$ML^{-1}T^{-2}$
P_i	Interstage pressure	$ML^{-1}T^{-2}$

(continued)

		Dimensions in MLT\$
P_r	Reduced pressure	—
P_s	Safe working pressure	$ML^{-1}T^{-2}$
P_v	Vapor pressure of liquid	$ML^{-1}T^{-2}$
P_1	Initial pressure	$ML^{-1}T^{-2}$
P_2	Final pressure	$ML^{-1}T^{-2}$
ΔP	Difference in system pressures ($P_1 - P_2$)	$ML^{-1}T^{-2}$
ΔP_f	Pressure drop due to friction	$ML^{-1}T^{-2}$
p	Cost of power, pumping	$\$M^{-1}L^{-2}T^2$
Q	Volumetric flow rate	L^3T^{-1}
R	Shear stress on surface, pipes	$ML^{-1}T^{-2}$
R	Universal gas constant	$L^2T^{-2}\theta^{-1}$
S	Basic allowable stress for pipe material	$ML^{-1}T^{-2}$
T	Temperature, absolute	θ
T_c	Critical temperature	θ
T_r	Reduced temperature	—
T_1	Initial temperature	θ
T_2	Final temperature	θ
t	Pipe wall thickness	L
t_m	Minimum required thickness	L
t_p	Pressure design thickness	L
u	Fluid velocity	LT^{-1}
v	Volume per unit mass	$M^{-1}L^3$
W	Work done	L^2T^{-2}
Z	Compressibility factor	—
z	Height above datum	L
Δz	Difference in elevation ($z_1 - z_2$)	L
γ	Ratio of specific heats	—
γ_T	Temperature coefficient	—
η	Pump efficiency	—
η_t	Turbine efficiency	—
ρ	Fluid density	ML^{-3}
ρ_0	Density of water at 15 °C	ML^{-3}
ρ_1	Fluid density of flowing fluid at process conditions	ML^{-3}
μ	Viscosity of fluid	$ML^{-1}T^{-1}$
σ_d	Design stress	$ML^{-1}T^{-2}$
σ_s	Safe working stress	$ML^{-1}T^{-2}$
Re	Reynolds number	—
$NPSH_{avail}$	Net positive suction head available at the pump suction	L
$NPSH_{reqd}$	Net positive suction head required at the pump suction	L

20.14 Problems

20.1. Select suitable valve types for the following applications:

1. Isolating a heat exchanger.
2. Manual control of the water flow into a tank used for making up batches of sodium hydroxide solution.
3. The valves need to isolate a pump and provide emergency manual control on a bypass loop.
4. Isolation valves in the line from a vacuum column to the steam ejectors producing the vacuum.
5. Valves in a line where cleanliness and hygiene are an essential requirement.

State the criterion used in the selection for each application.

20.2 A supply of nitrogen is required as an inert gas for blanketing and purging vessels. After generation, the nitrogen is compressed and stored in a bank of cylinders at a pressure of 5 barg. The inlet pressure to the compressor is 0.5 barg, and the temperature is 20 °C. Calculate the maximum power required to compress 100 m³/h. A single-stage reciprocating compressor will be used.

20.3. Hydrogen chloride gas, produced by burning chlorine with hydrogen, is required at a supply pressure of 600 kN/m² gauge. The pressure can be achieved by either operating the burner under pressure or by compressing the hydrogen chloride gas. For a production rate of hydrogen chloride of 10,000 kg/h, compare the power requirement of compressing the hydrogen supply to the burner with that to compress the product hydrogen chloride. The chlorine feed is supplied at the required pressure from a vaporizer. Both the hydrogen and chlorine feeds are essentially pure. Hydrogen is supplied to the burner in 1% excess of the stoichiometric requirement. A two-stage centrifugal compressor will be used for both duties. Take the polytropic efficiency for both compressors as 70%. The hydrogen supply pressure is 120 kN/m² and the temperature 25 °C. The hydrogen chloride is cooled to 50 °C after leaving the burner. Assume that the compressor intercooler cools the gas to 50 °C for both duties.

Which process would you select and why?

20.4. Estimate the work required to compress ethylene from 32 MPa to 250 MPa in a two-stage reciprocating compressor where the gas is initially at 30 °C and leaves the intercooler at 30 °C.

20.5. Crude dichlorobenzene is pumped from a storage tank to a distillation column.

The tank is blanketed with nitrogen, and the pressure above the liquid surface is held constant at 0.1 bar gauge pressure. The minimum depth of liquid in the tank is 1 m.

The distillation column operates at a pressure of 500 mmHg (500 mm of mercury, absolute). The feed point to the column is 12 m above the base of the tank. The tank and column are connected by a 50 mm internal diameter commercial steel pipe, 200 m long. The pipe run from the tank to the column contains the following valves and fittings: 20 standard radius 90-degree elbows; two gate valves to isolate the pump (operated fully open); an orifice plate and a flow-control valve.

If the maximum flow rate required is 20,000 kg/h, calculate the pump motor rating (power) needed. Take the pump efficiency as 70% and allow for a pressure drop of 0.5 bar across the control valve and a loss of 10 velocity heads across the orifice. Density of dichlorobenzene 1300 kg/m³, viscosity 1.4 cp.

20.6. A liquid is contained in a reactor vessel at 115 bar absolute pressure. It is transferred to a storage vessel through a 50 mm internal diameter commercial steel pipe. The storage vessel is nitrogen blanketed, and pressure above the liquid surface is kept constant at 1500 N/m² gauge. The total run of pipe between the two vessels is 200 m. The miscellaneous losses due to entry and exit losses, fittings, valves, etc., amount to 800 equivalent pipe diameters. The liquid level in the storage vessel is at an elevation 20 m *below* the level in the reactor.

A turbine is fitted in the pipeline to recover the excess energy that is available over that required to transfer the liquid from one vessel to the other. Estimate the power that can be taken from the turbine when the liquid transfer rate is 5000 kg/h. Take the efficiency of the turbine as 70%.

The properties of the fluid are density 895 kg/m³, viscosity 0.76 mNm⁻²s.

20.7. A process fluid is pumped from the bottom of one distillation column to another using a centrifugal pump. The line is standard commercial steel pipe 75 mm internal diameter. From the column to the pump inlet,

the line is 25 m long and contains six standard elbows and a fully open gate valve. From the pump outlet to the second column, the line is 250 m long and contains 10 standard elbows, four gate valves (operated fully open), and a flow-control valve. The fluid level in the first column is 4 m above the pump inlet. The feed point of the second column is 6 m above the pump inlet. The operating pressure in the first column is 1.05 bara and that of the second column 0.3 barg.

Determine the operating point on the pump characteristic curve when the flow is such that the pressure drop across the control valve is 35 kN/m^2 . The physical properties of the fluid are density 875 kg/m^3 , viscosity $1.46 \text{ mN m}^{-2}\text{s}$.

Also, determine the NPSH at this flow rate if the vapor pressure of the fluid at the pump suction is 25 kN/m^2 . Pump characteristics:

Flow rate, m^3/h	0.0	18.2	27.3	36.3	45.4	54.5	63.6
Head, m of liquid	32.0	31.4	30.8	29.0	26.5	23.2	18.3

- 20.8.** Revisiting Example 20.8, suppose the flow was controlled using a plug-disk globe valve and the initial design in the example assumed that the valve is fully open. What range of flow rates can be achieved if the valve can be throttled down to quarter open? When the valve is quarter open, what fraction of the pump work is lost across the valve?
- 20.9.** Estimate the shaft work required to pump 65 gal/min of sugar solution in water (specific gravity = 1.05) if the pump inlet pressure is 25 psig and the outlet pressure required is 155 psig.
- 20.10** A shell and tube cooler in an aromatics complex cools 26,200 lb/h of naphtha (specific gravity 0.78, viscosity 0.007 cP). The cooler has 347 tubes, 16 ft long, $\frac{3}{4}$ -inch diameter. If the naphtha is on the tube side, estimate the tube side pressure drop.
- 20.11** In a detergent-making process, 1400 gal/h of water flows through a 2-inch pipe system as follows: Exit from pump, 2 ft vertical, open gate valve, 14 ft vertical, 90-degree bend, 12 ft horizontal, quarter open globe valve, 20 ft horizontal, 90-degree bend, 6 ft horizontal, 90-degree bend, 12 ft vertical, 90-degree bend, 14 ft horizontal, 90-degree bend, 4 ft vertical, 90-degree bend, 28 ft horizontal, open gate valve, 3 ft horizontal, entry to tank containing 30 ft of liquid.
- If the pump and tank are both at grade level, estimate the head that the pump must deliver.
 - If the pump inlet pressure is 25 psig, what is the outlet pressure?
 - Estimate the pump shaft work.
 - If the pump is powered by an electric motor with 85% efficiency, what is the annual electricity consumption?
- 20.12** What size control valve would you select to regulate a flow of $4 \text{ m}^3/\text{h}$ of liquid of specific gravity 0.7 if the desired valve pressure drop is 140 kPa?

Appendices

APPENDIX A: GRAPHICAL SYMBOLS FOR PIPING SYSTEMS AND PLANT

Available at <http://www.elsevier.com/books-and-journals/book-companion/9780128211793>.

APPENDIX B: CORROSION CHARTS

Available at <http://www.elsevier.com/books-and-journals/book-companion/9780128211793>.

APPENDIX C: PHYSICAL PROPERTY DATA BANK

Available at <http://www.elsevier.com/books-and-journals/book-companion/9780128211793>.

APPENDIX D: CONVERSION FACTORS

An asterisk (*) denotes an exact relationship			
Length	*1 in.	:	25.4 mm
	*1 ft	:	0.3048 m
	*1 yd	:	0.9144 m
	1 mile	:	1.6093 km
	*1 Å (angstrom)	:	10^{-10} m
Time	*1 min	:	60 s
	*1 h	:	3.6 ks
	*1 day	:	86.4 ks
	1 year	:	31.5 Ms
Area	*1 in. ²	:	645.16 mm ²
	1 ft ²	:	0.092903 m ²
	1 yd ²	:	0.83613 m ²
	1 acre	:	4046.9 m ²
	1 mile ²	:	2.590 km ²
Volume	1 in. ³	:	16.387 cm ³
	1 ft ³	:	0.02832 m ³
	1 yd ³	:	0.76453 m ³

(Continued)

An asterisk (*) denotes an exact relationship—Cont'd			
Mass	1 UK gal	:	4546.1 cm ³
	1 US gal	:	3785.4 cm ³
	1 short (US) ton	:	907.18 kg
	1 oz	:	28.352 g
	*1 lb	:	0.45359237 kg
	1 cwt	:	50.8023 kg
Force	1 long (UK) ton	:	1016.06 kg
	1 short (US) ton	:	907.18 kg
	1 pdl	:	0.13826 N
	1 lbf	:	4.4482 N
	1 kgf	:	9.8067 N
	1 tonf	:	9.9640 kN
Temperature difference	*1 dyn	:	10 ⁻⁵ N
	*1 deg F (deg R)	:	5/9 deg C (deg K)
Energy (work, heat)	1 ft lbf	:	1.3558 J
	1 ft pdl	:	0.04214 J
	*1 cal (internat. table)	:	4.1868 J
	1 erg	:	10 ⁻⁷ J
	1 Btu	:	1.05506 kJ
	1 hp h	:	2.6845 MJ
	*1 kW h	:	3.6 MJ
	1 therm	:	105.51 MJ
	1 thermie	:	4.1855 MJ
	1 Btu/ft ³	:	37.259 kJ/m ³
Velocity	1 ft/s	:	0.3048 m/s
	1 mile/h	:	0.44704 m/s
Volumetric flow	1 ft ³ /s	:	0.028316 m ³ /s
	1 ft ³ /h	:	7.8658 cm ³ /s
	1 UK gal/h	:	1.2628 cm ³ /s
	1 US gal/h	:	1.0515 cm ³ /s
Mass flow	1 lb/h	:	0.12600 g/s
	1 ton/h	:	0.28224 kg/s
Mass per unit area	1 lb/in. ²	:	703.07 kg/m ²
	1 lb/ft ²	:	4.8824 kg/m ²
Density	1 ton/sq mile	:	392.30 kg/km ²
	1 lb/in ³	:	27.680 g/cm ³
	1 lb/ft ³	:	16.019 kg/m ³
	1 lb/UK gal	:	99.776 kg/m ³
	1 lb/US gal	:	119.83 kg/m ³

(Continued)

An asterisk (*) denotes an exact relationship—Cont'd			
Pressure	1 lbf/in. ² (1 psi)	:	6.8948 kN/m ²
	1 ksi (1000 psi)	:	6.8948 MN/m ²
	1 tonf/in. ²	:	15.444 MN/m ²
	1 lbf/ft ²	:	47.880 N/m ²
	*1 standard atm	:	101.325 kN/m ²
	*1 atm (1 kgf/cm ²)	:	98.0665 kN/m ²
	*1 bar	:	10 ⁵ N/m ²
	1 ft water	:	2.9891 kN/m ²
	1 in. water	:	249.09 N/m ²
	1 in. Hg	:	3.3864 kN/m ²
	1 mmHg (1 torr)	:	133.32 N/m ²
Power (heat flow)	1 hp (British)	:	745.70 W
	1 hp (metric)	:	735.50 W
	1 erg/s	:	10 ⁻⁷ W
	1 ft lbf/s	:	1.3558 W
	1 Btu/h	:	0.29307 W
	1 ton of refrigeration	:	3516.9 W
Moment of inertia	1 lb ft ²	:	0.042140 kg m ²
Momentum	1 lb ft/s	:	0.13826 kg m/s
Angular momentum	1 lb ft ² /s	:	0.042140 kg m ² /s
Viscosity, dynamic	*1 P (Poise)	:	0.1 N s/m ²
	1 lb/ft h	:	0.41338 mN s/m ²
	1 lb/ft s	:	1.4882 N s/m ²
Viscosity, kinematic	*1 S (Stokes)	:	10 ⁻⁴ m ² /s
	1 ft ² /h	:	0.25806 cm ² /s
Surface energy	1 erg/cm ²	:	10 ⁻³ J/m ²
(surface tension)	(1 dyn/cm)	:	(10 ⁻³ N/m)
Mass flux density	1 lb/h ft ²	:	1.3562 g/s m ²
Heat flux density	1 Btu/h ft ²	:	3.1546 W/m ²
	*1 kcal/h m ²	:	1.163 W/m ²
Heat transfer coefficient	1 Btu/h ft ² F	:	5.6783 W/m ² K
Specific enthalpy (latent heat, etc.)	*1 Btu/lb	:	2.326 kJ/kg
Specific heat capacity	*1 Btu/lb °F	:	4.1868 kJ/kg K
Thermal	1 Btu/h ft °F	:	1.7307 W/m K
conductivity	1 kcal/h m °C	:	1.163 W/m K

Note: Where temperature difference is involved K = °C.

Taken from Mullin, J. W.: The Chemical Engineer No. 211 (Sept. 1967), 176. SI units in chemical engineering.

APPENDIX E: DESIGN PROJECTS (SHORTER PROBLEM STATEMENTS)

Available at <http://www.elsevier.com/books-and-journals/book-companion/9780128211793>.

Projects include: acetaminophen, acetic acid, acetophenone, acrolein, acrylic acid, alendronate, amlodipine besylate (NorvascTM), ammonia, argon recovery, aspartame, aspirin, benzene reduction, biopulping, black liquor recovery, bromine, carbon nanotubes, cellulose acetate, chemi-mechanical pulping, chlor-alkali, chlorine-free pulp bleaching, chloroform, ciprofloxacin, citalopram hydrobromide (CelexaTM), 3-R citronellol, Cleve's acid, clopidogrel (PlavixTM), cocoa processing, crude oil distillation, cyclohexanone with phenol, cyclosporin A, dextrans, dicalcium phosphate, dicyclopentadiene, 2,6-dimethyl naphthalene, doxycycline, erythorbic acid, ethanol by fermentation, ethylene by oxidative dehydrogenation, ethylene by steam cracking, ethylene from ethanol, fexofenadine (AllegraTM), Fischer-Tropsch catalyst, fluconazole, fluoxetine hydrochloride, fluticasone propionate (FloventTM), folic acid, fuel processor, gas to liquids by Fischer-Tropsch synthesis, granulocyte colony stimulating factor, guaifenesin (ActifedTM, RobitussinTM), hydrocracking, hydrogen by steam reforming of methane, hydrogen for fuel cells, ibuprofen (AdvilTM, MotrinTM), insect repellant, isomerization of naphtha, Kraft pulping, krypton recovery, lactic acid by fermentation, lansoprazole (PrevacidTM), linear alkylbenzenes, lisinopril, loratidine, low-fat snacks, d-malic acid, mannitol, margarine, methanol to olefins, methylene chloride, moisturizing lotion, monosodium glutamate, 2,6-naphthalene dicarboxylic acid, natural gas liquefaction, natural gas liquids recovery, niacinamide, nitric acid, nitrobenzene, s-ofloxacin, omepra-zole (PrilosecTM), oxygen (small scale portable/medical), paroxetine (PaxilTM), phenol, phosgene, phosphoric acid, polylactic acid, potassium permanganate, propylene by catalytic dehydrogenation, propylene by steam cracking, propylene glycol by fermentation, propylene oxide, pseudoephedrine, pyridine, riboflavin, risperidone (Risperdal), salicylic acid USP, setraline hydrochloride (ZolofTM), simvastatin (ZocorTM), sulfur dioxide treatment, sulfur recovery, sumatriptan (ImigranTM), tocopherol, toxic waste disposal, urea DSM process, venlafaxine (EffexorTM), xenon recovery & zeolite synthesis.

APPENDIX F: DESIGN PROJECTS (LONGER PROBLEM STATEMENTS)

Available at <http://www.elsevier.com/books-and-journals/book-companion/9780128211793>.

Projects include: 2-ethylhexanol, acrylonitrile, aniline, chlorine (from hydrogen chloride), chloro-benzenes, hydrogen from fuel oil, methyl ethyl ketone & urea.

APPENDIX G: EQUIPMENT SPECIFICATION (DATA) SHEETS

Available at <http://www.elsevier.com/books-and-journals/book-companion/9780128211793>.

APPENDIX H: TYPICAL SHELL AND TUBE HEAT EXCHANGER TUBE-SHEET LAYOUTS

Available at <http://www.elsevier.com/books-and-journals/book-companion/9780128211793>.

APPENDIX I: MATERIAL SAFETY DATA SHEET

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